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(54) **OFF-ROAD UTILITY VEHICLE**

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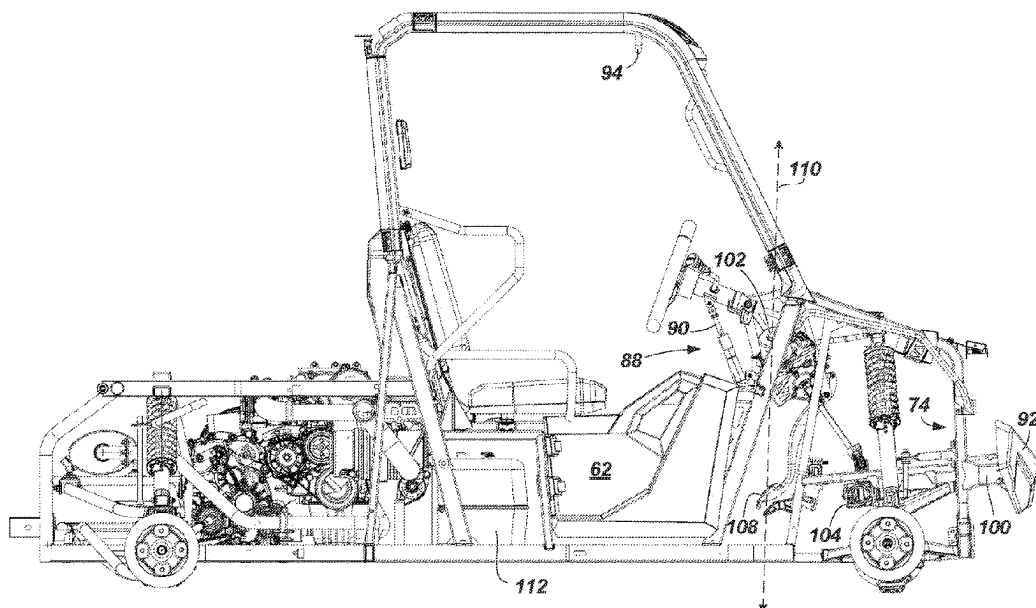
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(57)

ABSTRACT

An off-highway recreational vehicle having a longitudinal centerline includes a frame and a rear suspension. The rear suspension includes a left A-arm, wherein the left A-arm has a pivot axis, the pivot axis being on a right-hand side of the longitudinal centerline.

16 Claims, 134 Drawing Sheets



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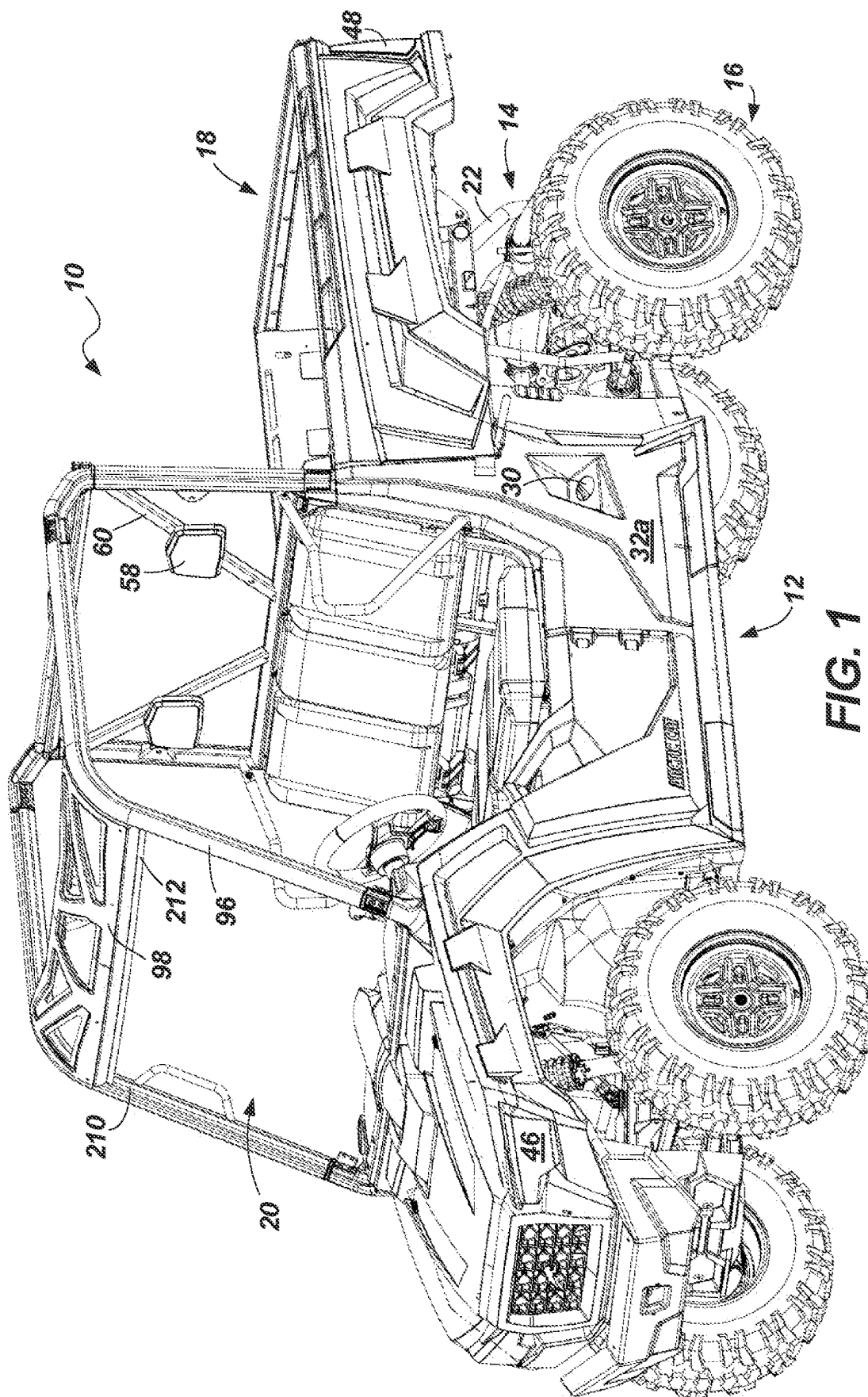
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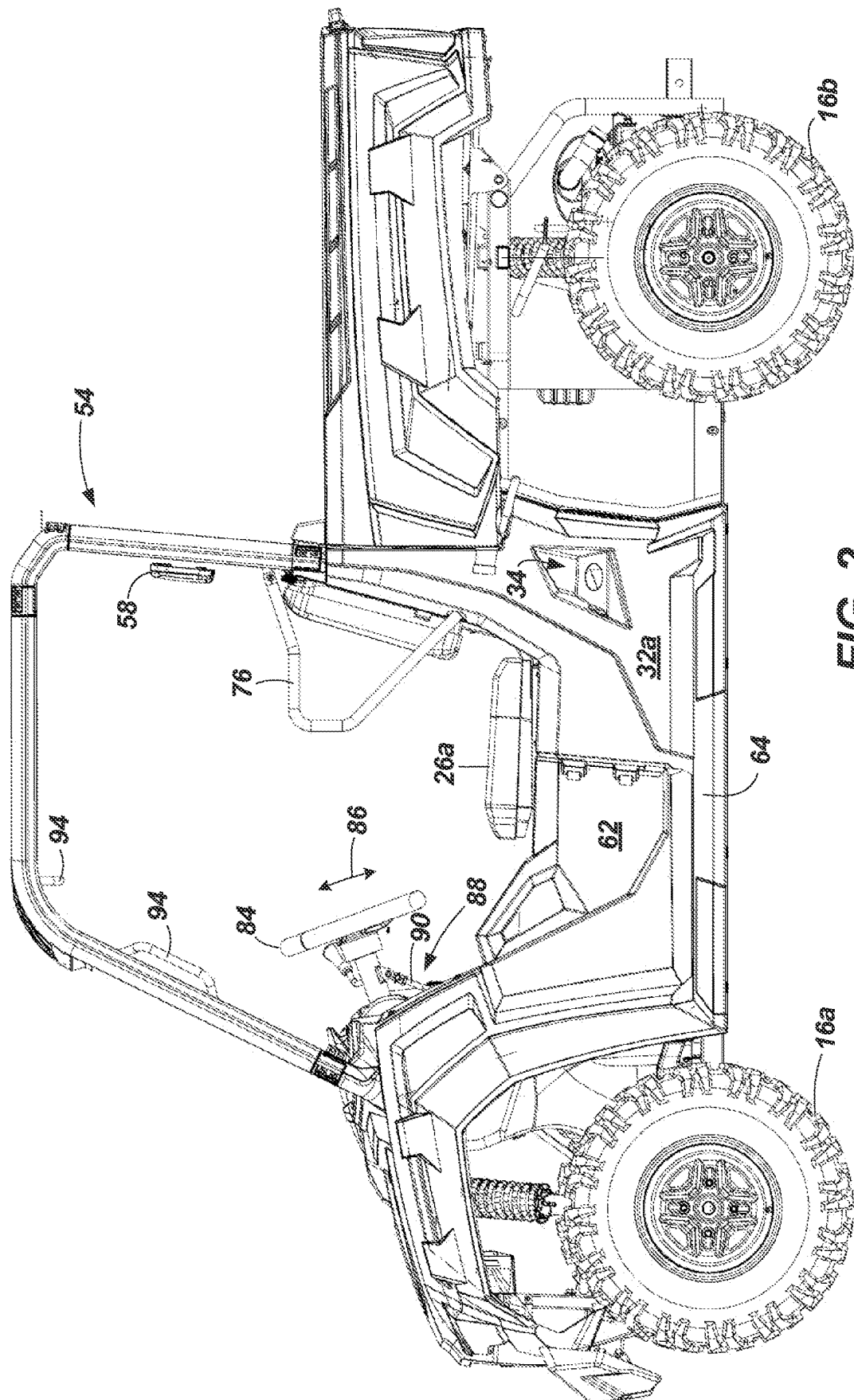
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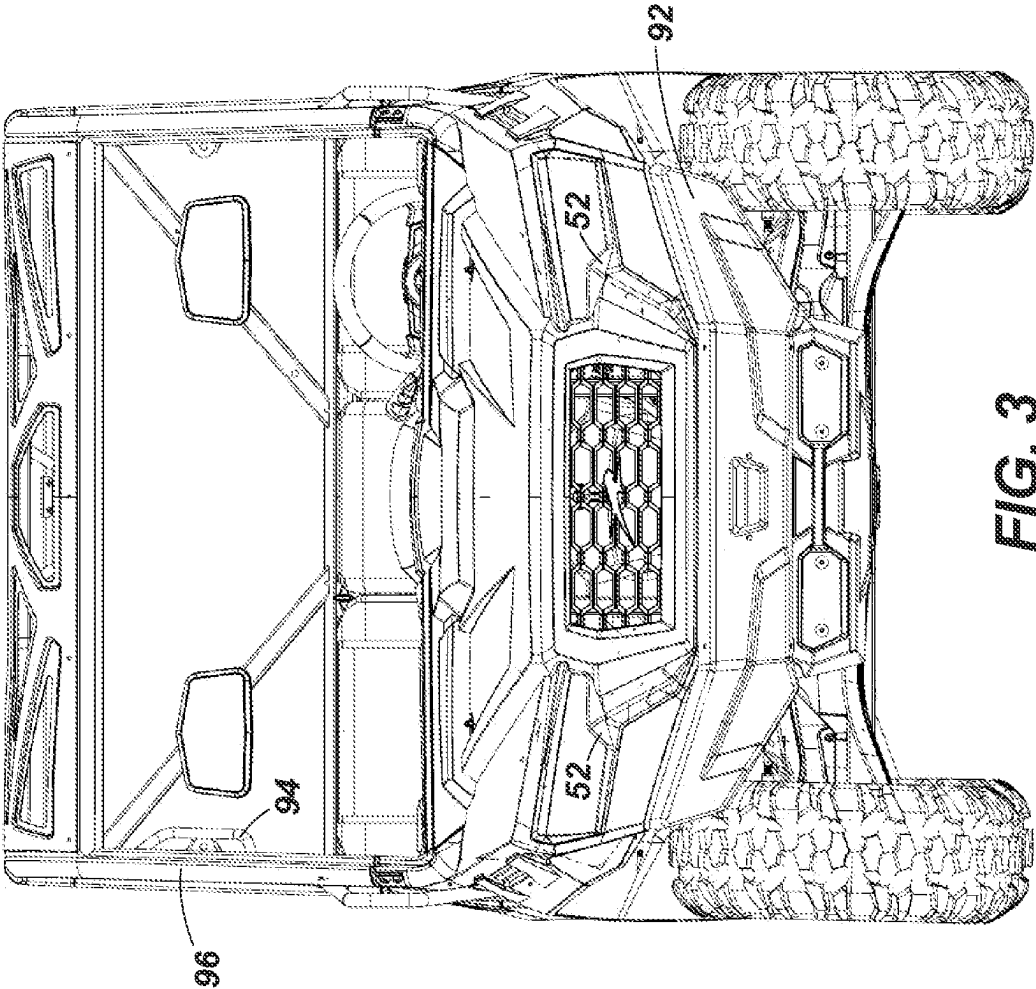
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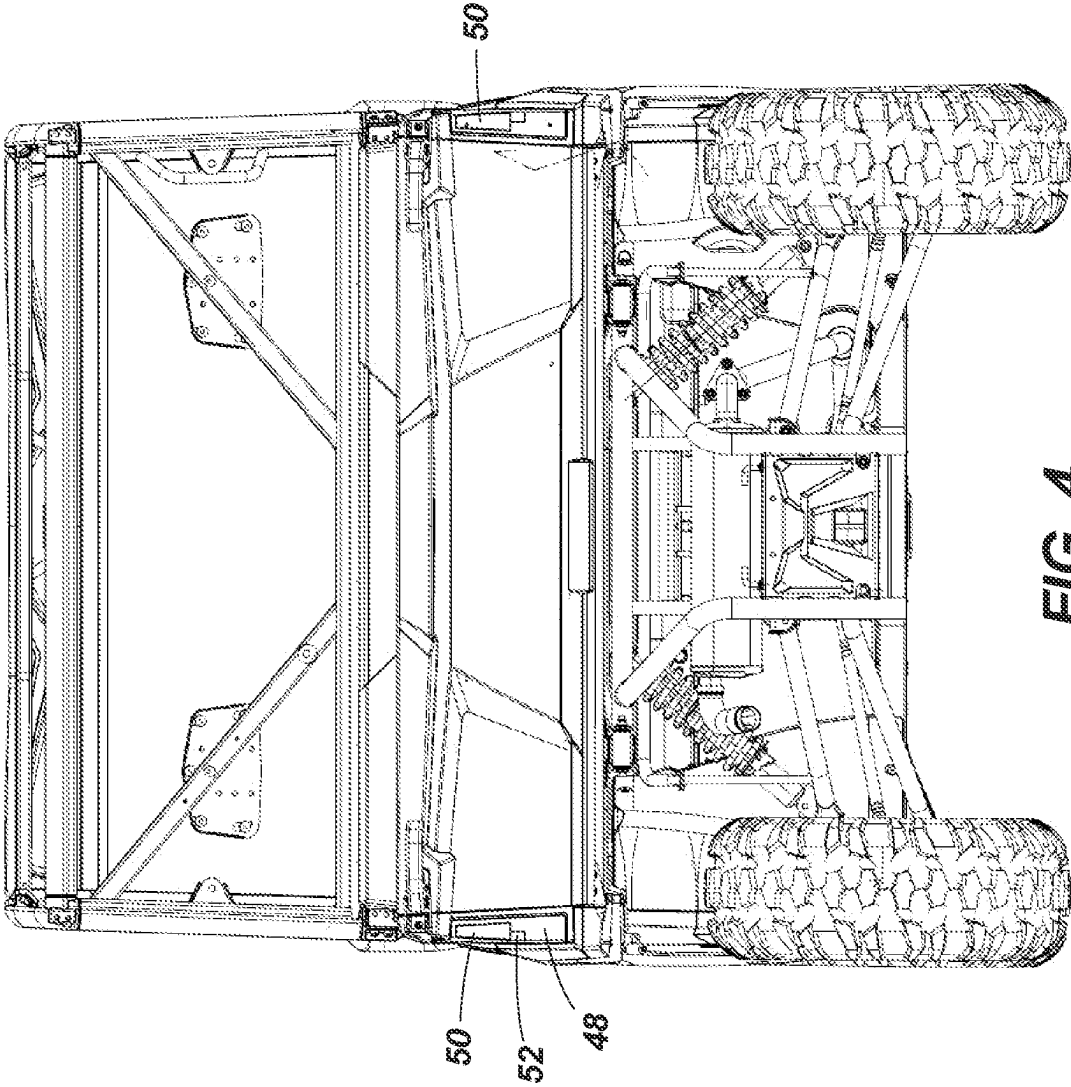
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**FIG. 2**





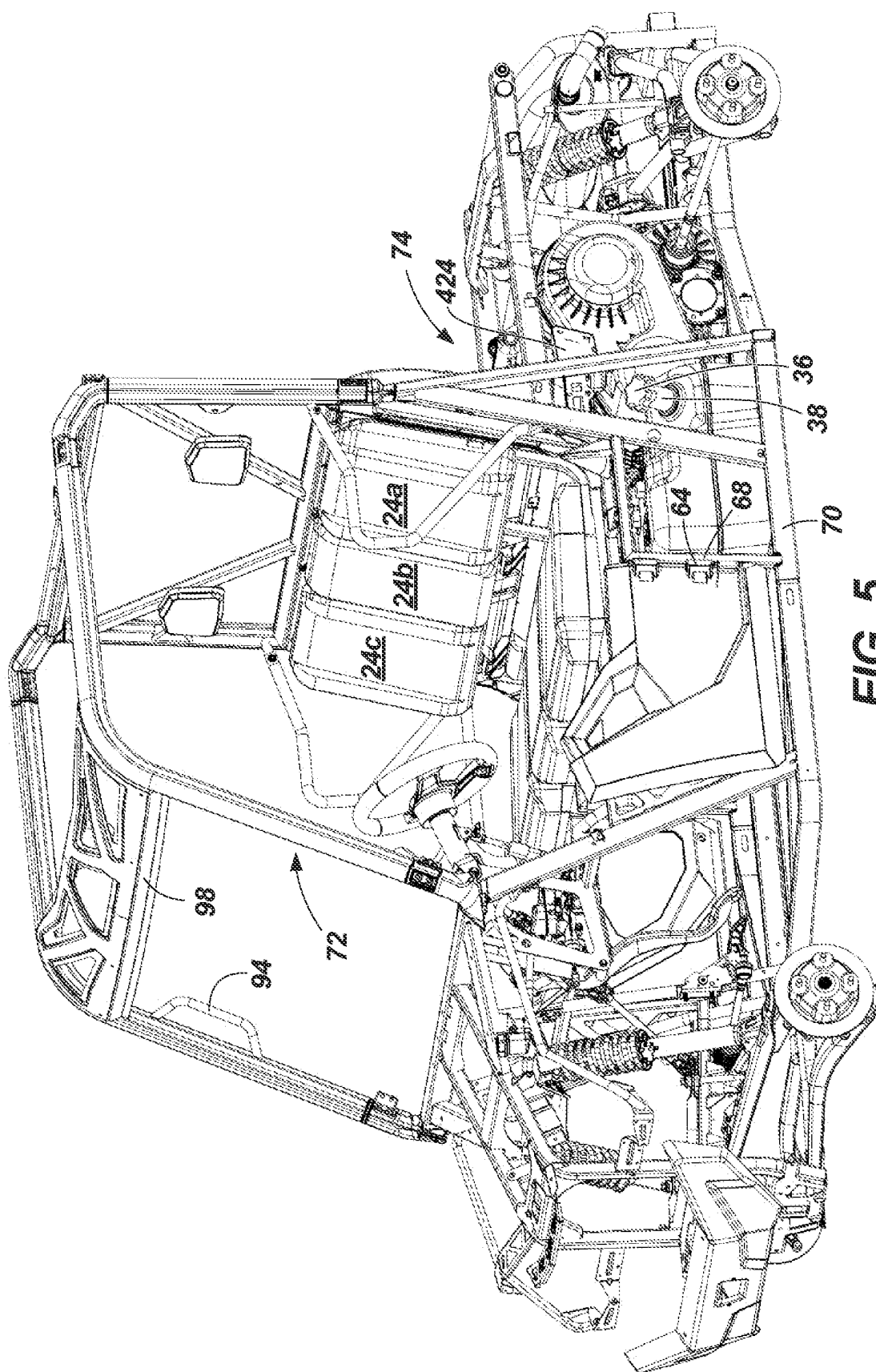
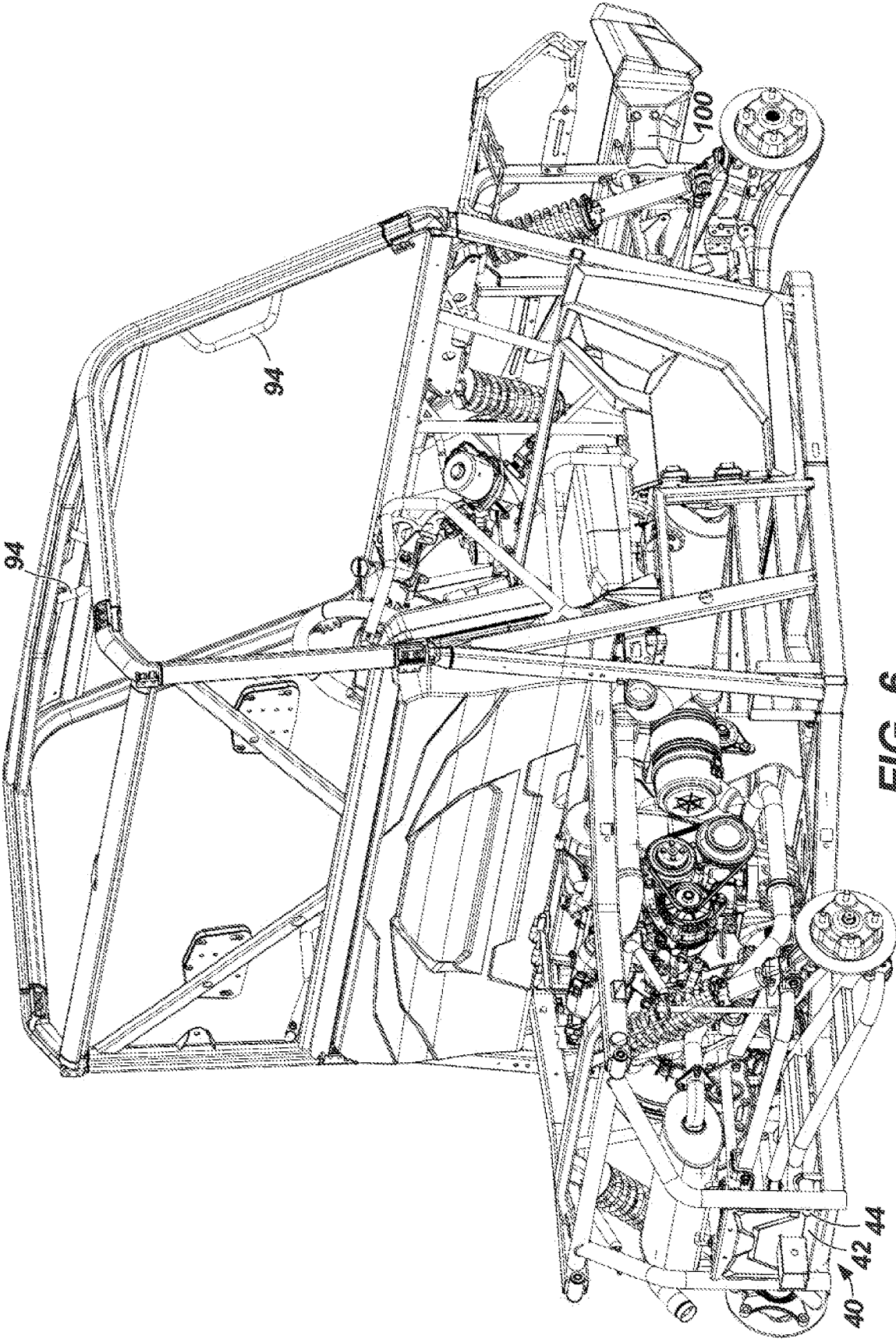
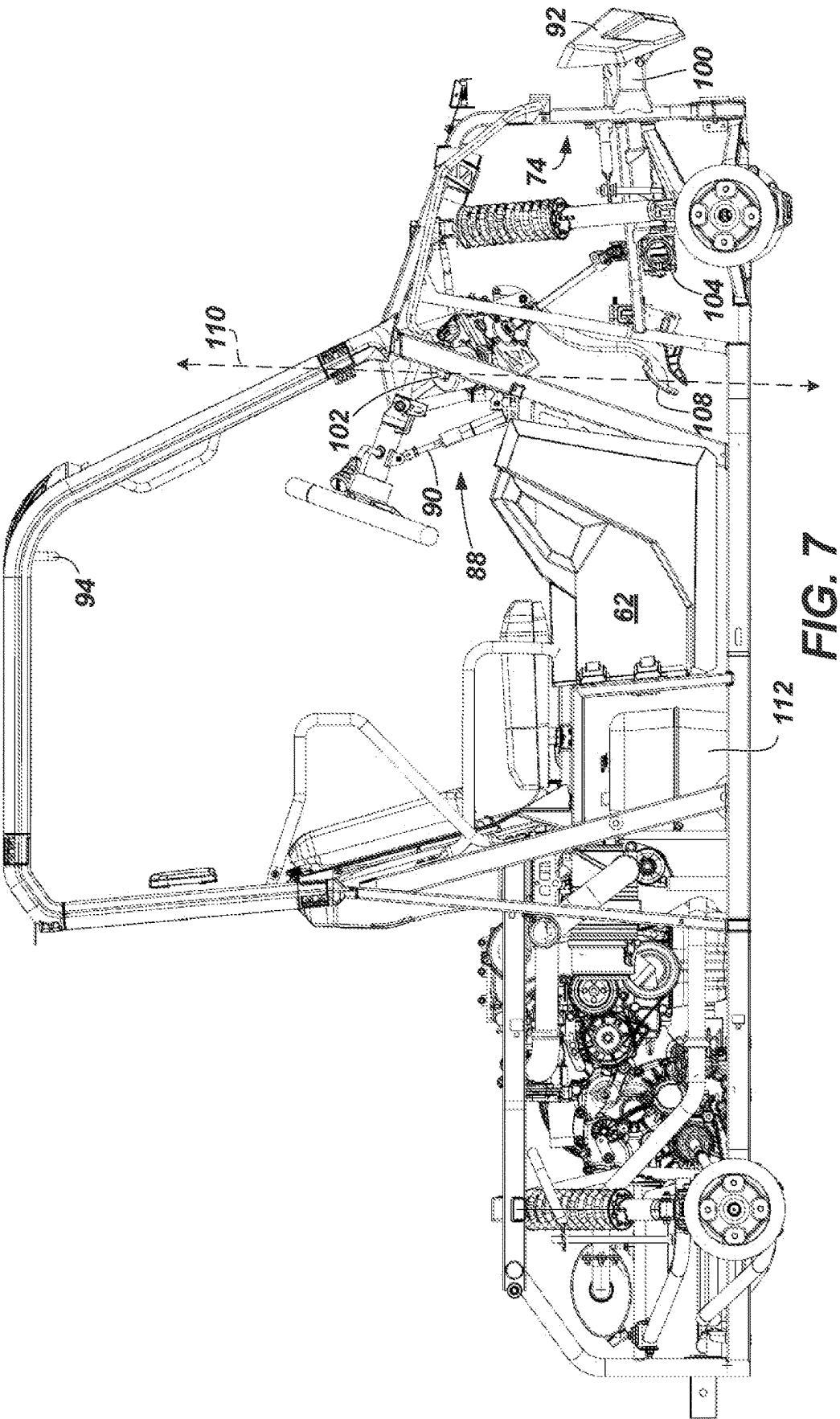
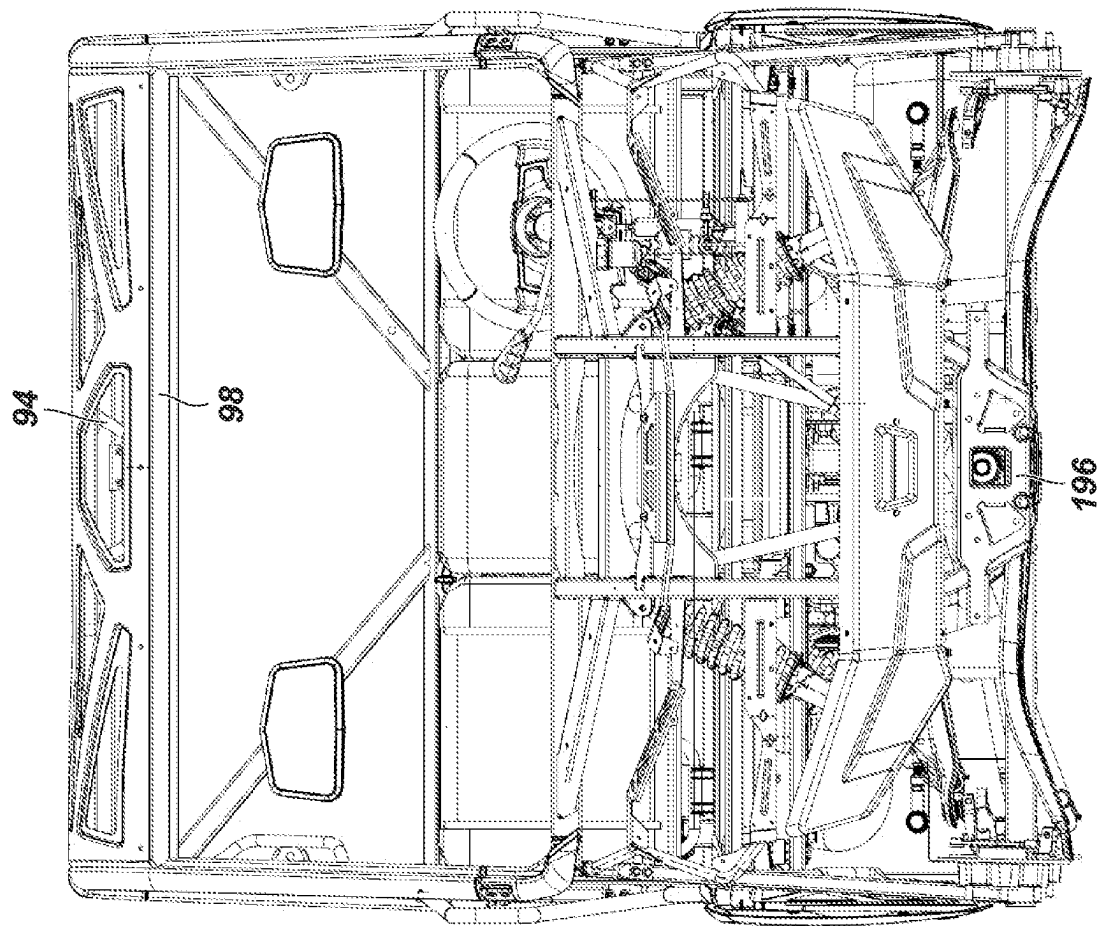


FIG. 5







F/G. 8

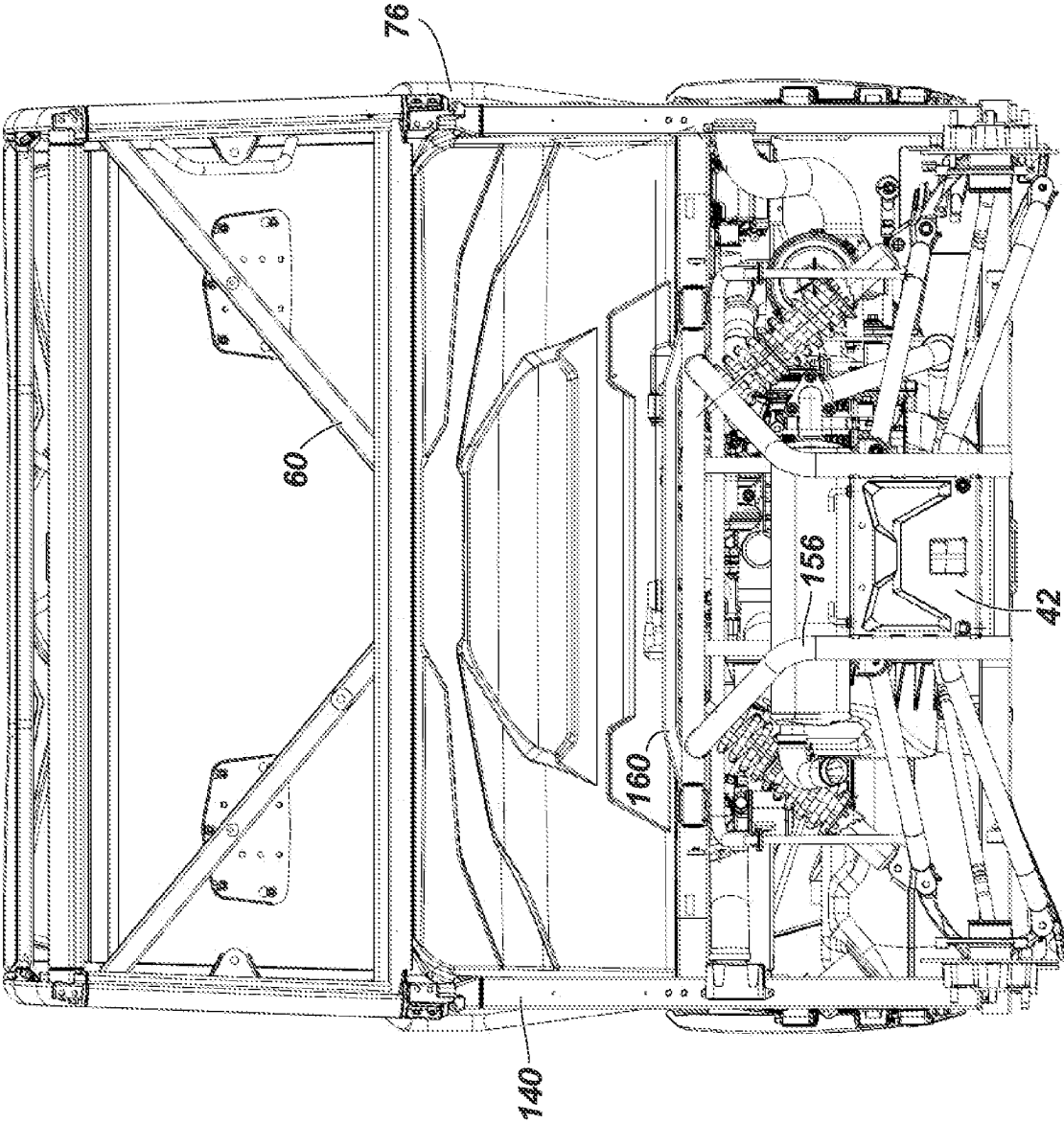


FIG. 9

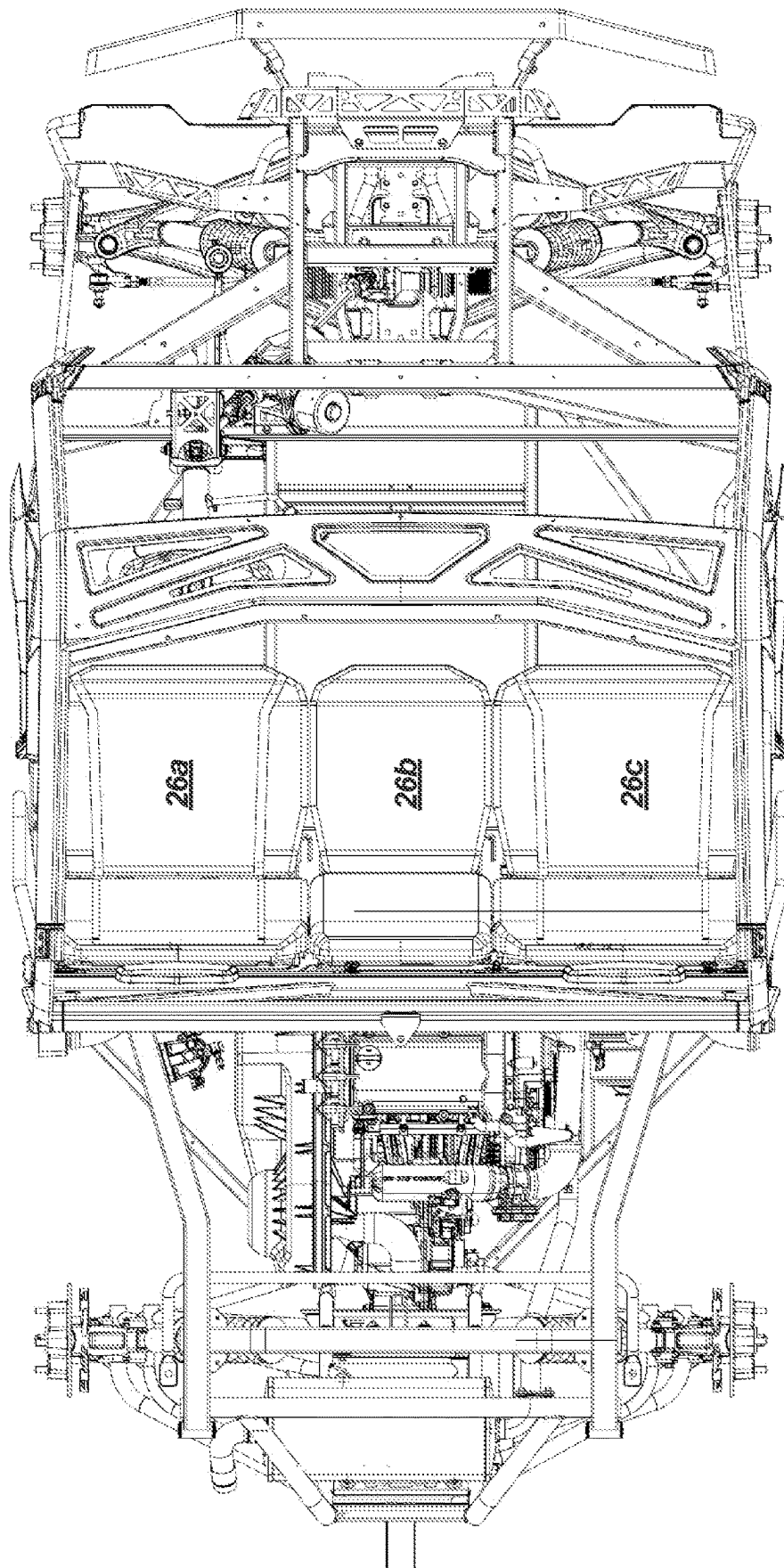
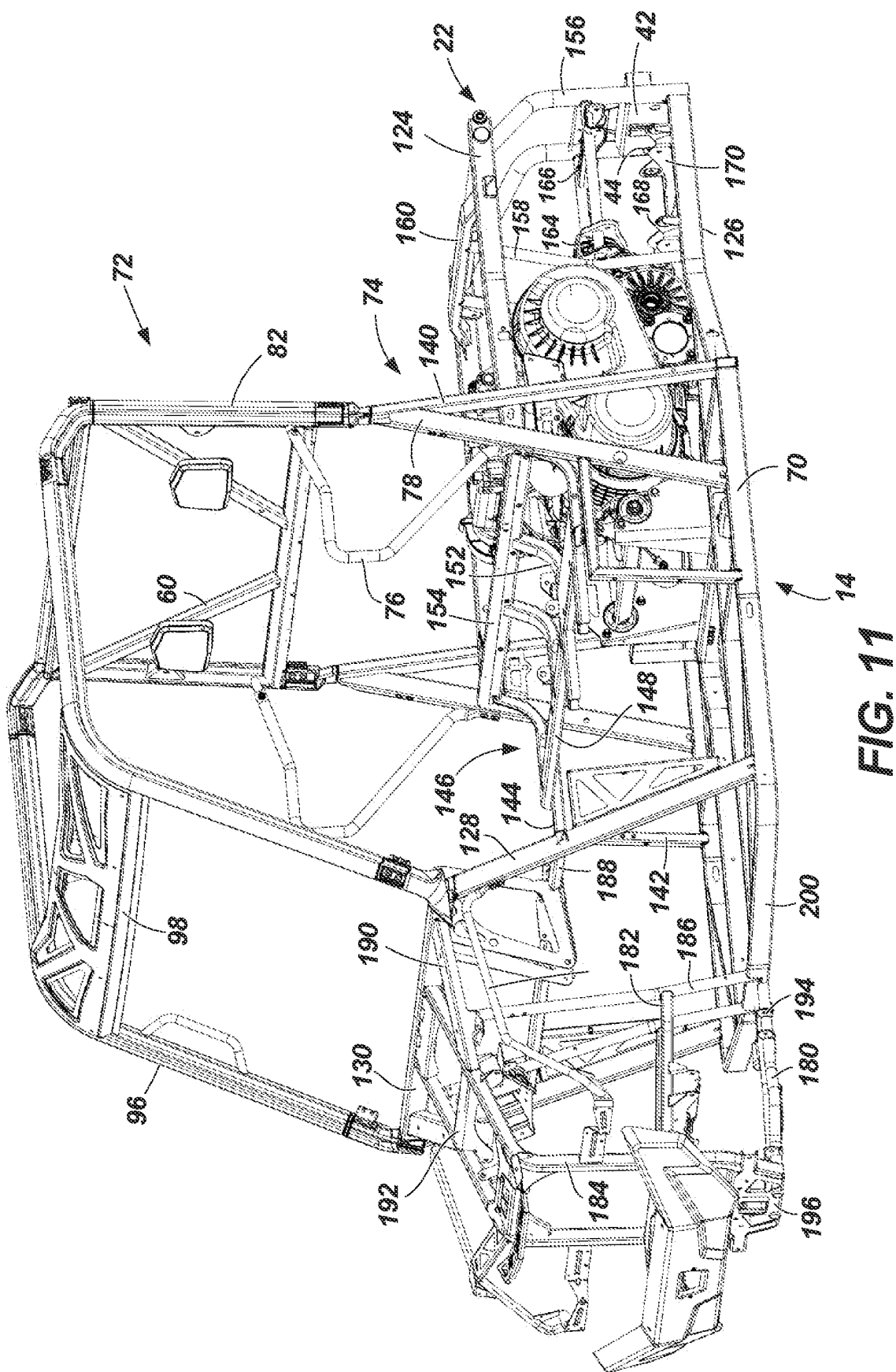


FIG. 10



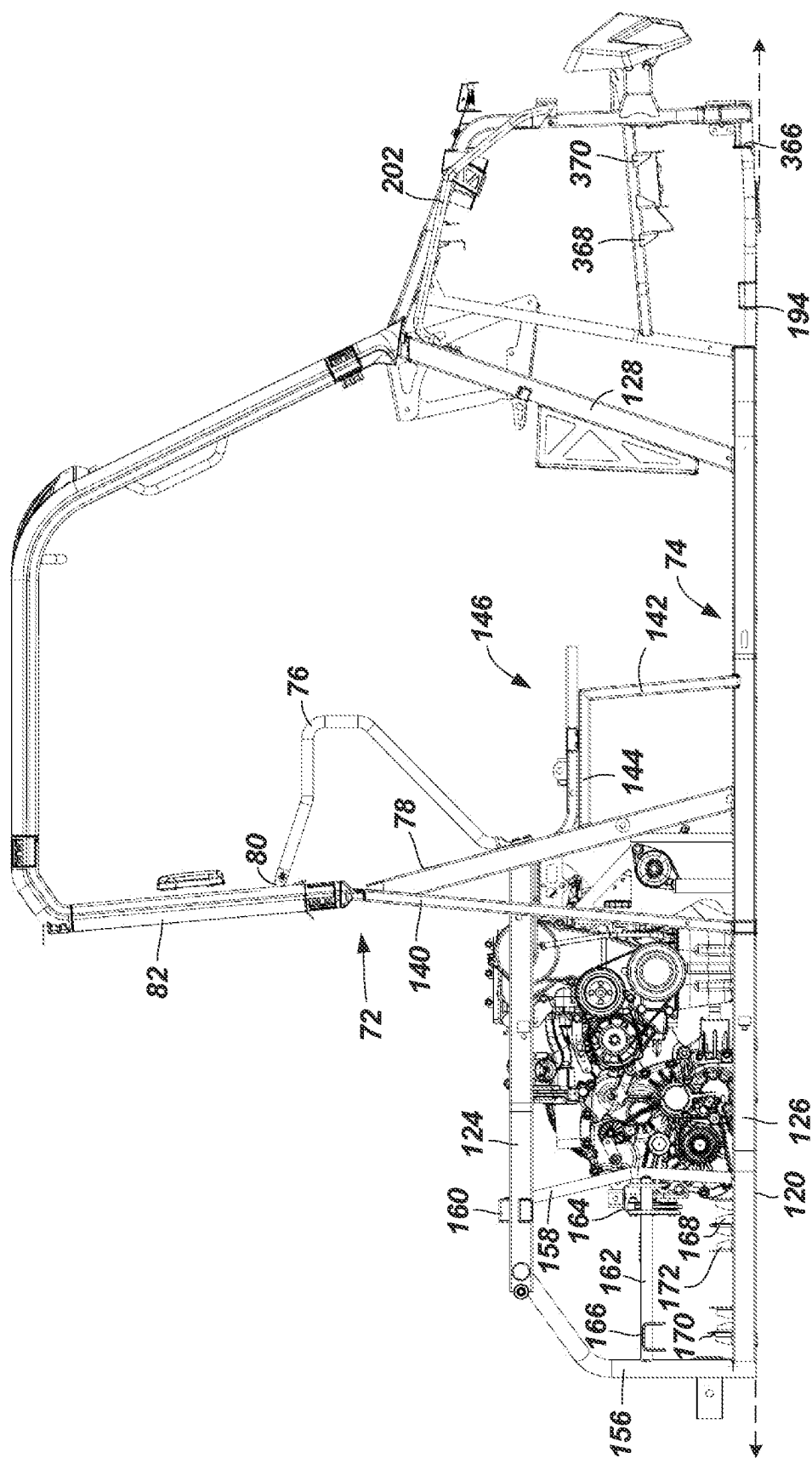


FIG. 12

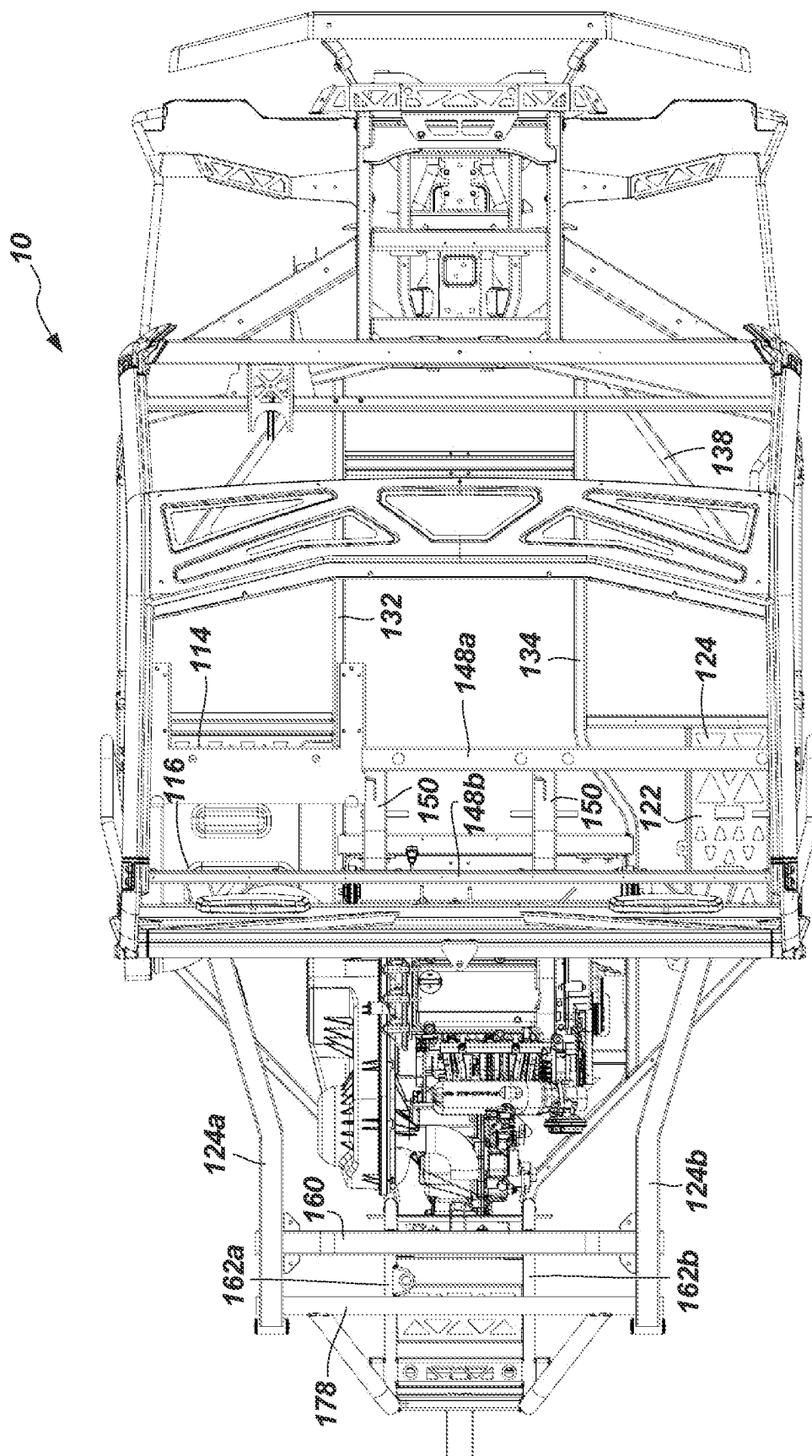


FIG. 13

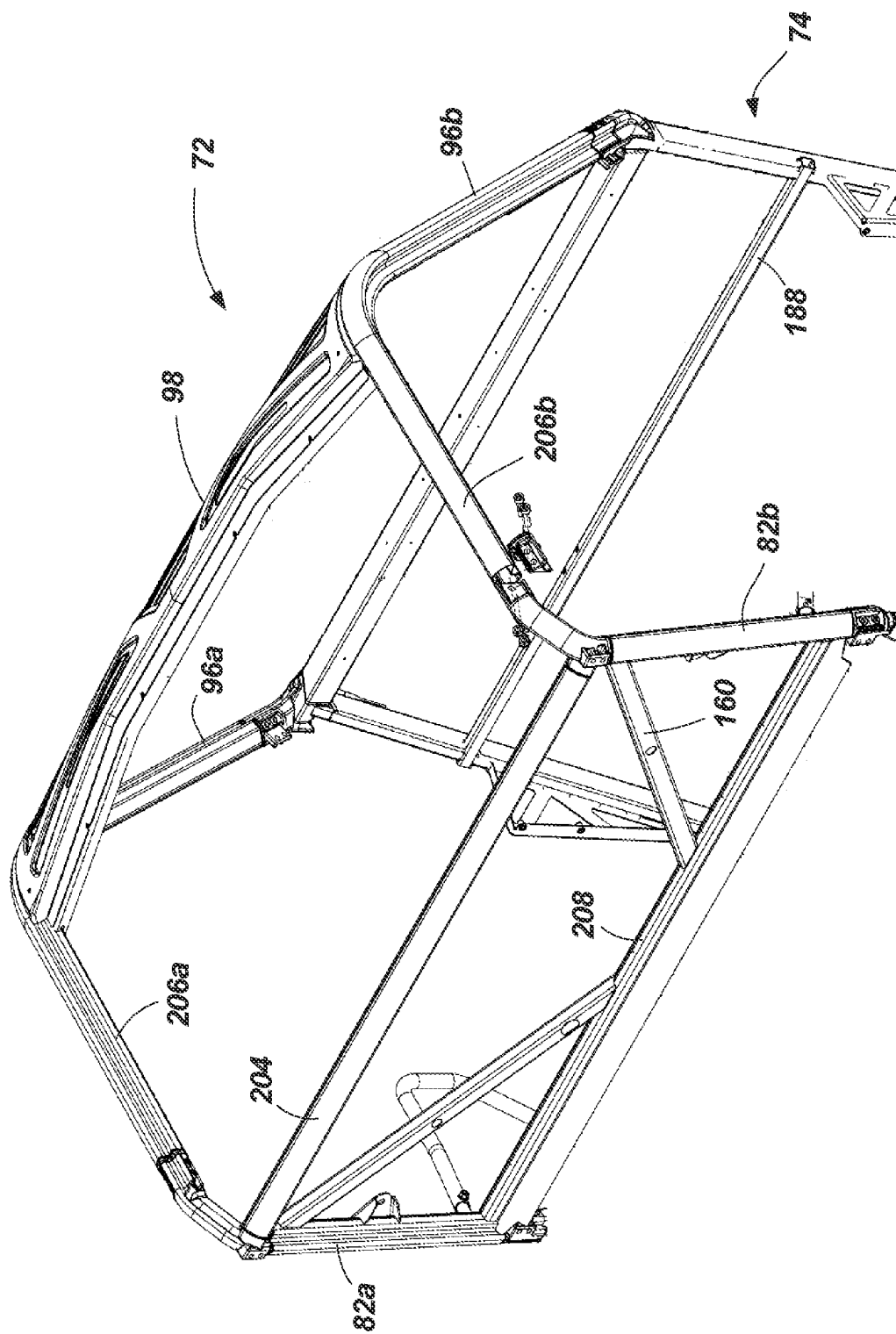


FIG. 14

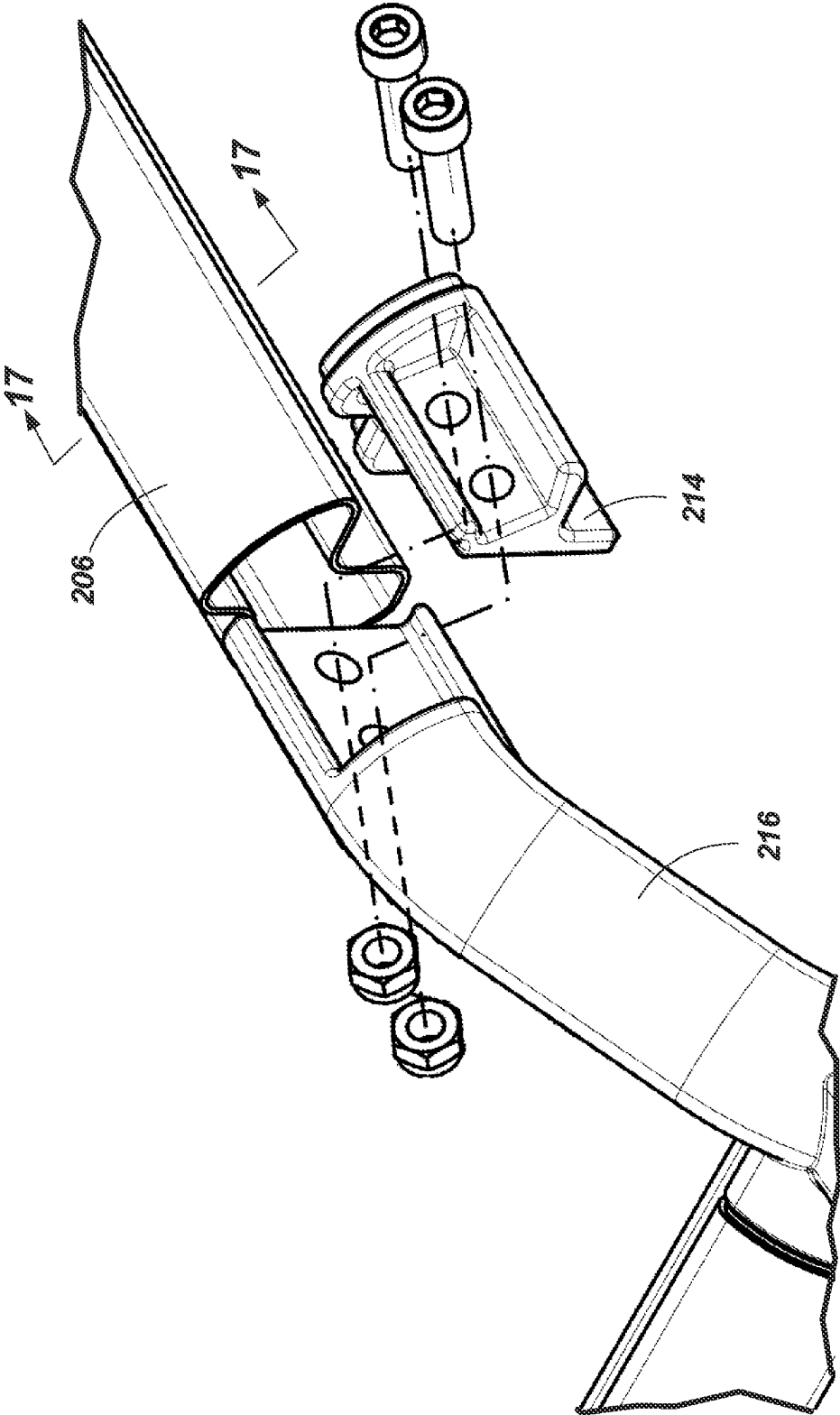


FIG. 15

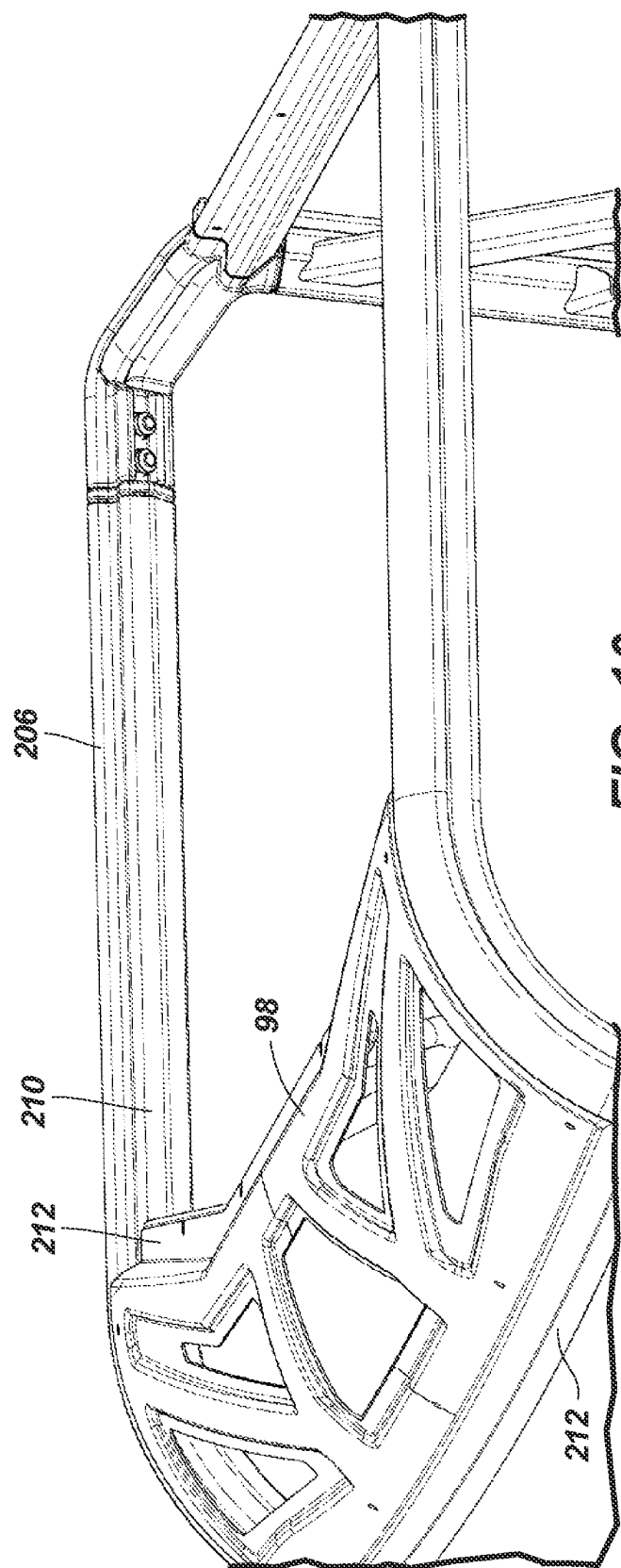


FIG. 16

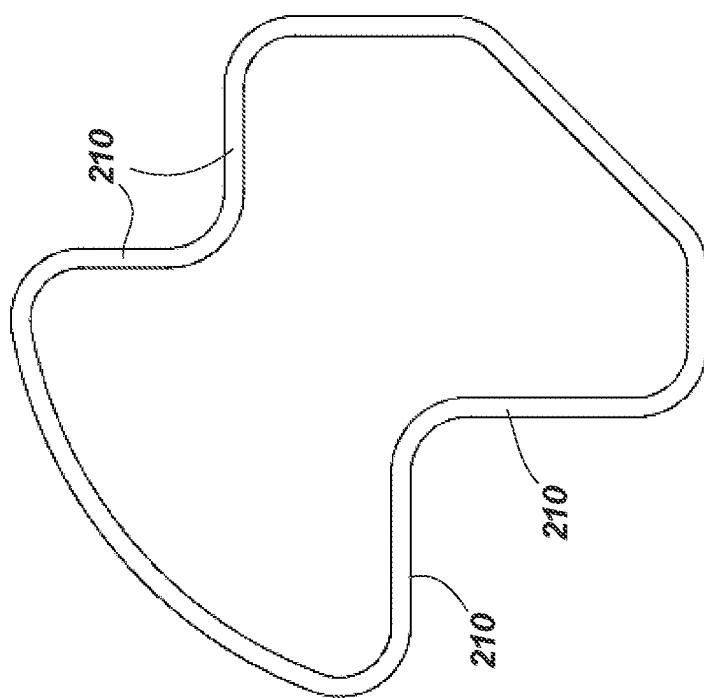
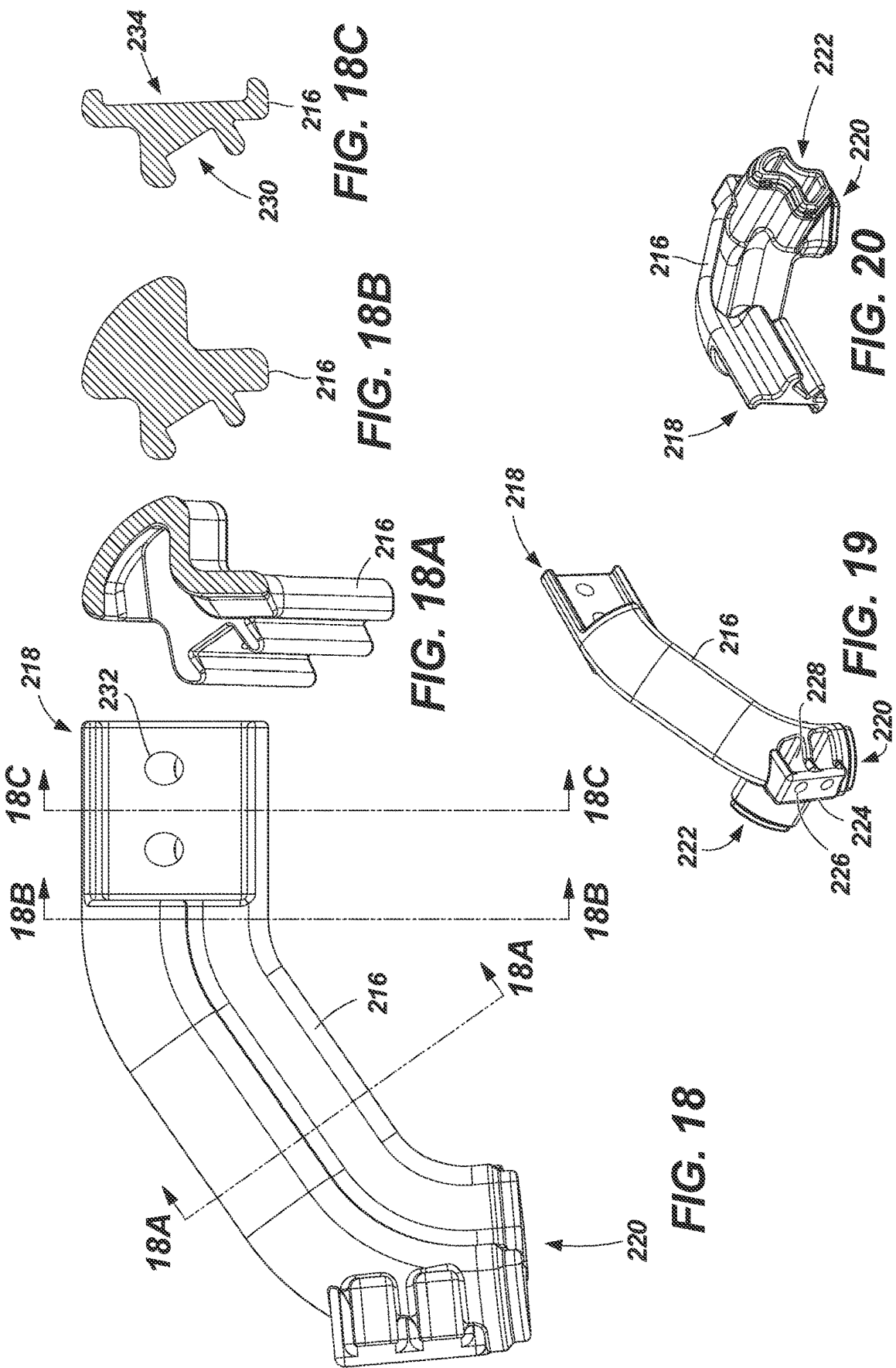


FIG. 17



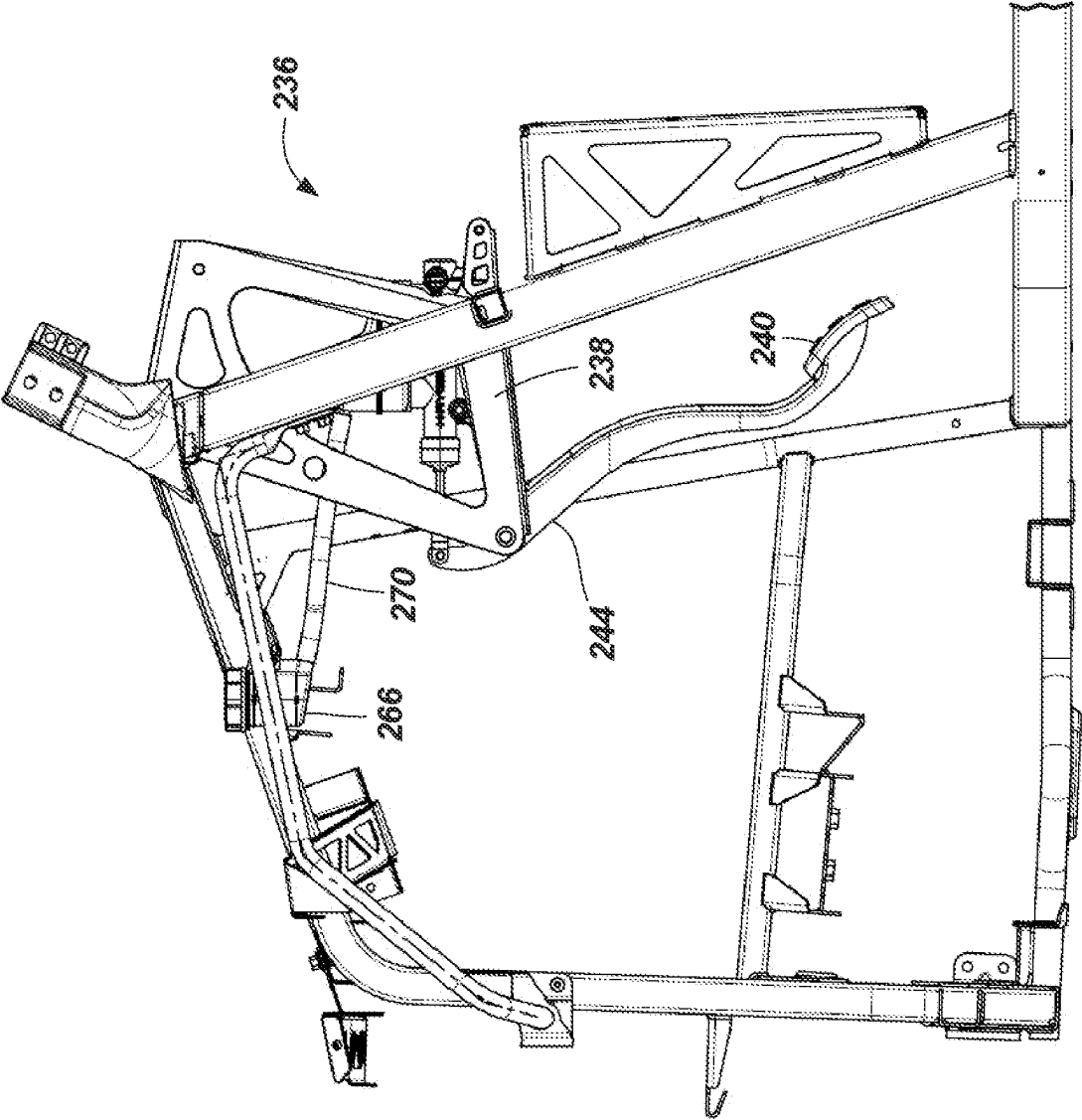


FIG. 21

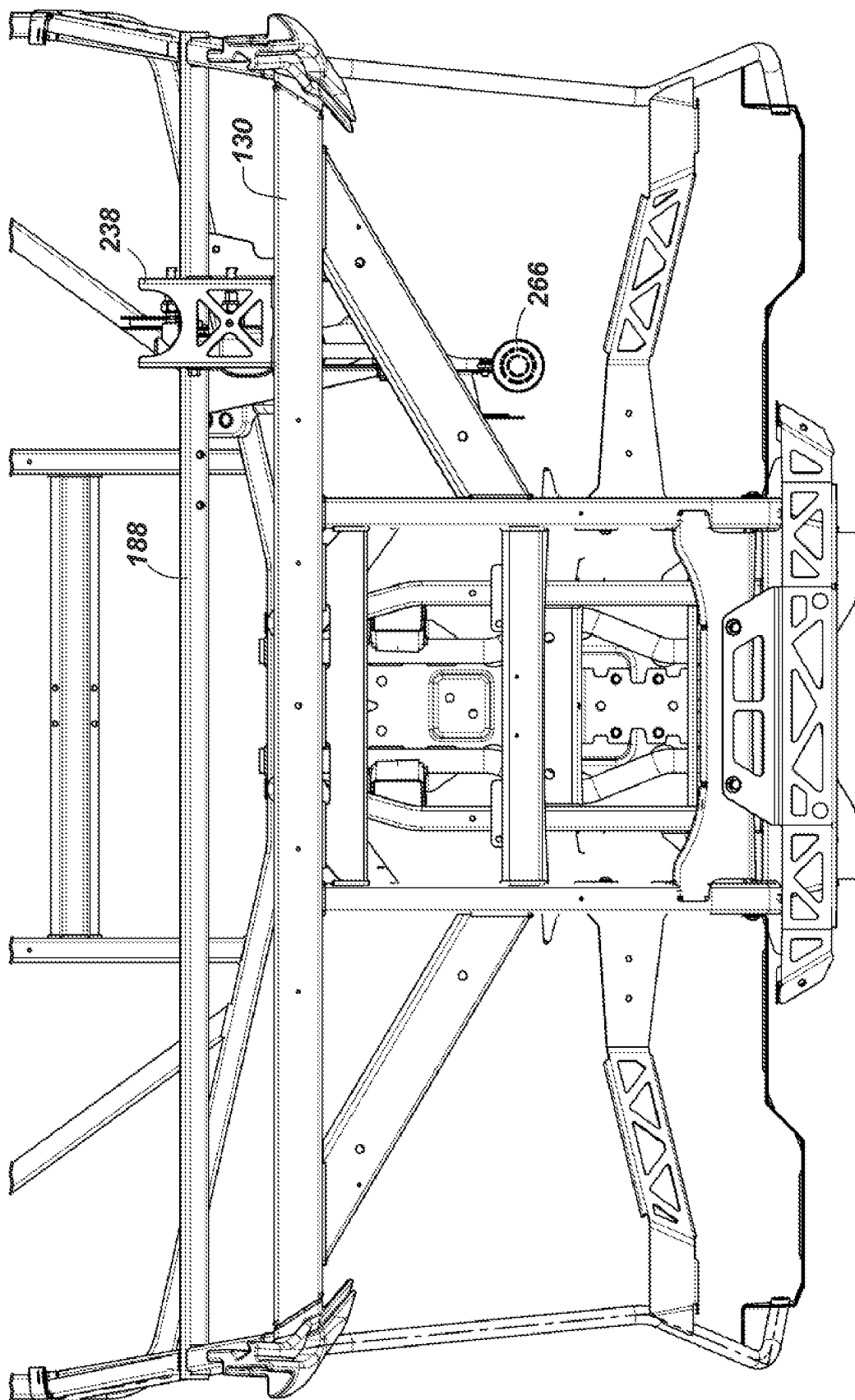


FIG. 22

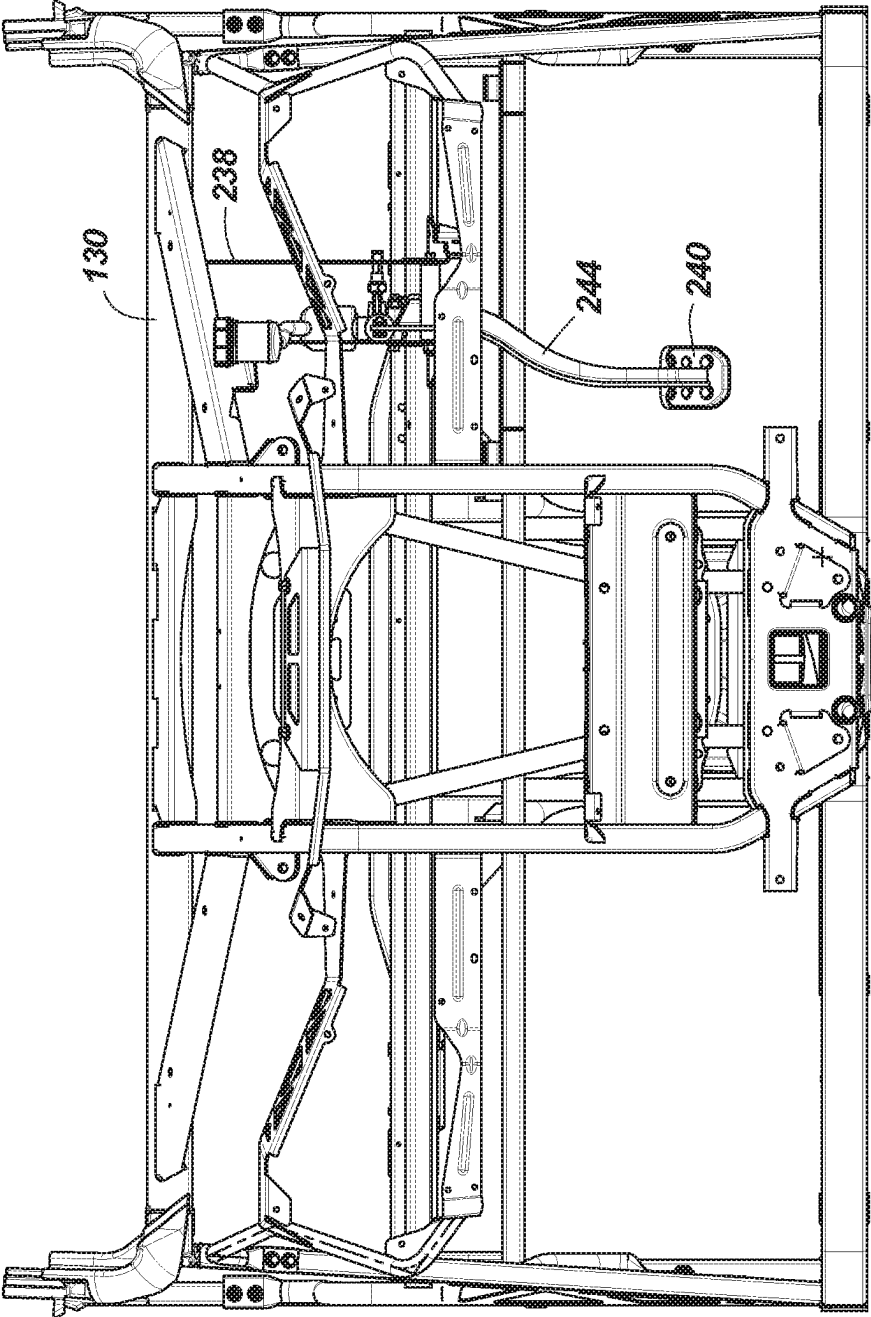


FIG. 23

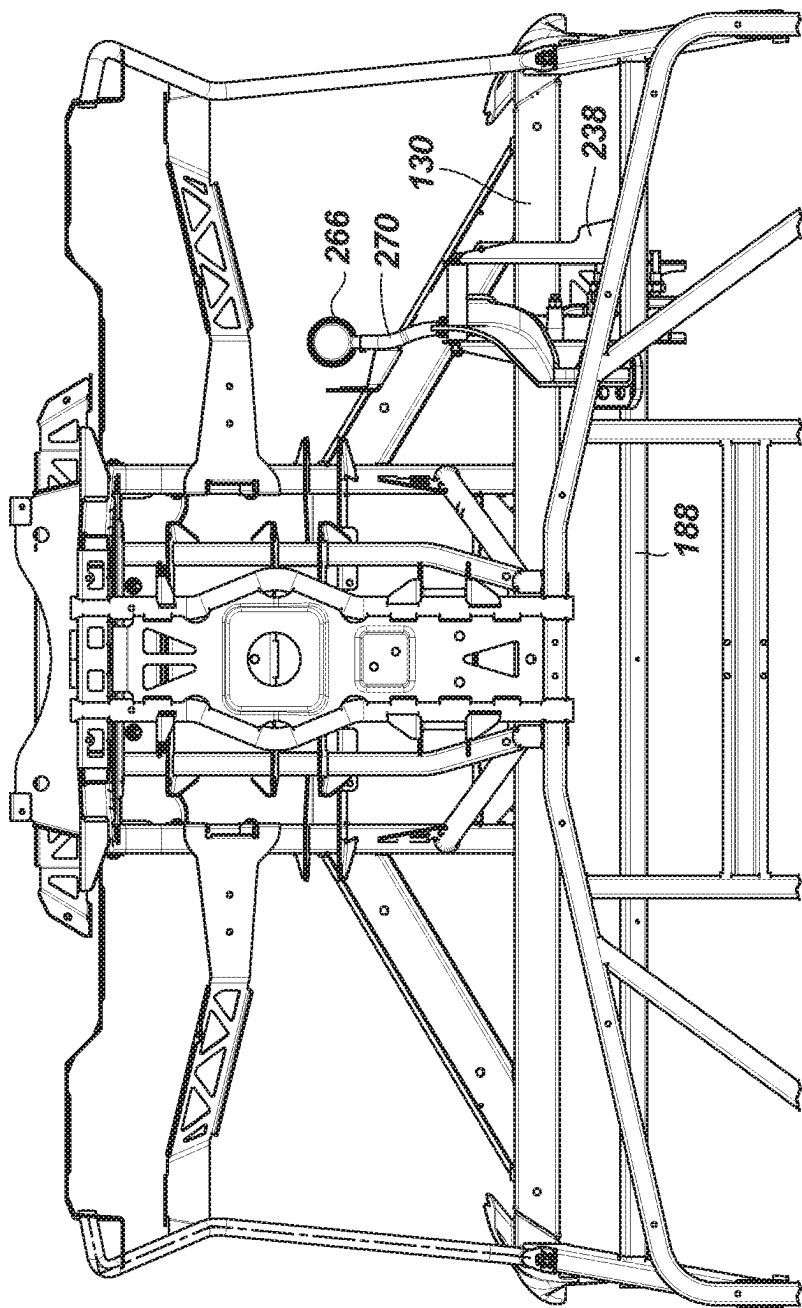
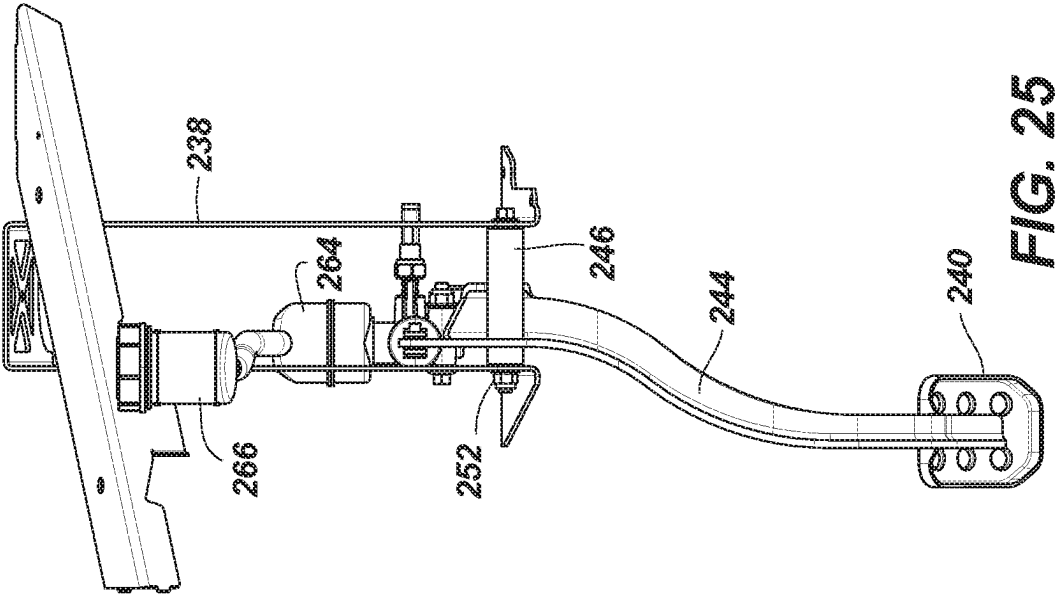
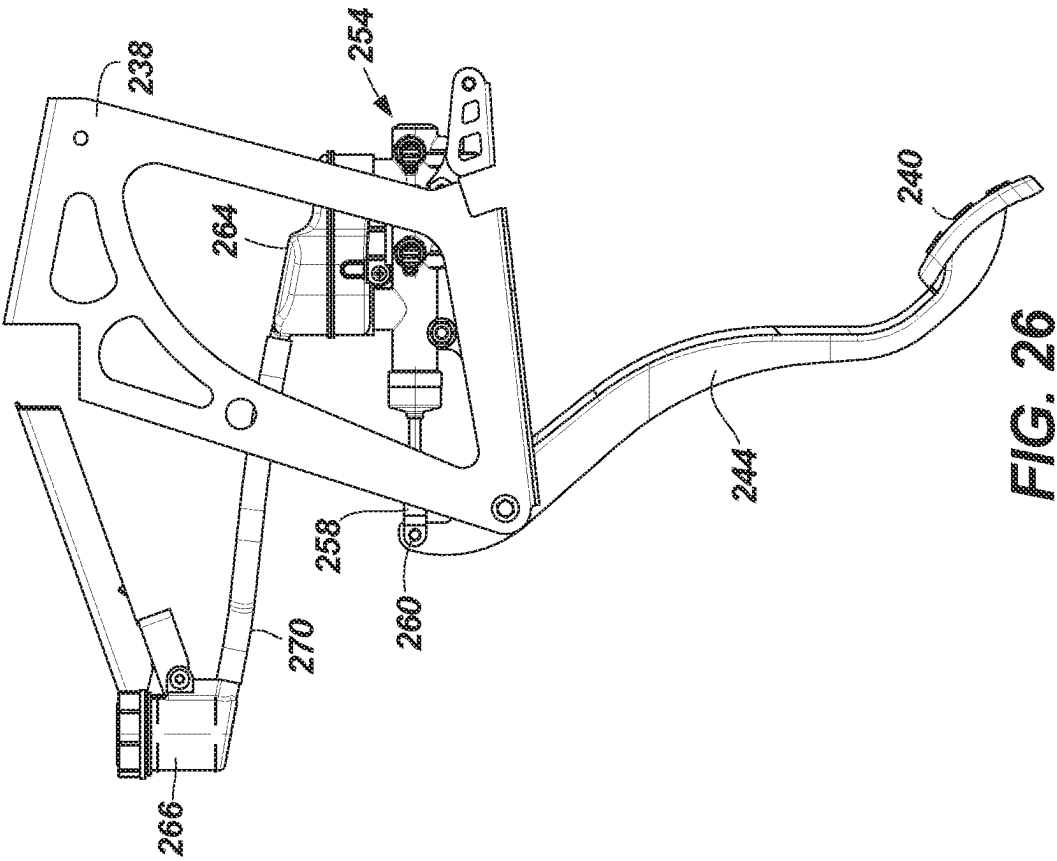
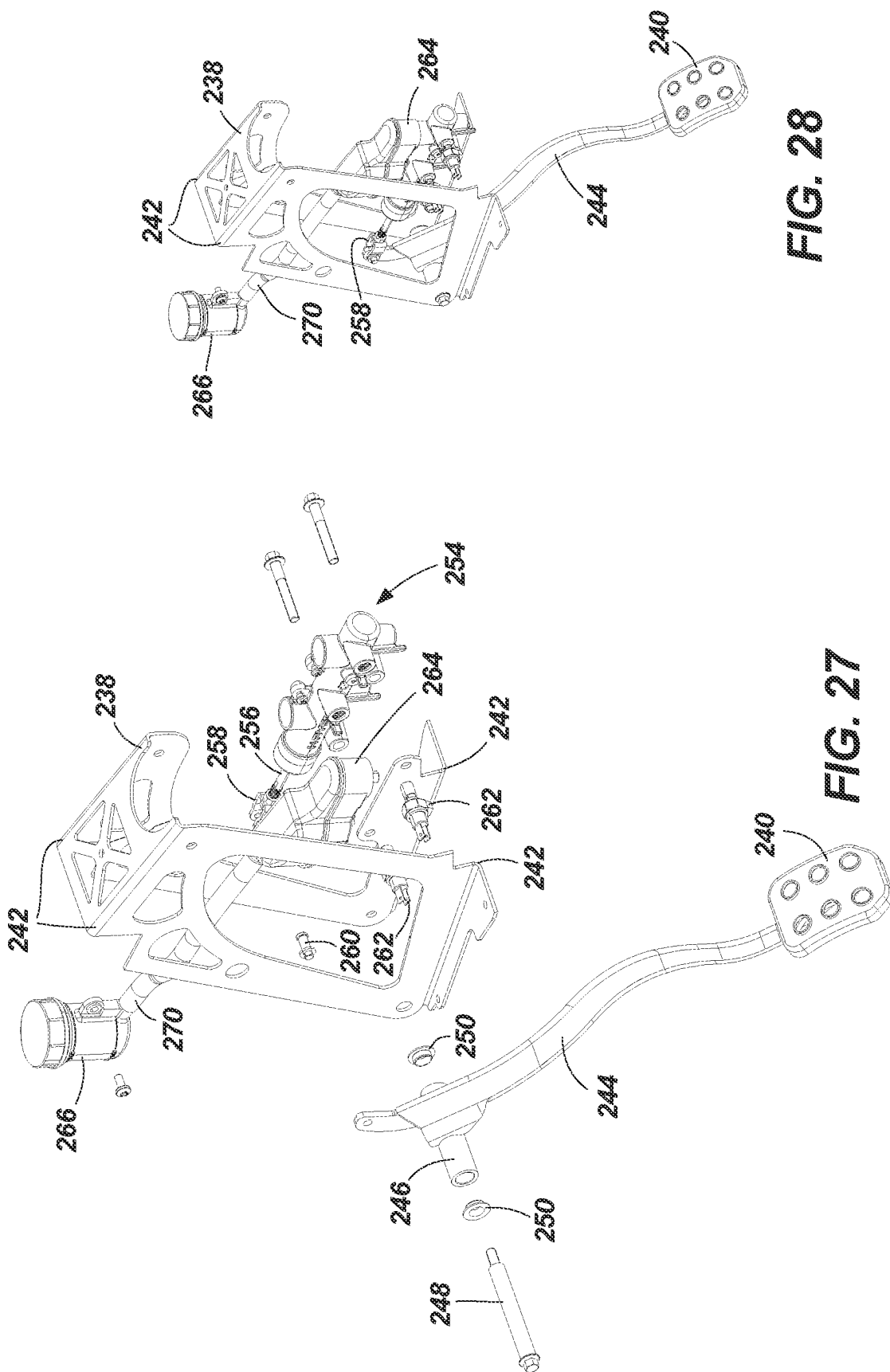


FIG. 24





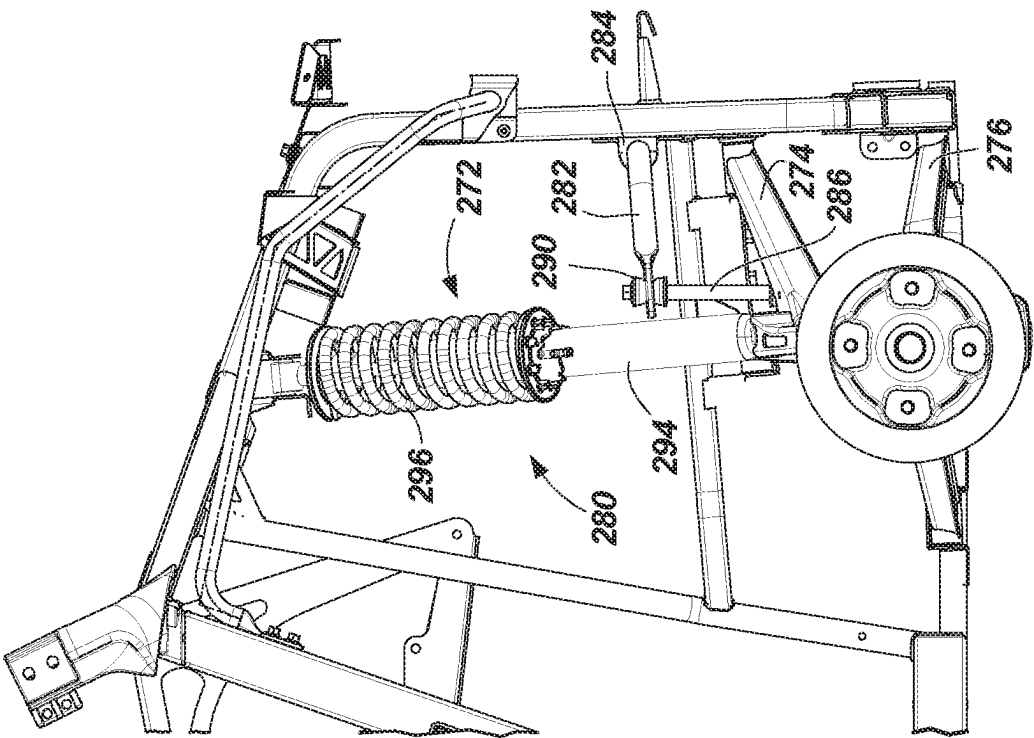


FIG. 29

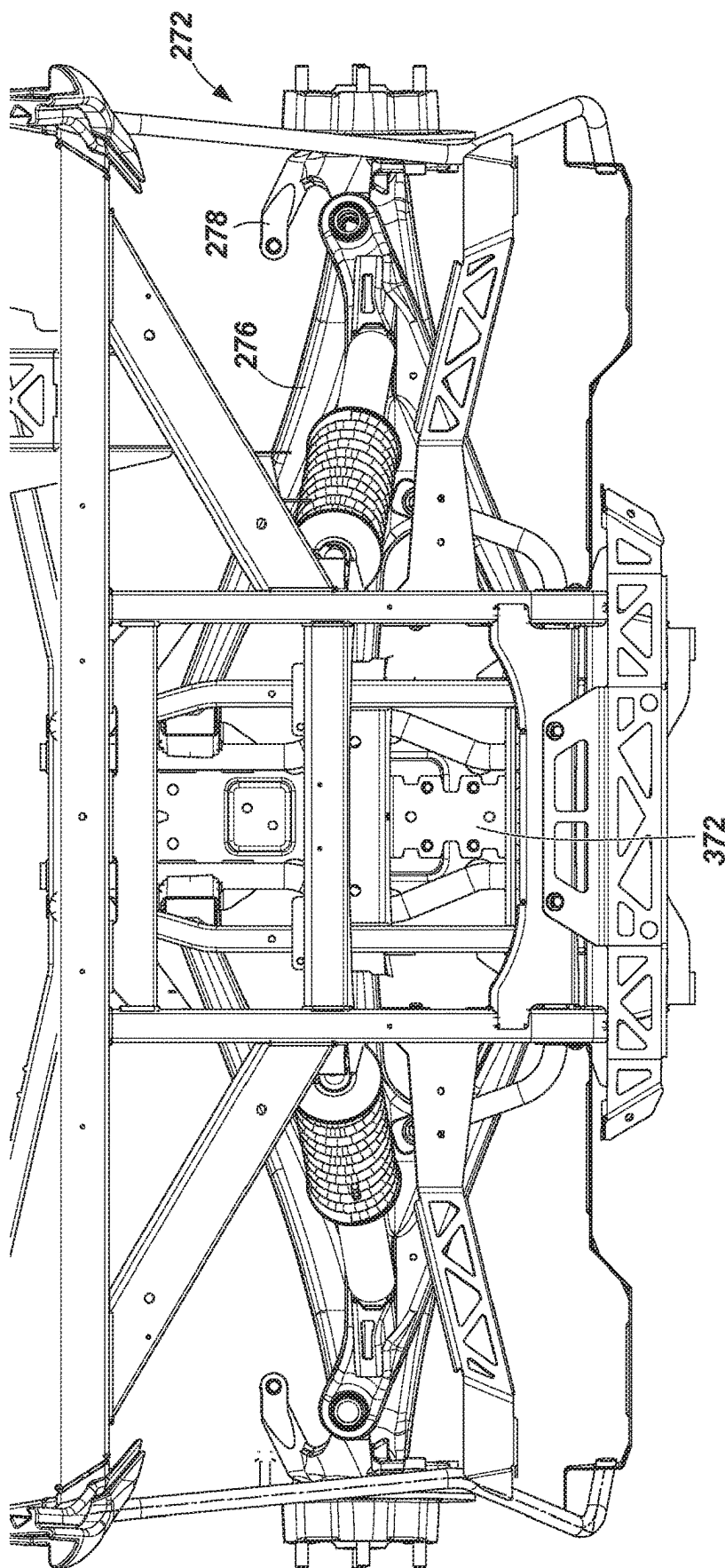


FIG. 30

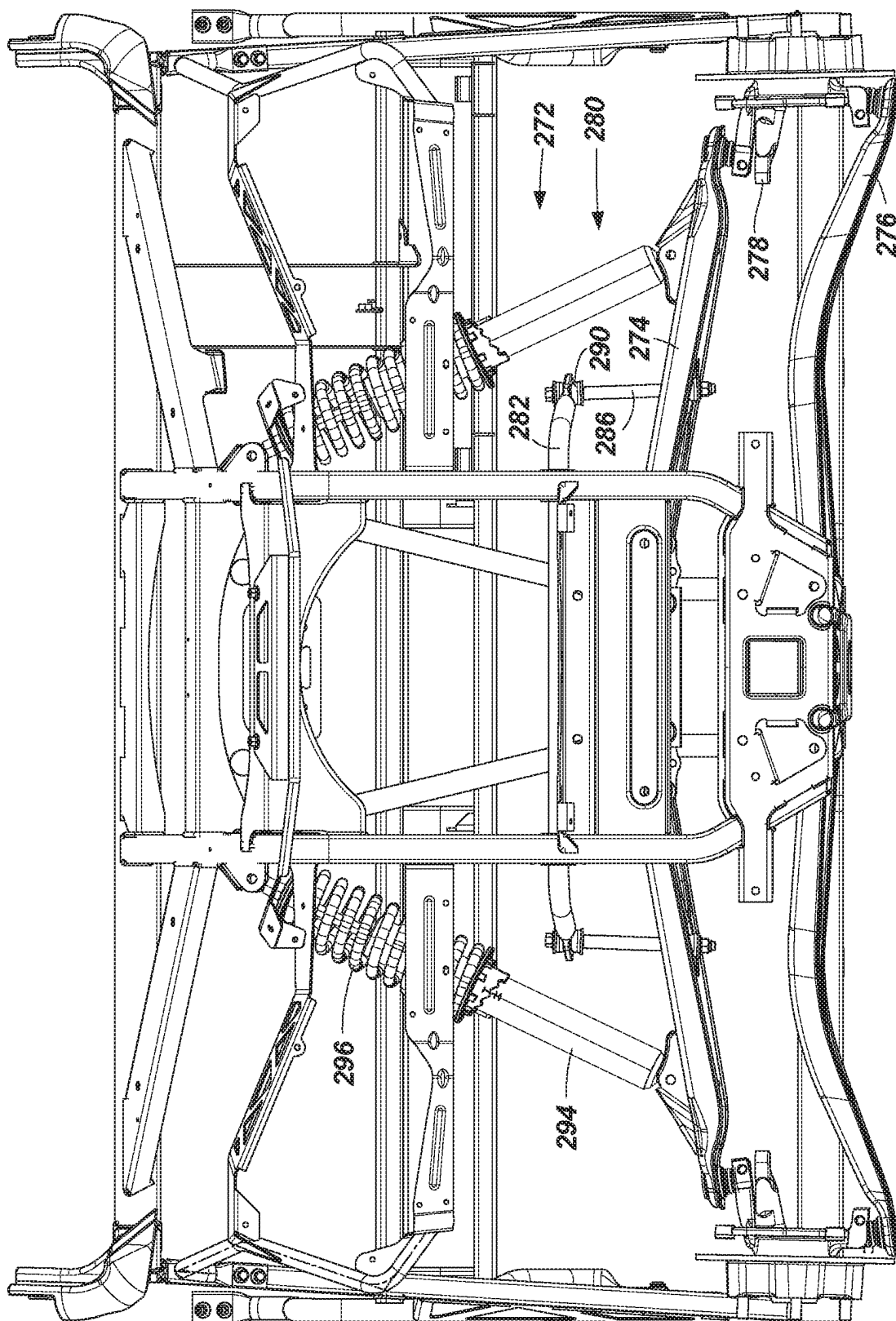


FIG. 31

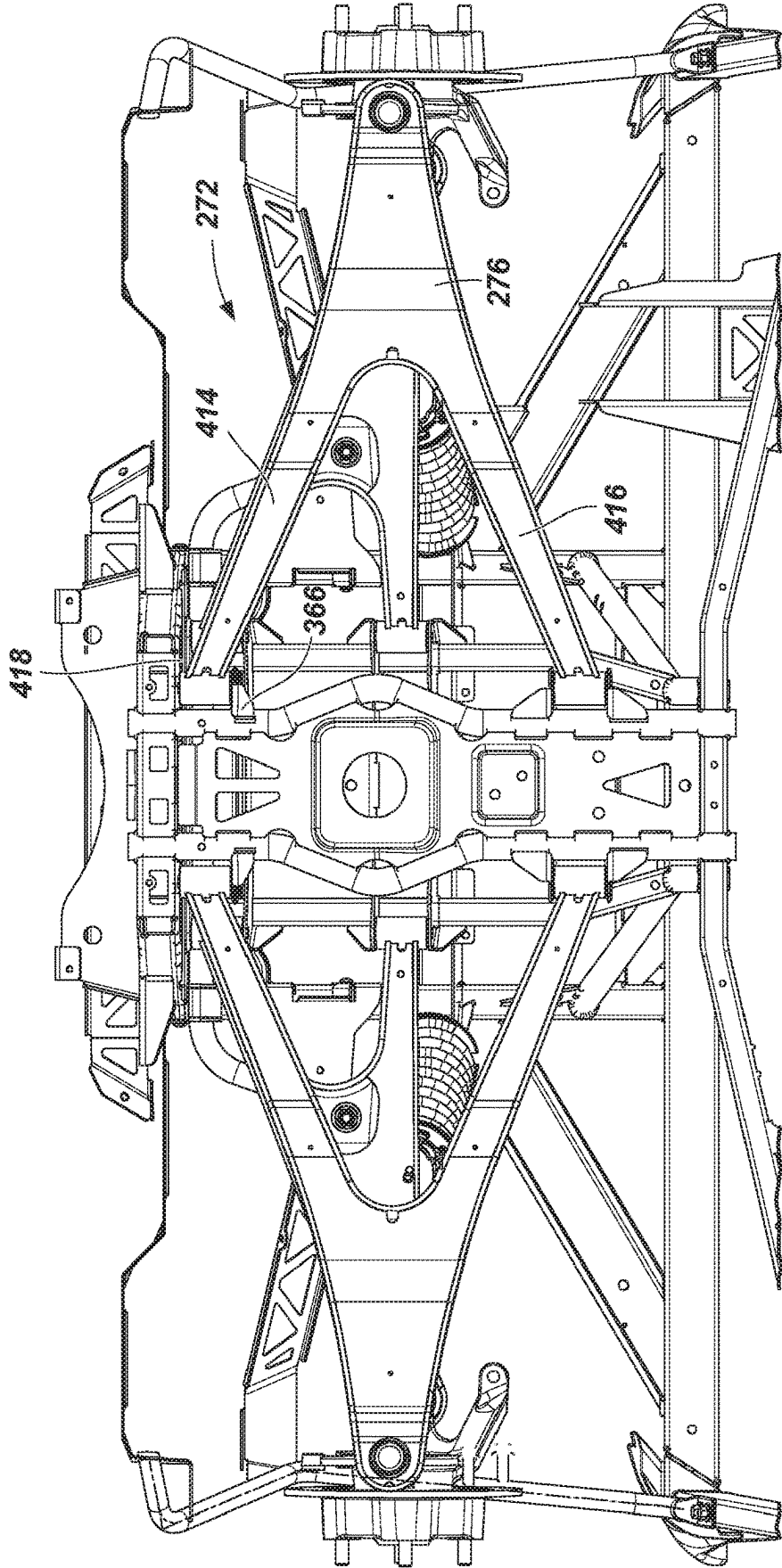


FIG. 32

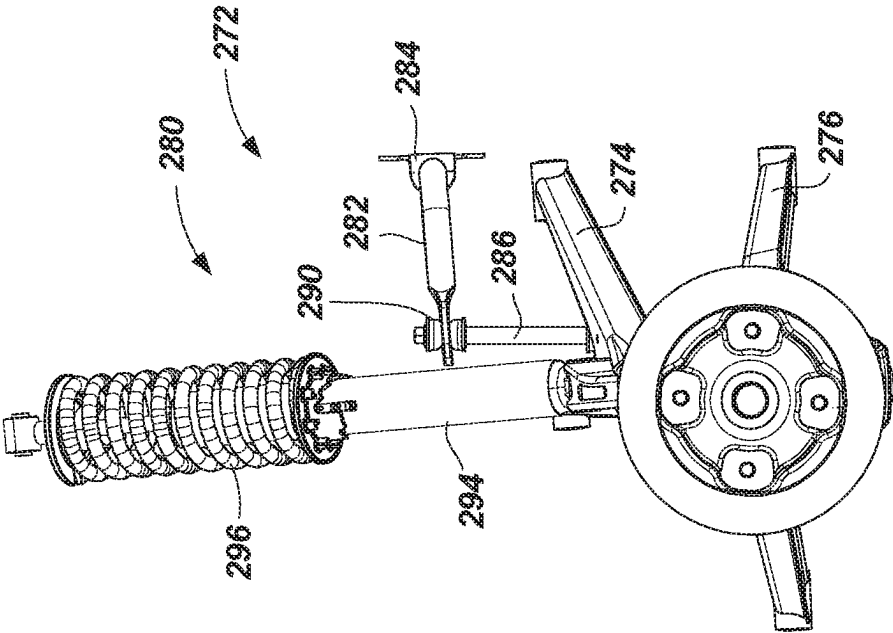


FIG. 33

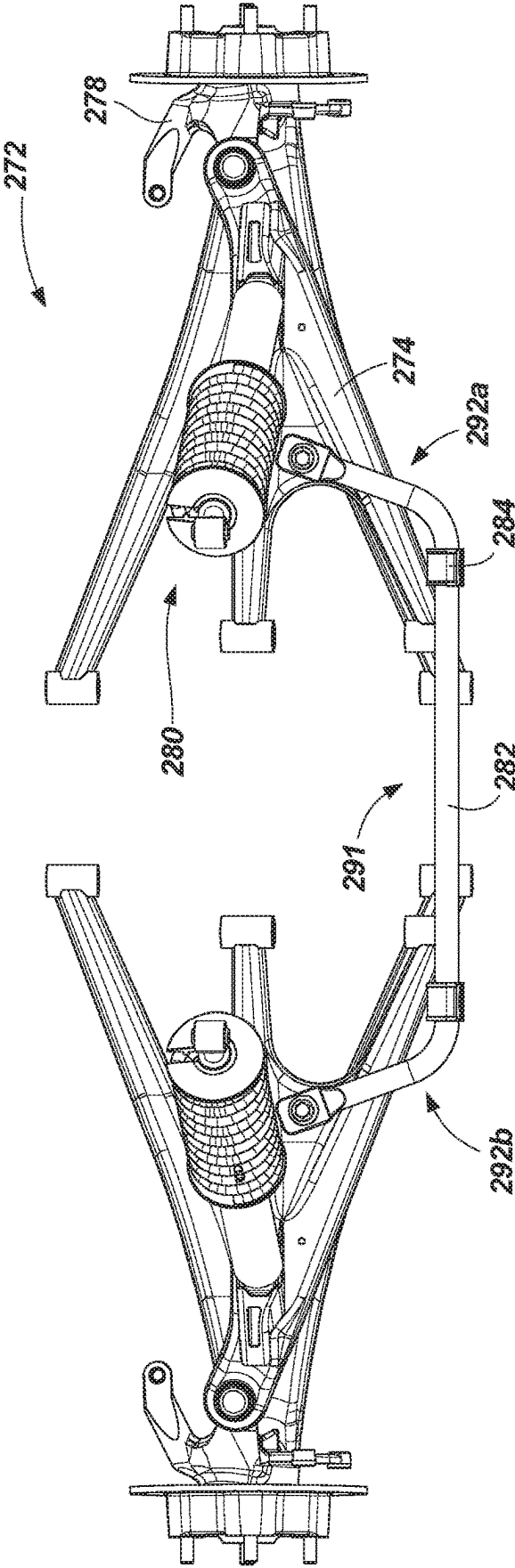


FIG. 34

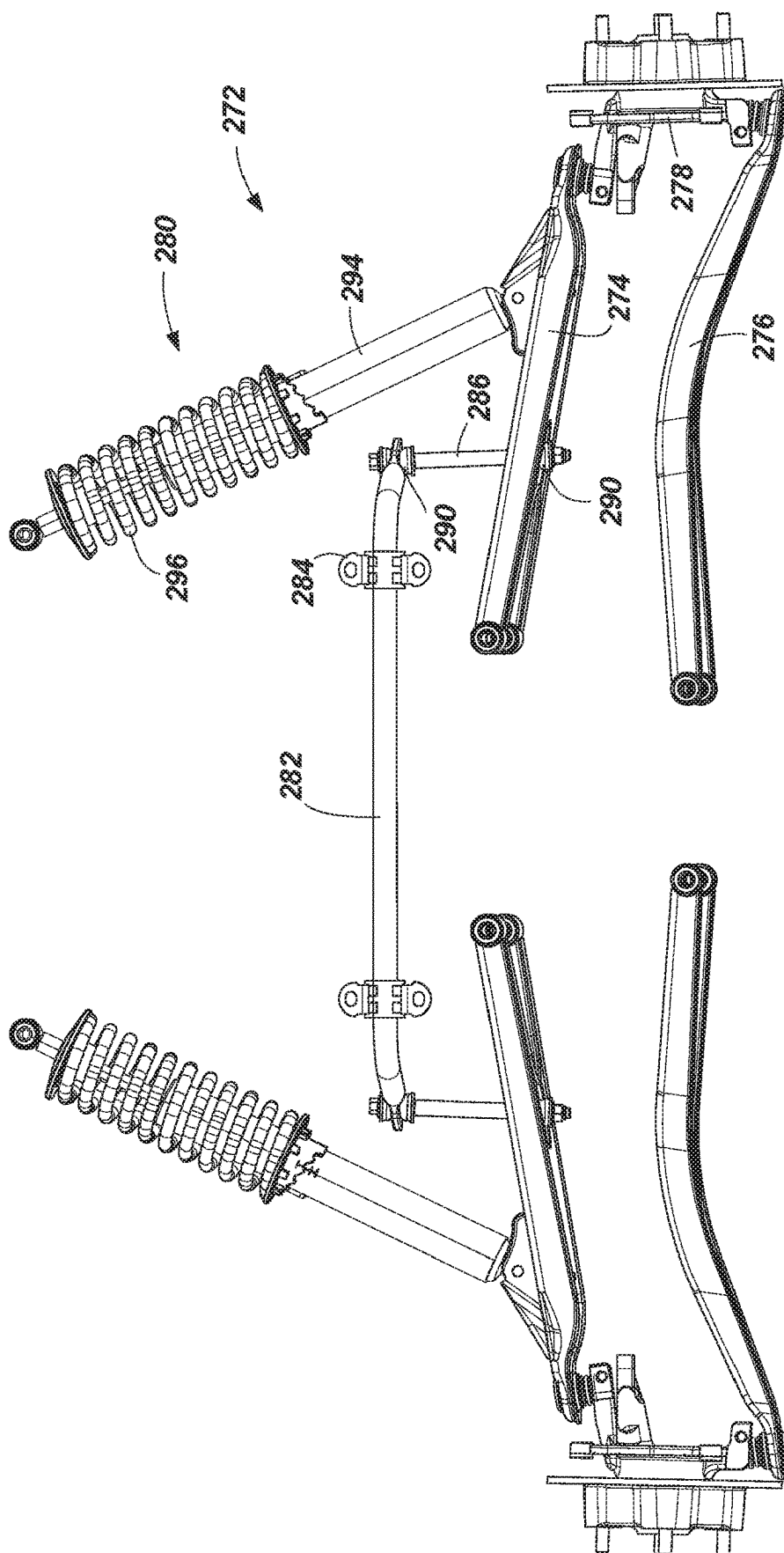


FIG. 35

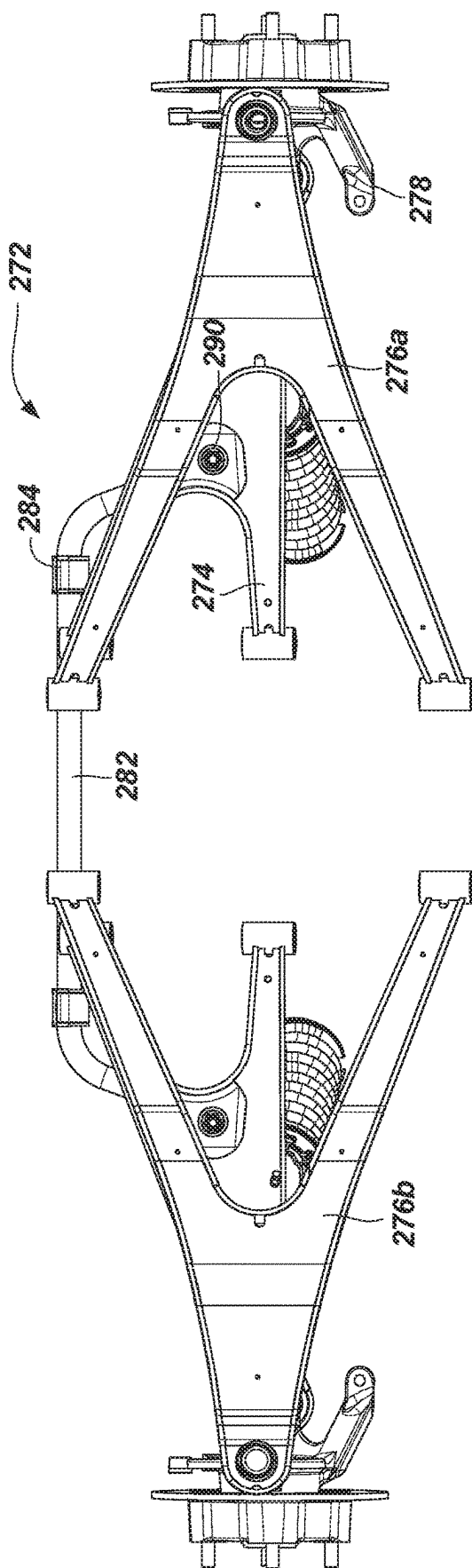
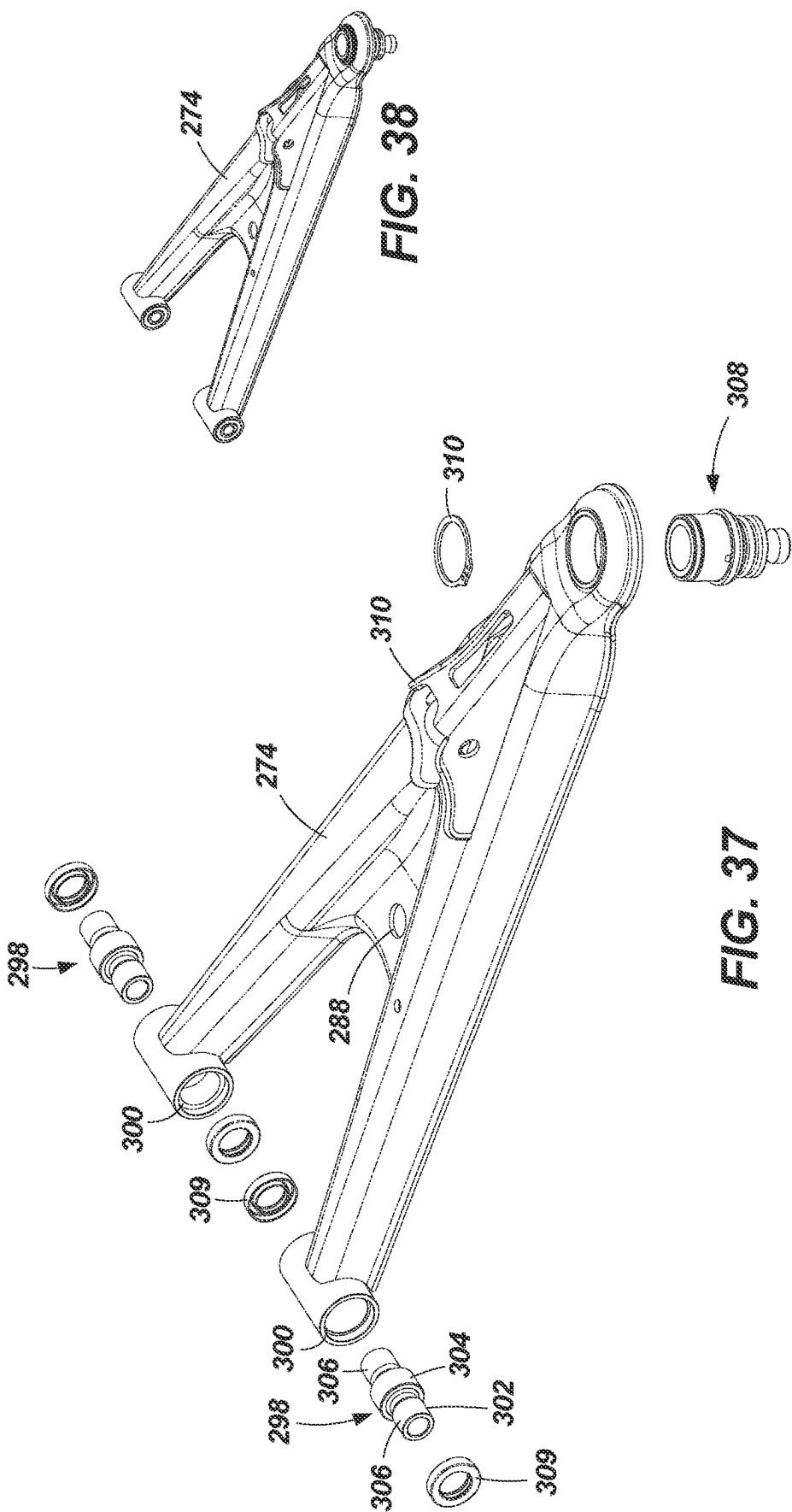


FIG. 36



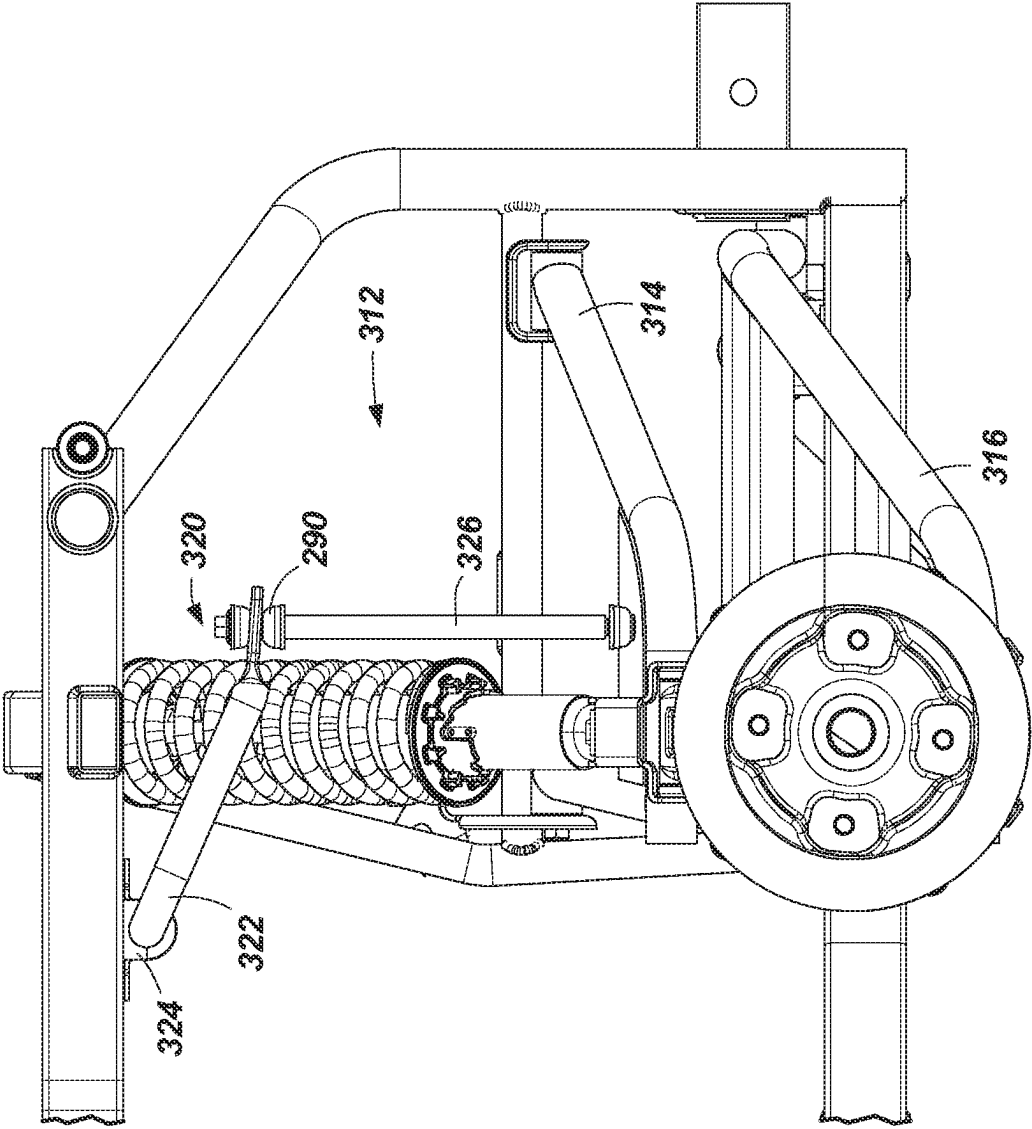


FIG. 39

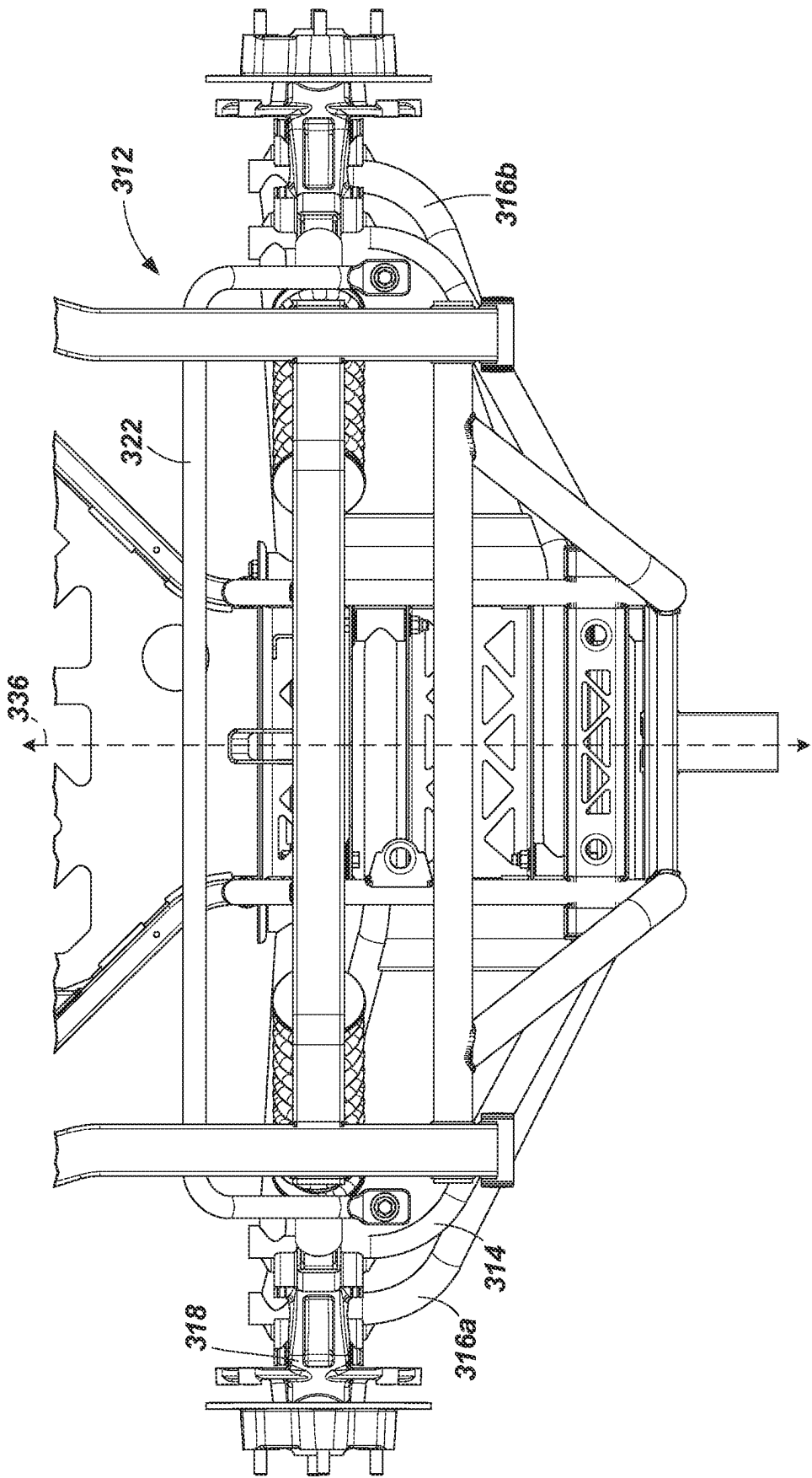


FIG. 40

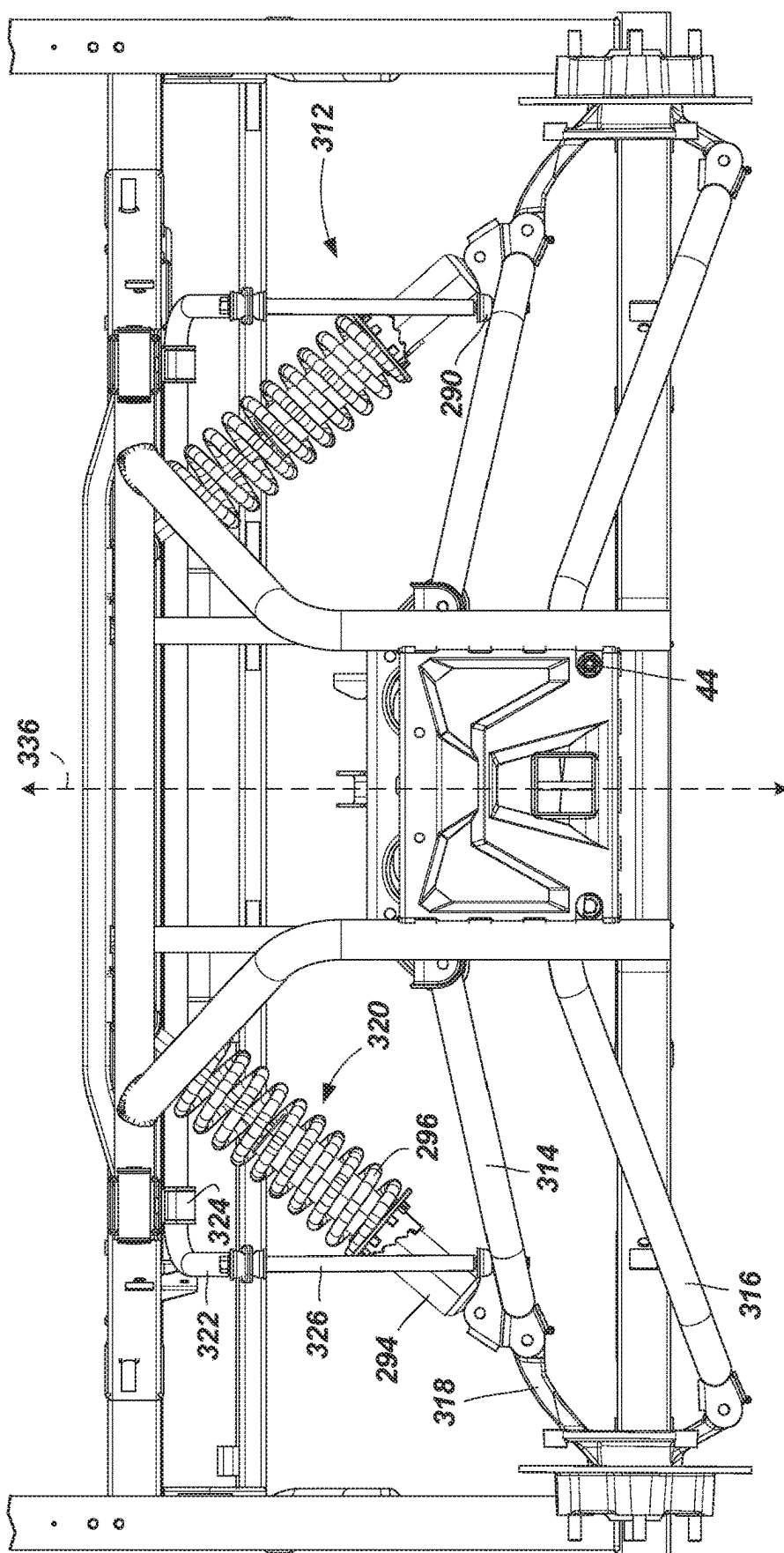


FIG. 41

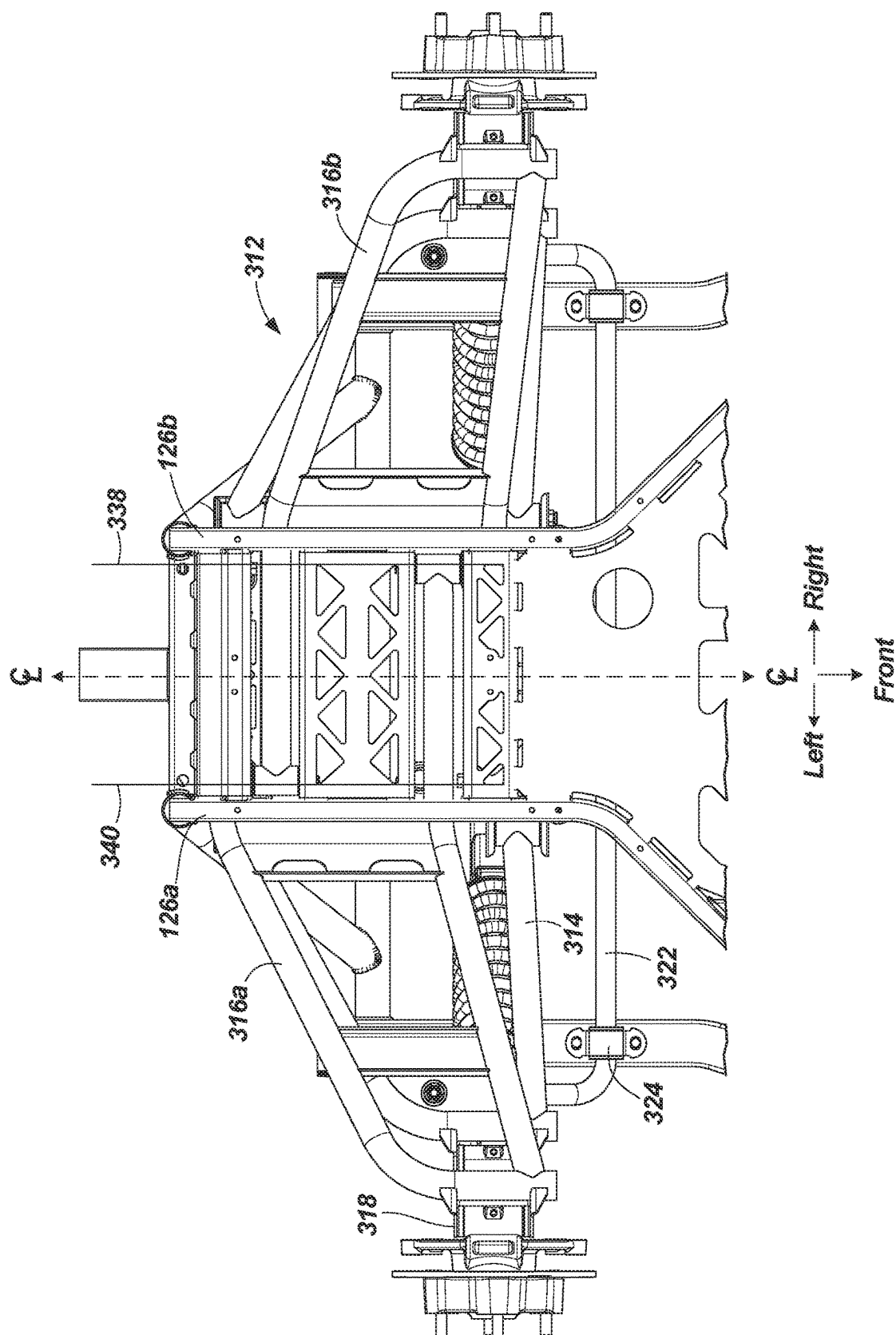


FIG. 42

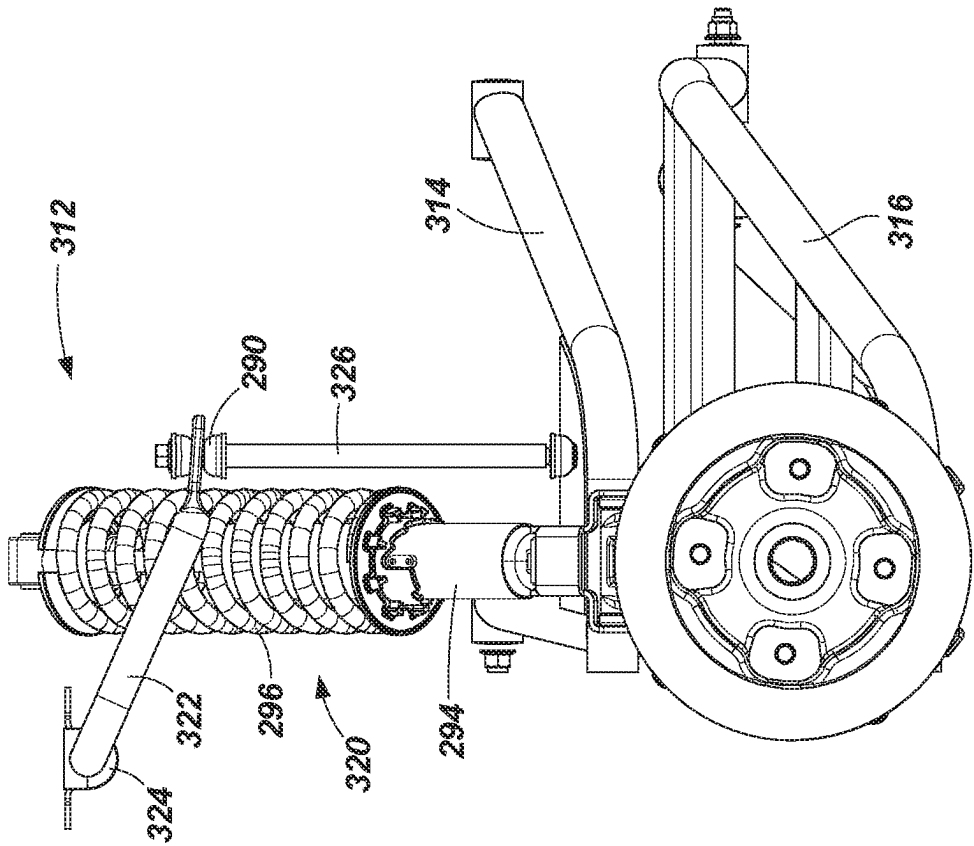


FIG. 43

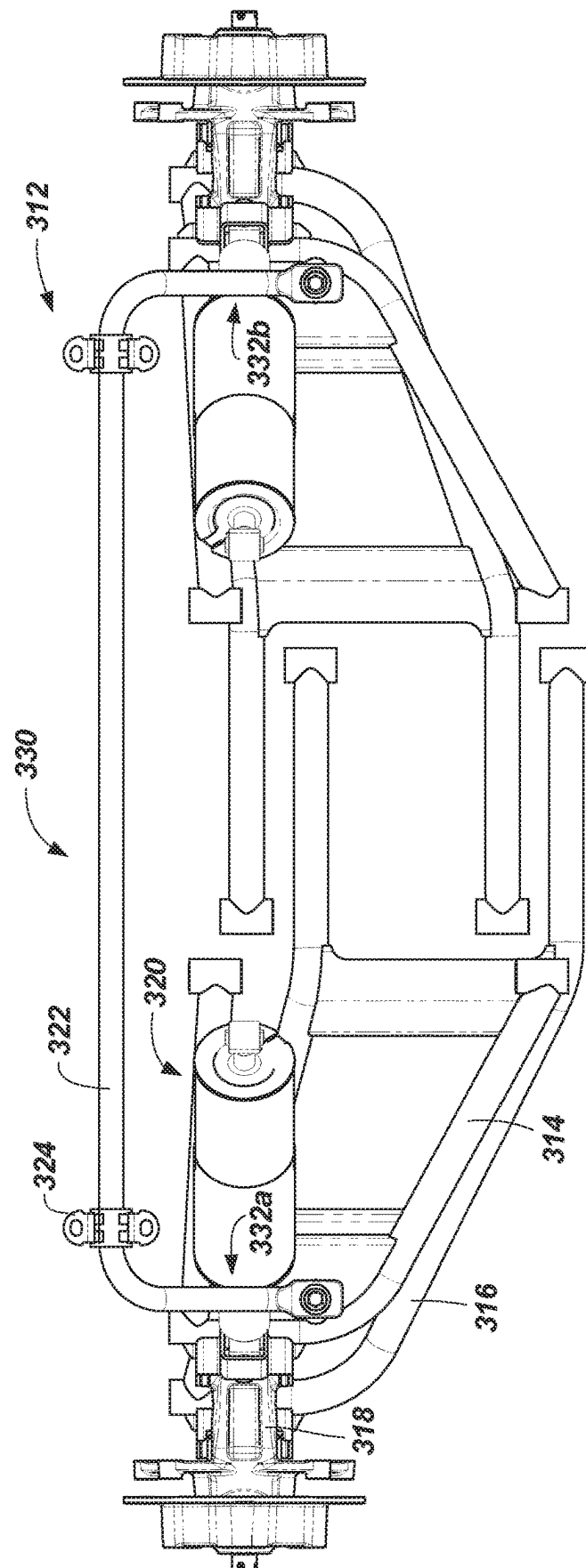


FIG. 44

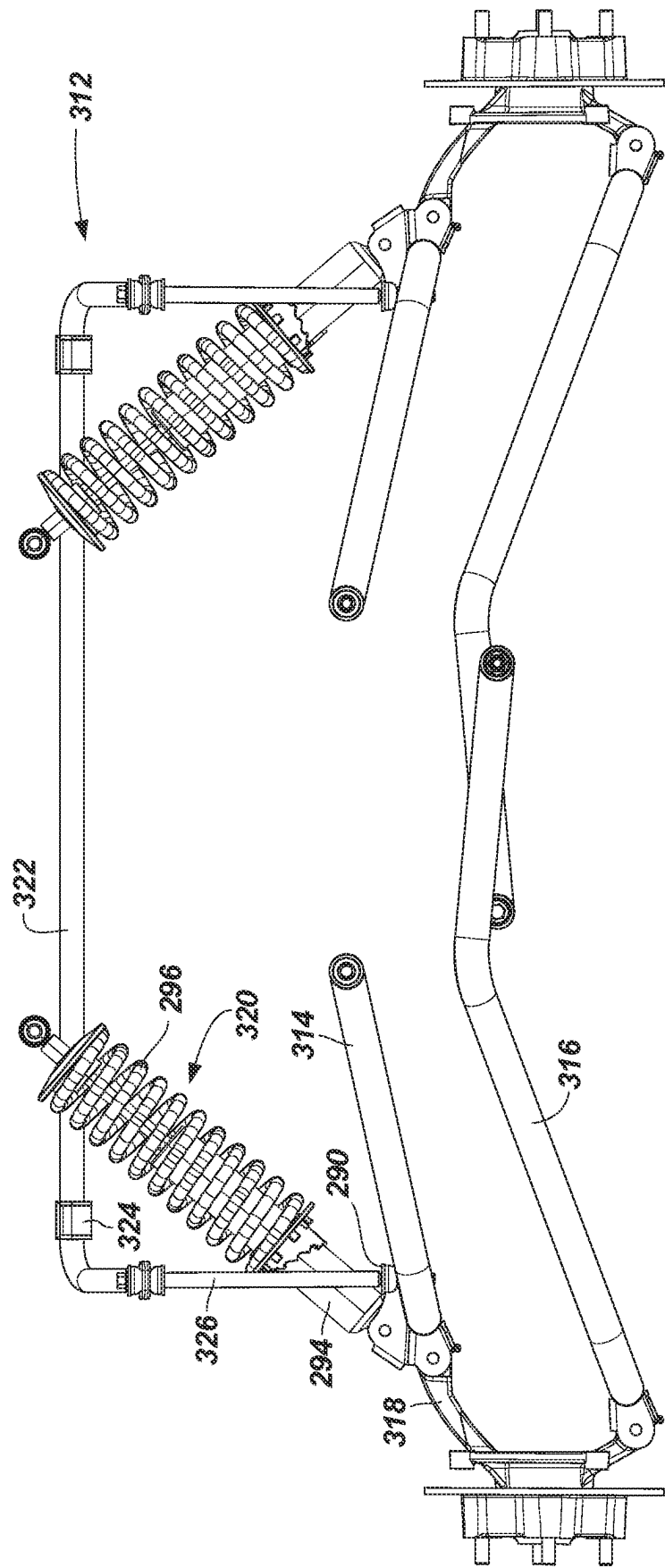


FIG. 45

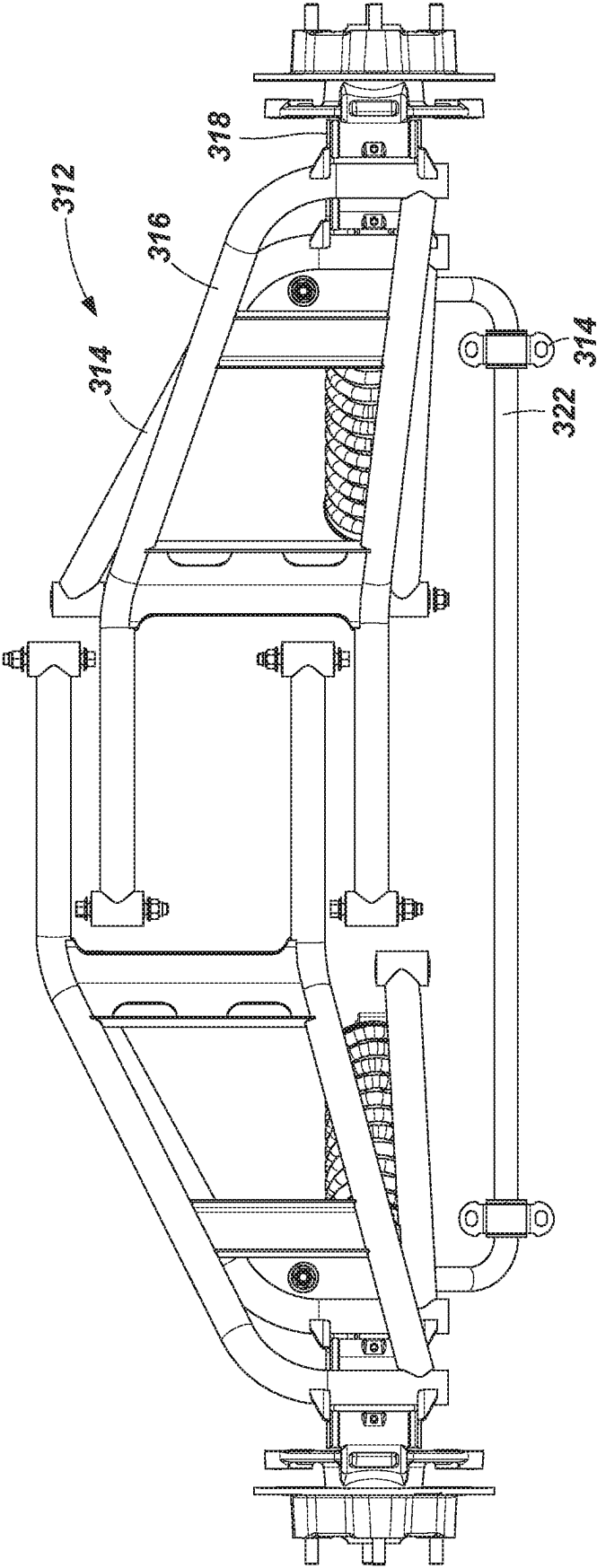


FIG. 46

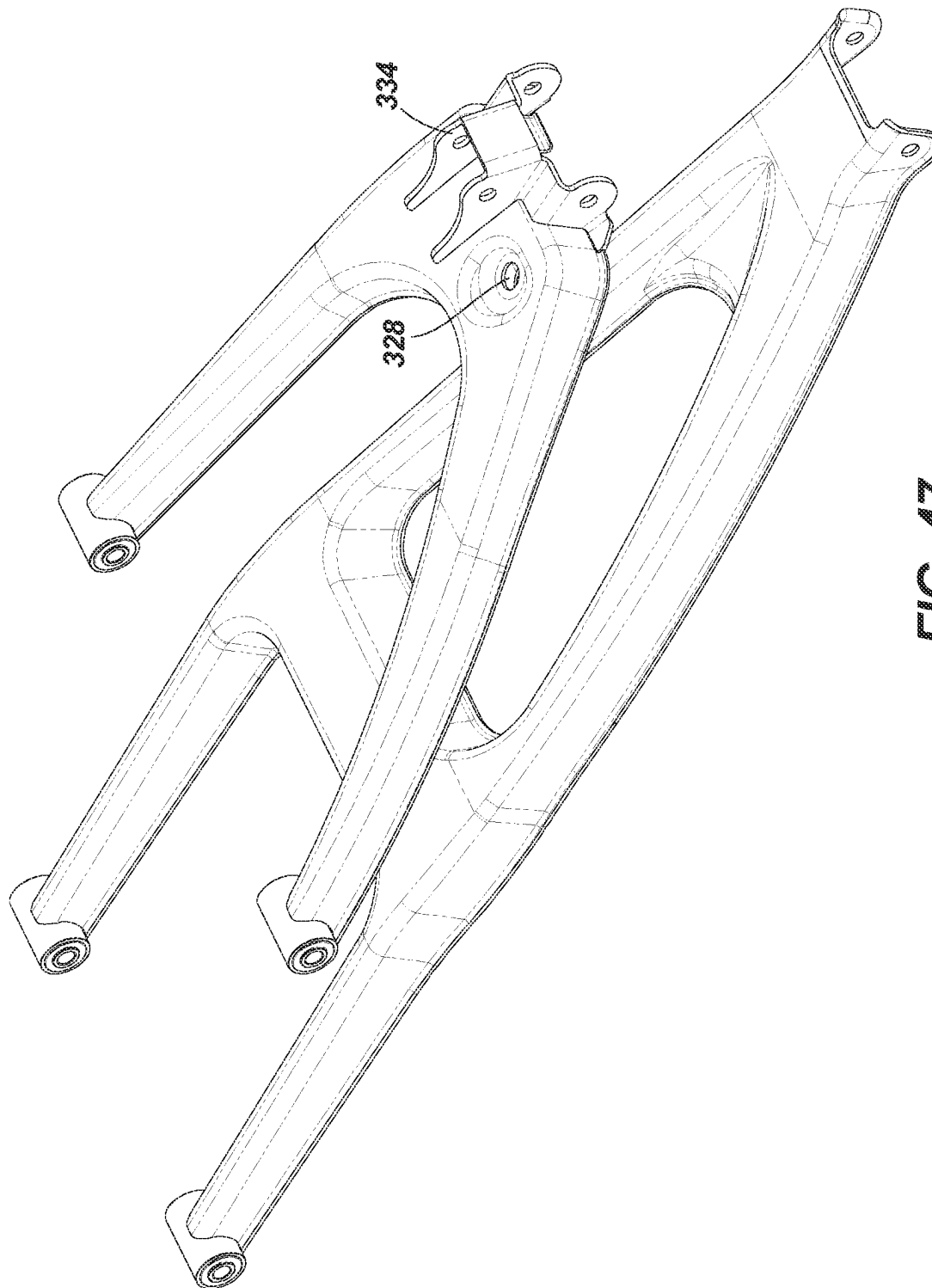


FIG. 47

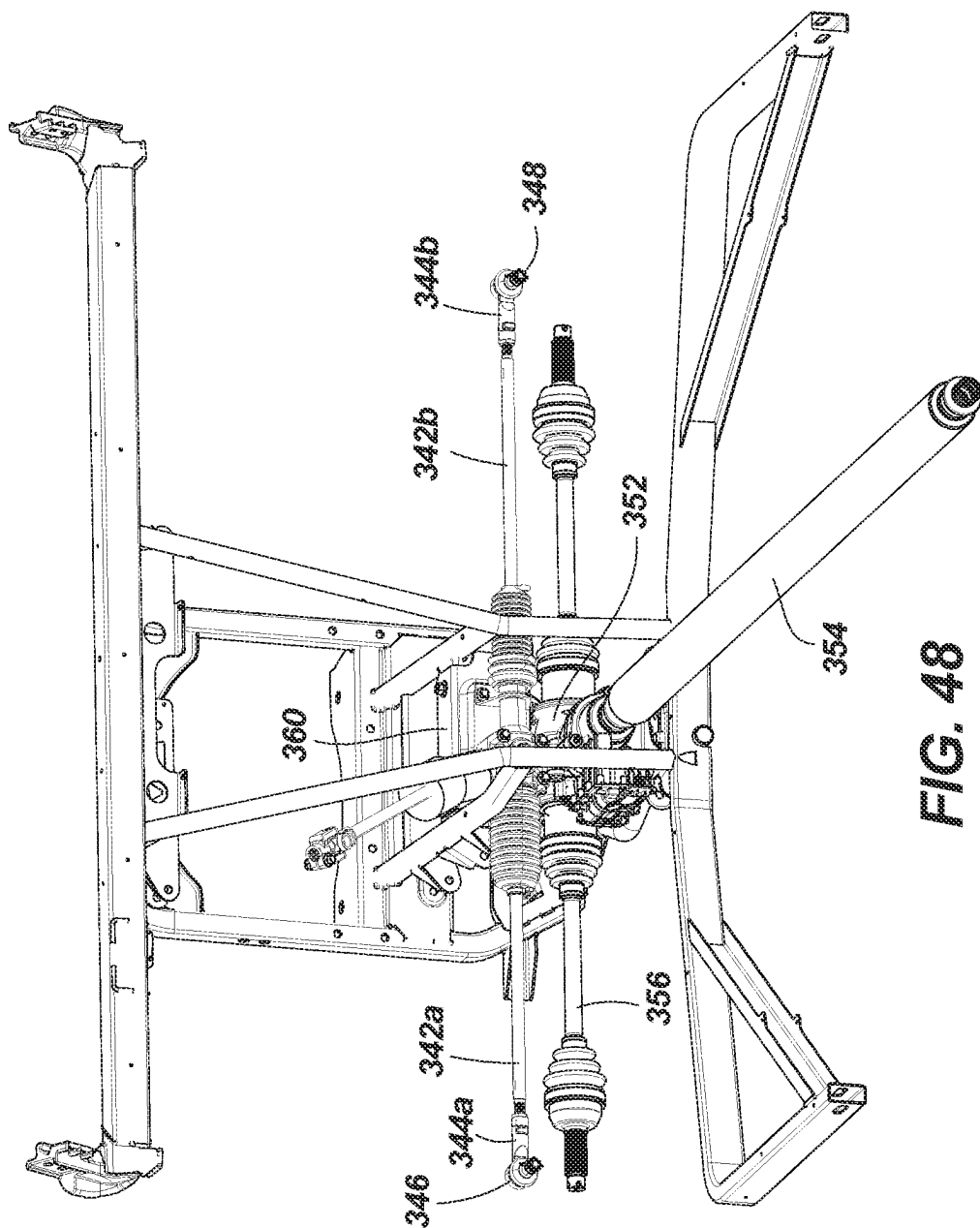


FIG. 48

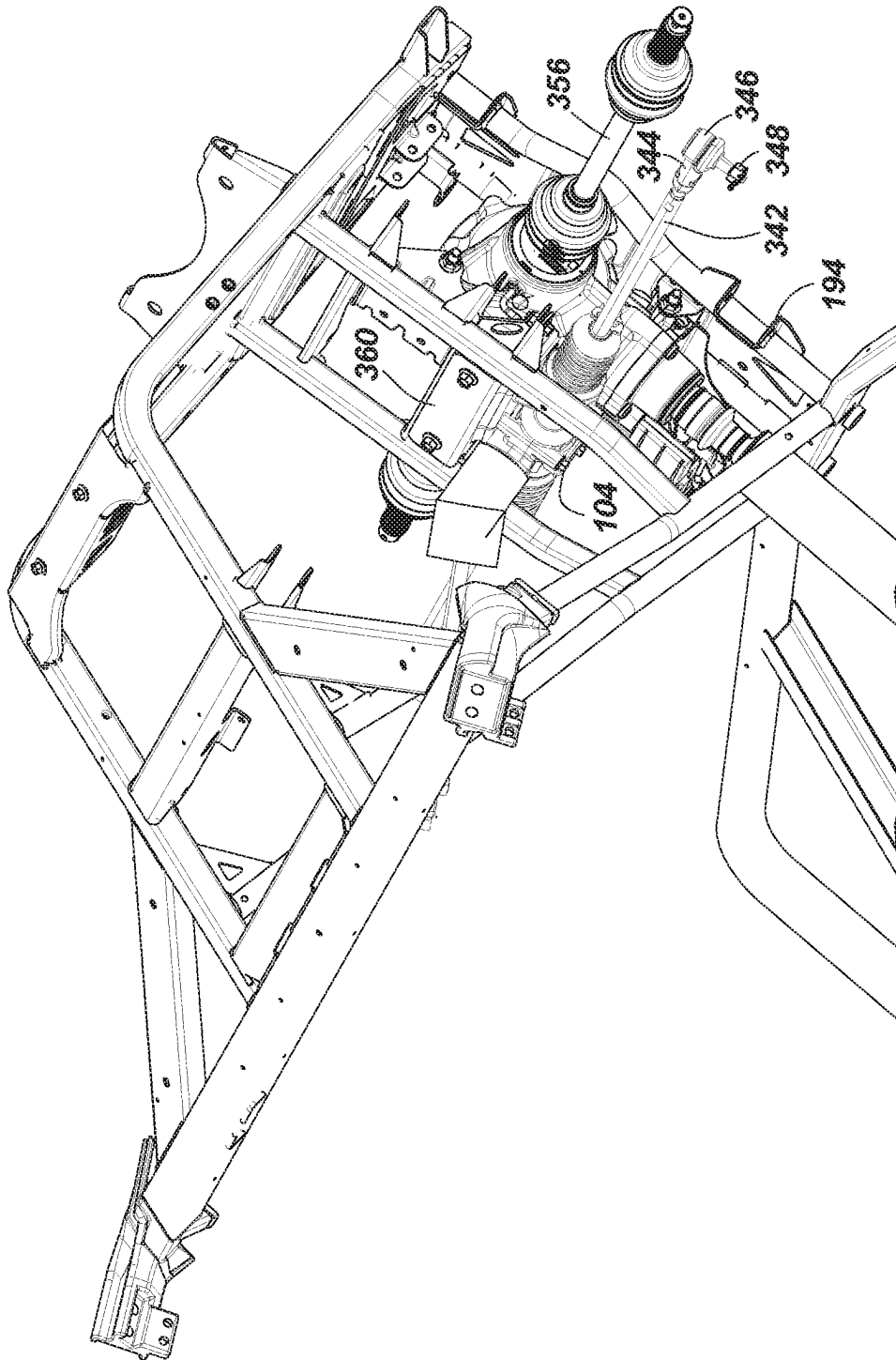


FIG. 49

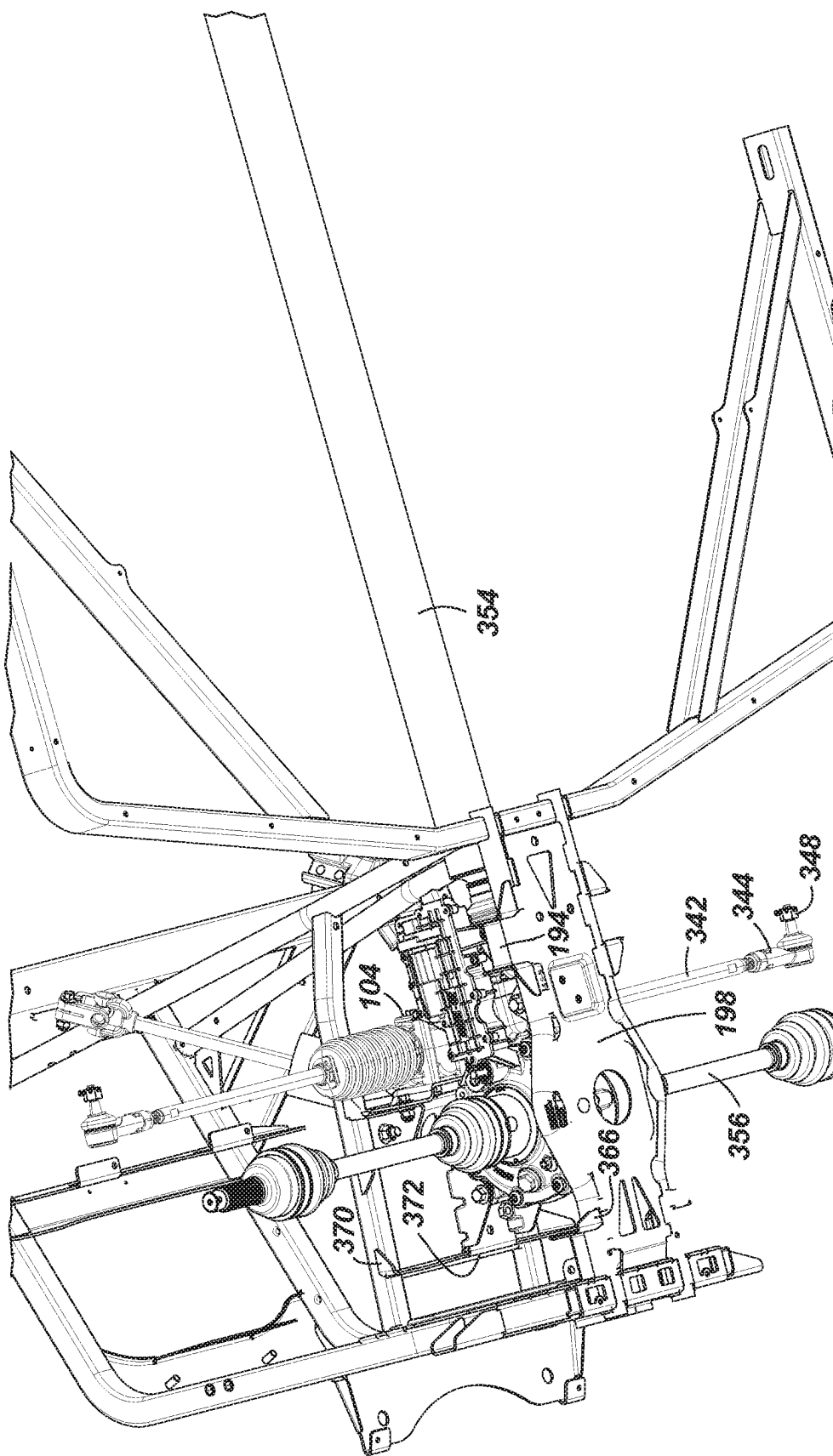


FIG. 50

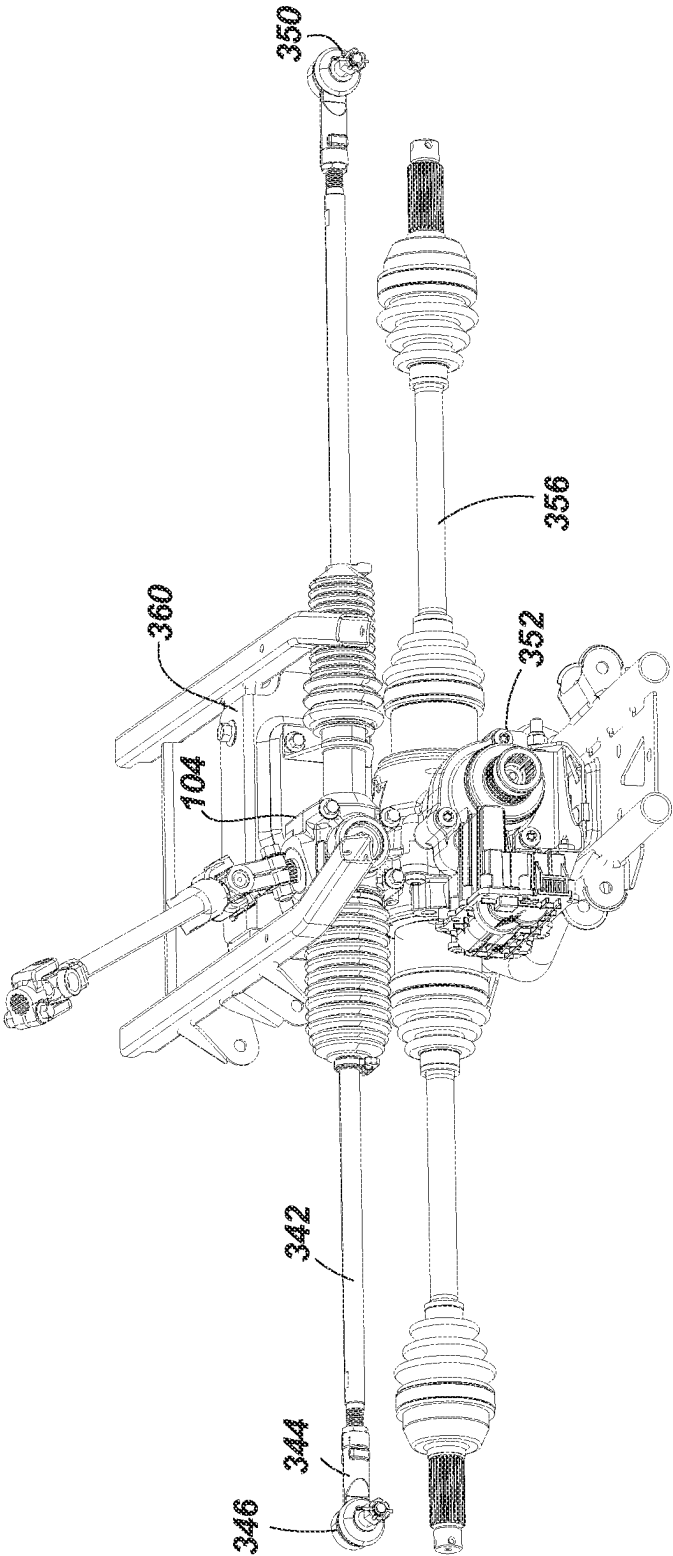


FIG. 51

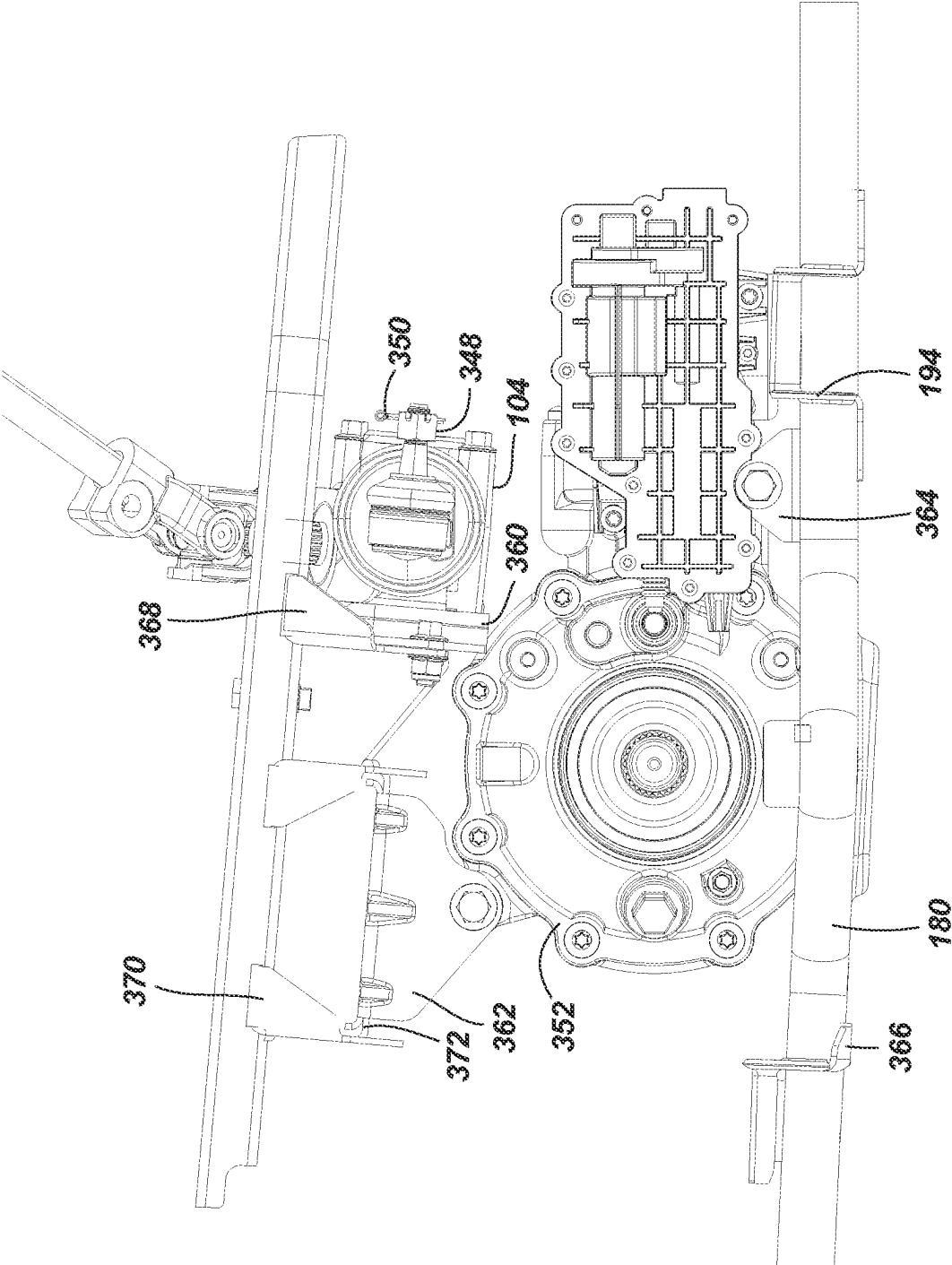


FIG. 52

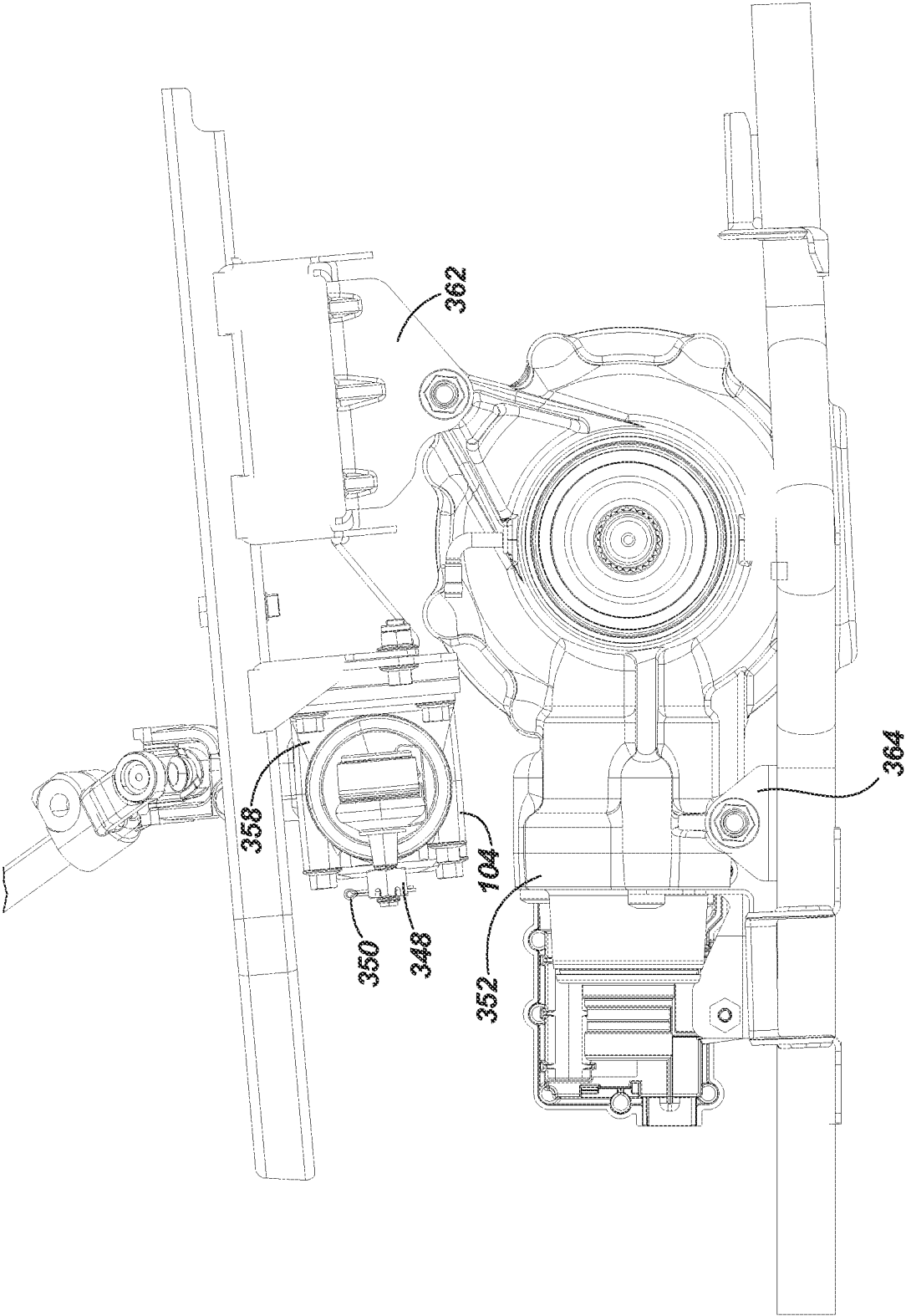


FIG. 53

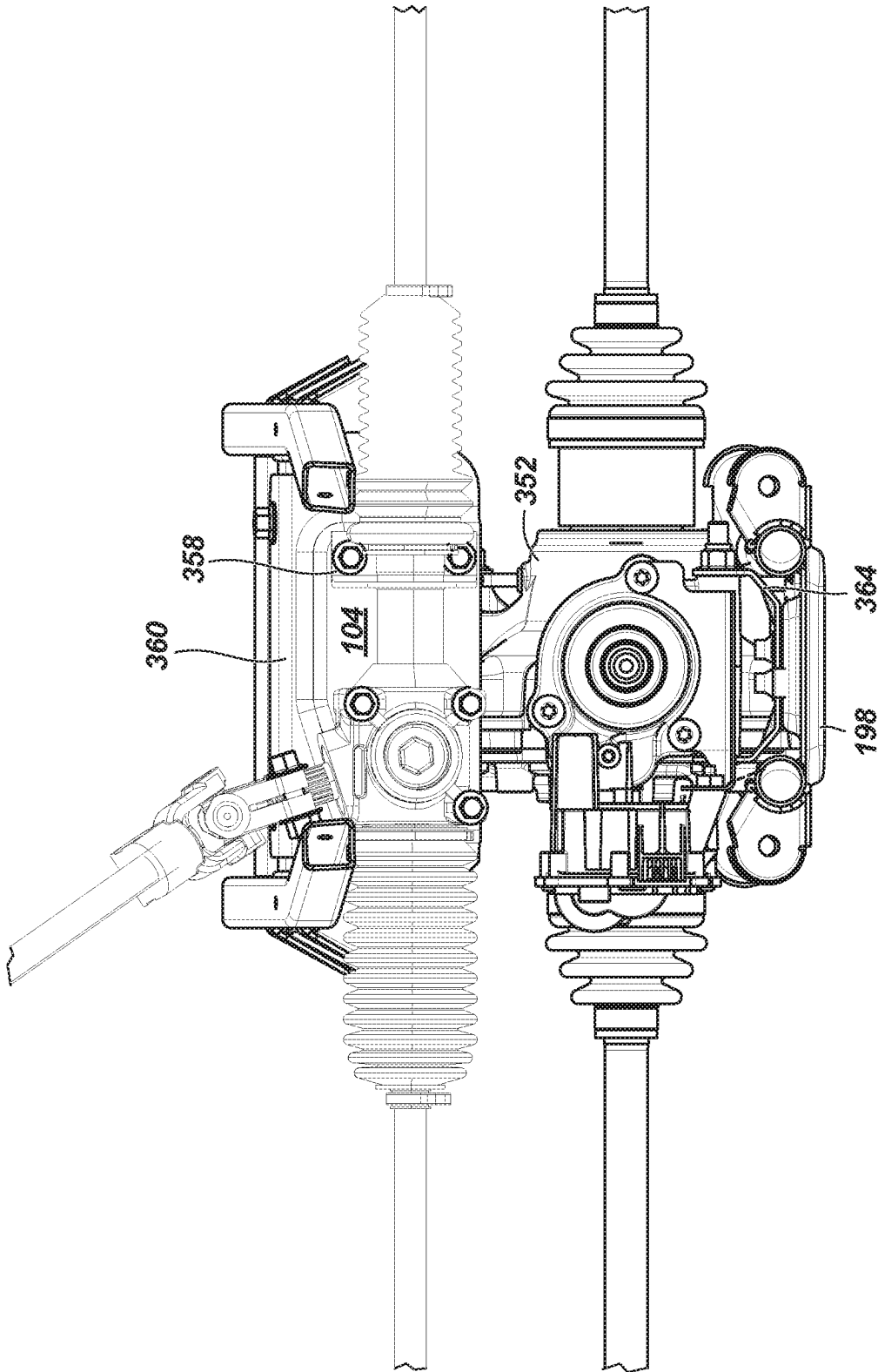
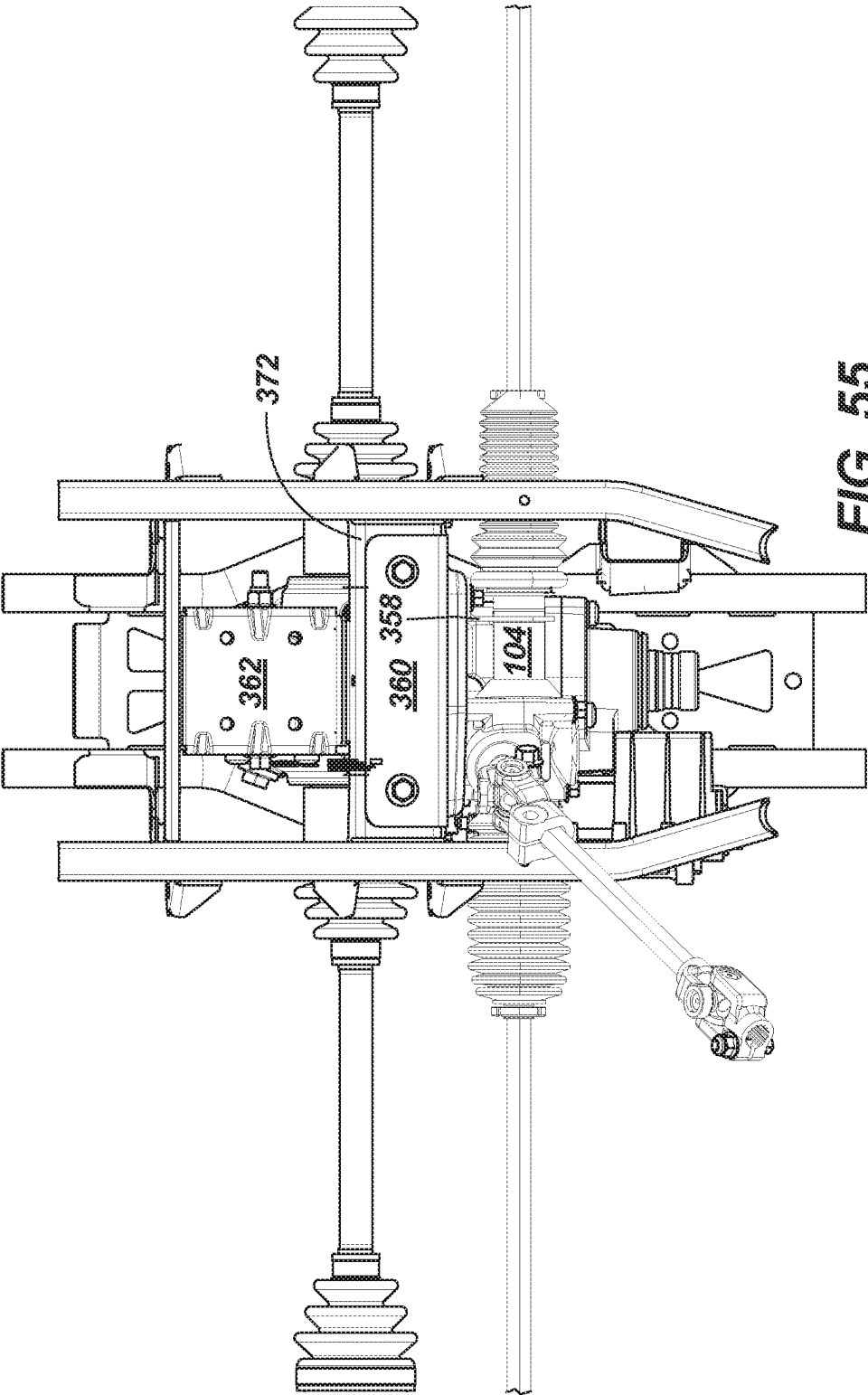


FIG. 54



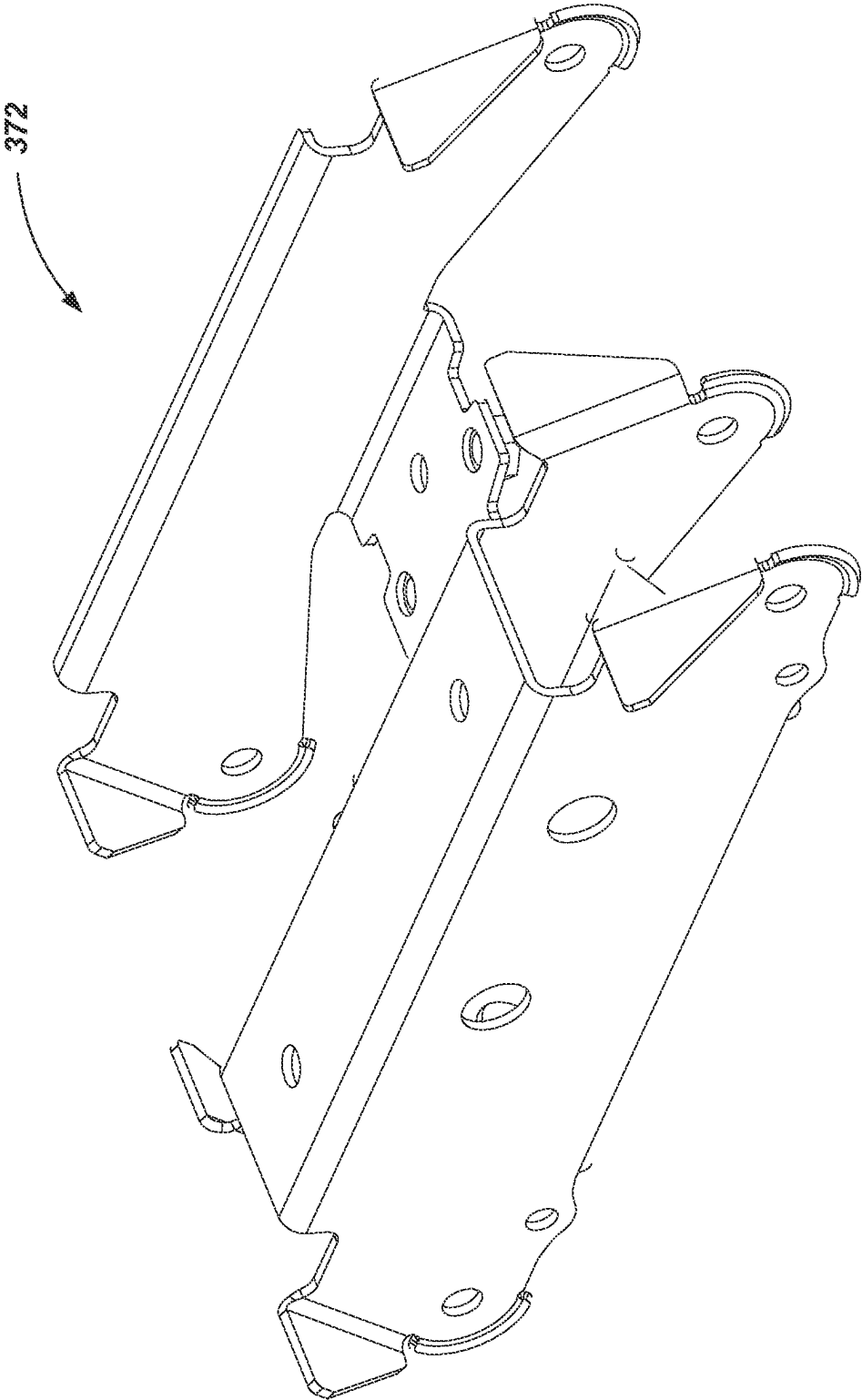
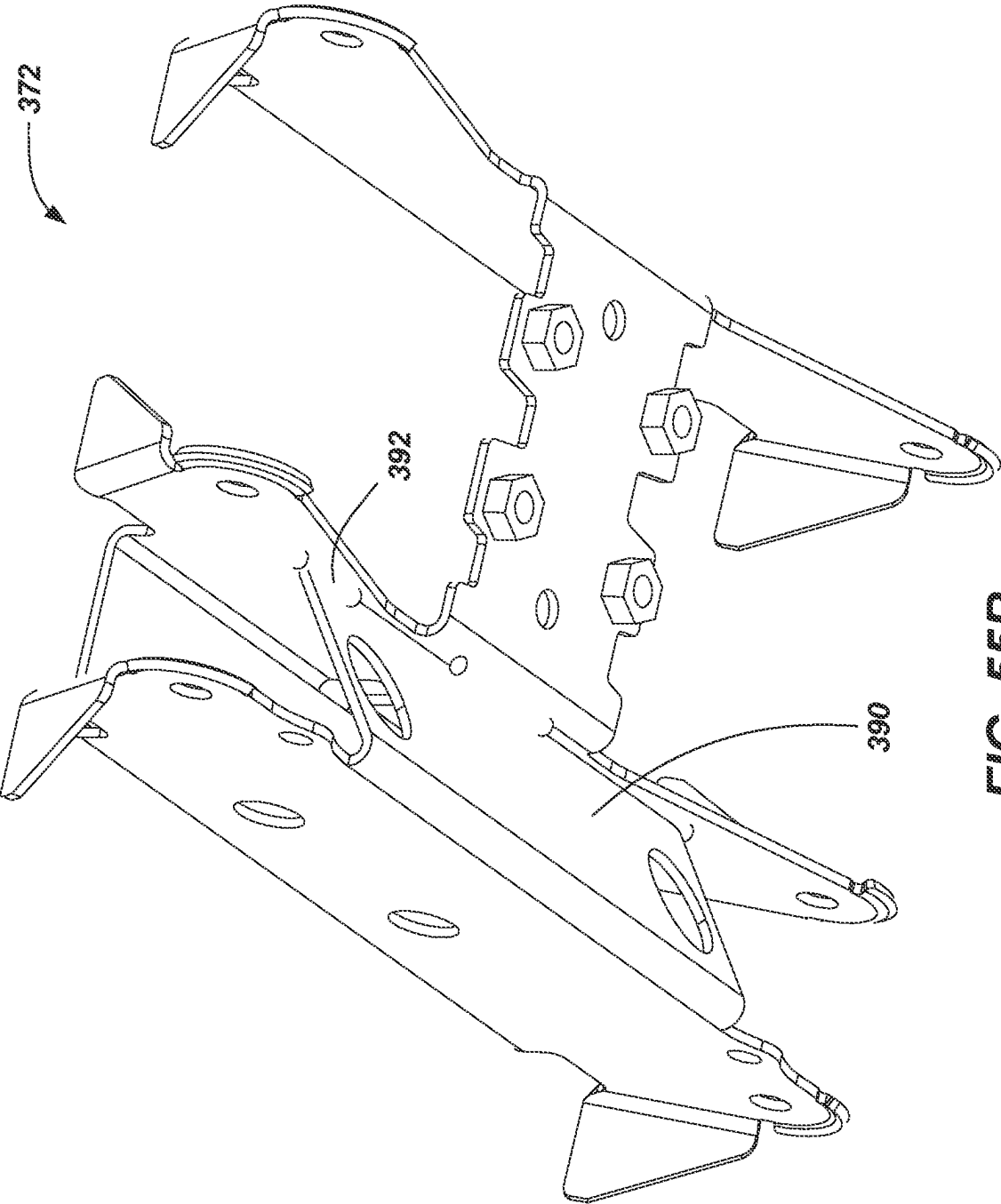
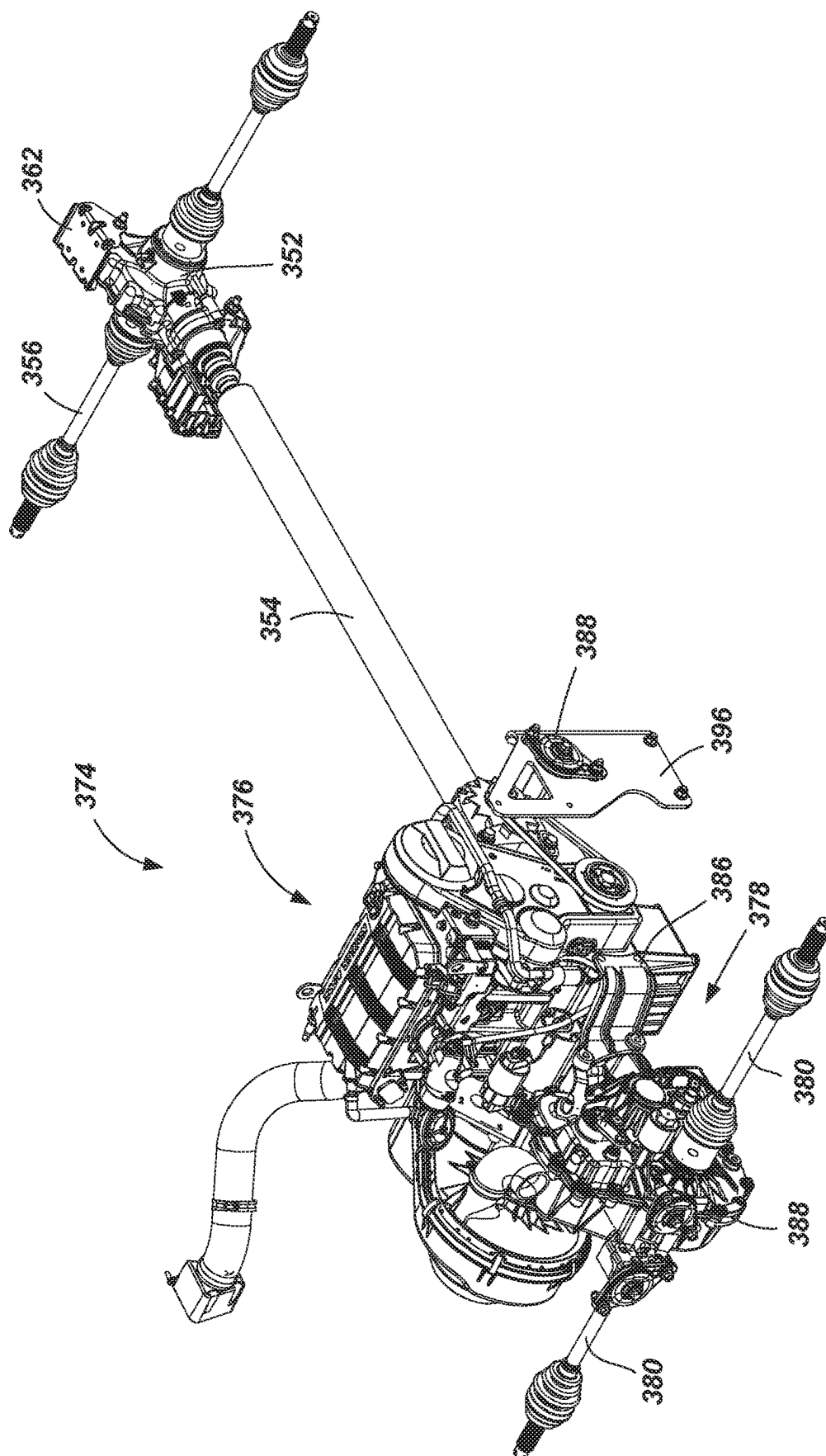


FIG. 55A





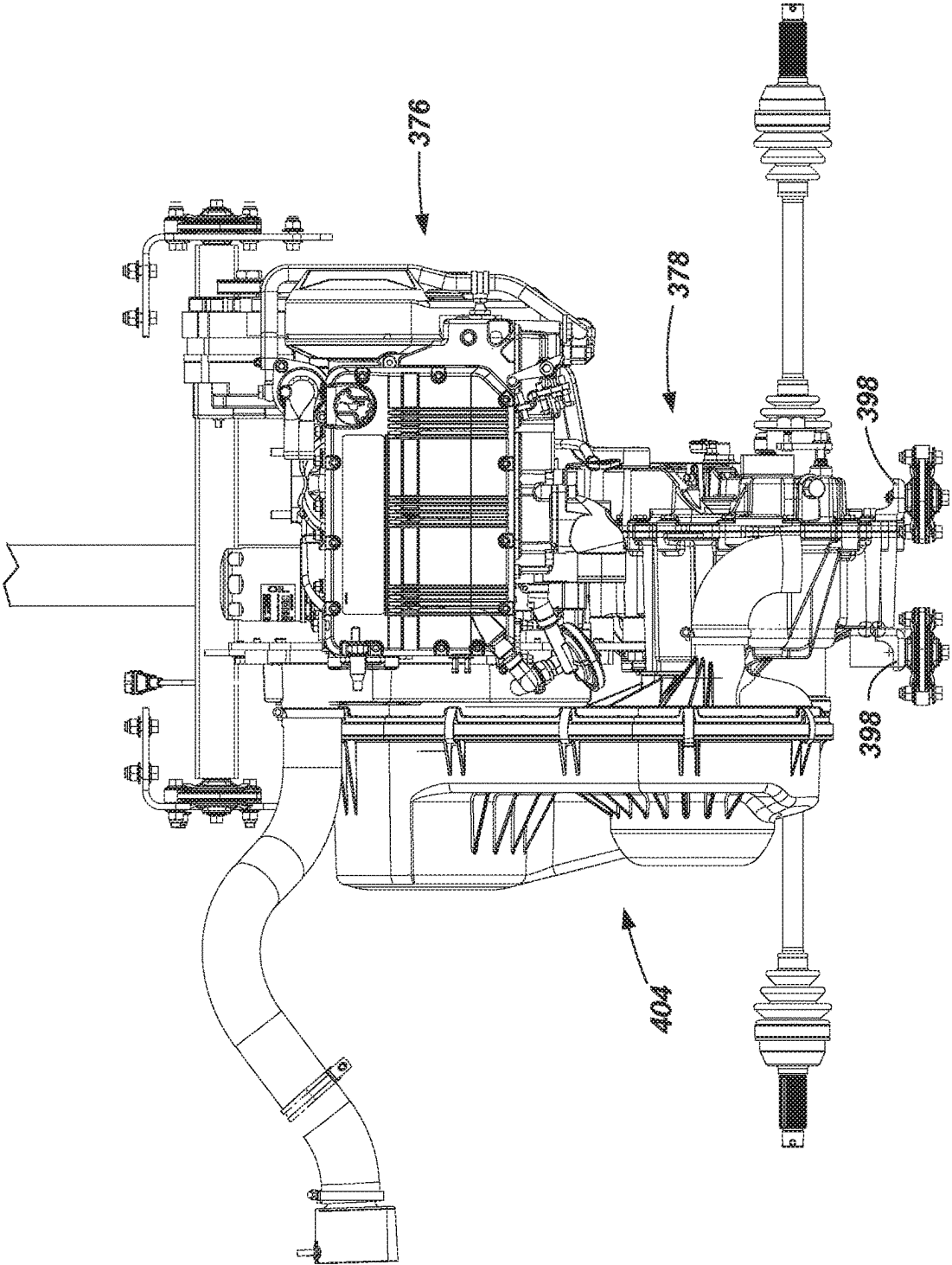


FIG. 57

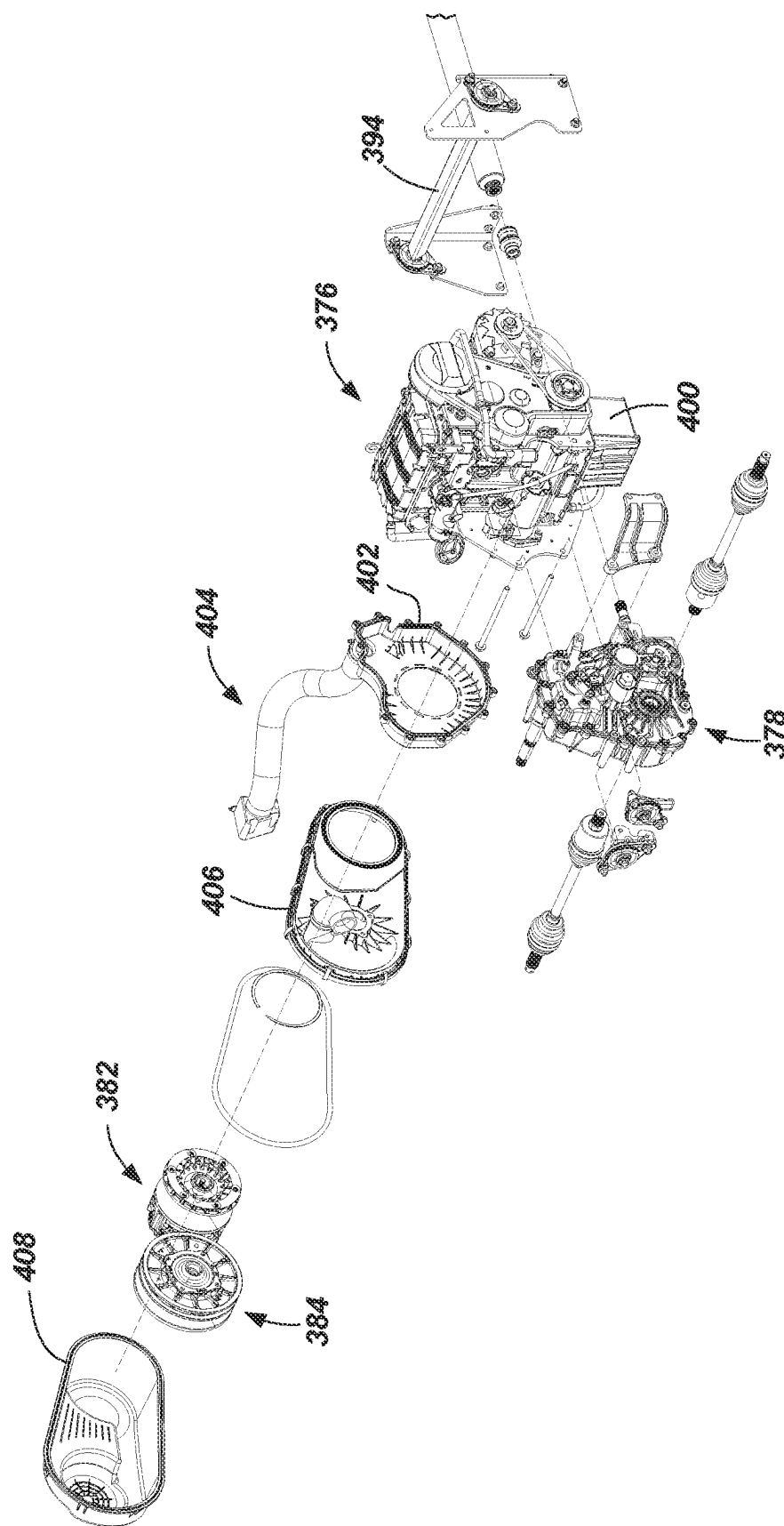
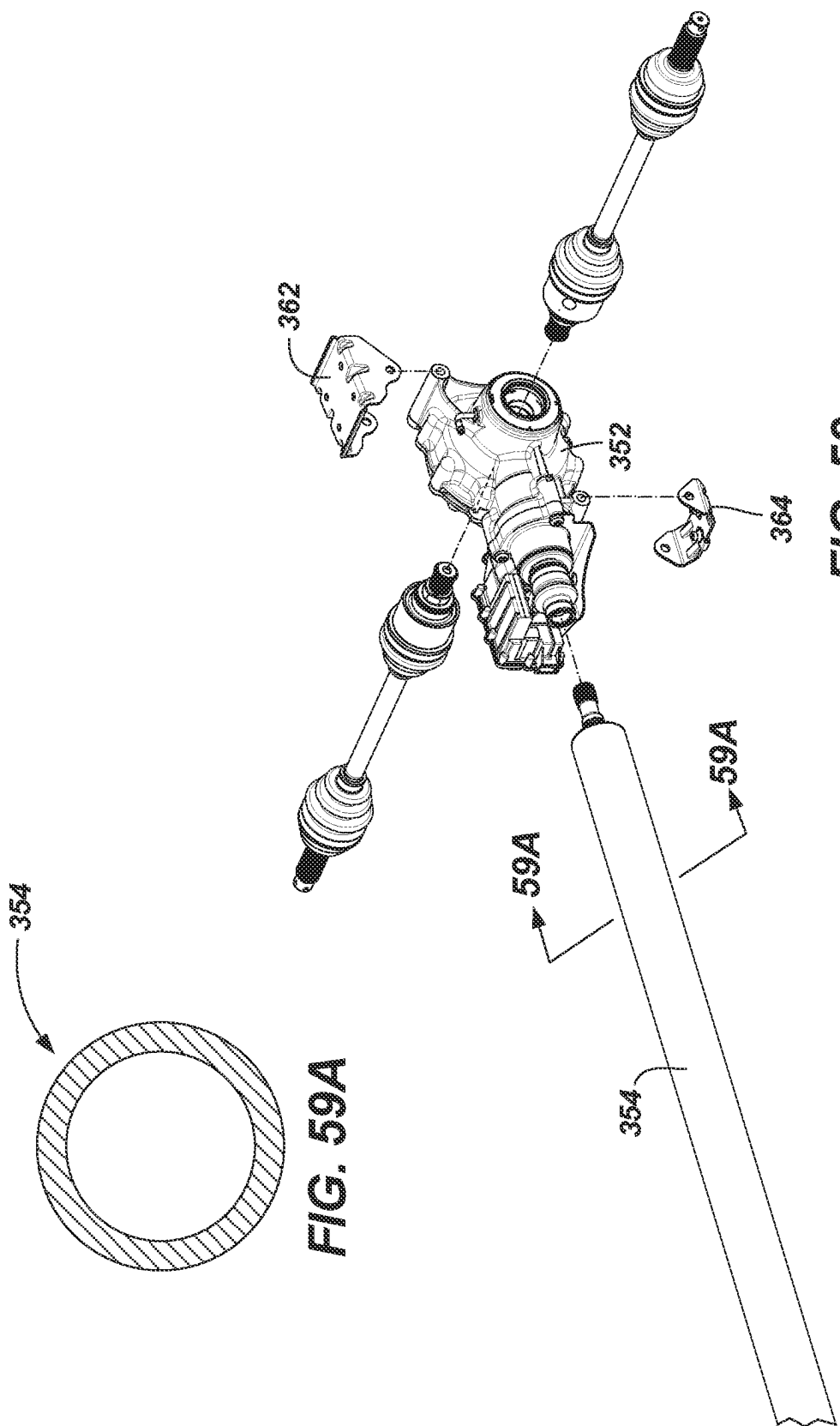
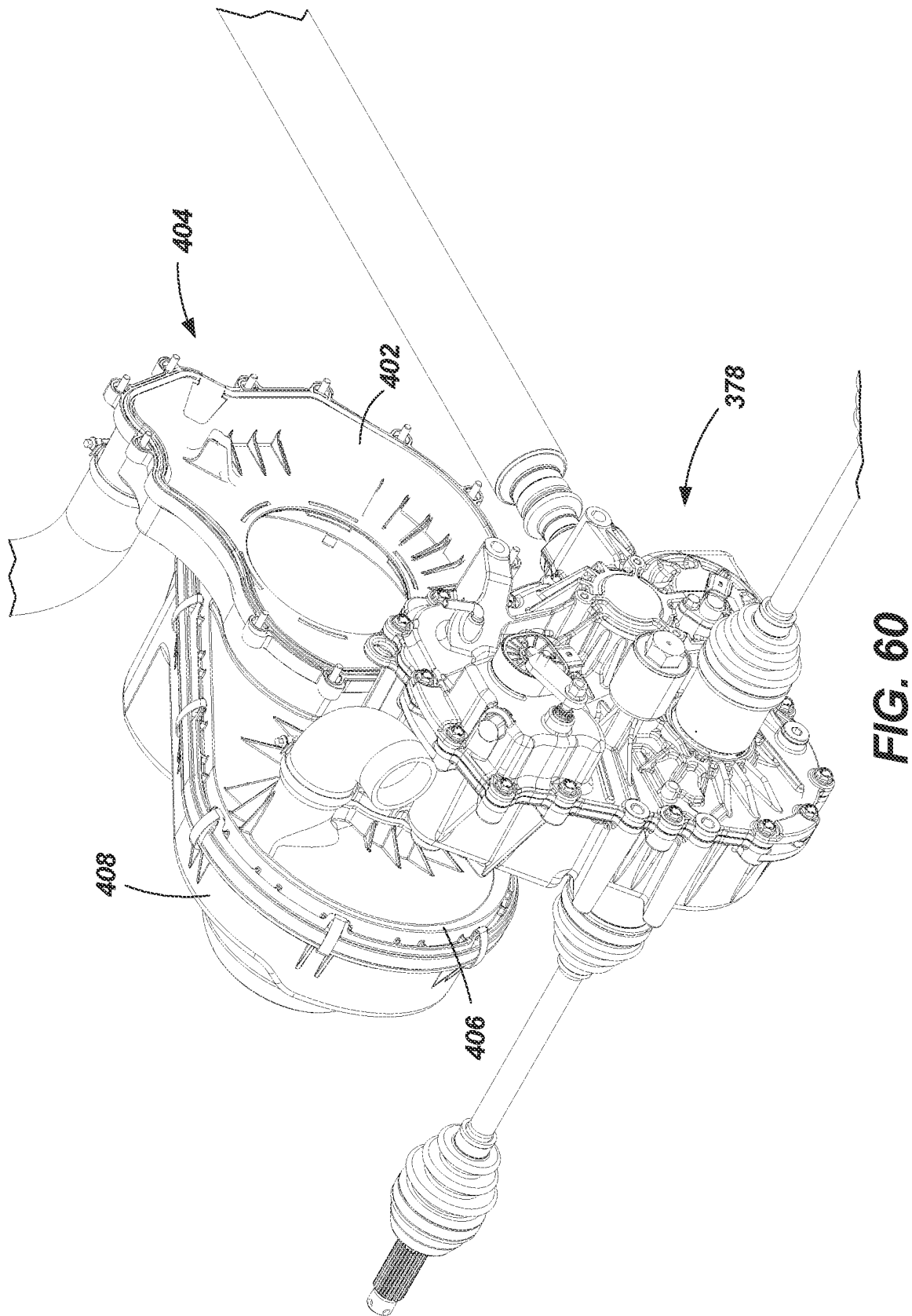
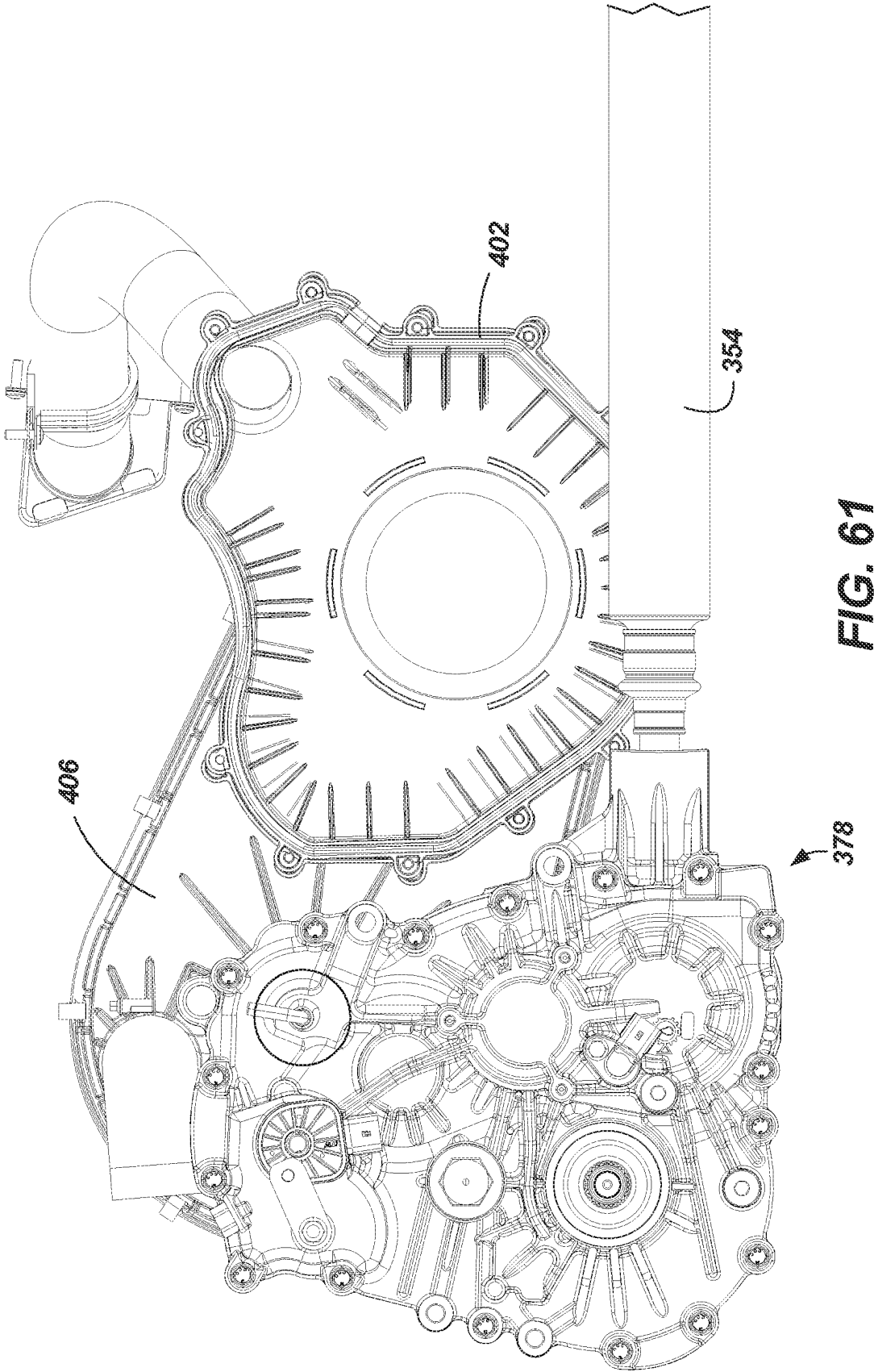


FIG. 58







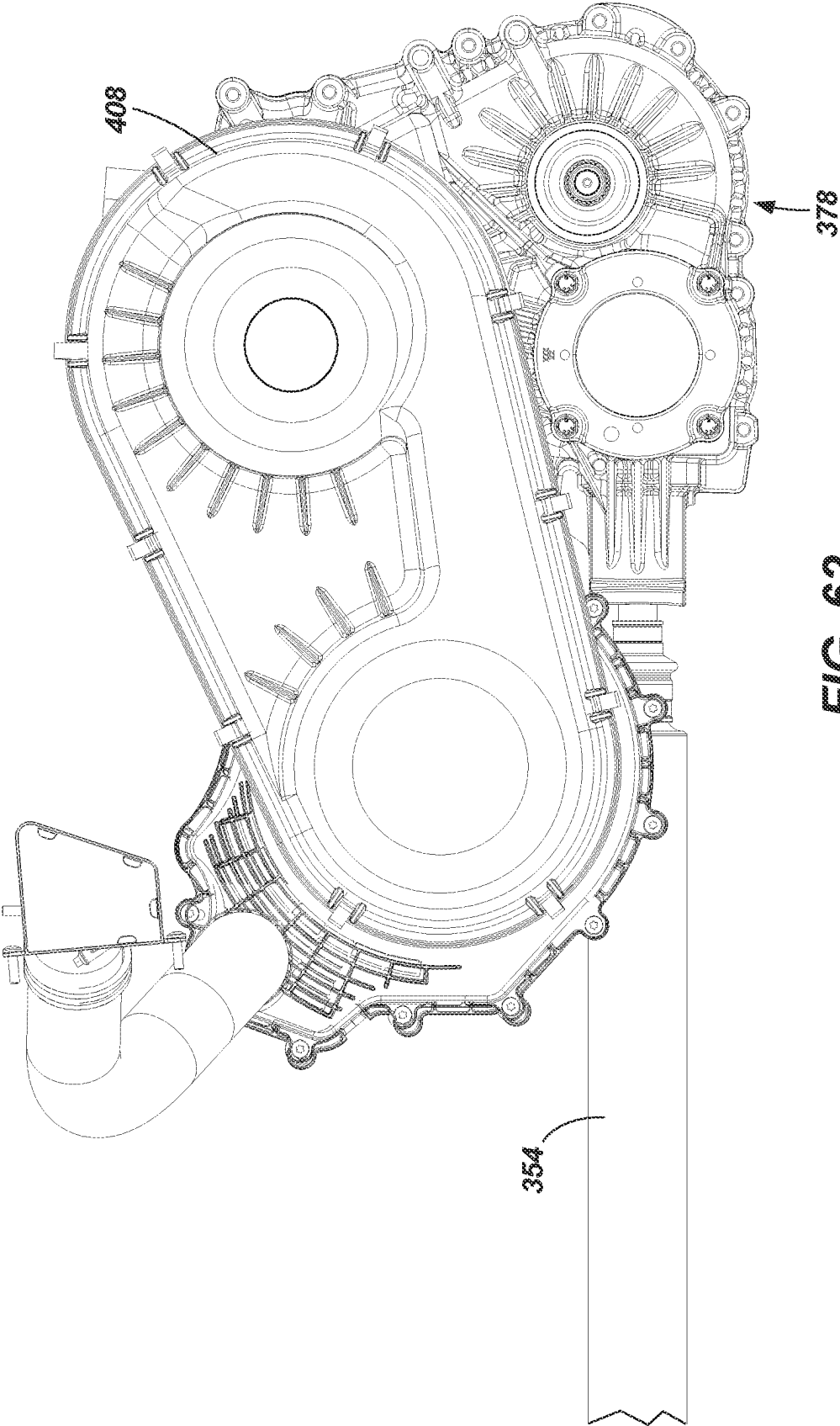


FIG. 62

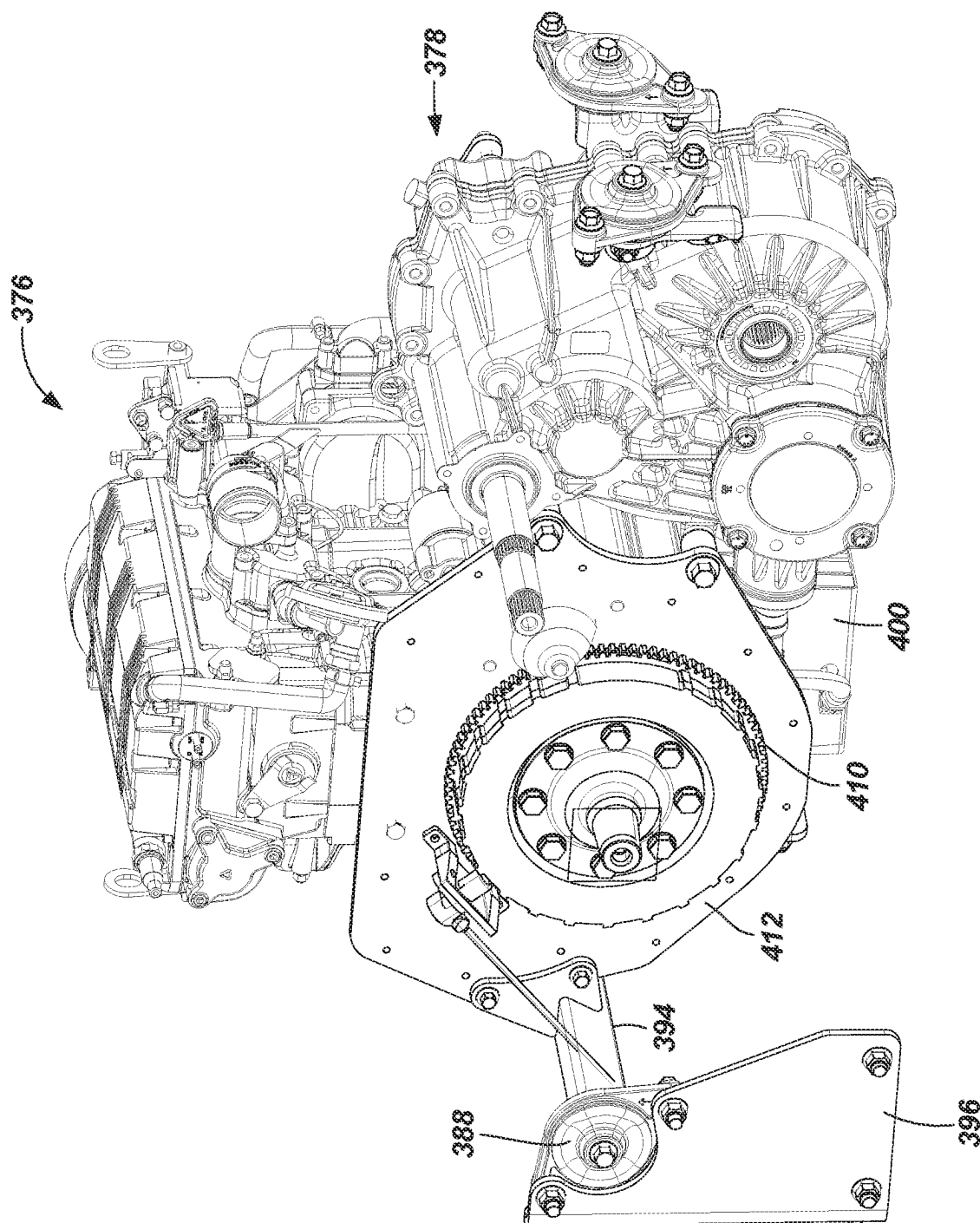


FIG. 63

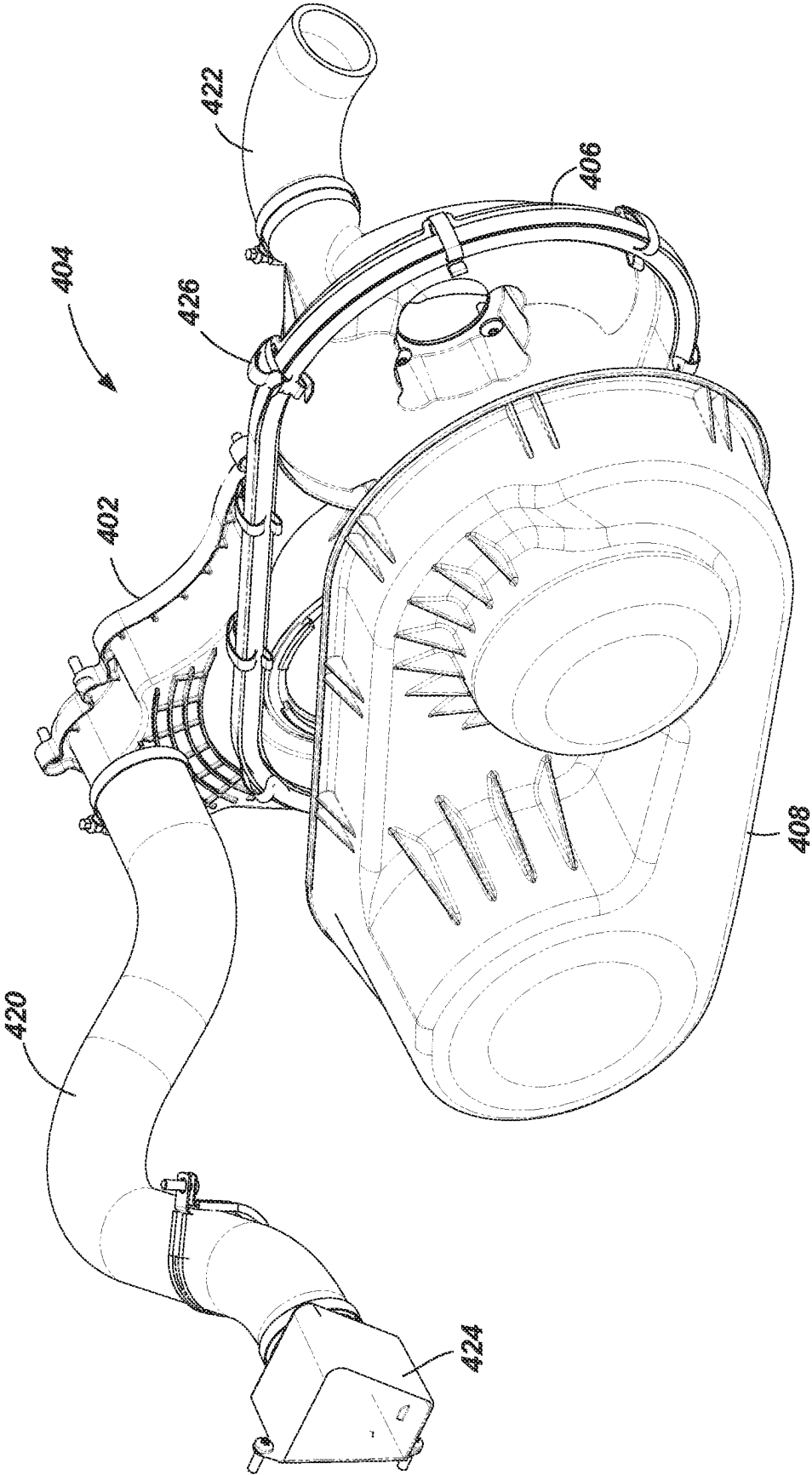


FIG. 64

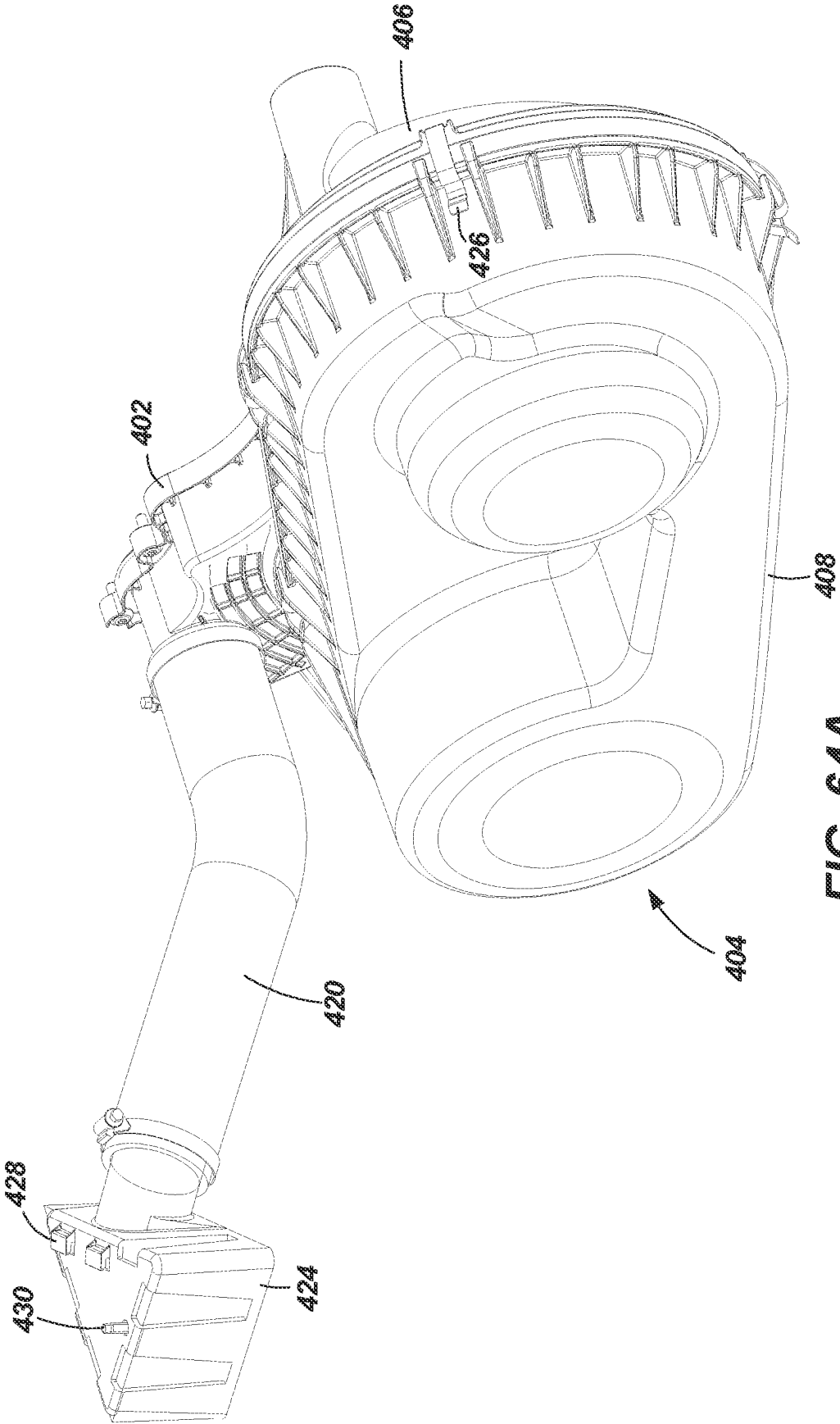


FIG. 64A

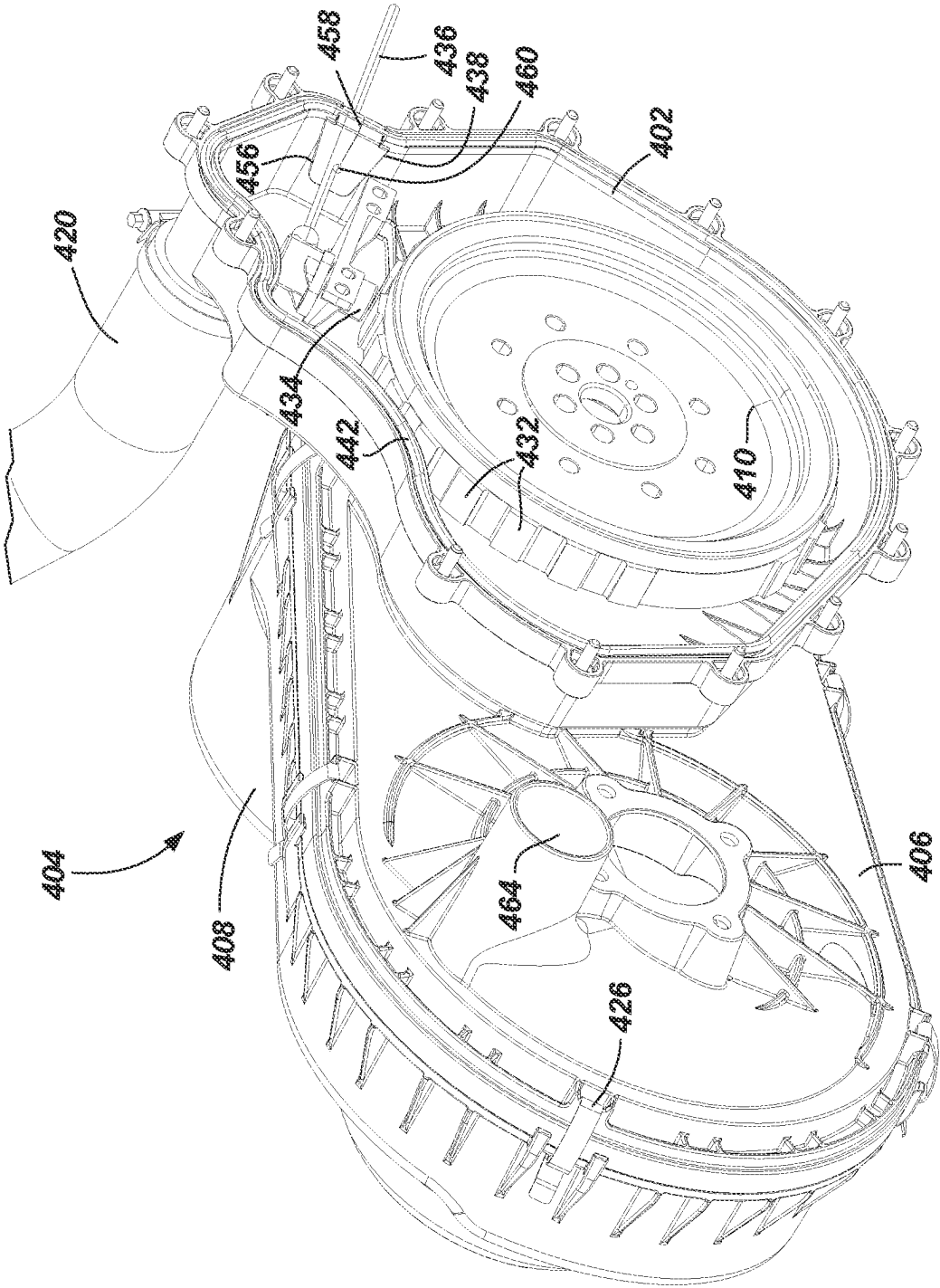


FIG. 64B

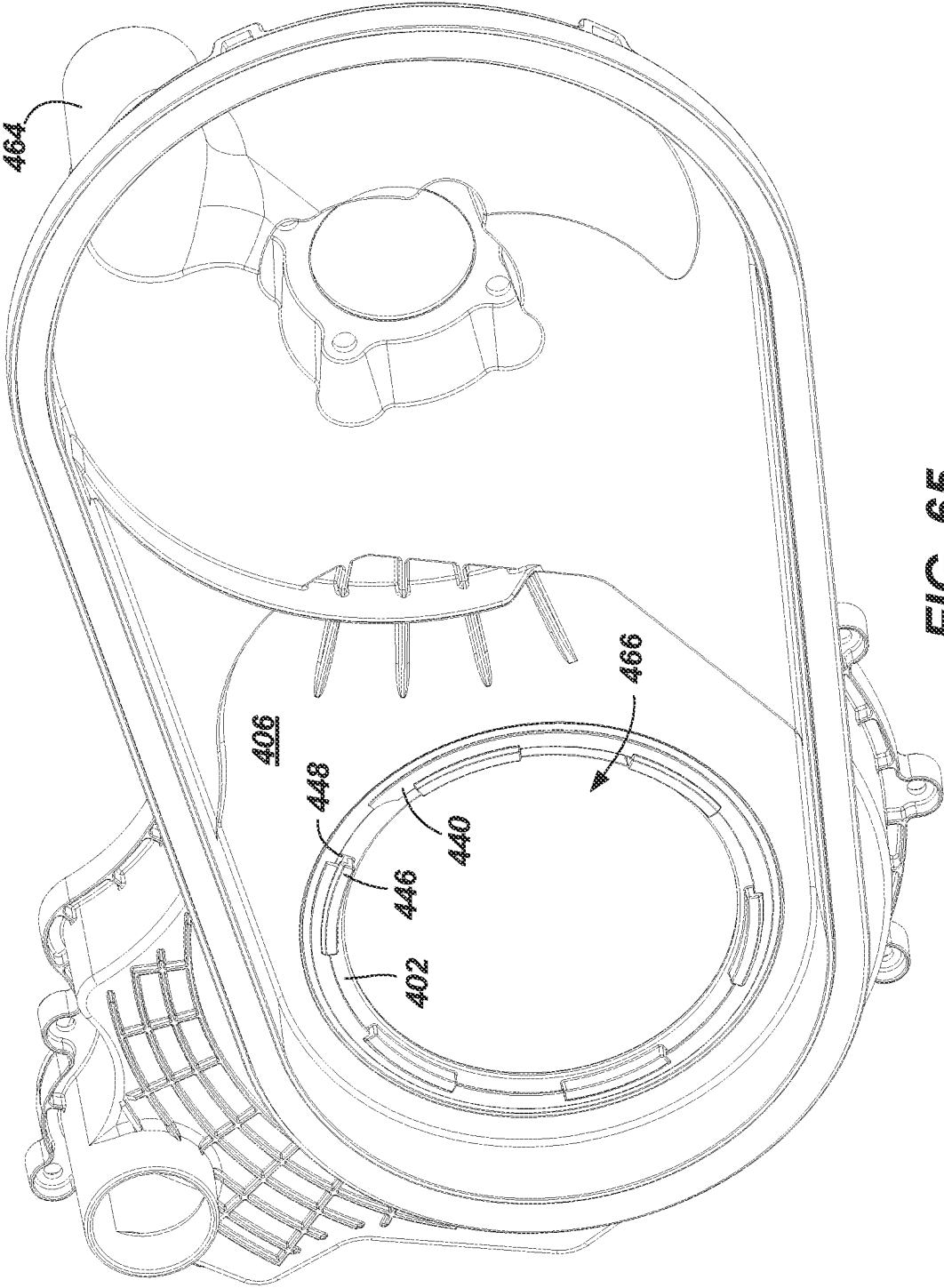


FIG. 65

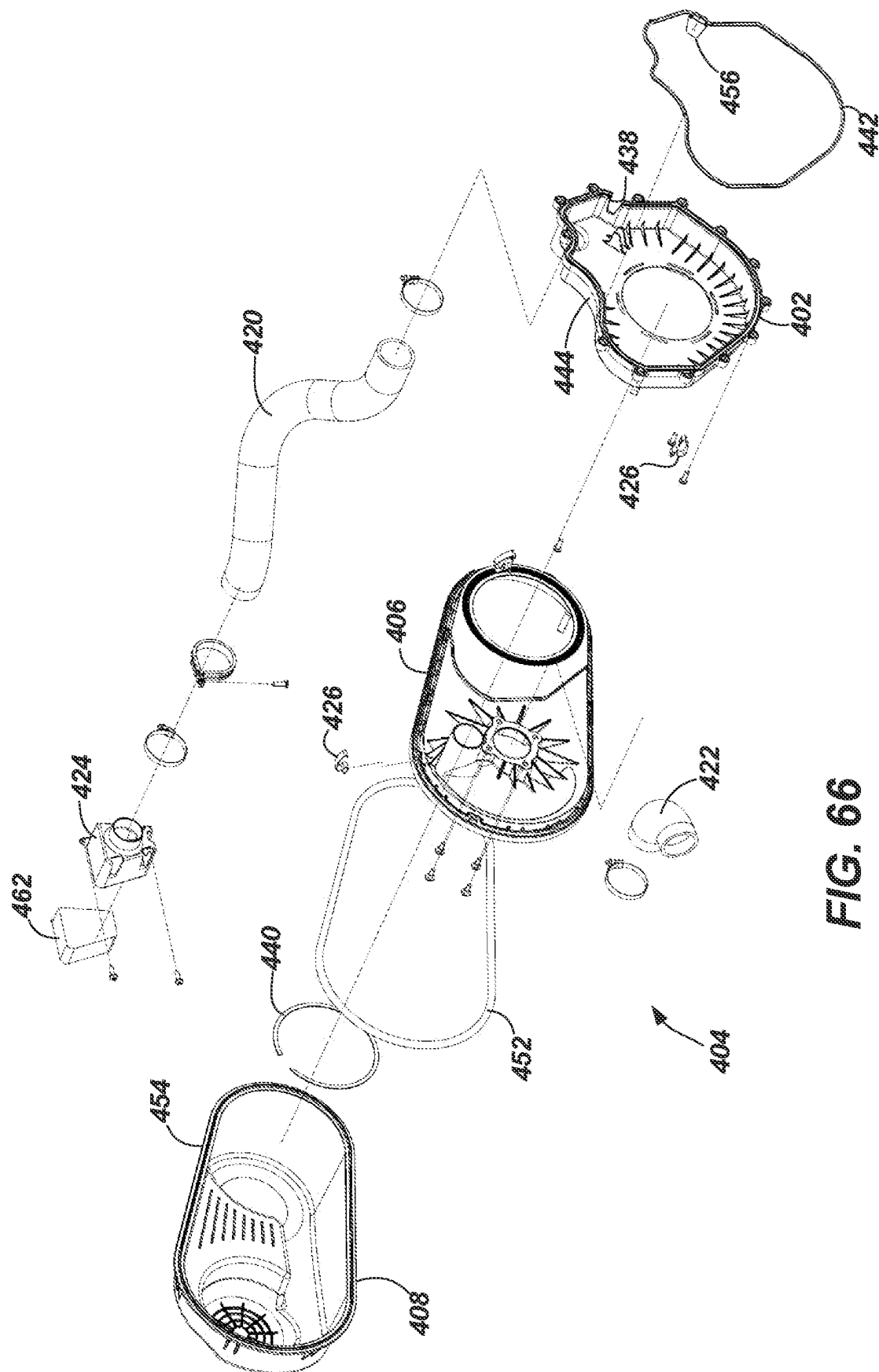


FIG. 66

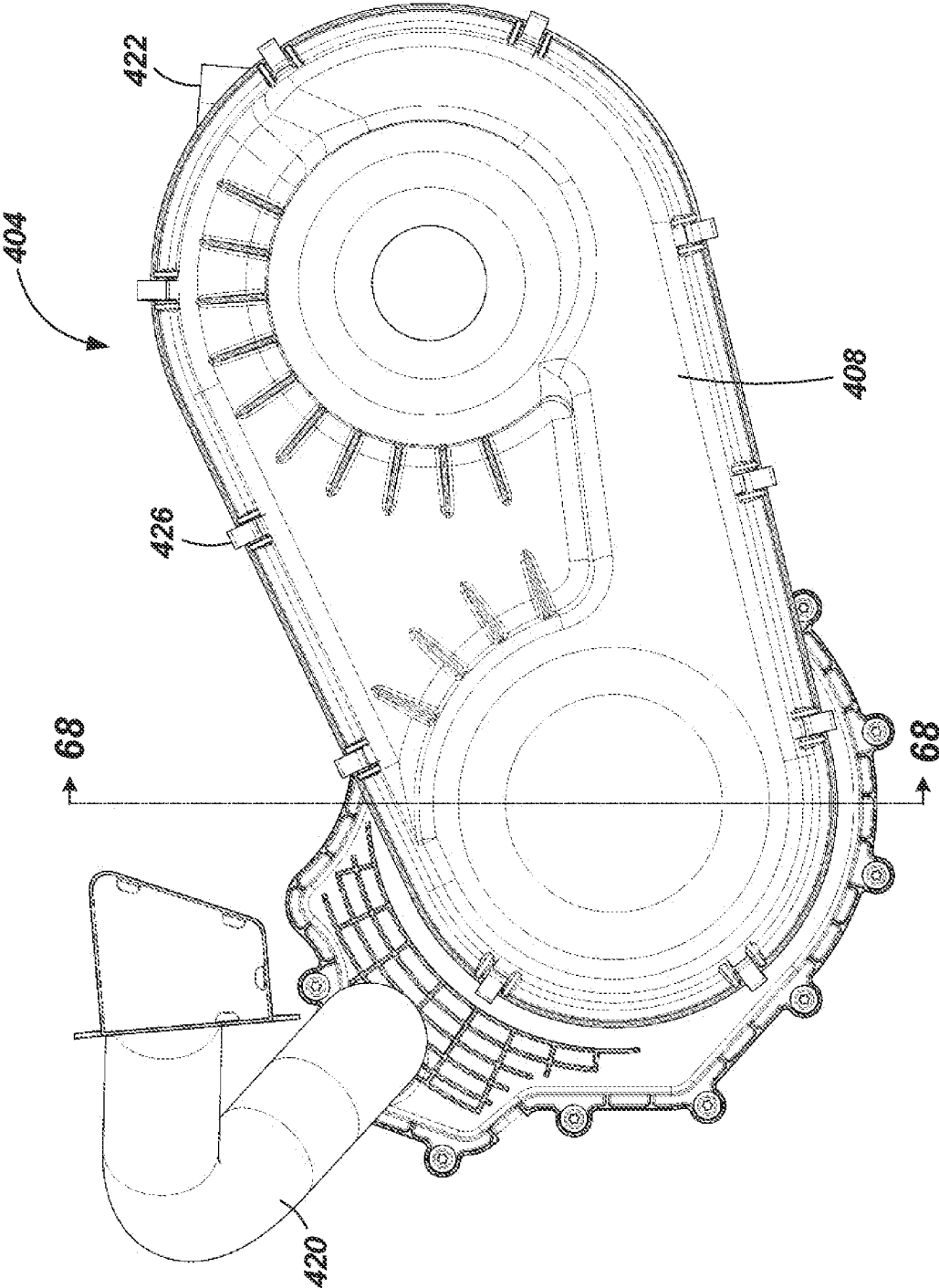


FIG. 67

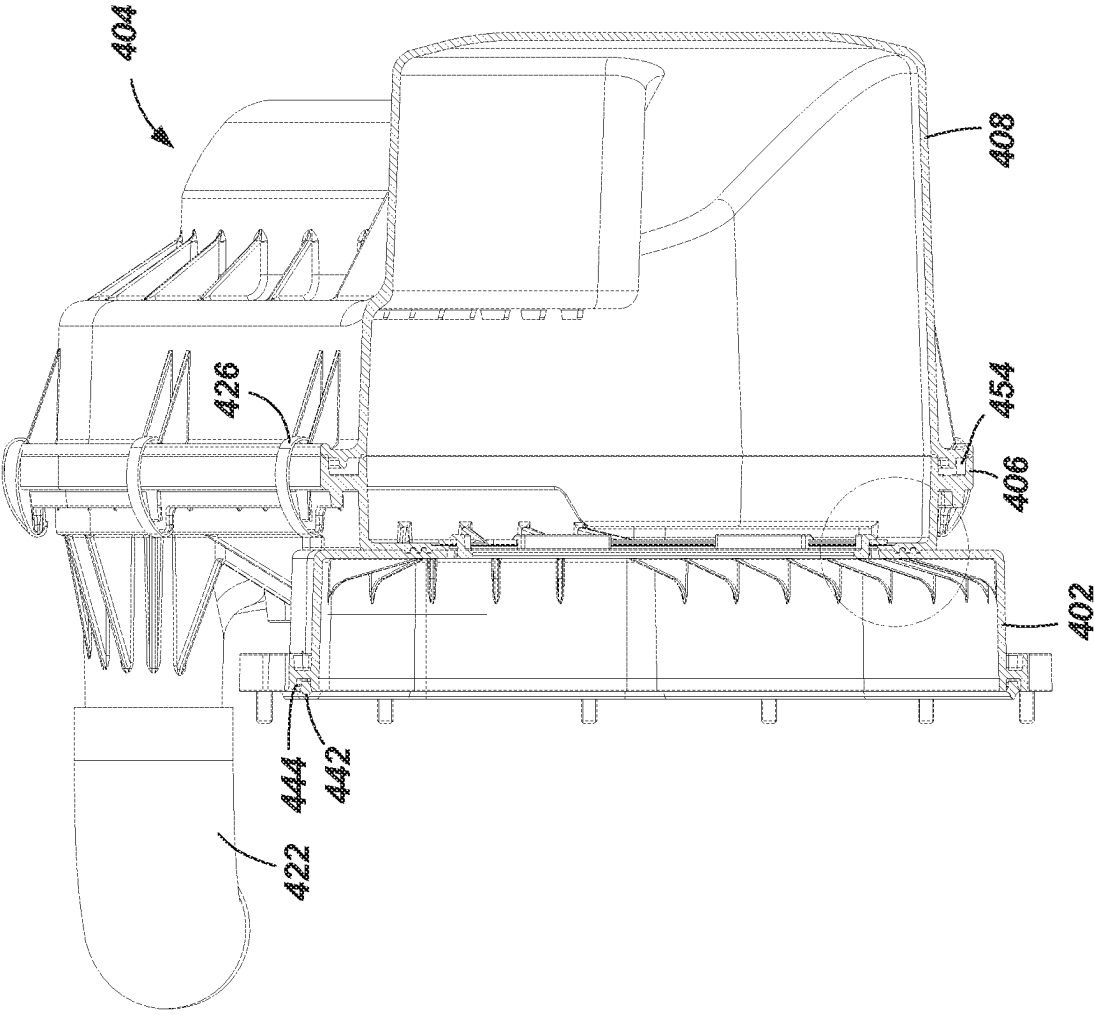
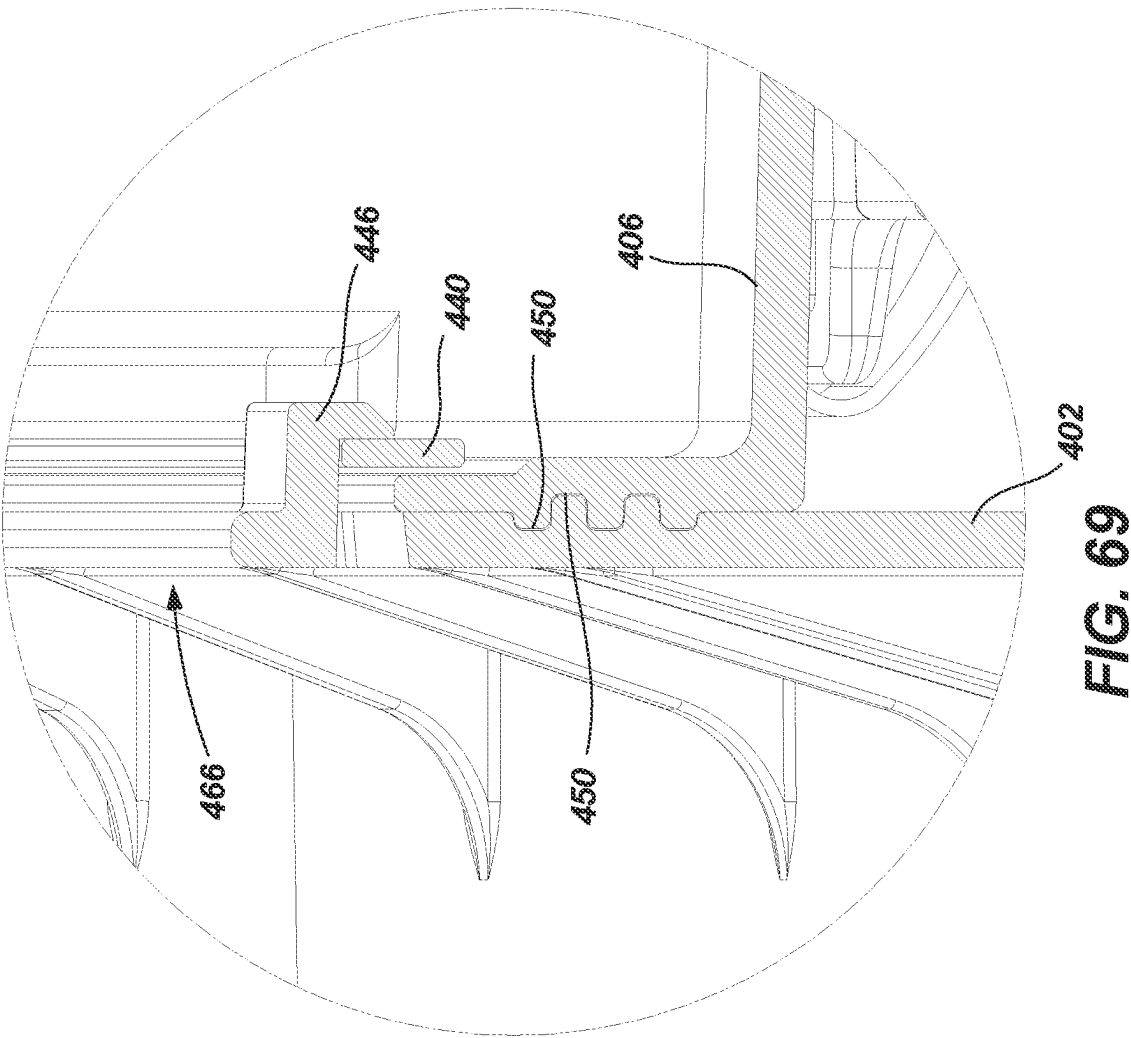


FIG. 68



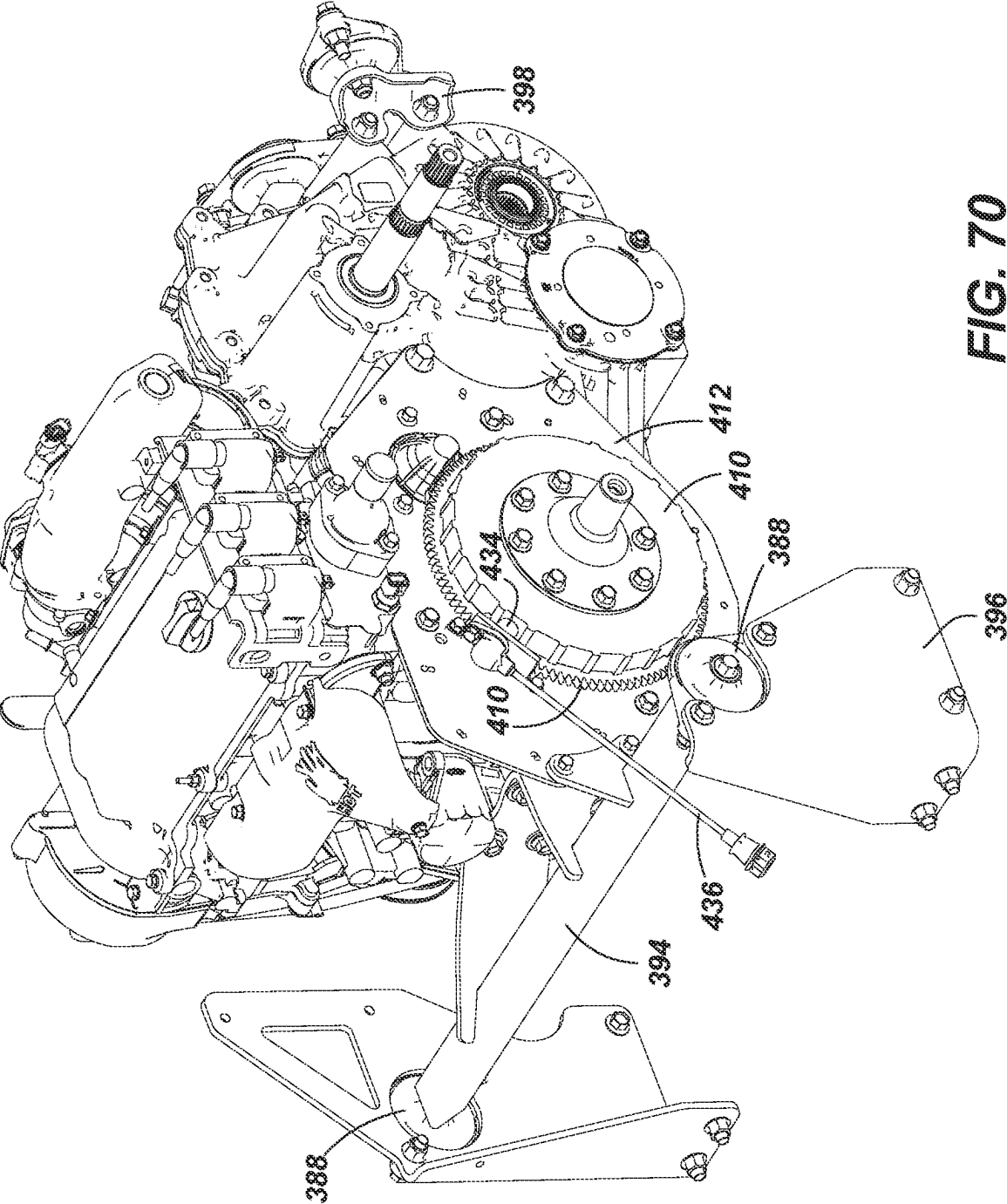


FIG. 70

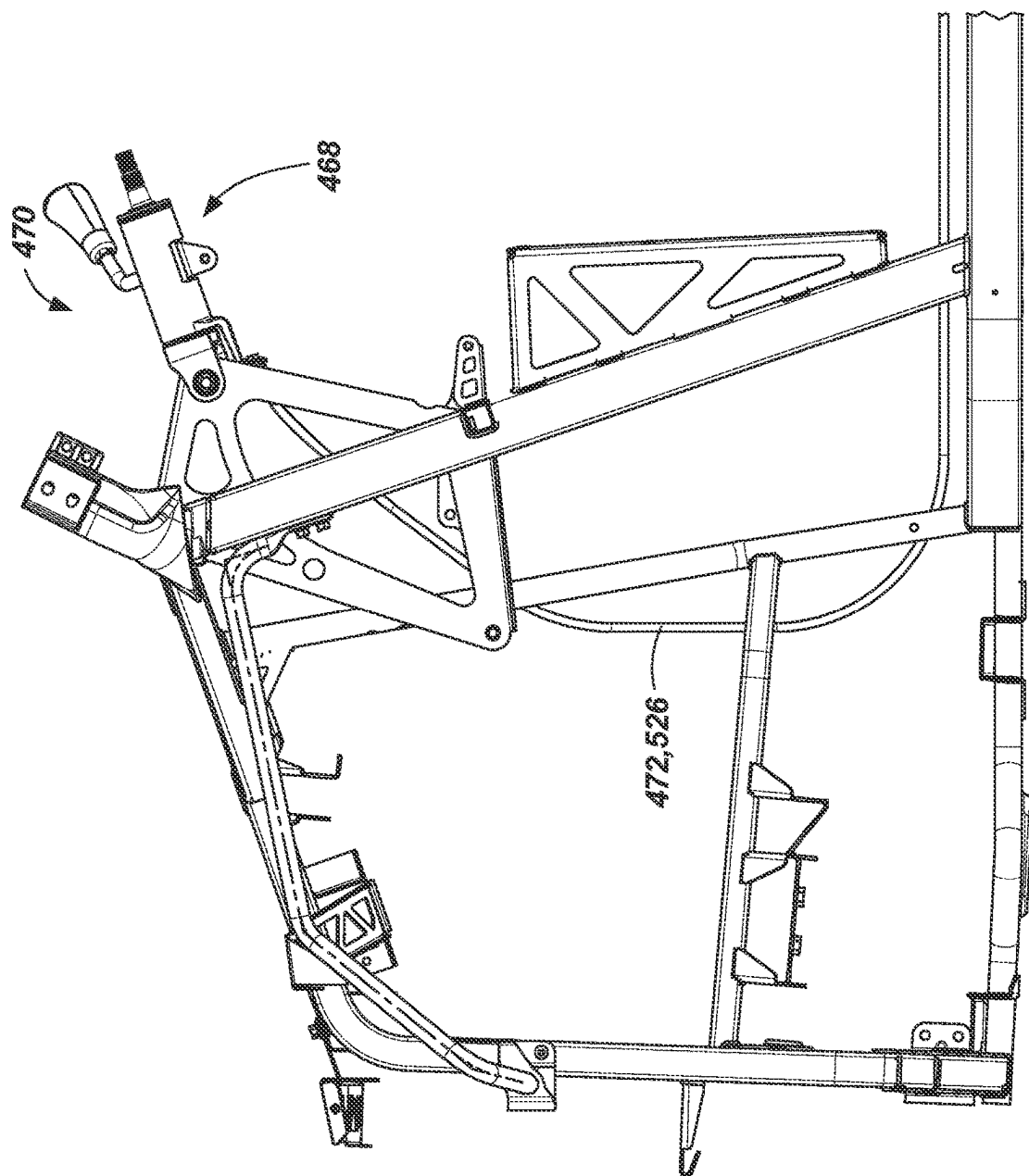
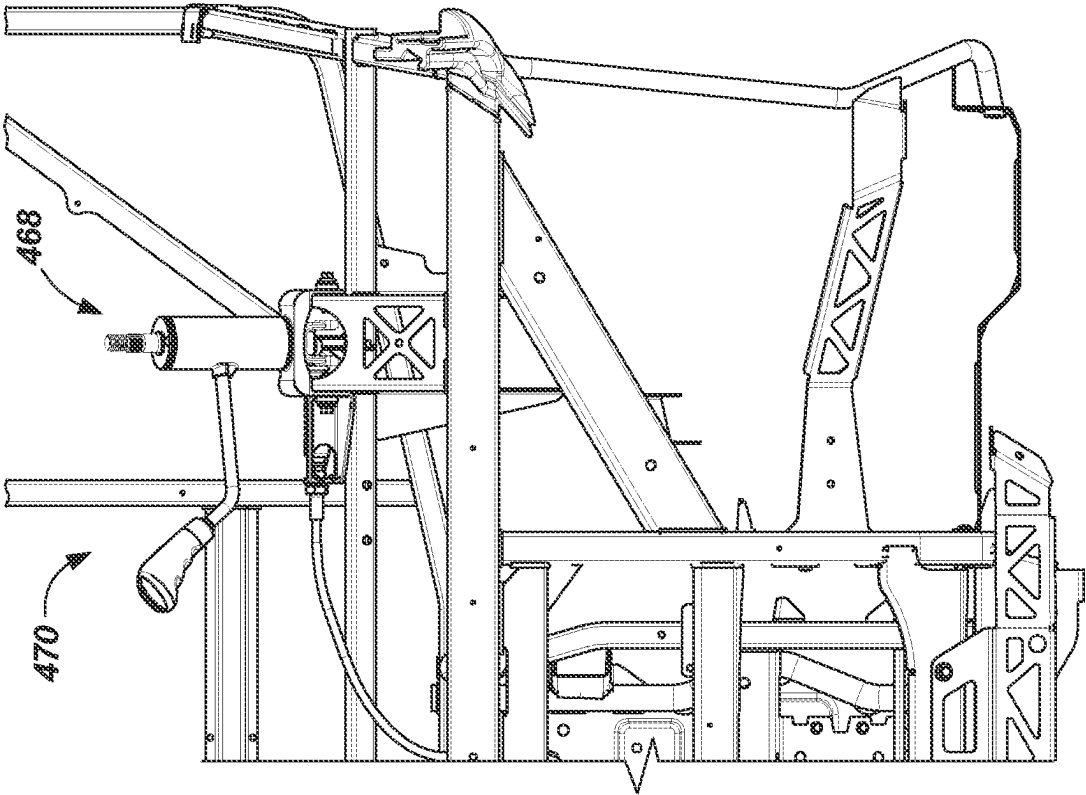


FIG. 71



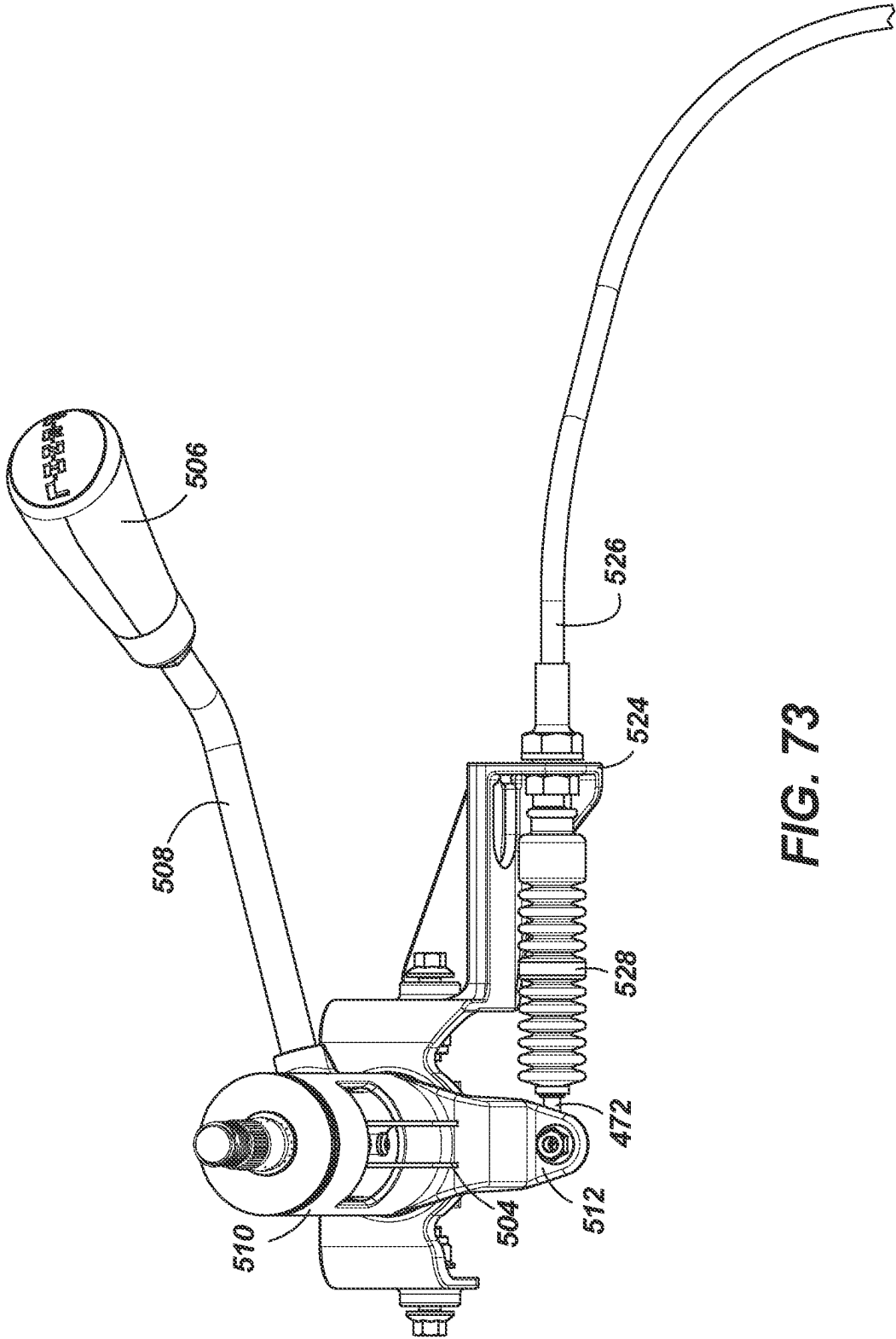
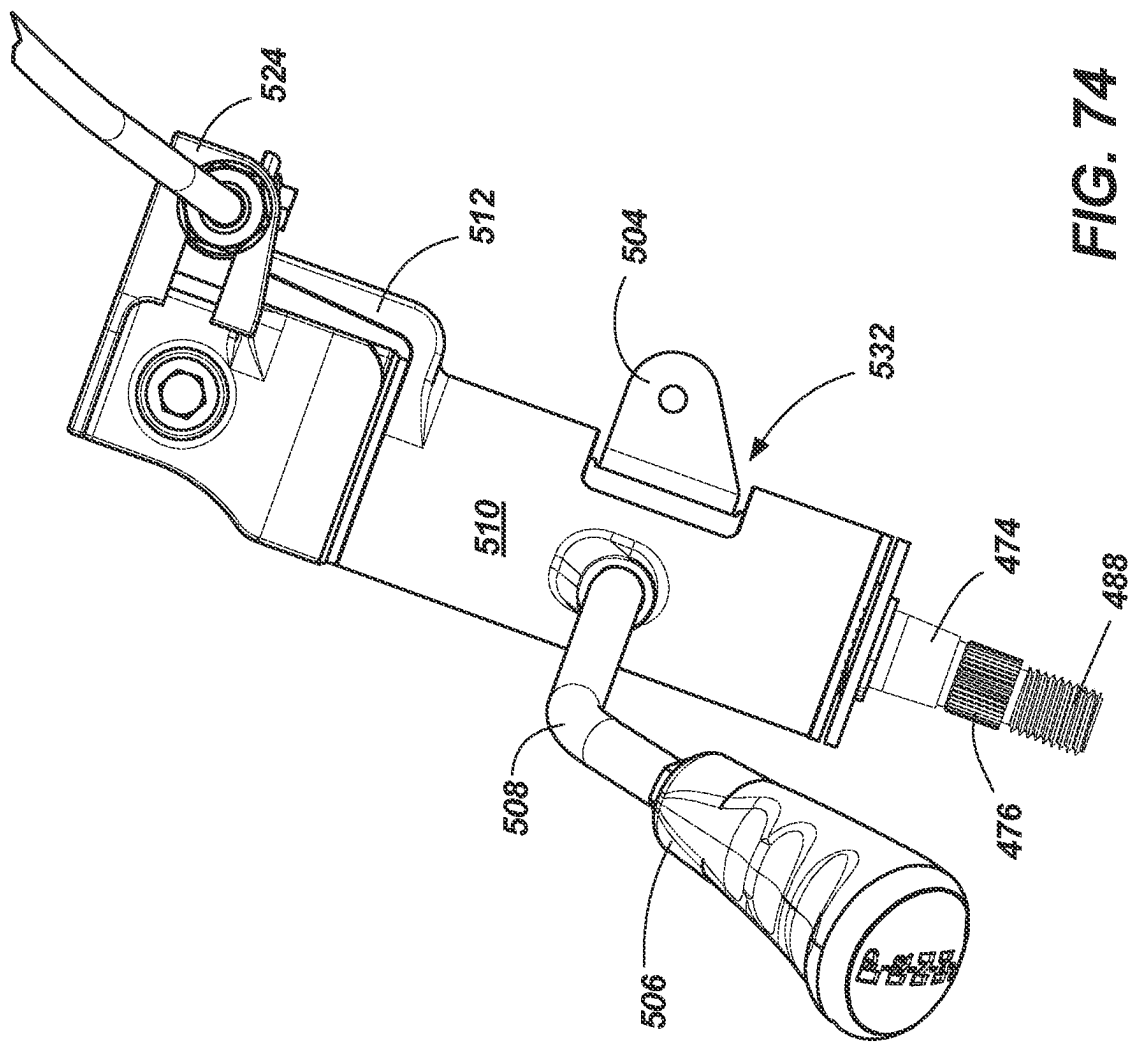


FIG. 73



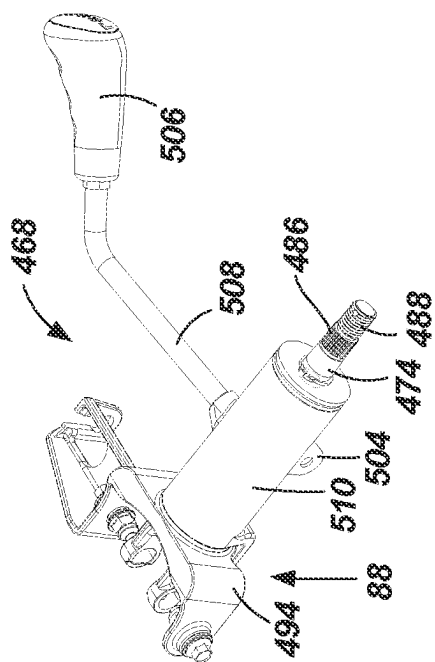


FIG. 76

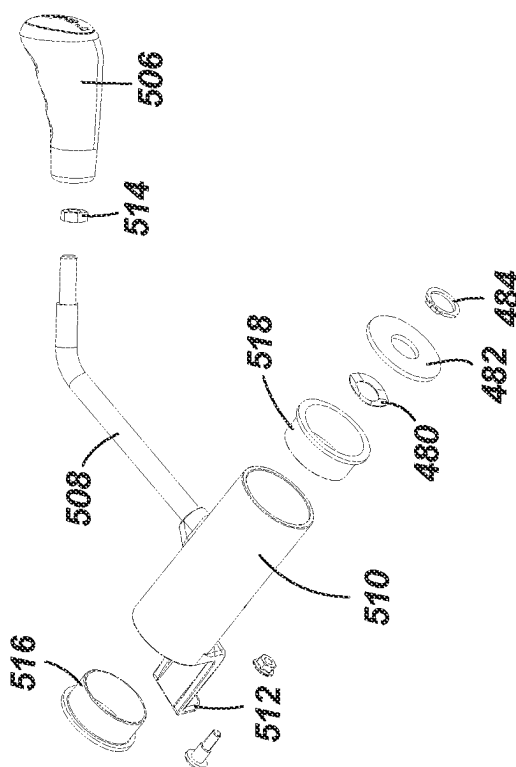
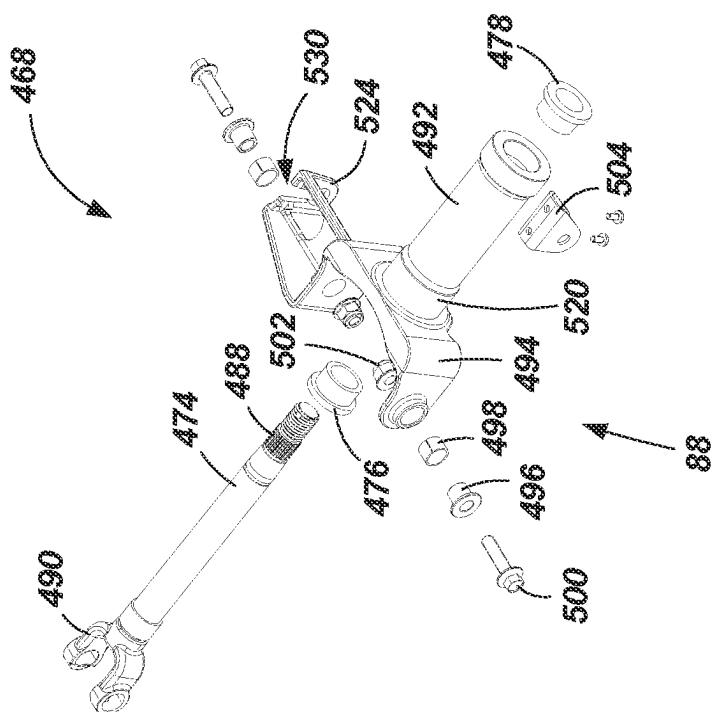


FIG. 75



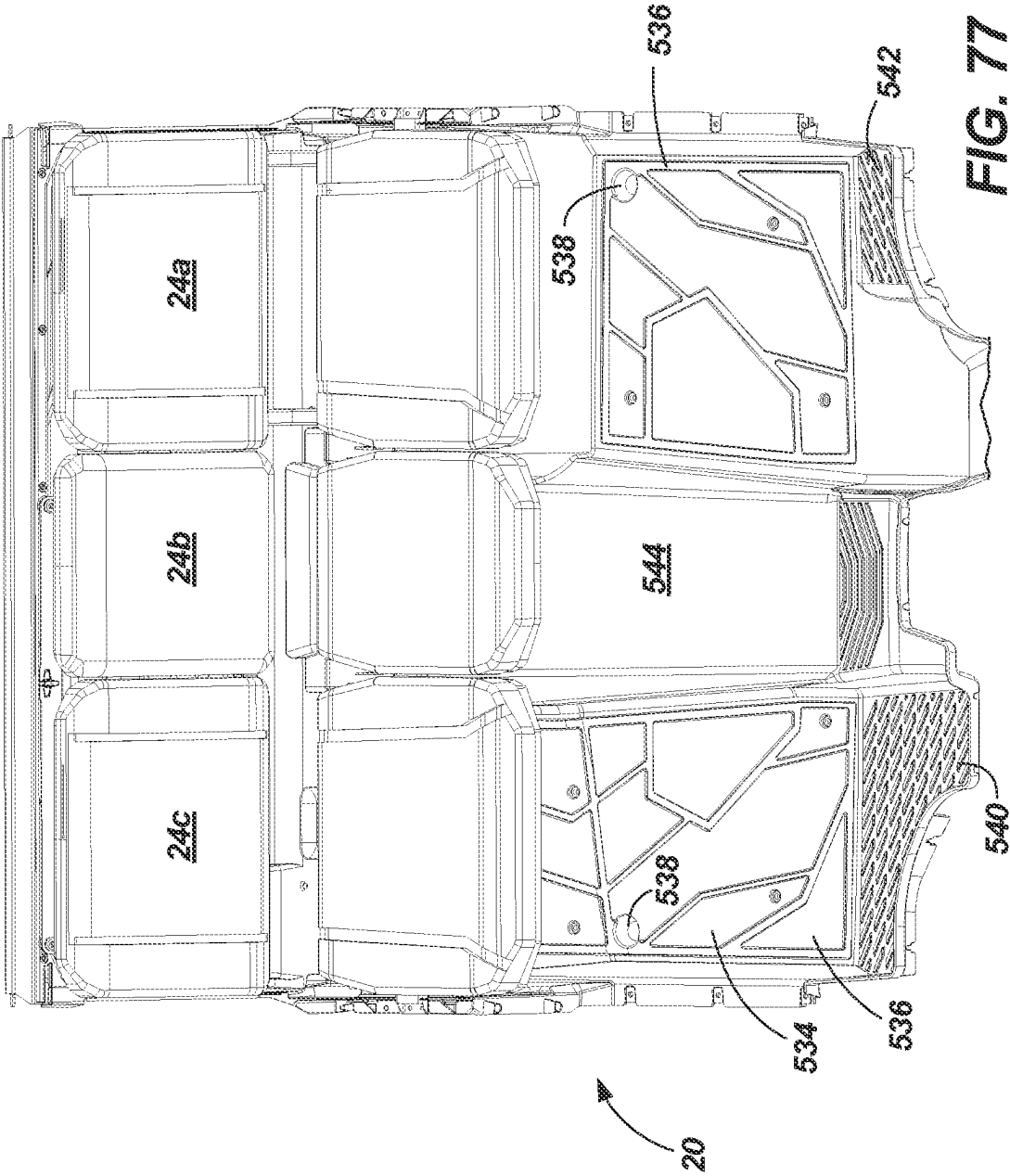


FIG. 77

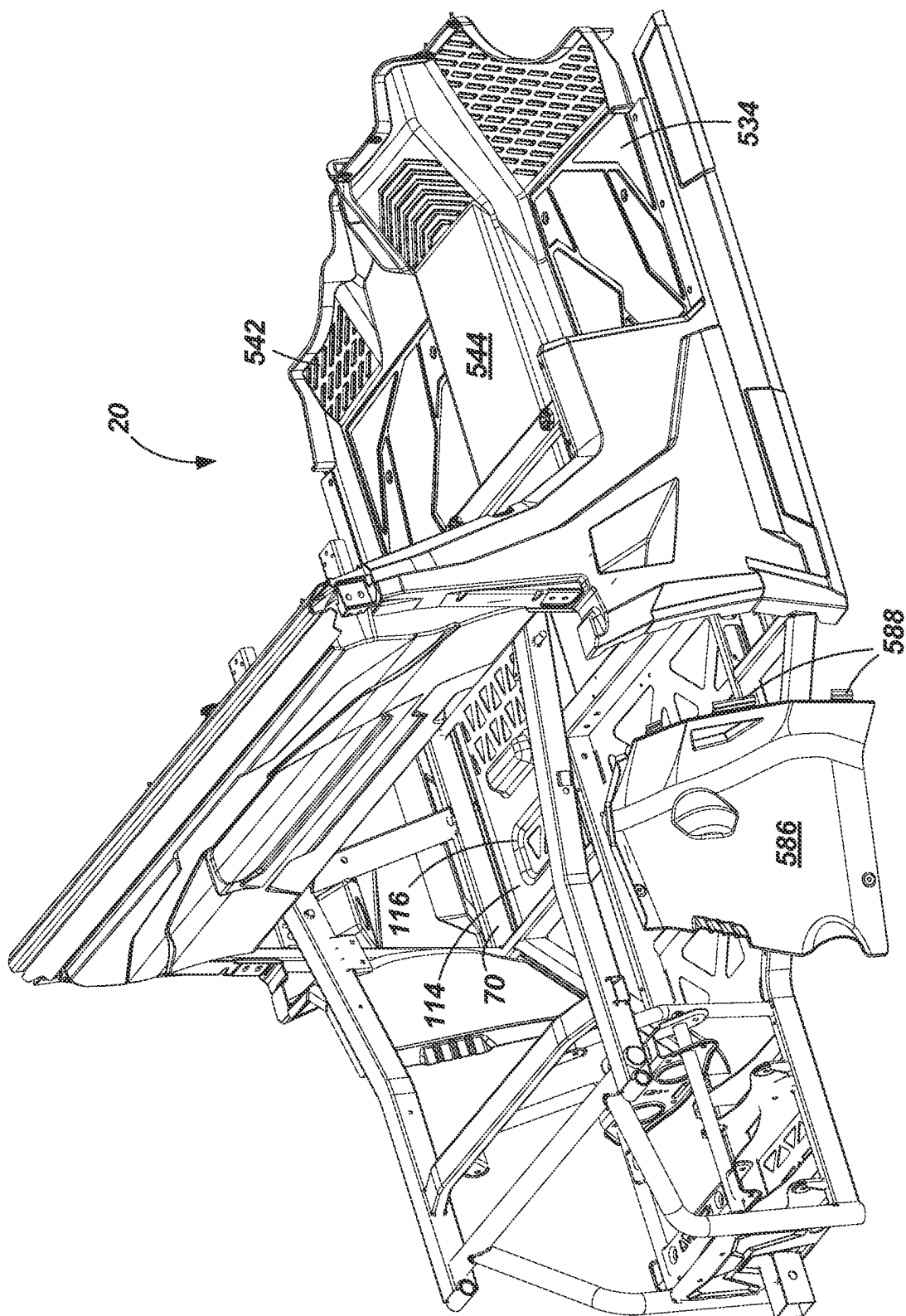


FIG. 78

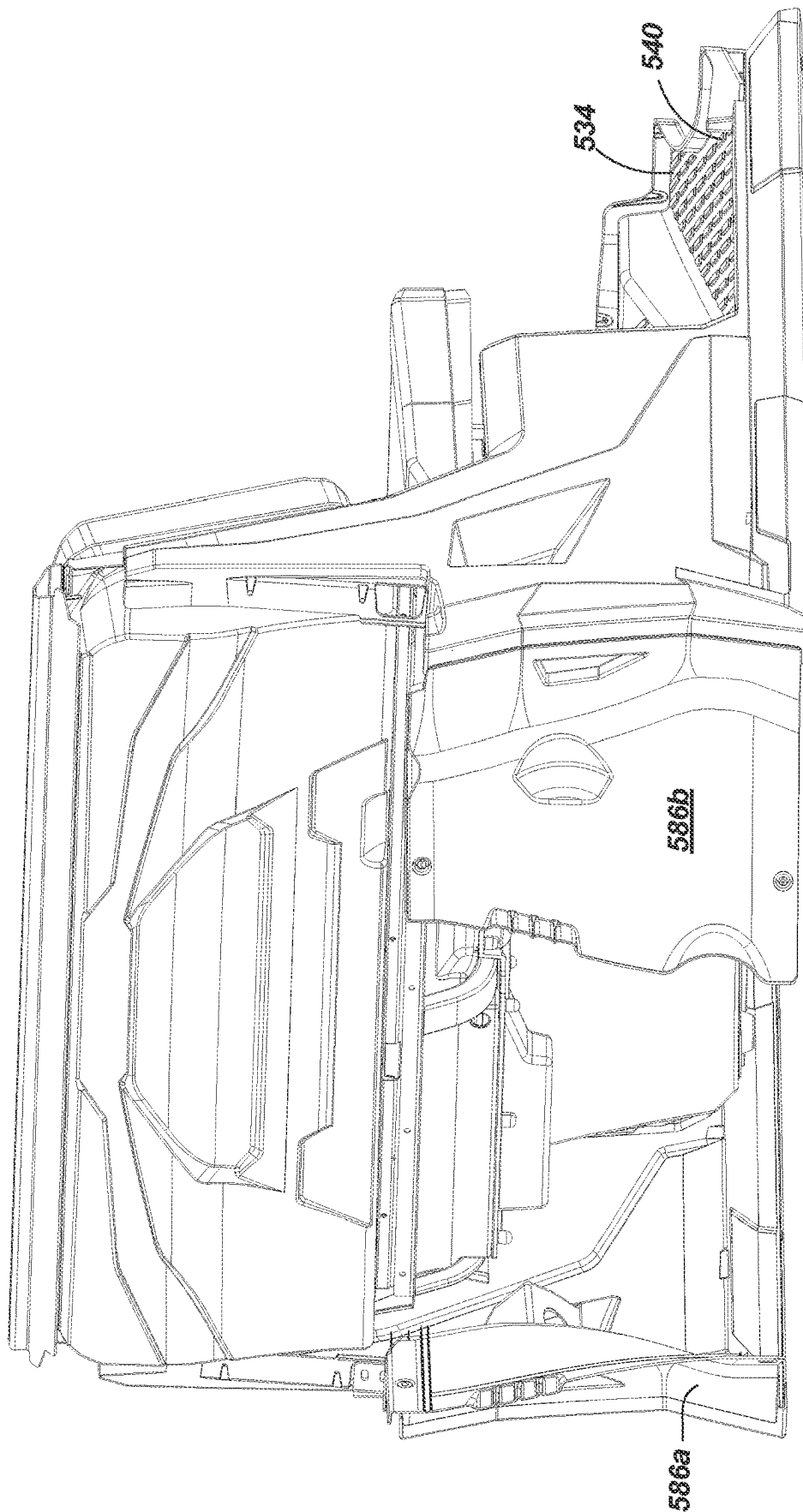


FIG. 79

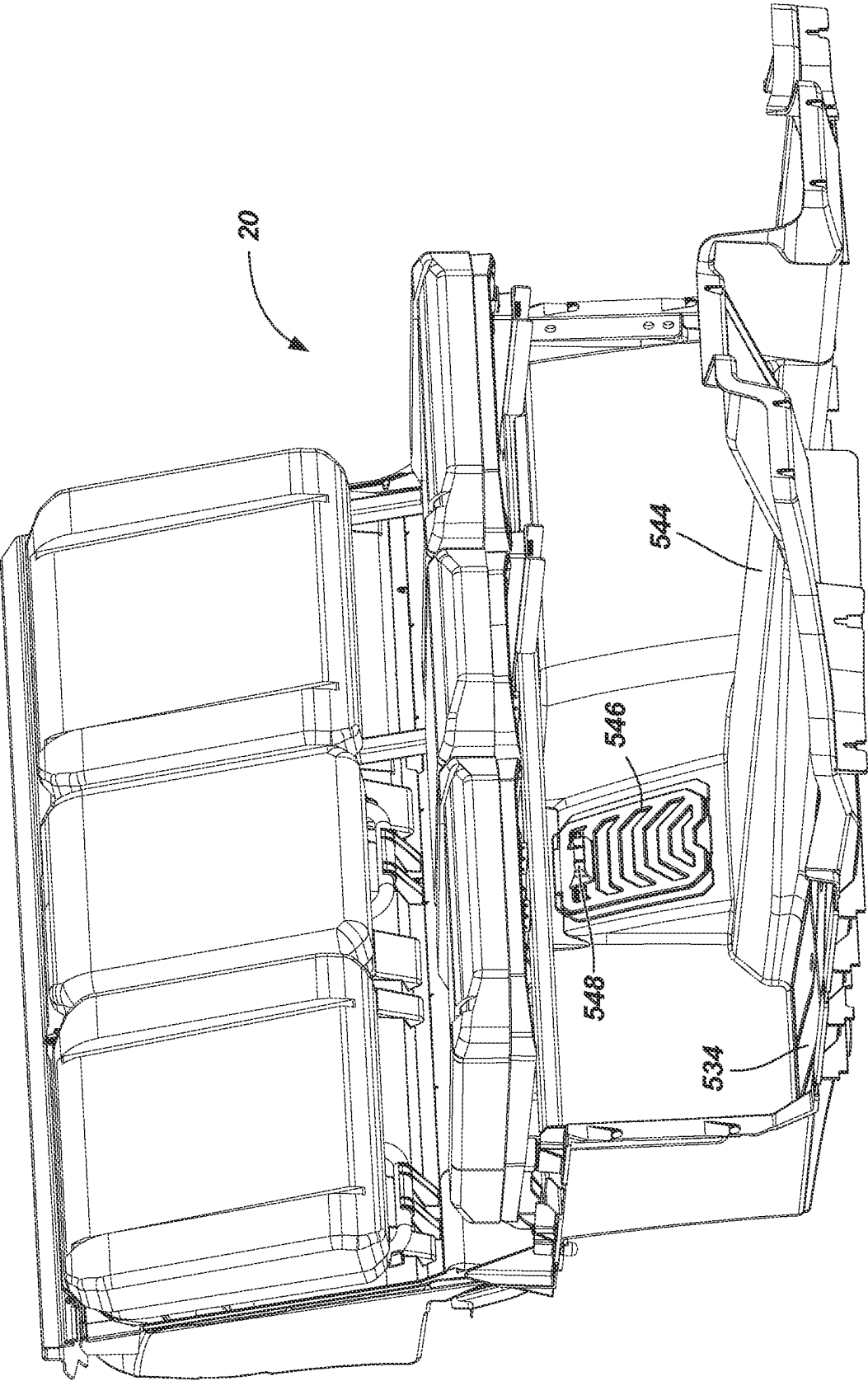
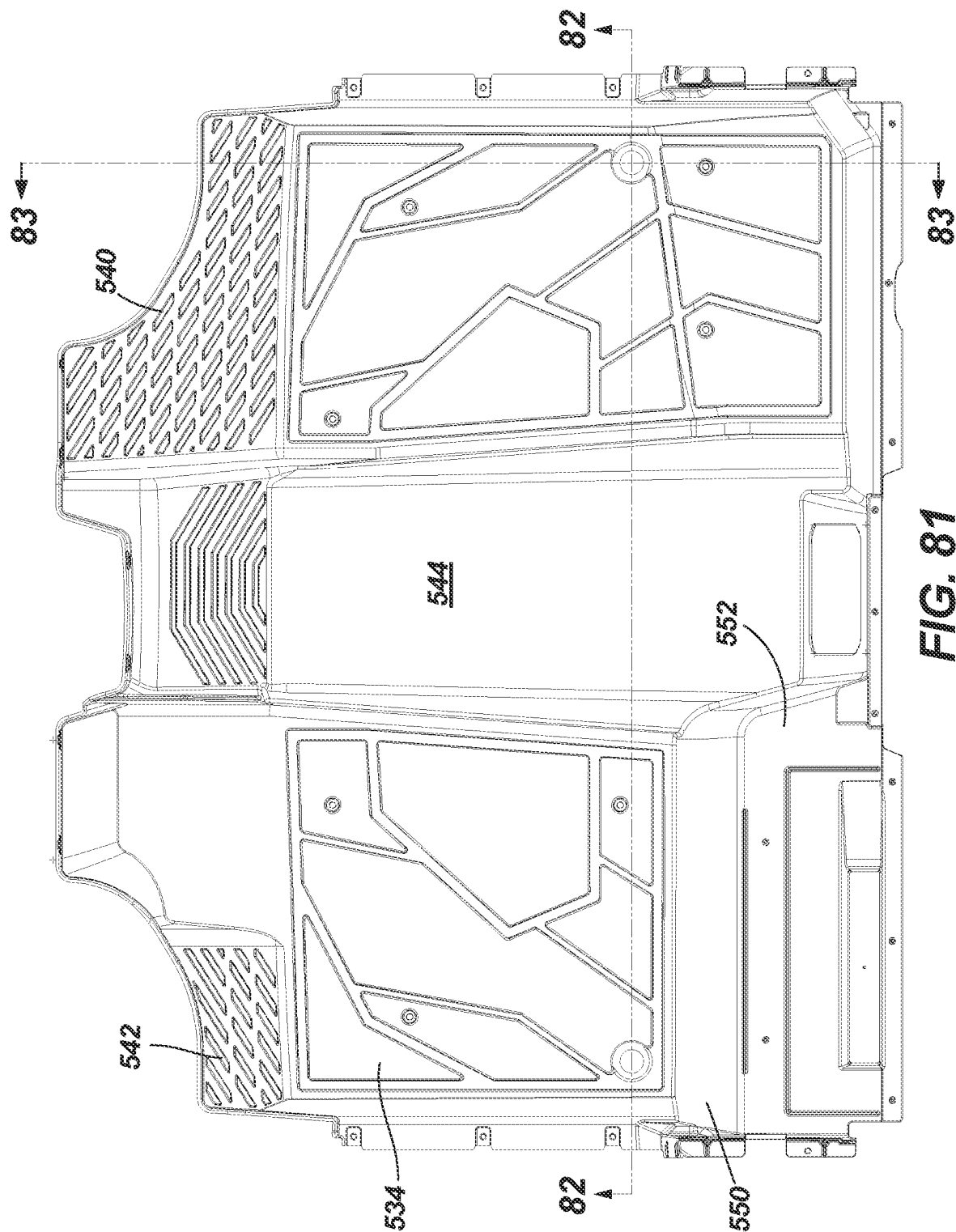


FIG. 80



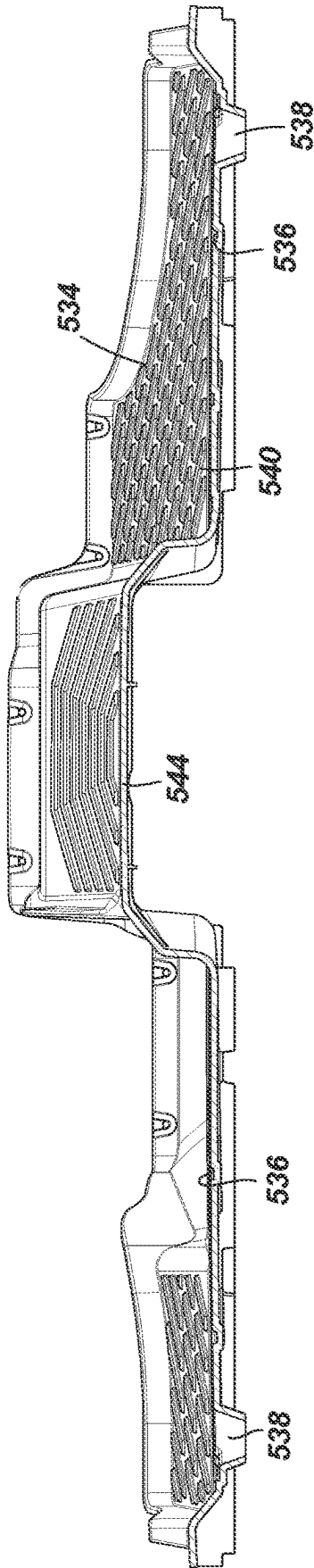


FIG. 82

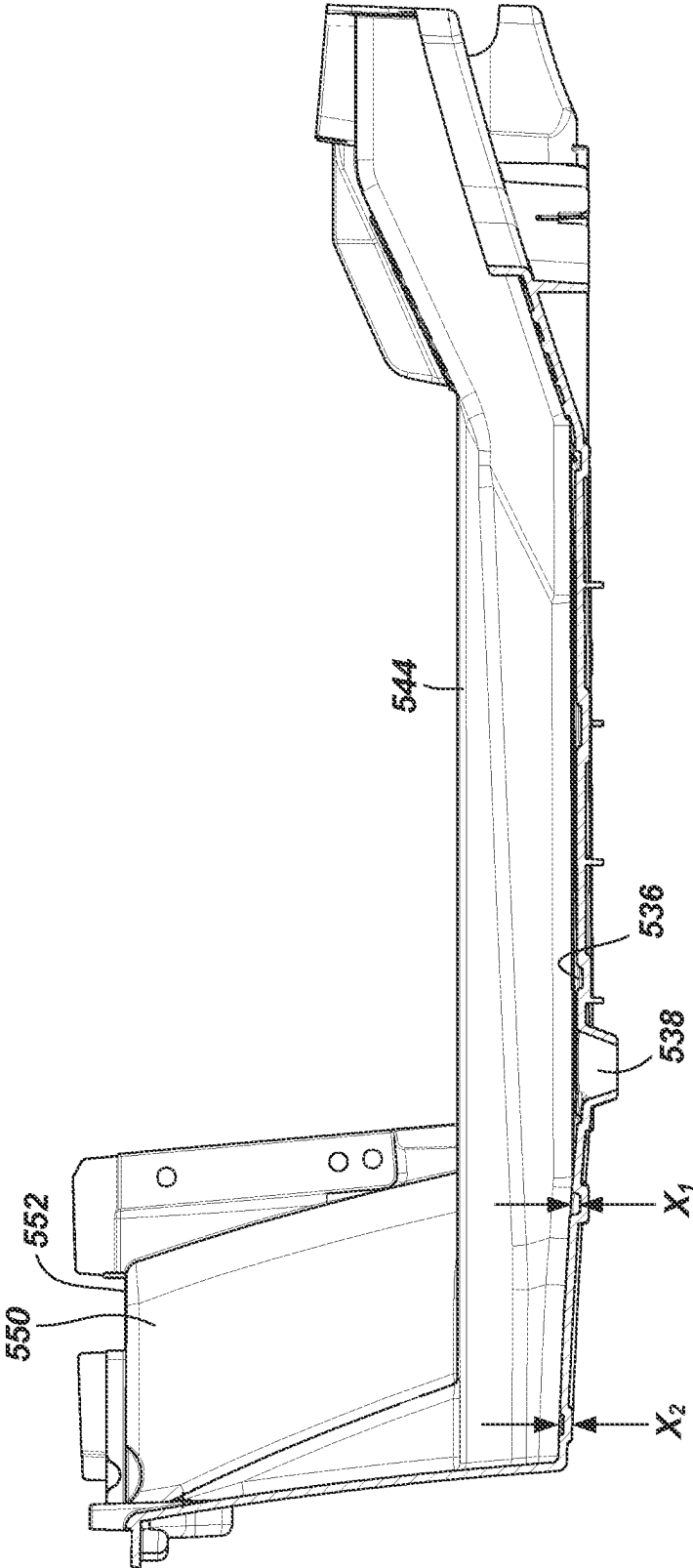


FIG. 83

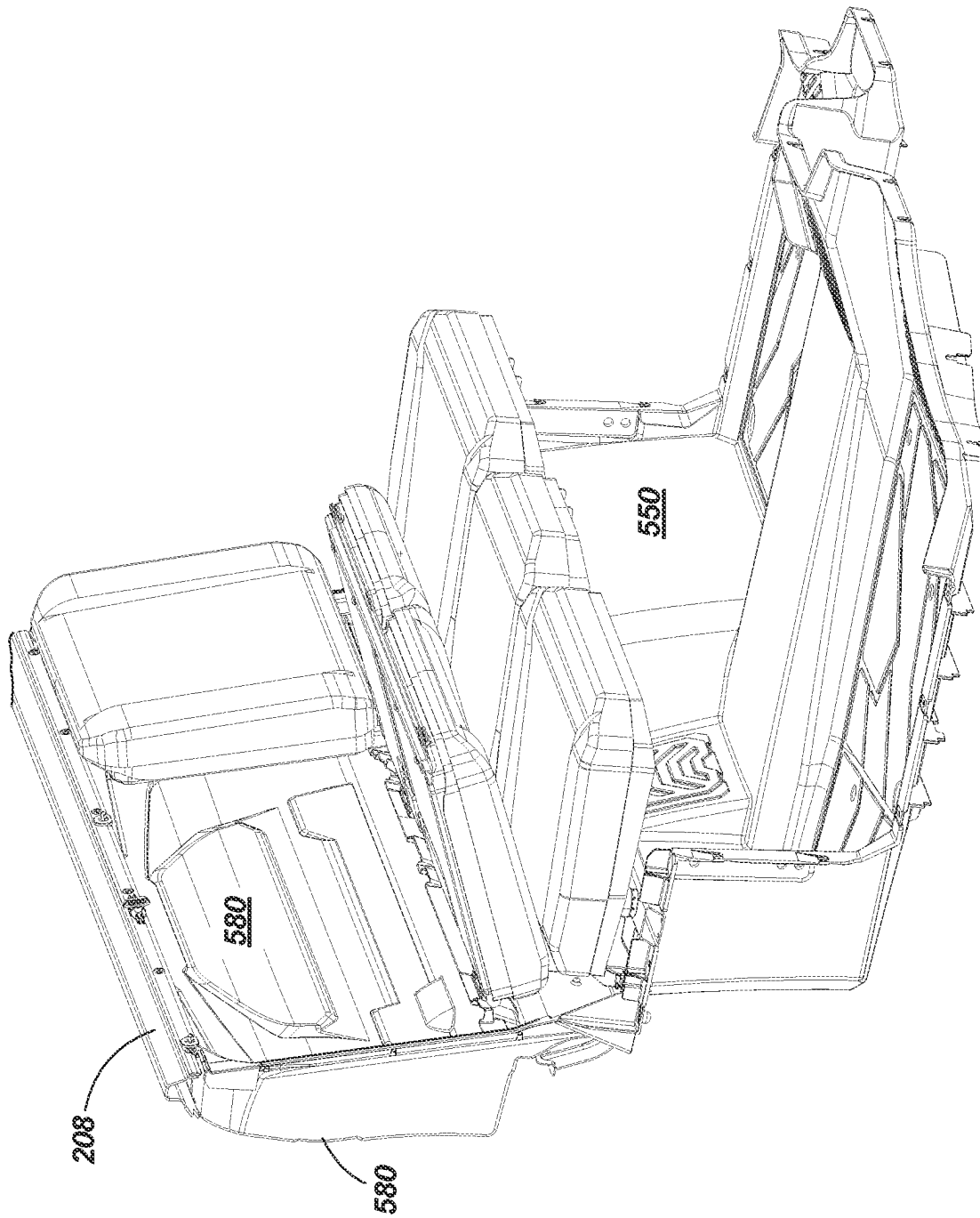


FIG. 84

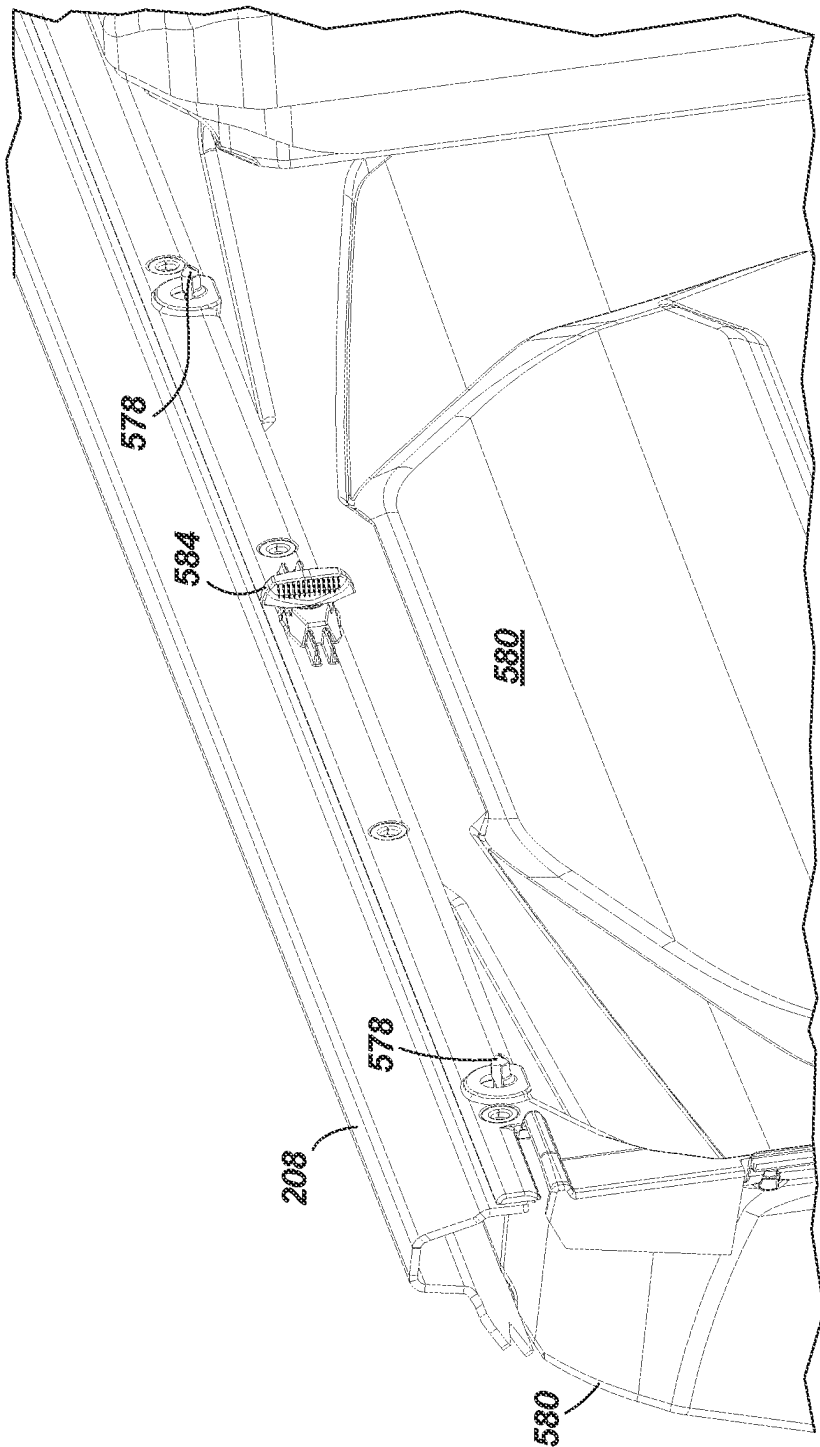
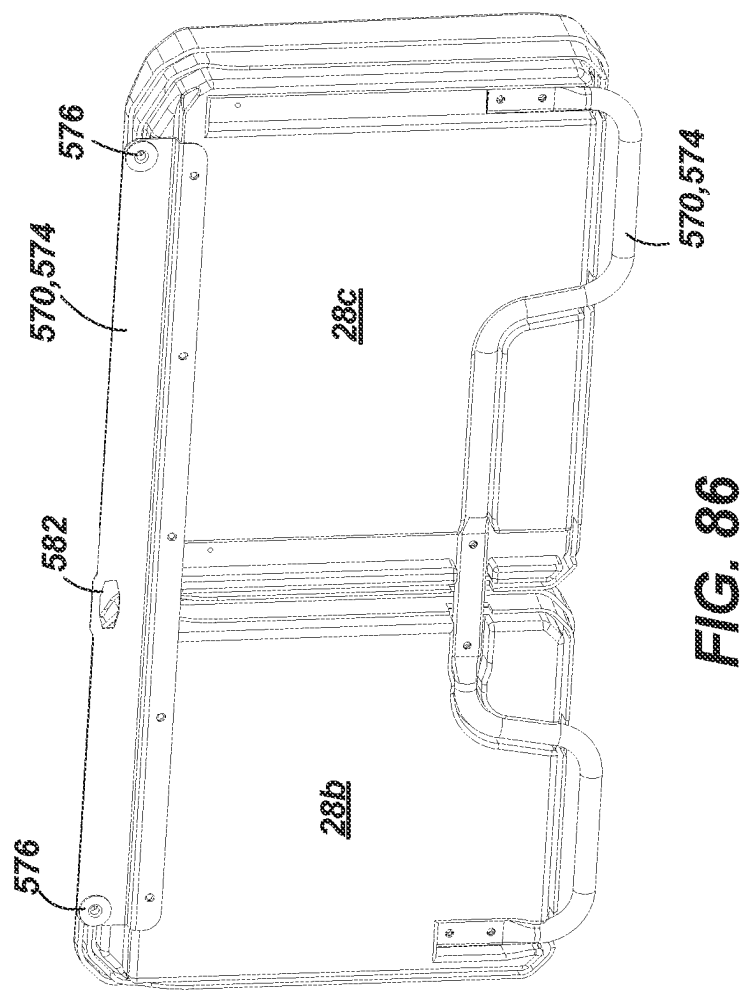
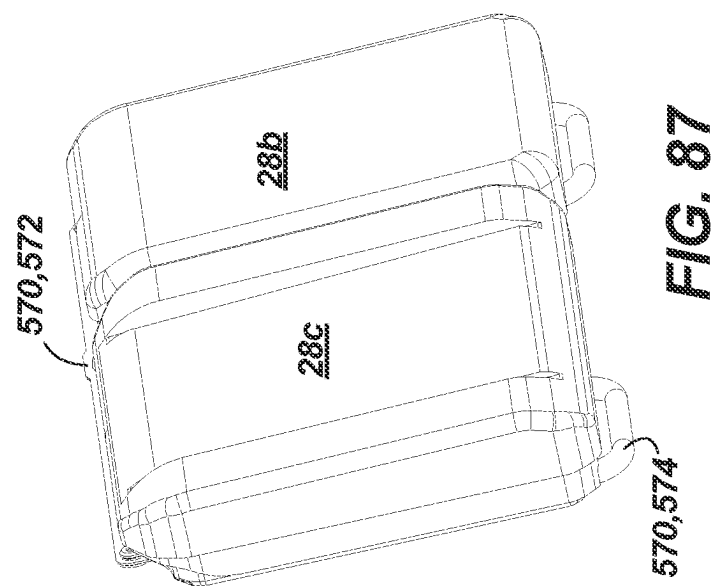
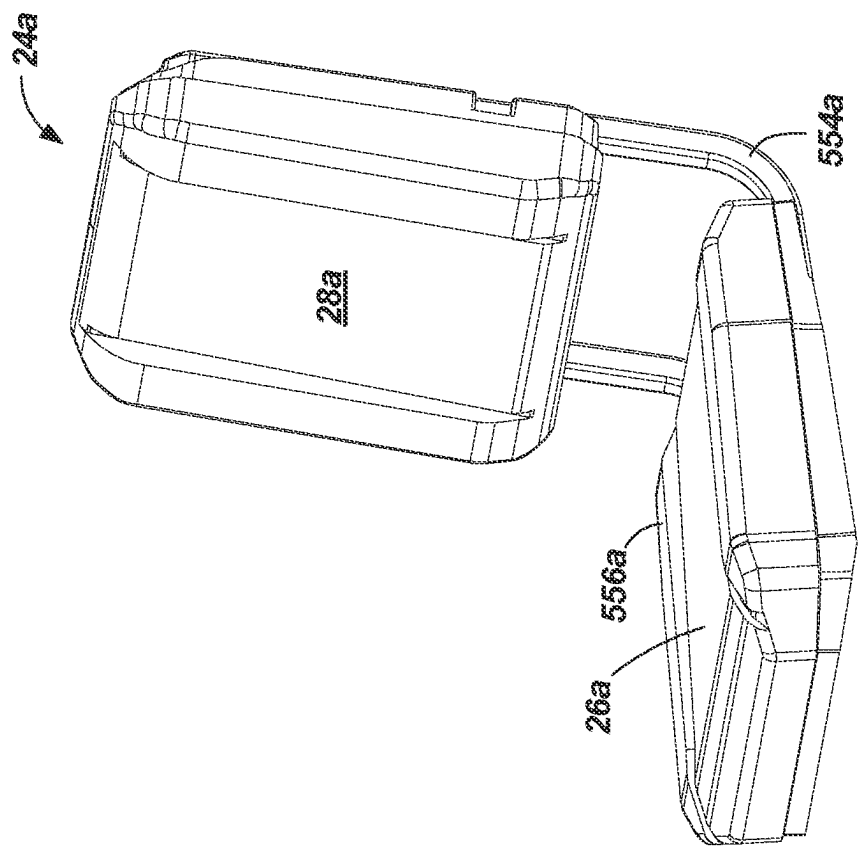
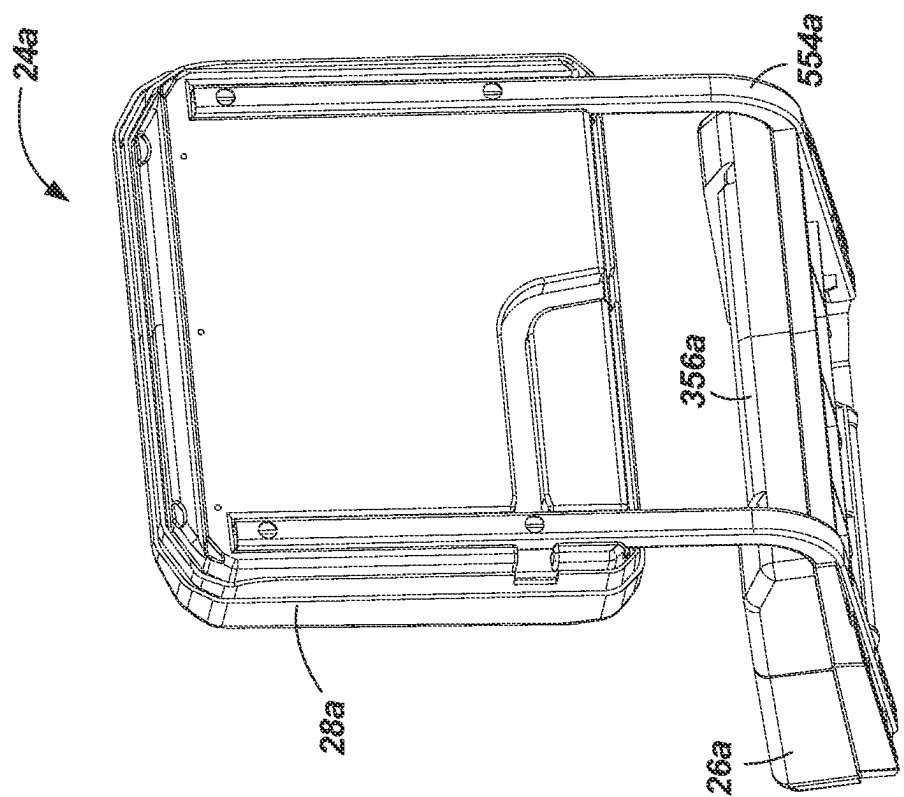
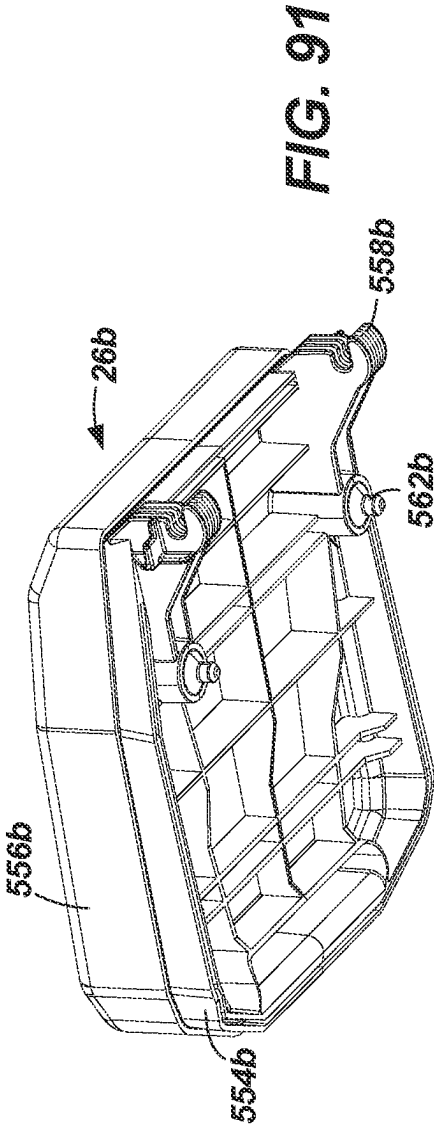
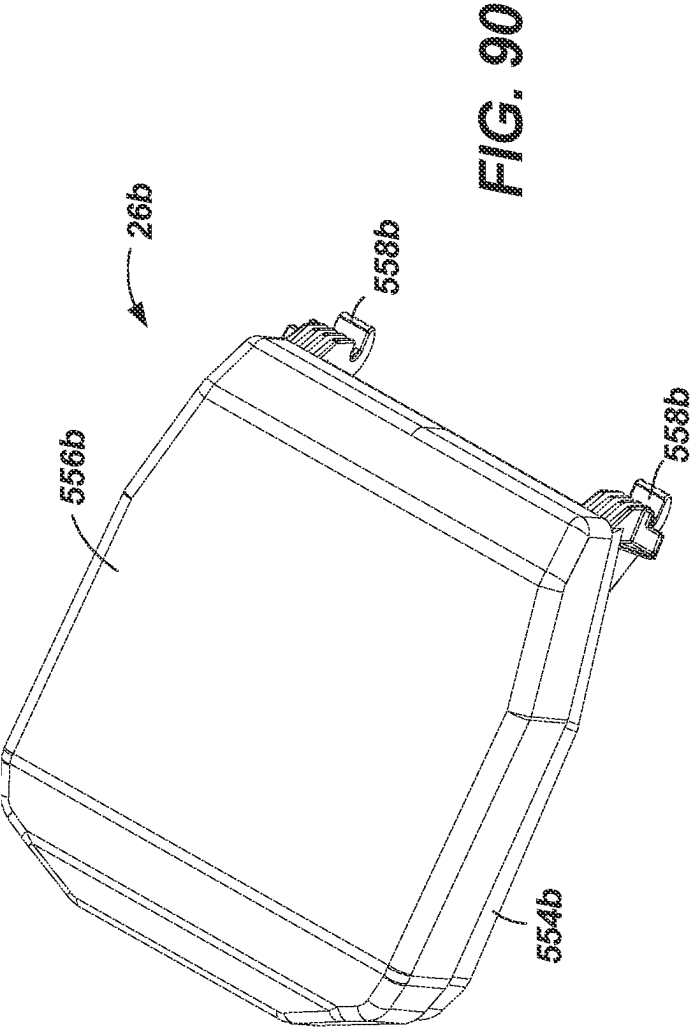
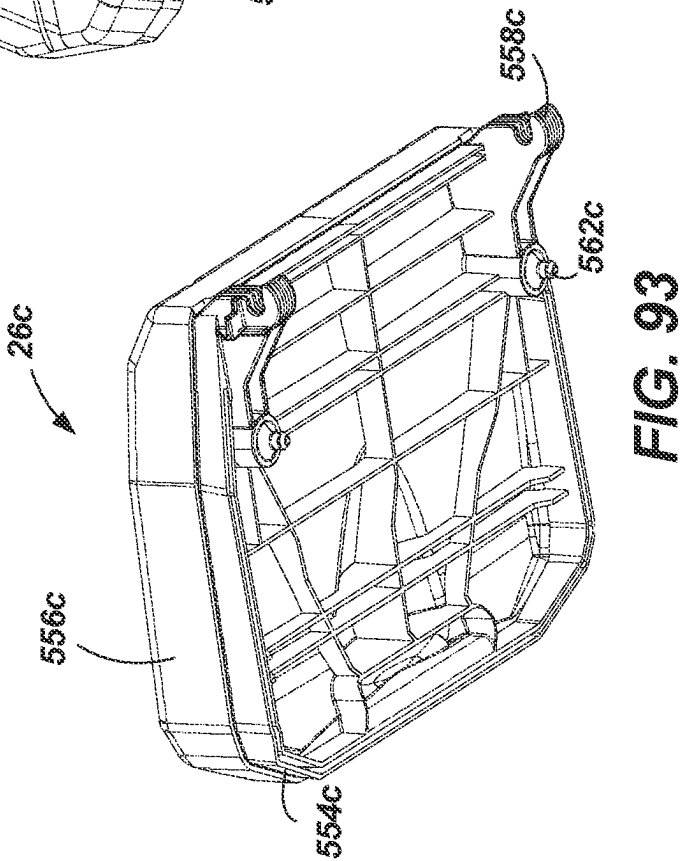
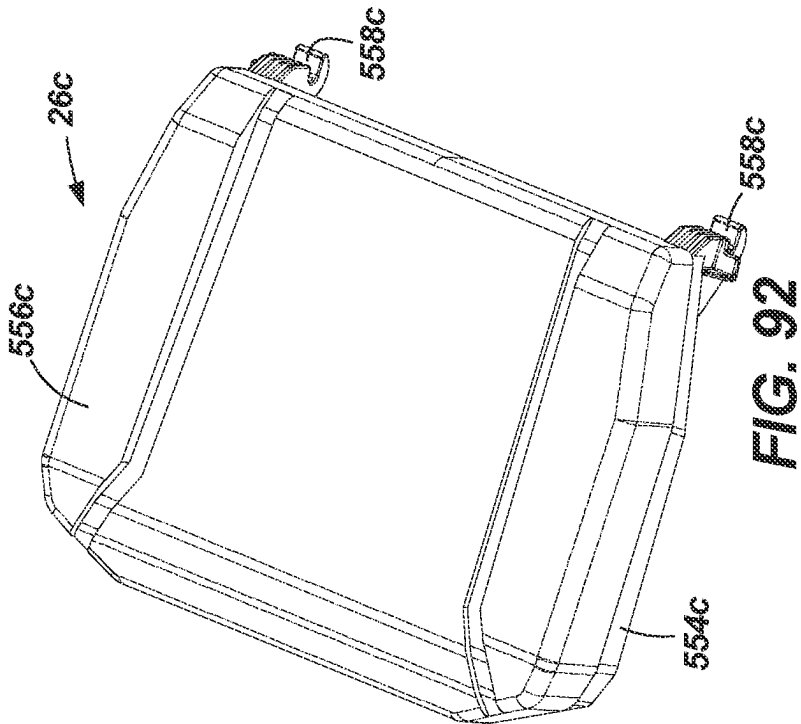


FIG. 85









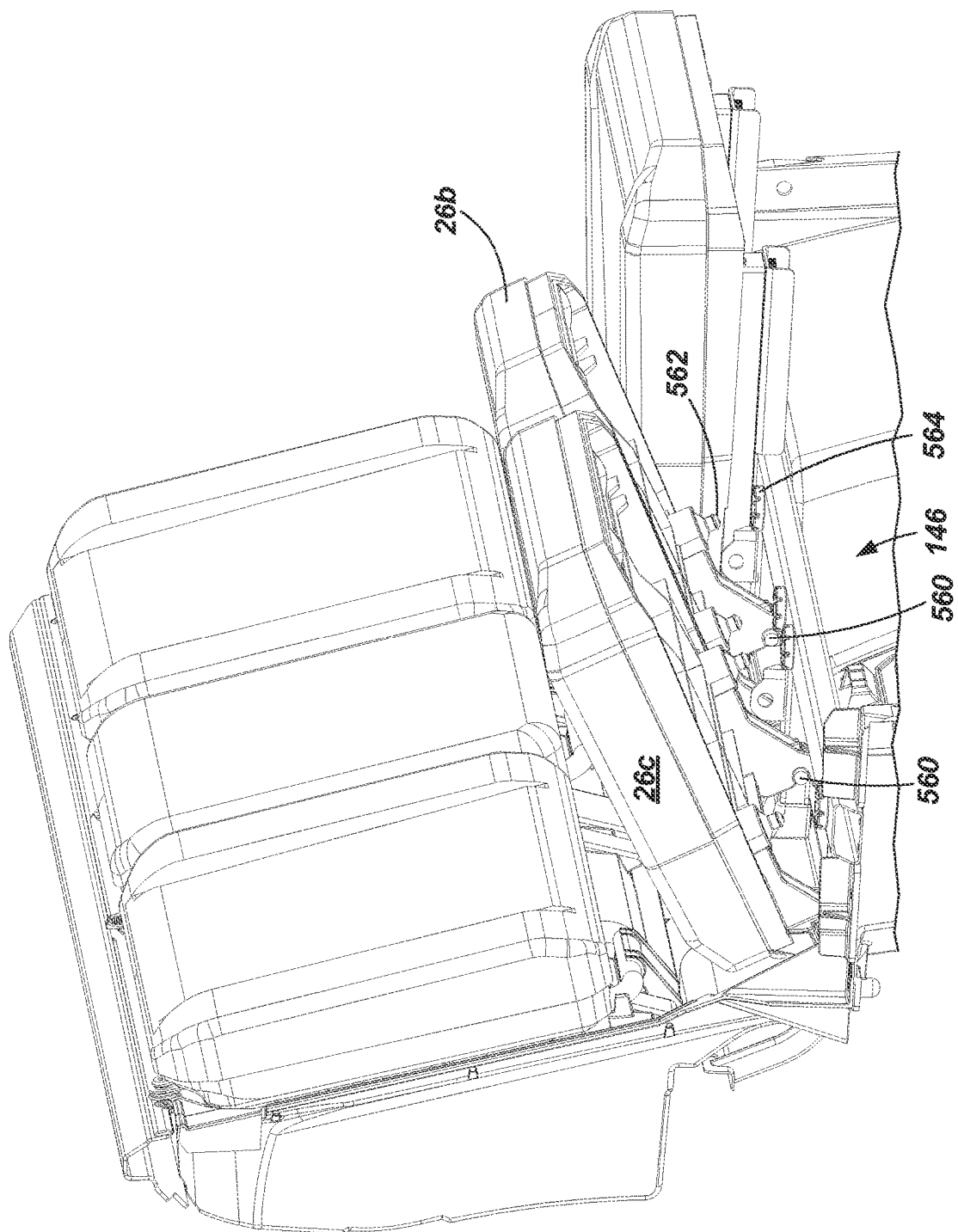


FIG. 94

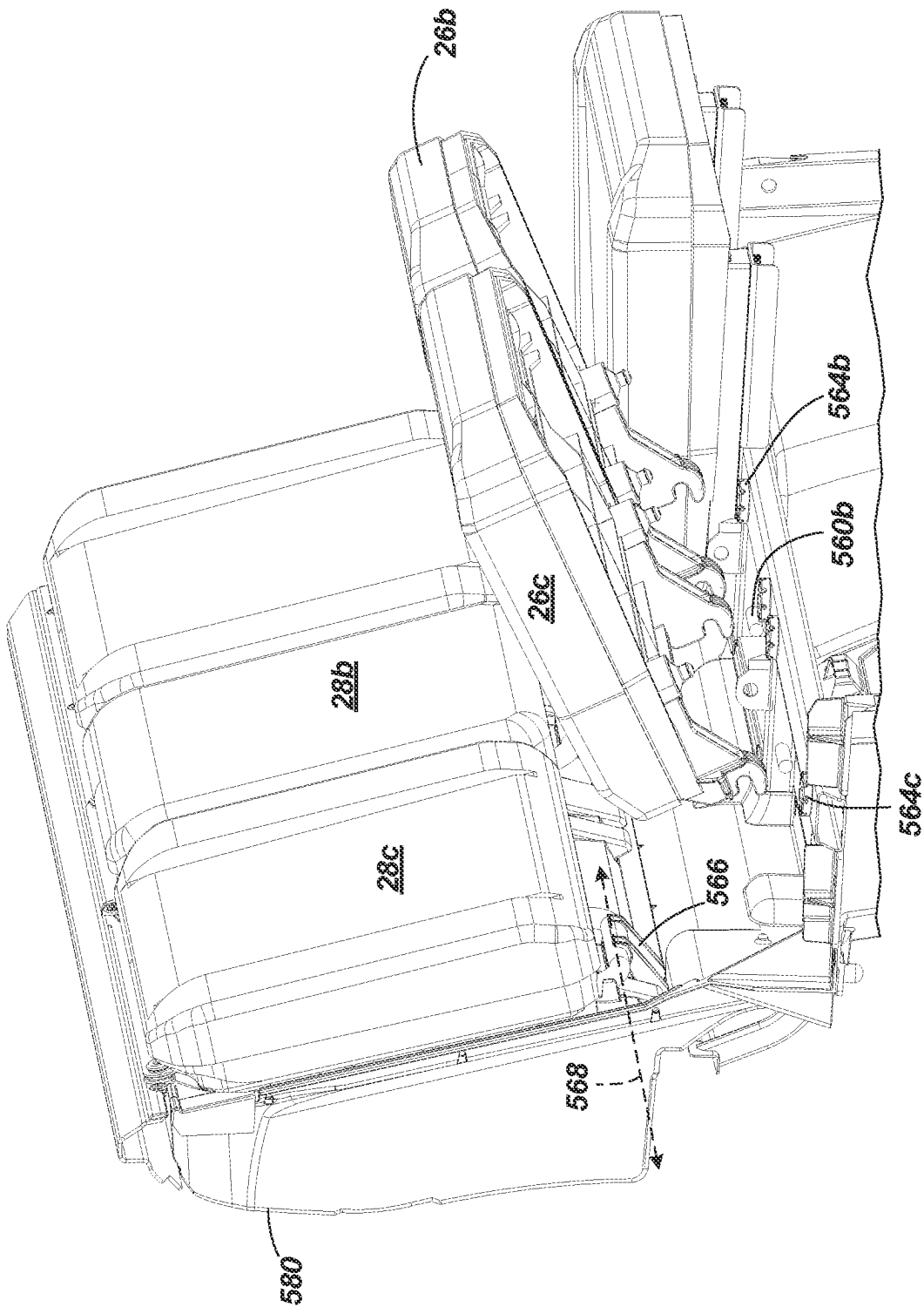
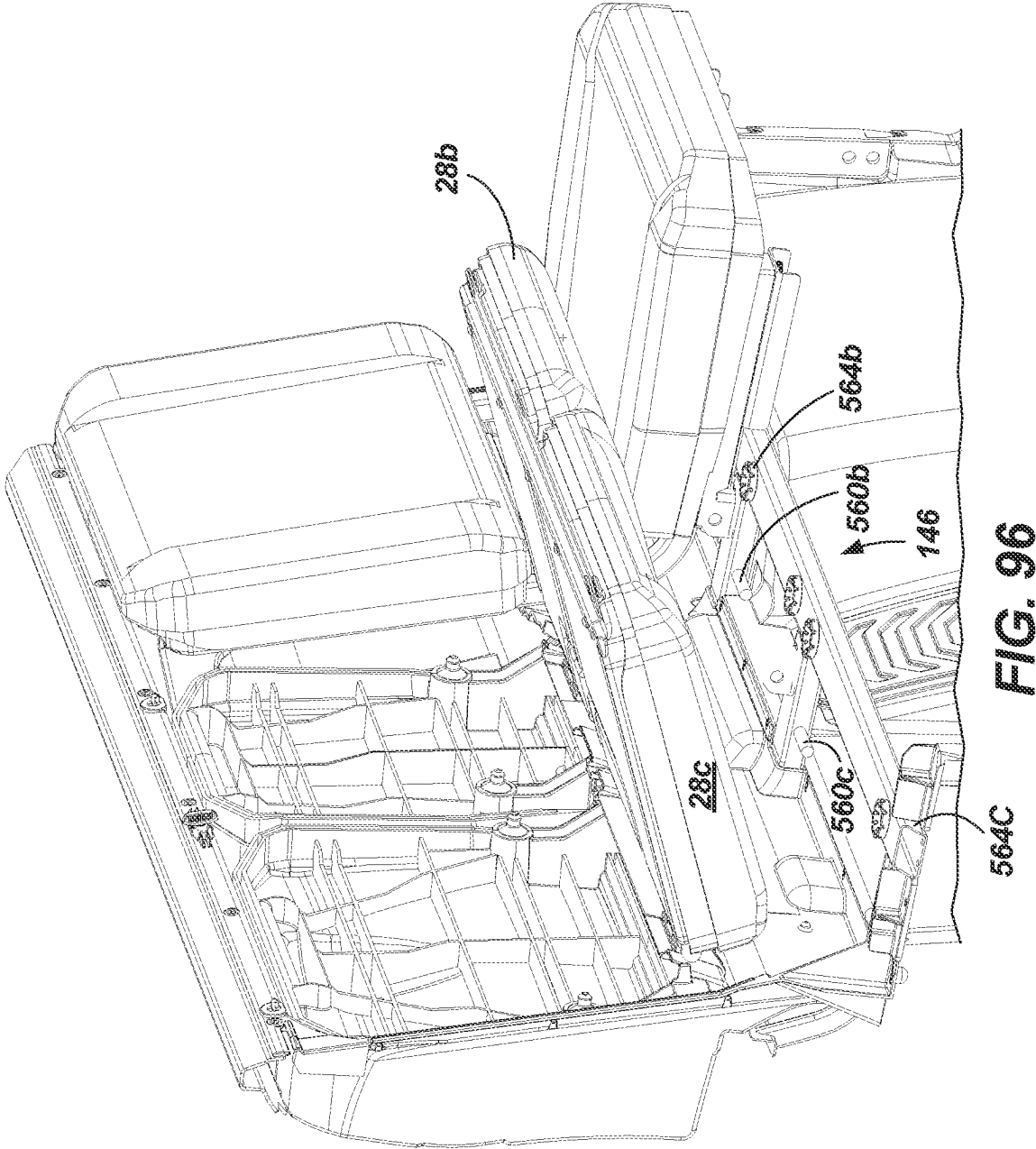
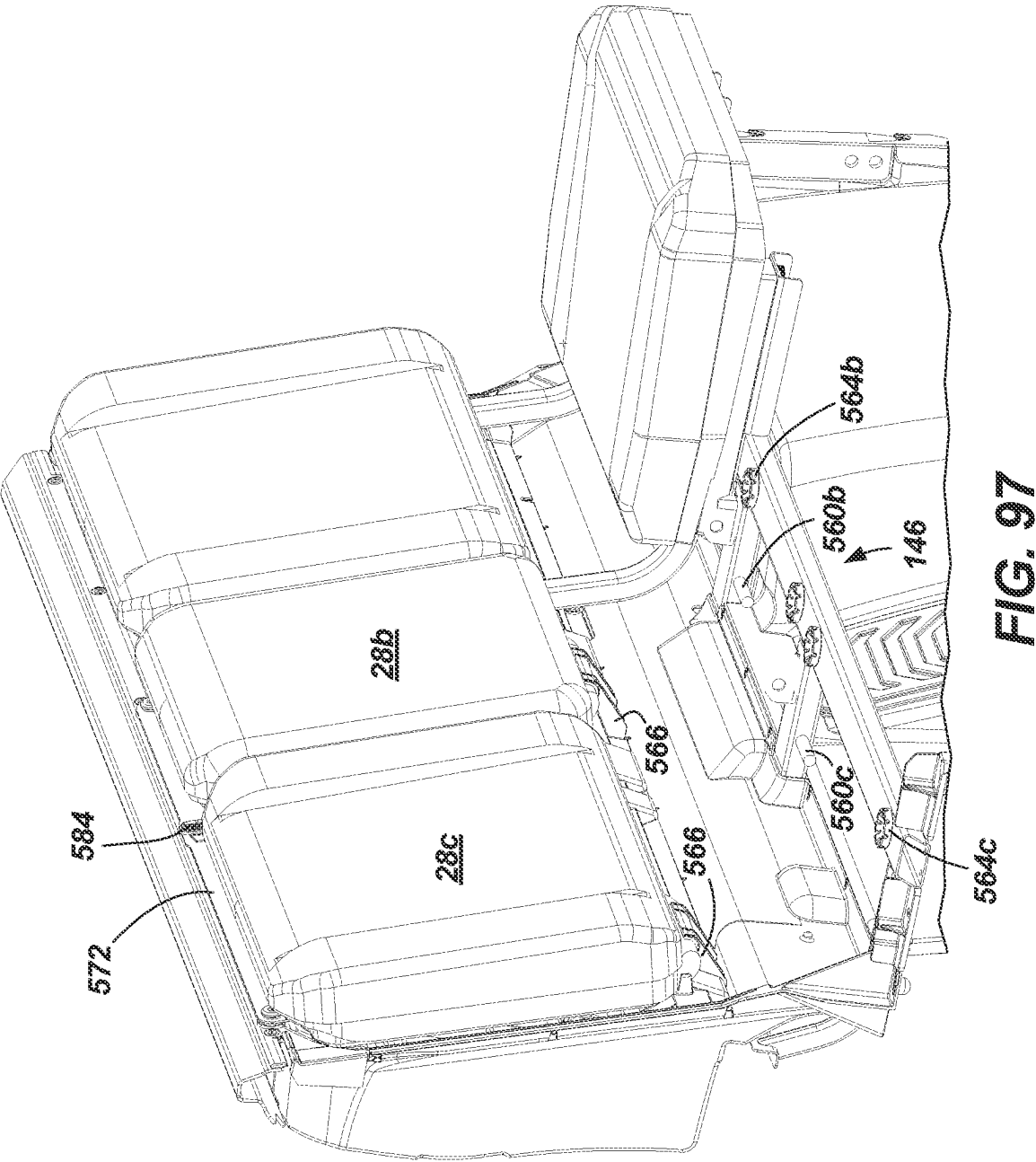


FIG. 95





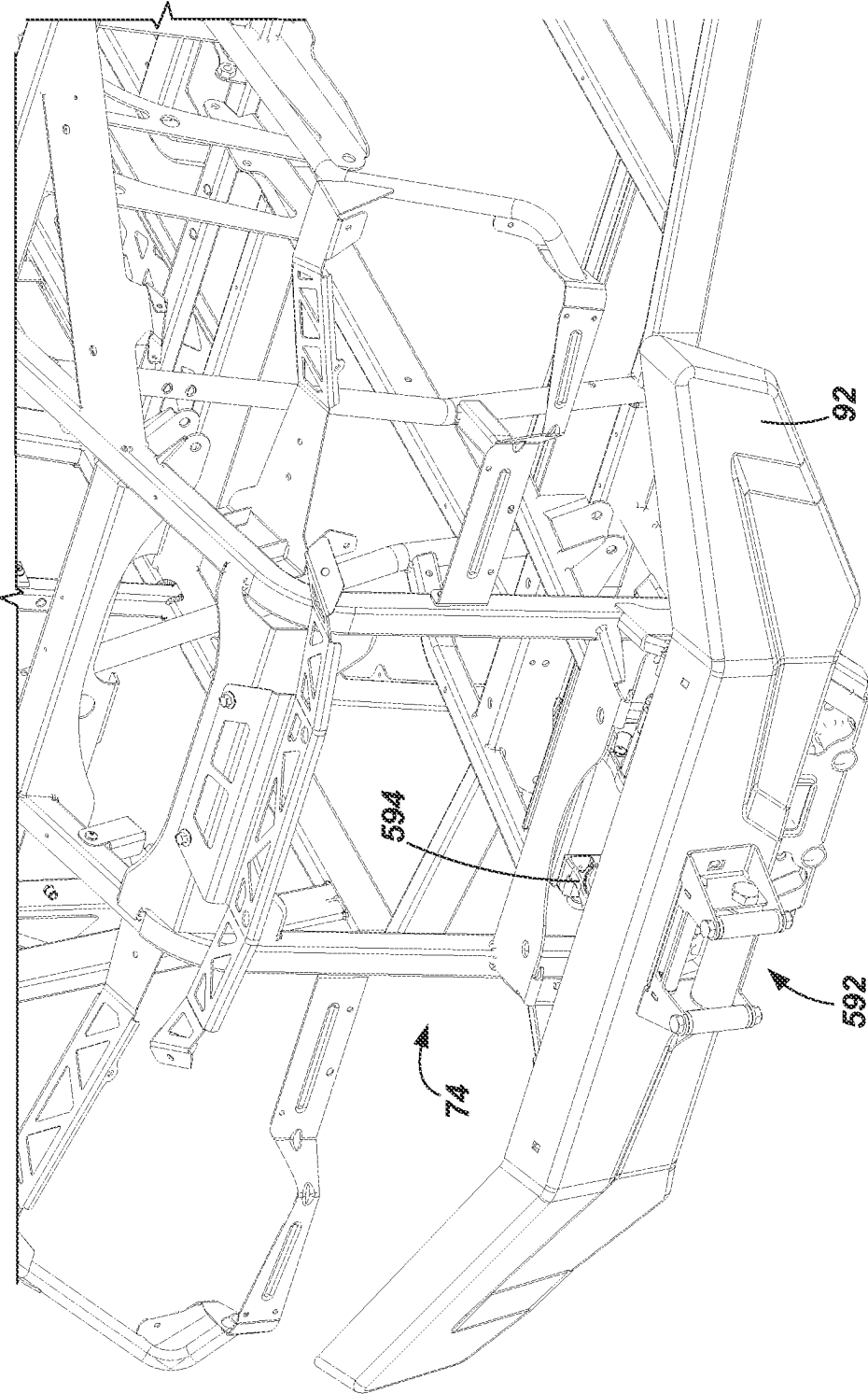


FIG. 98

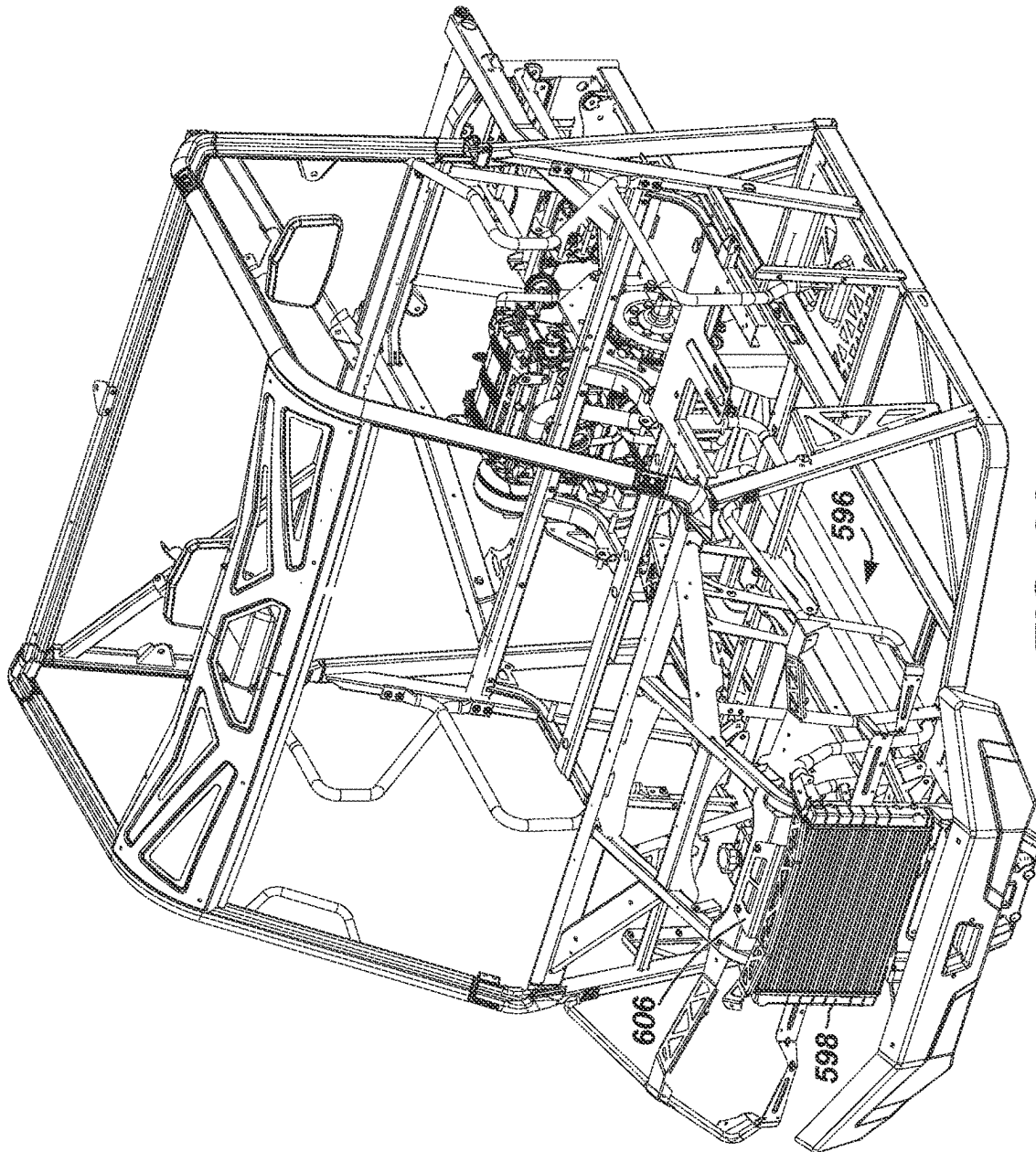
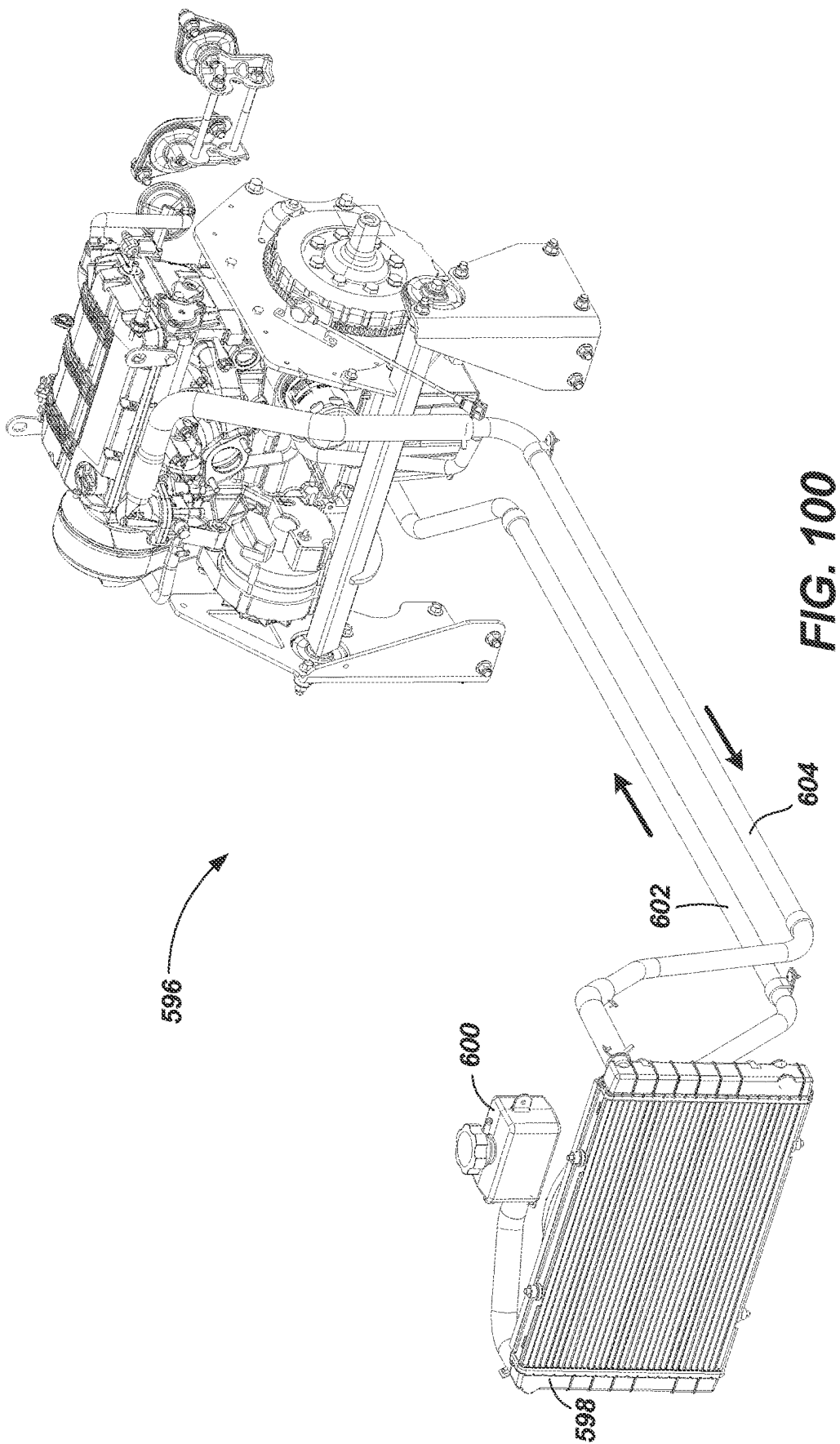
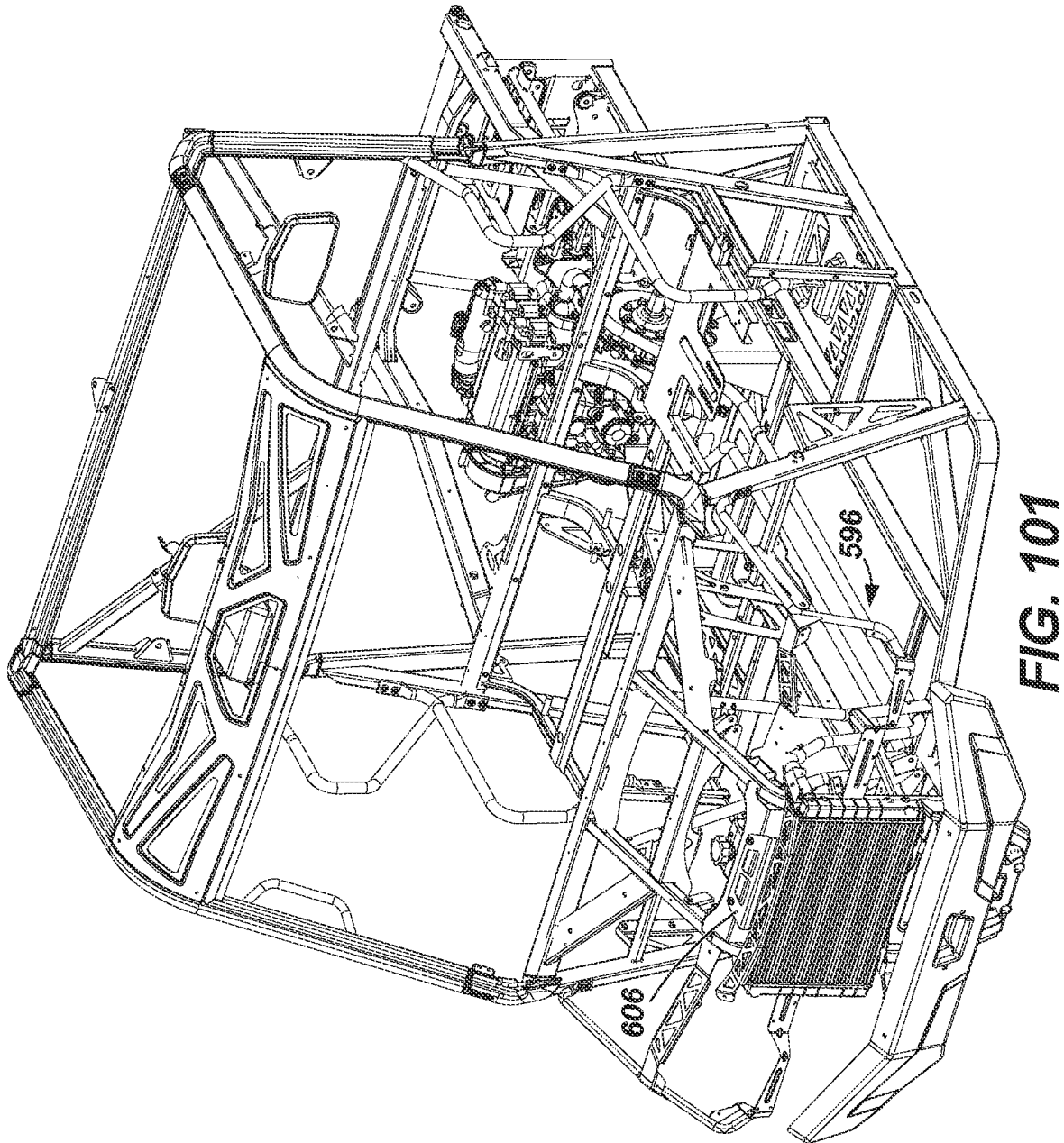
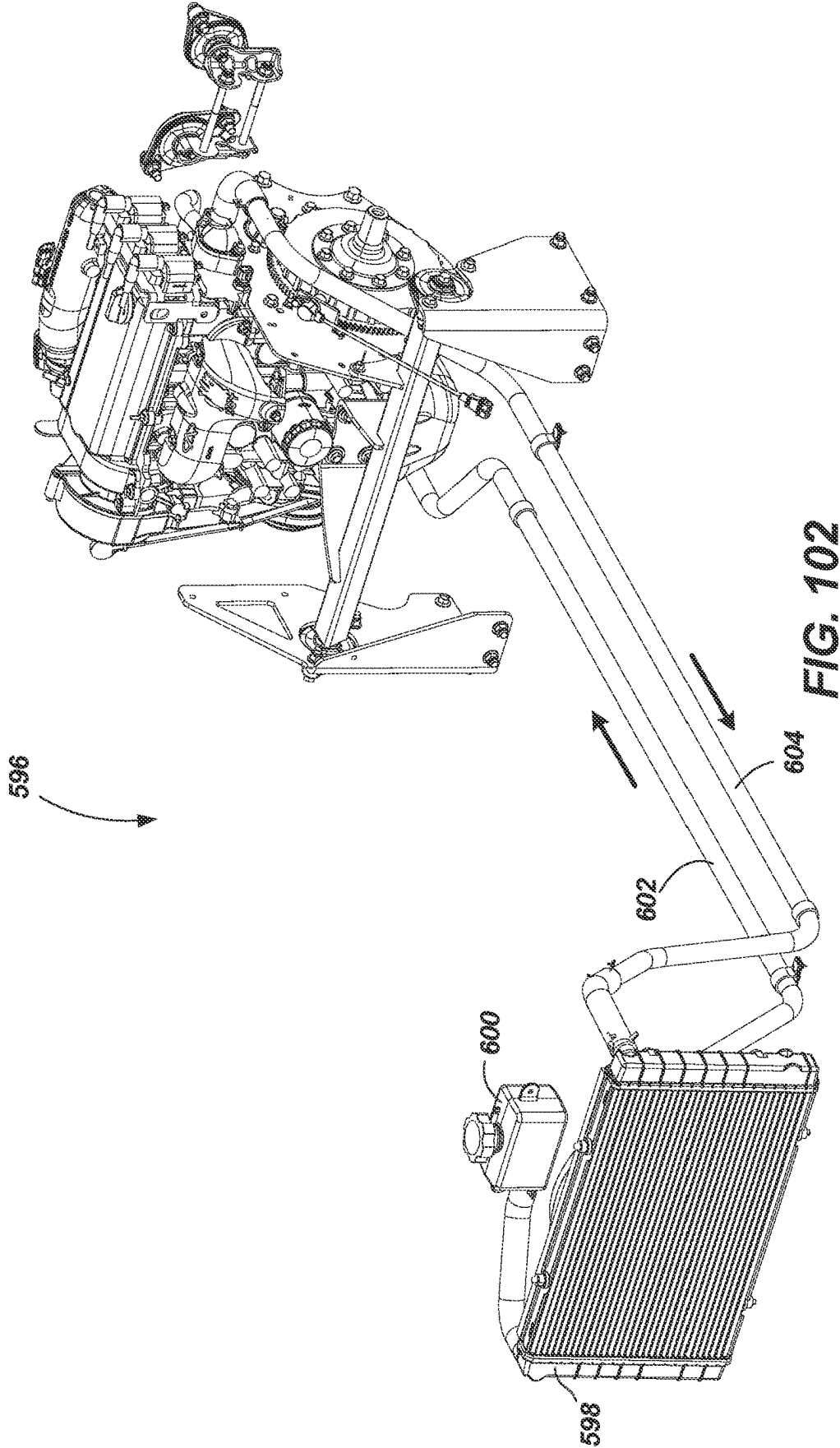


FIG. 99







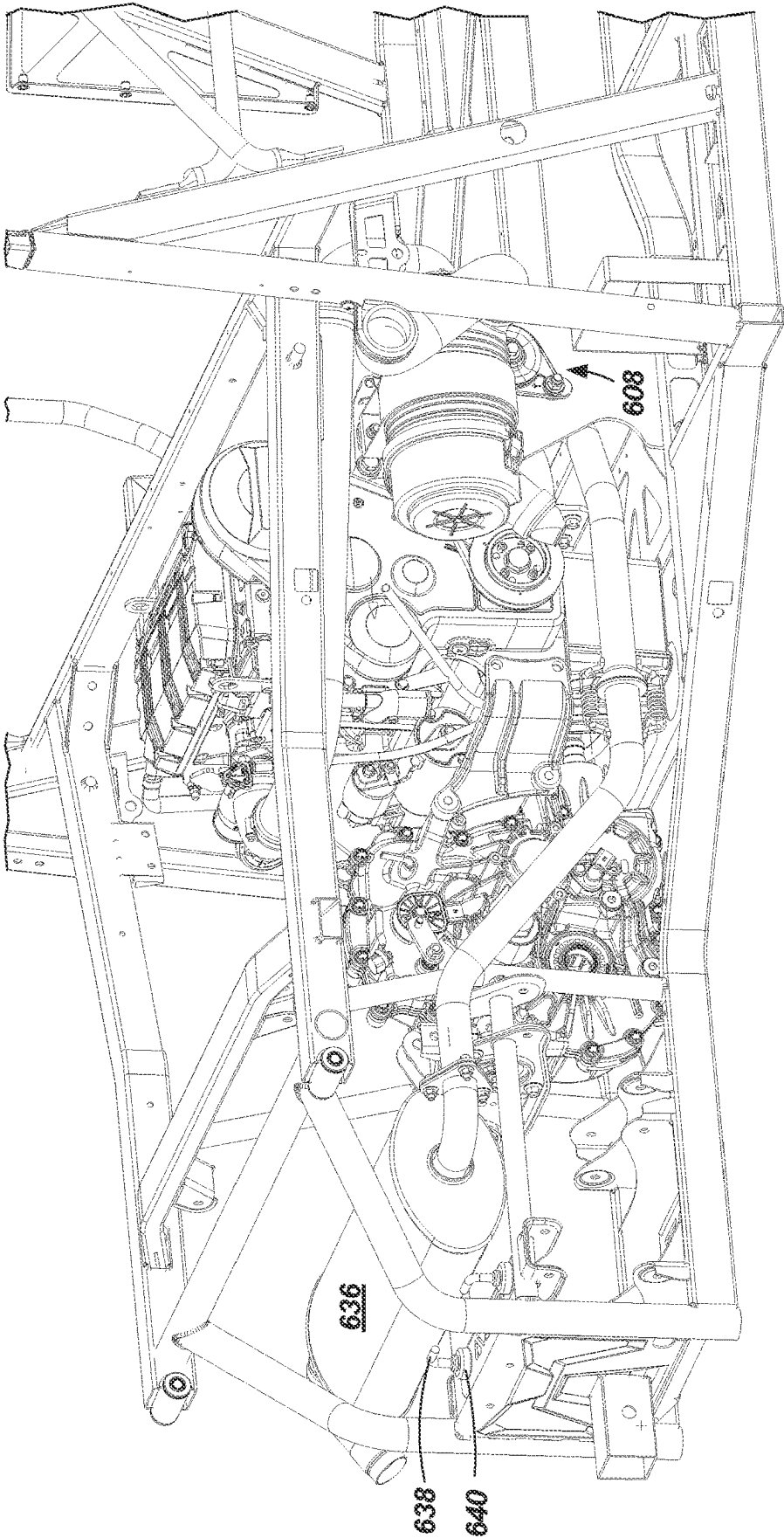


FIG. 103

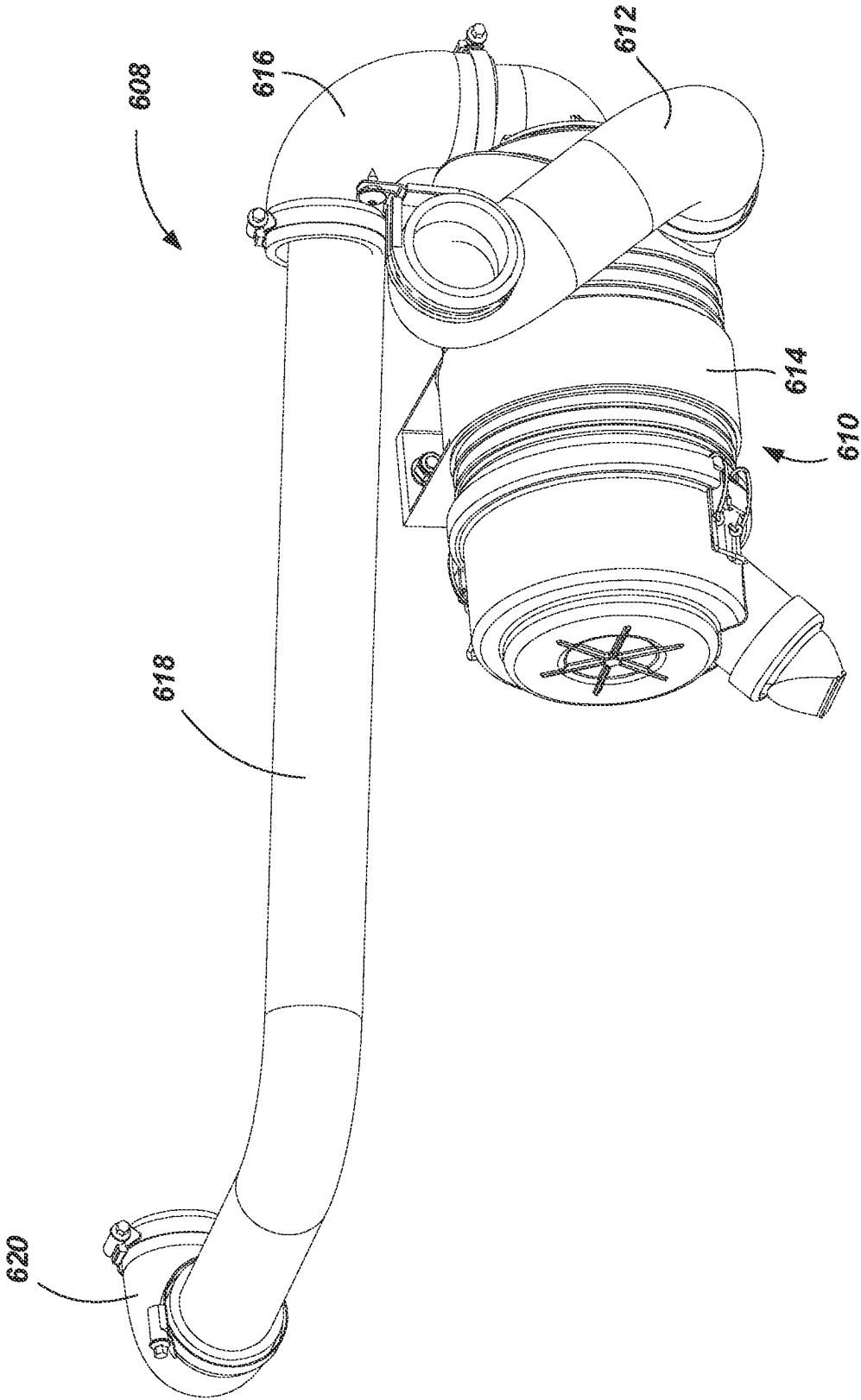


FIG. 104

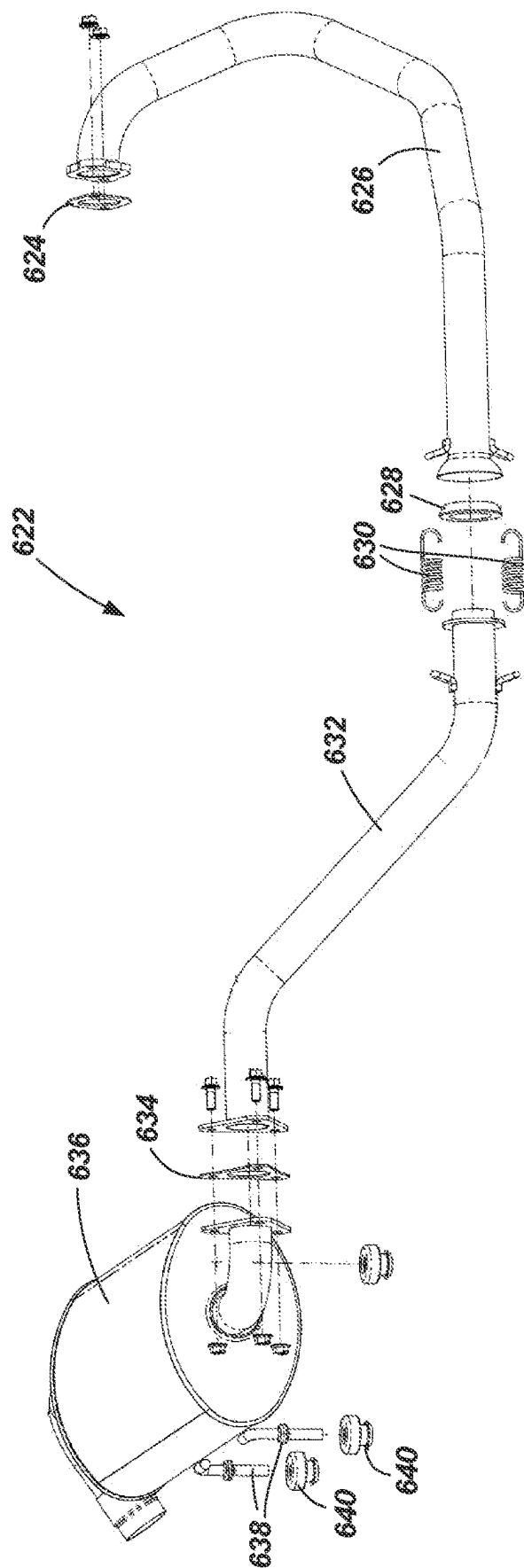


FIG. 105

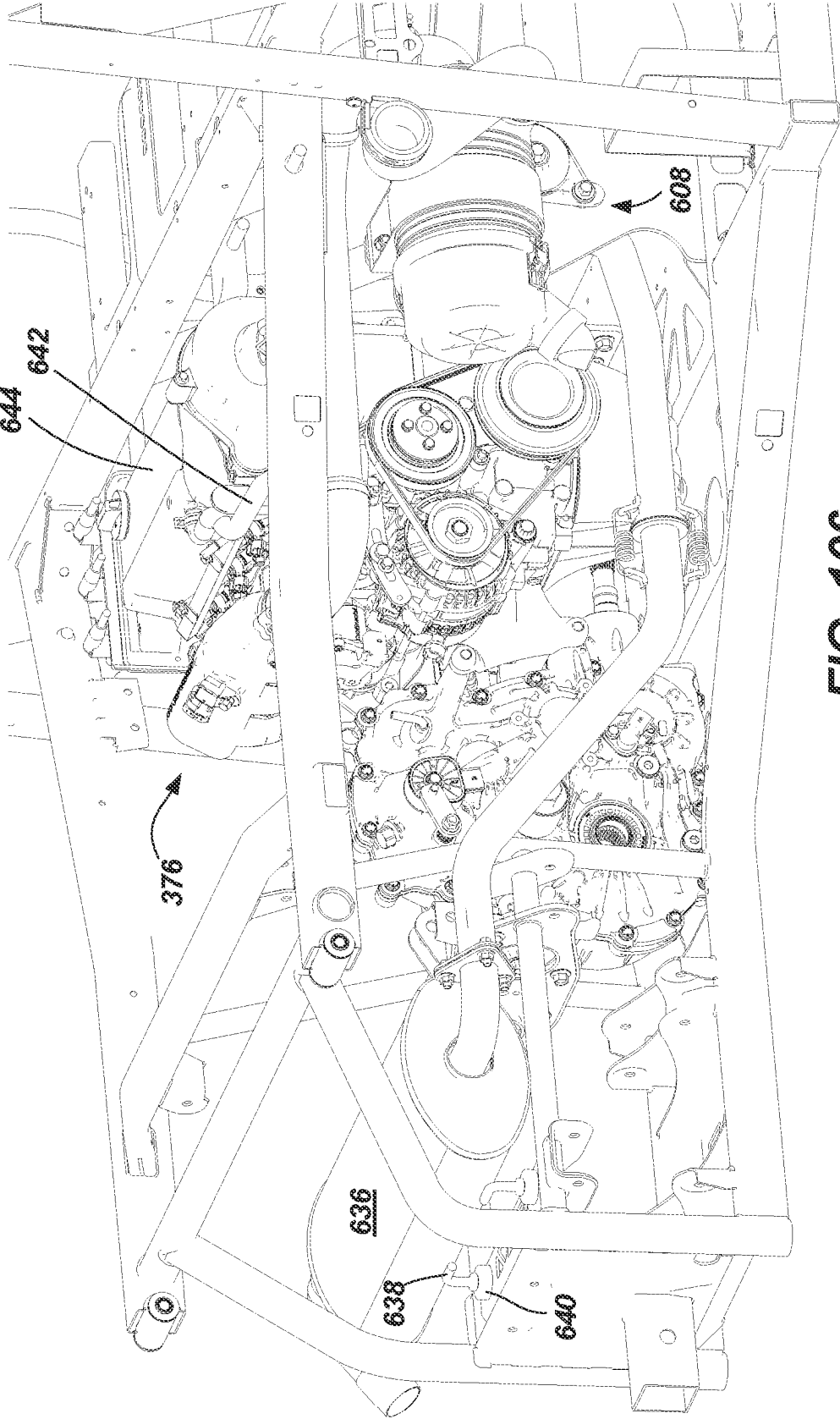


FIG. 106

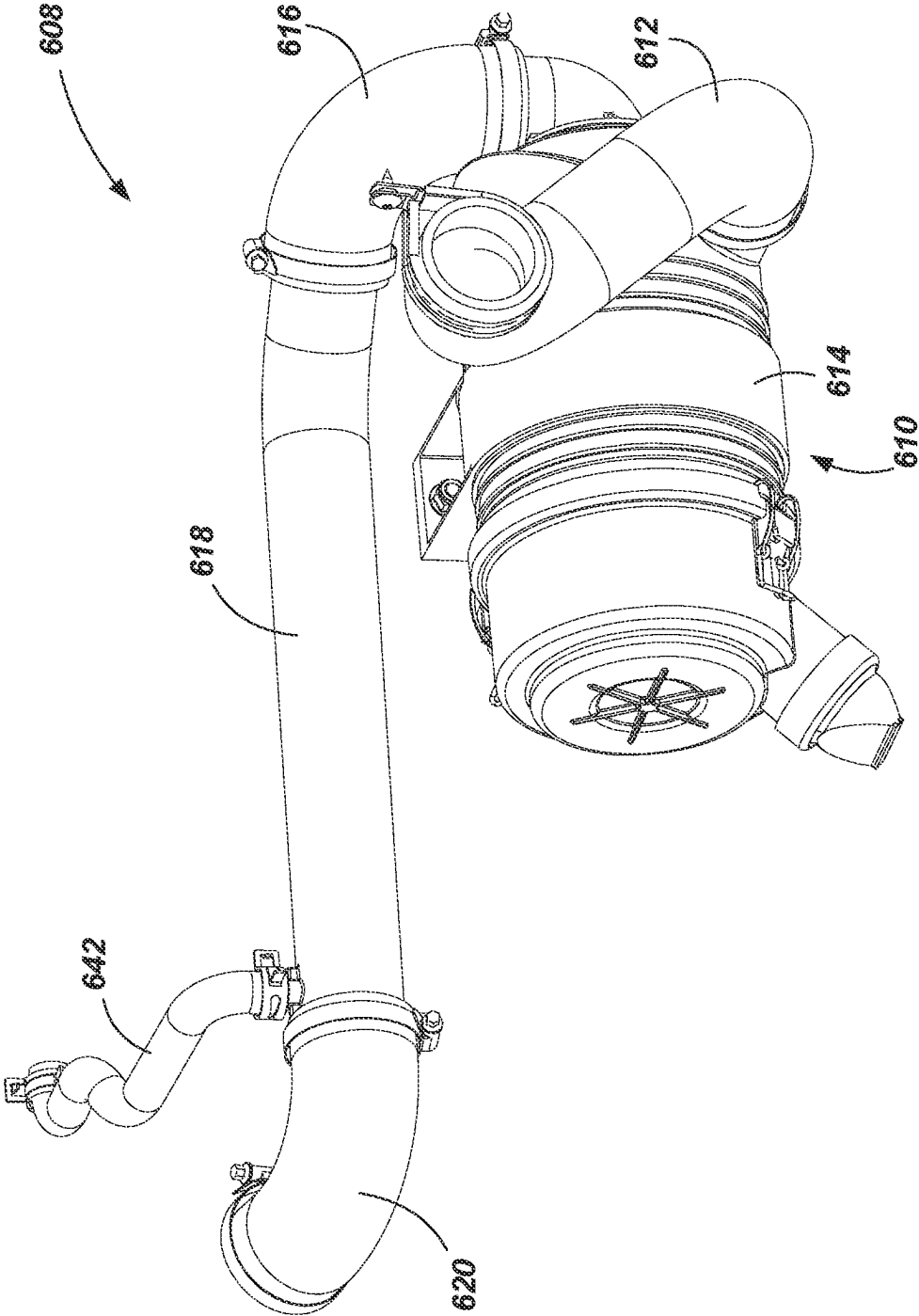
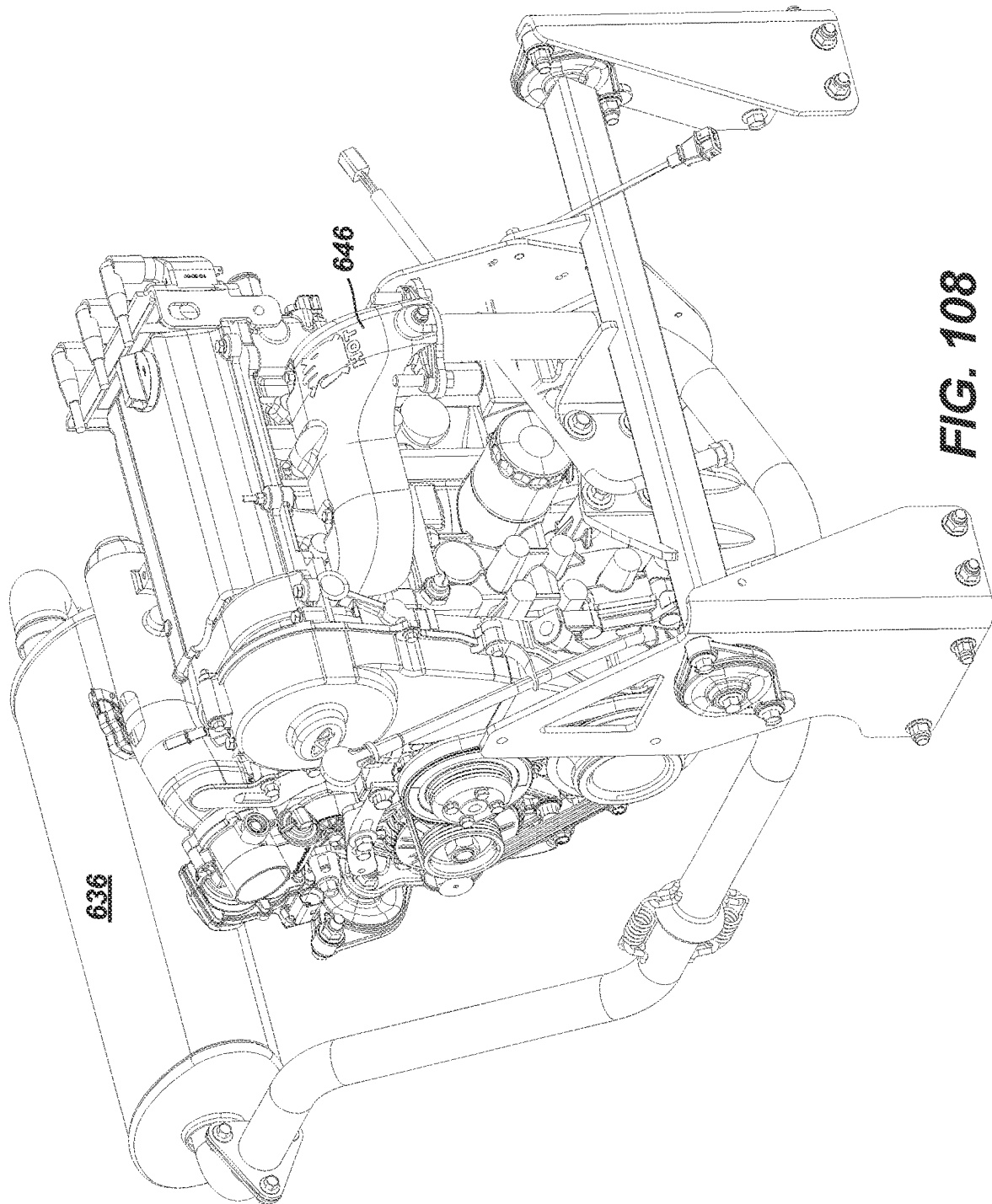


FIG. 107



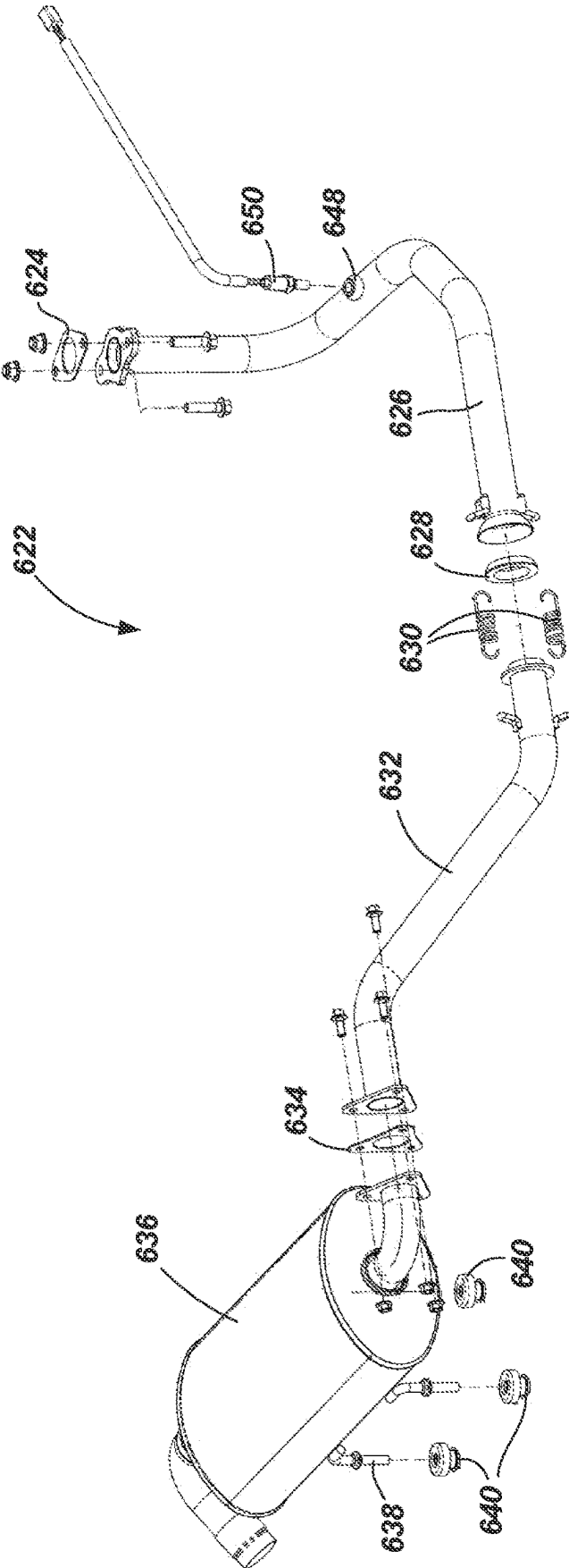


FIG. 109

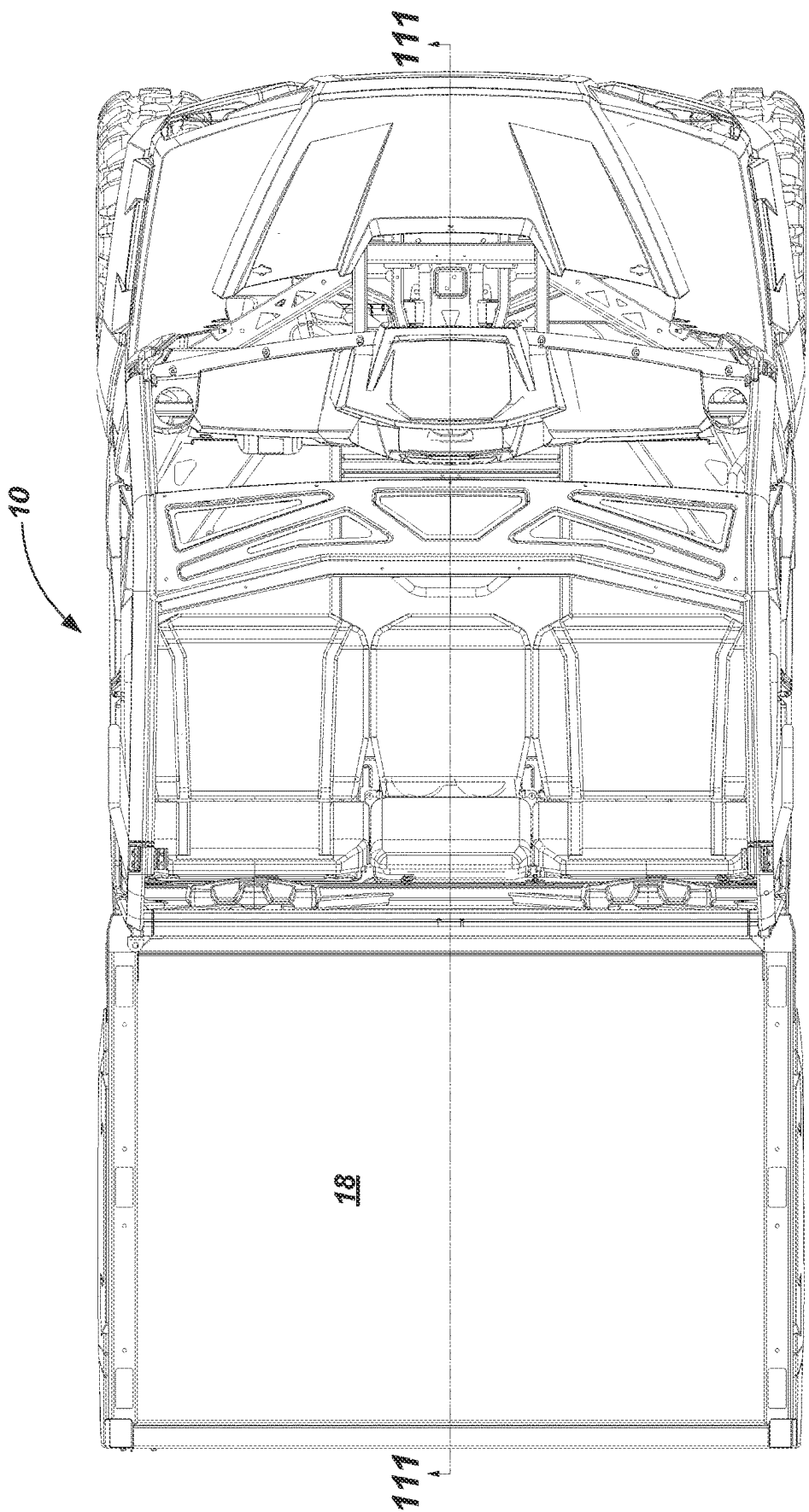


FIG. 110

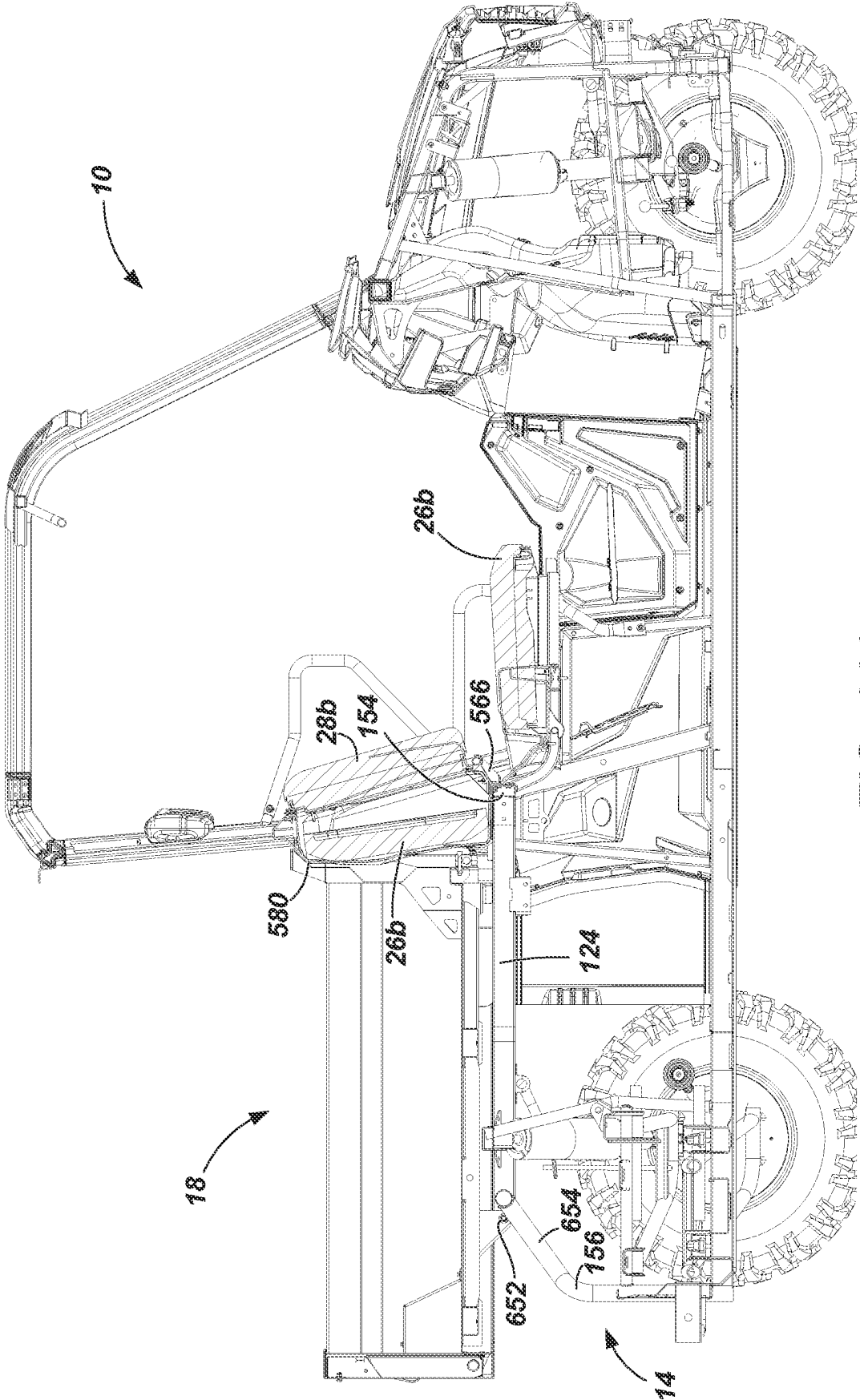


FIG. 111

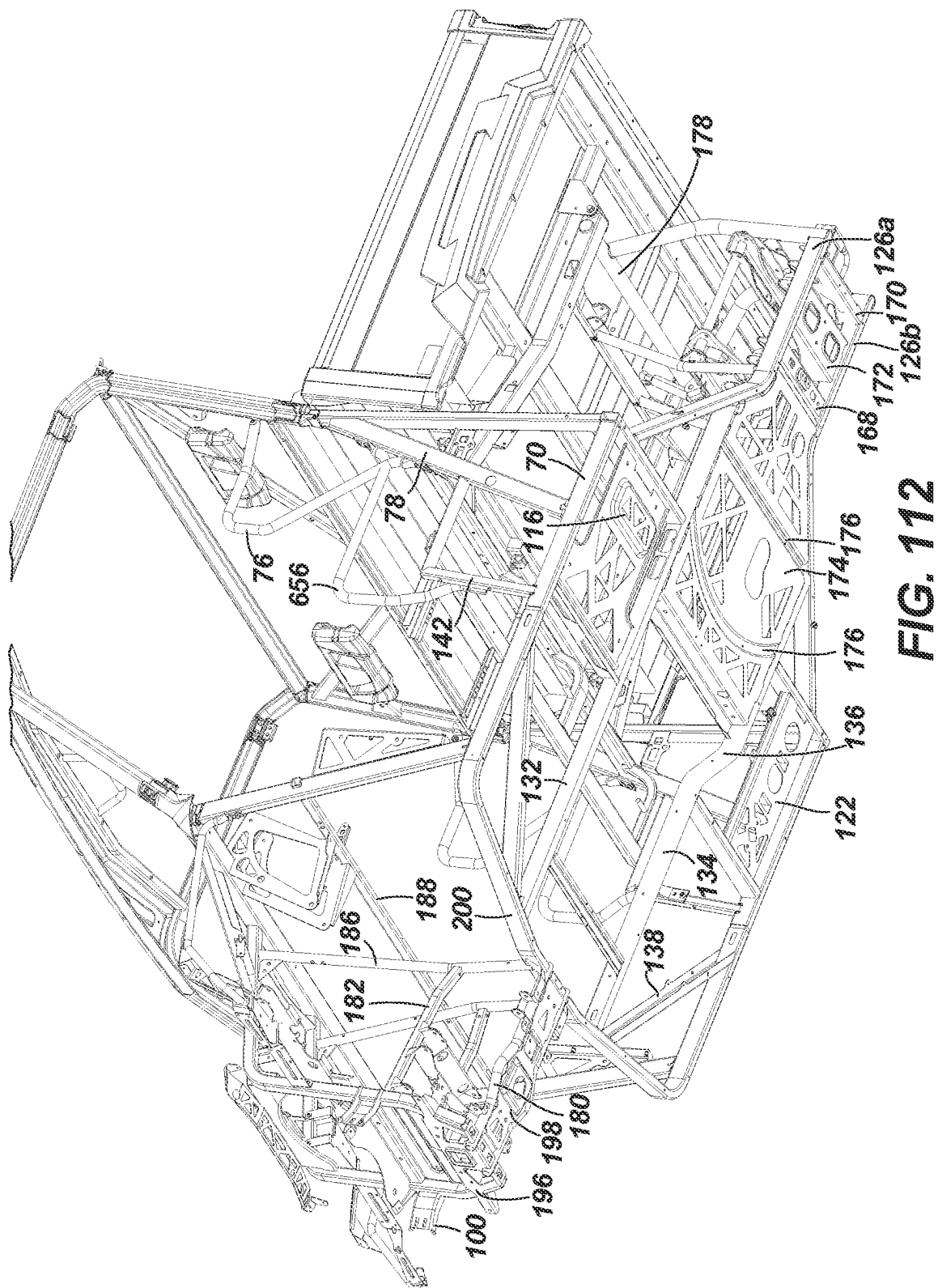


FIG. 112

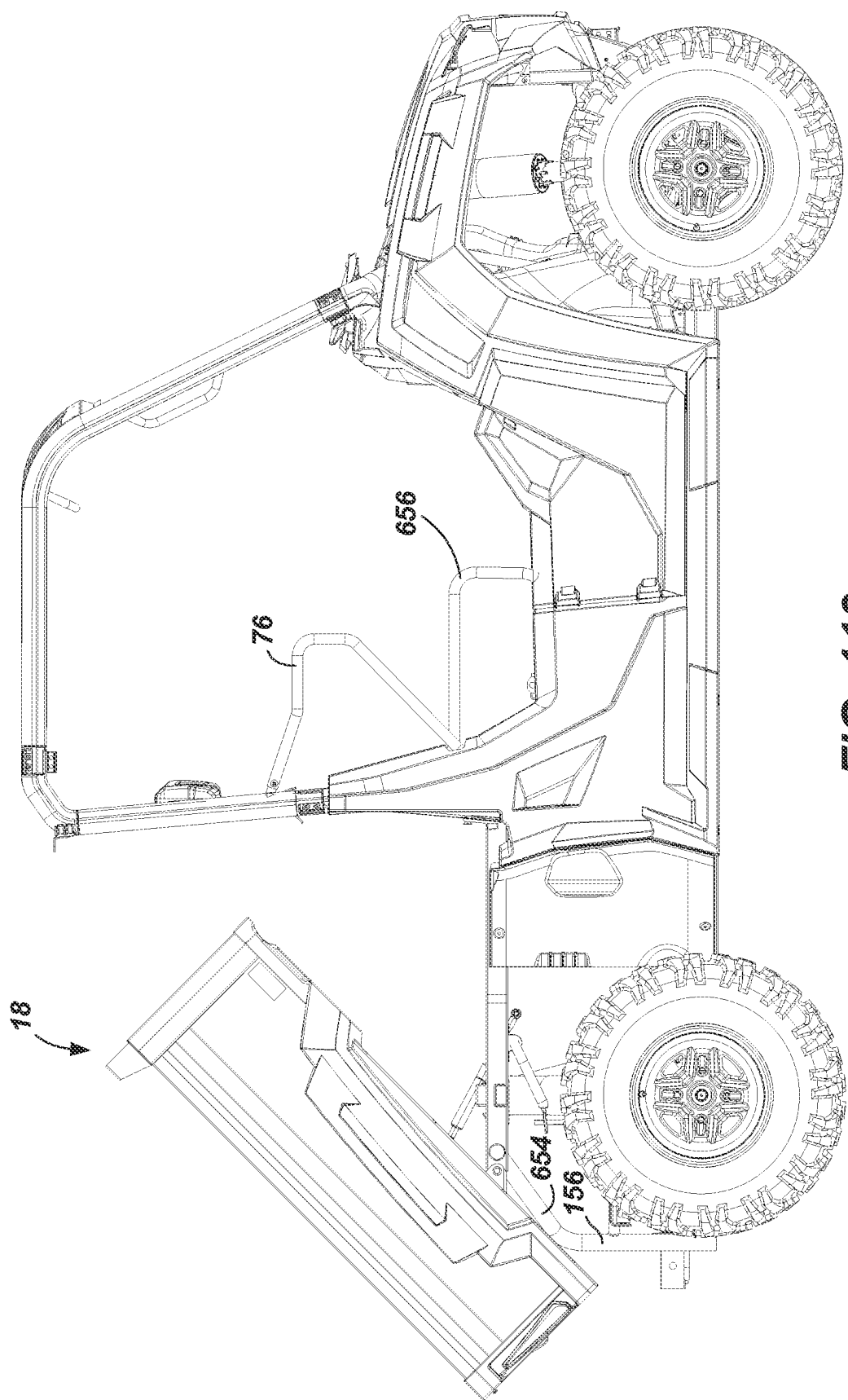


FIG. 113

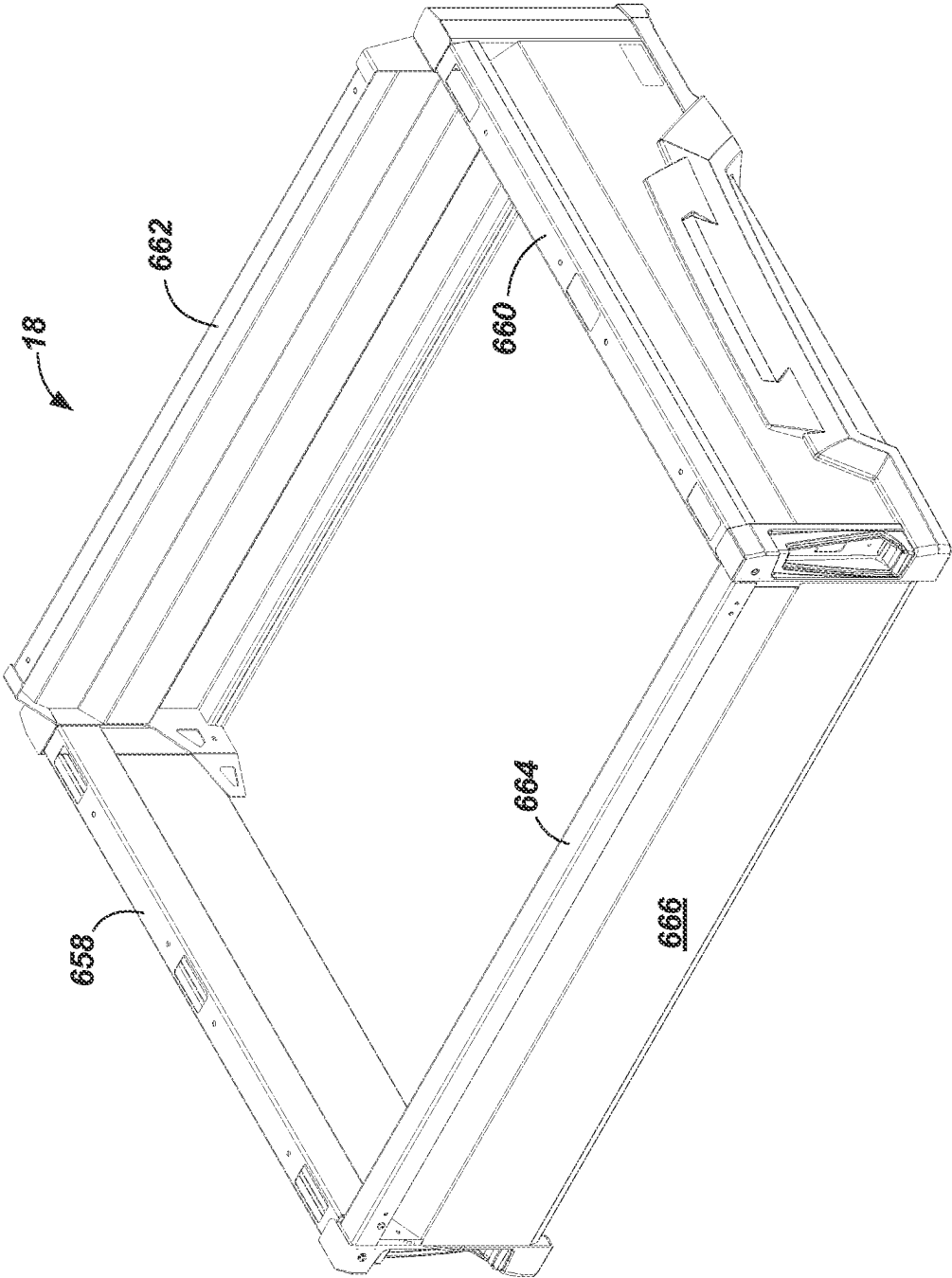
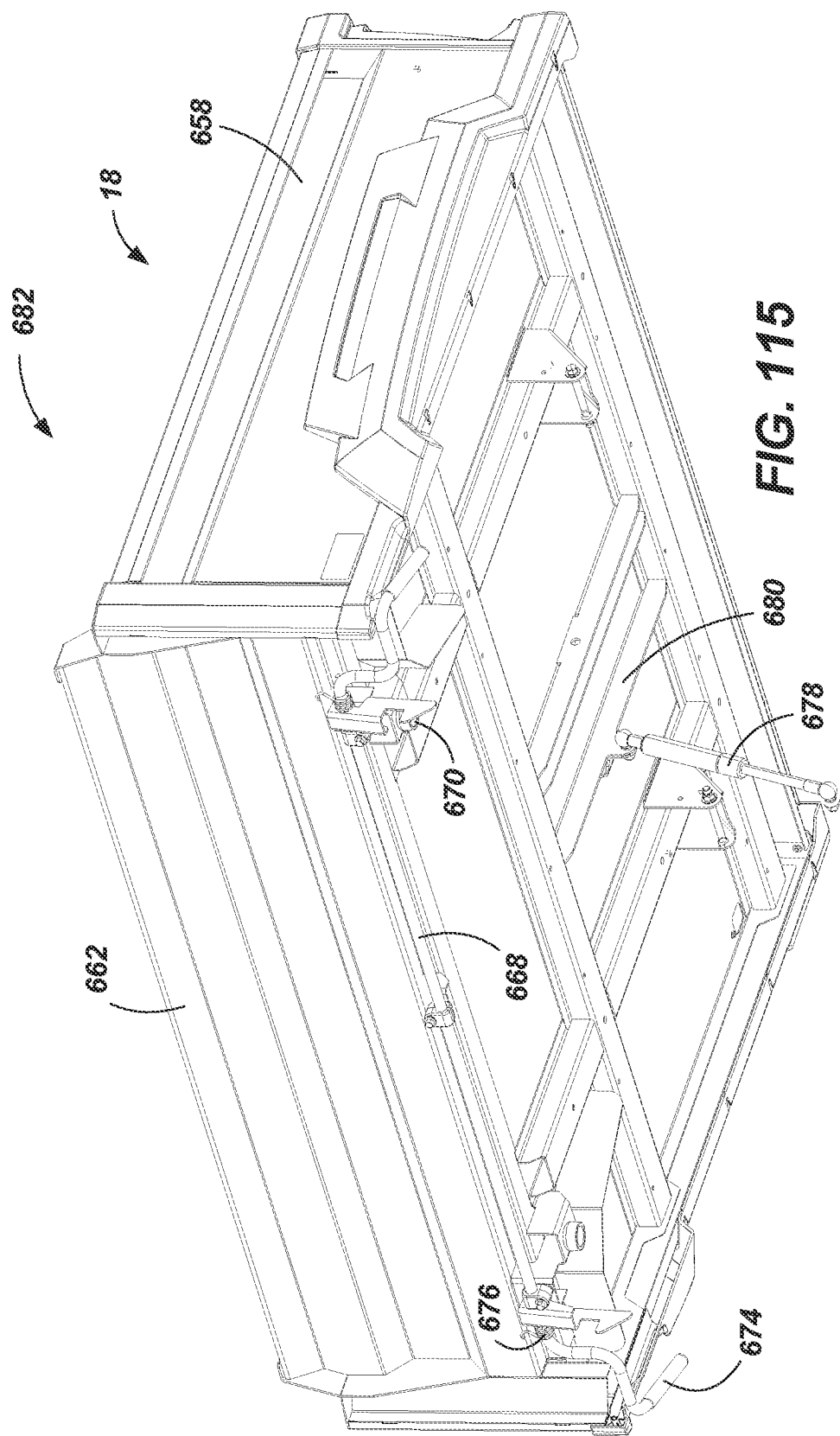
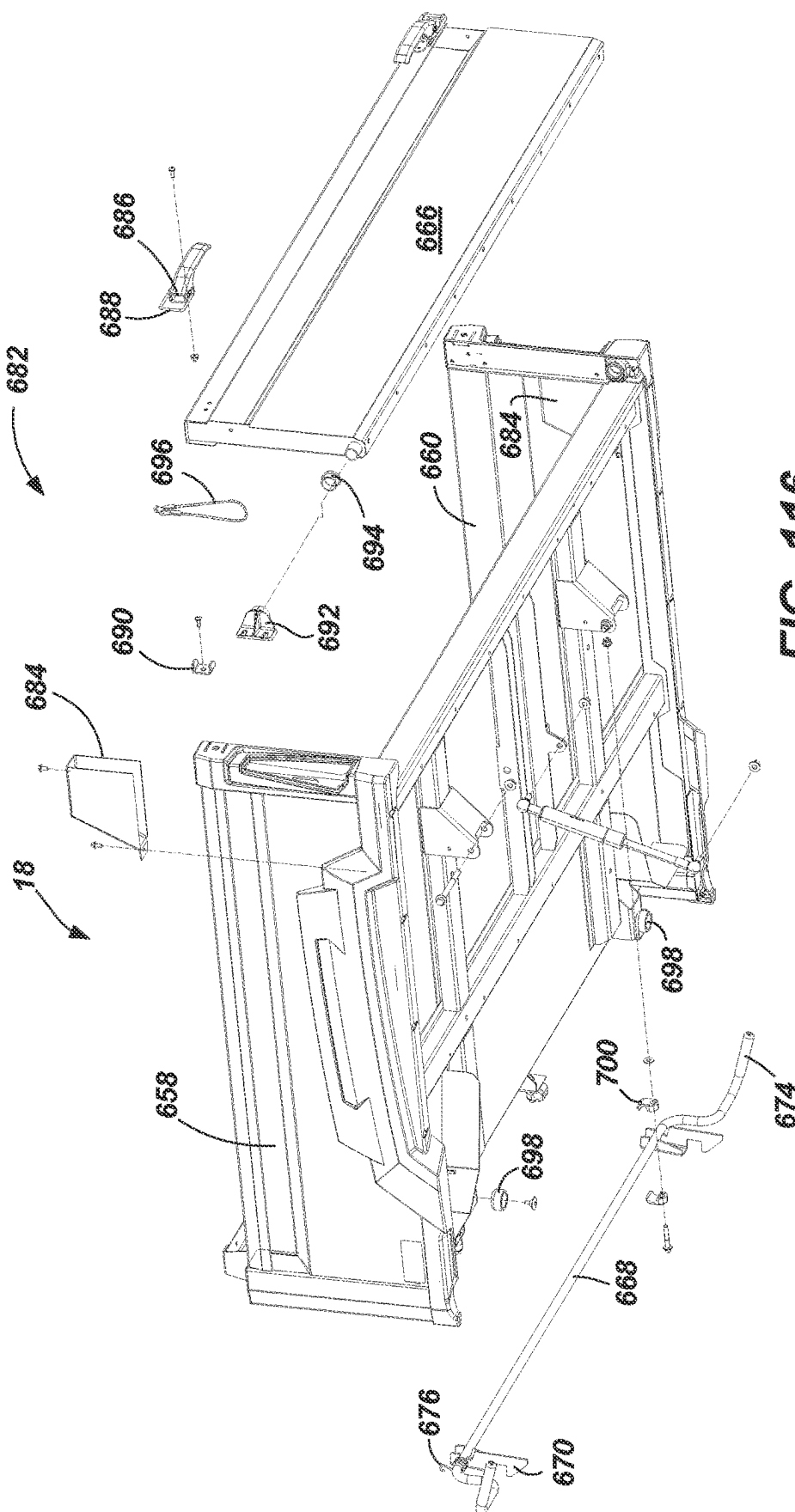
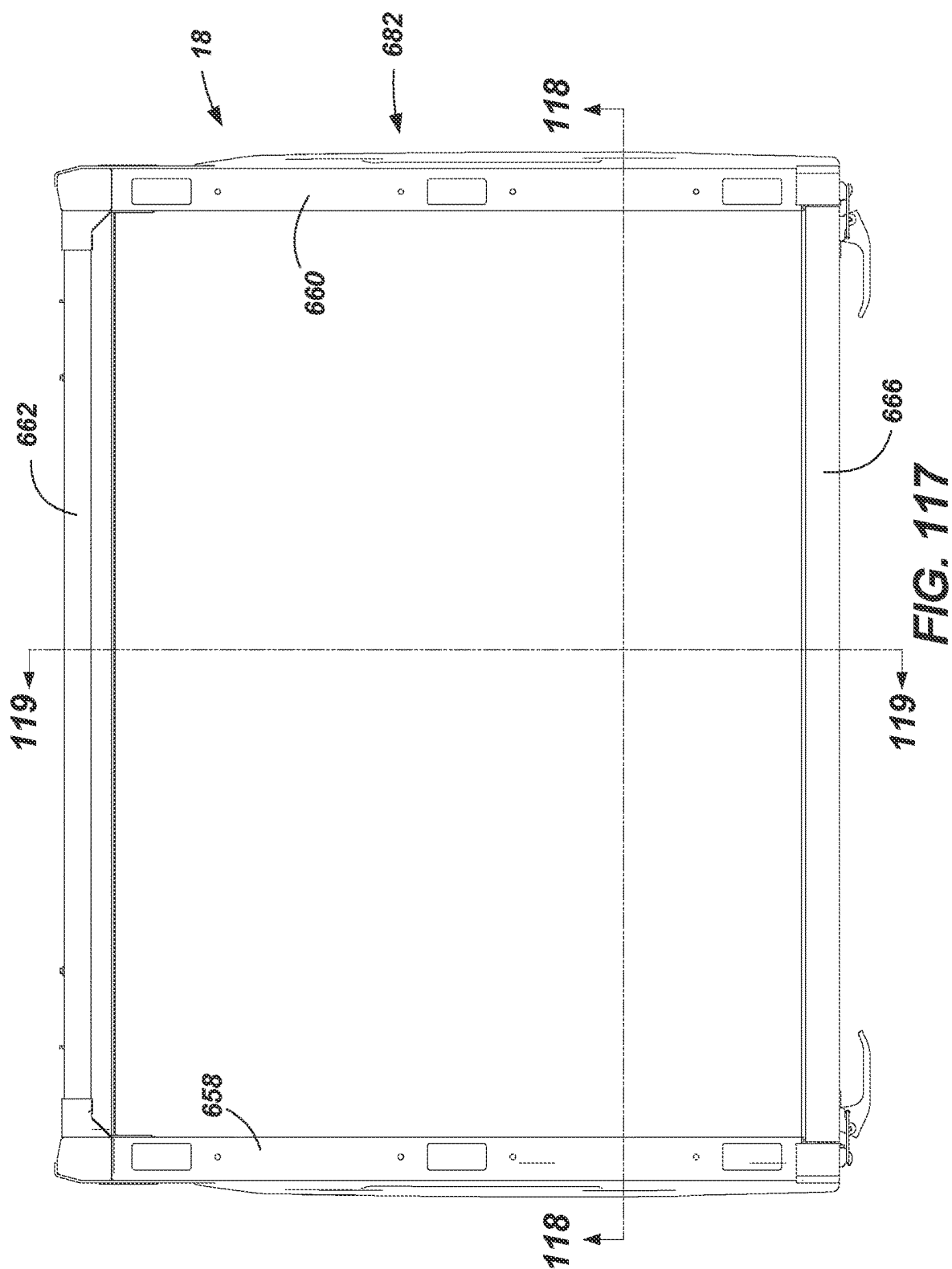


FIG. 114







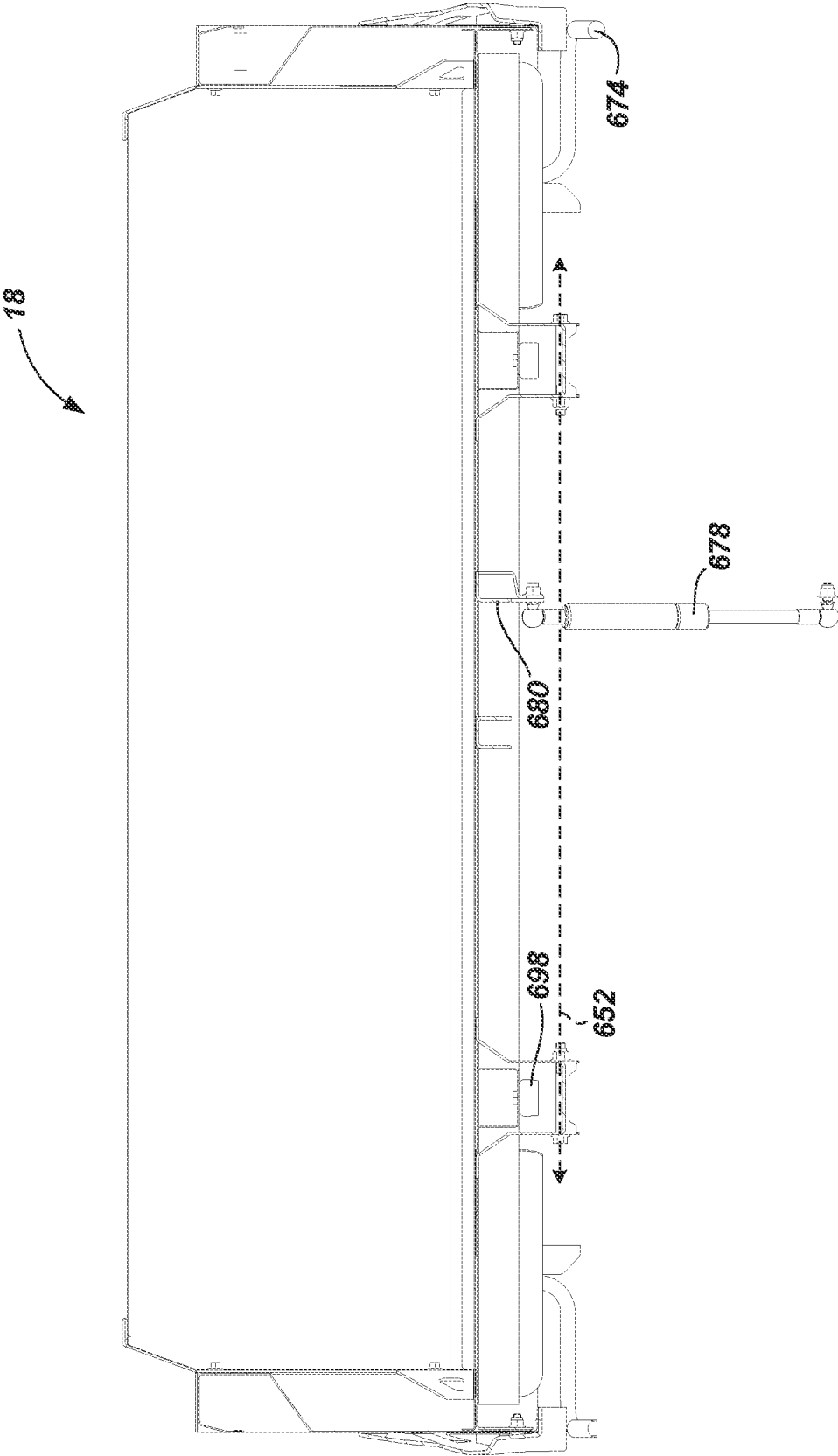


FIG. 118

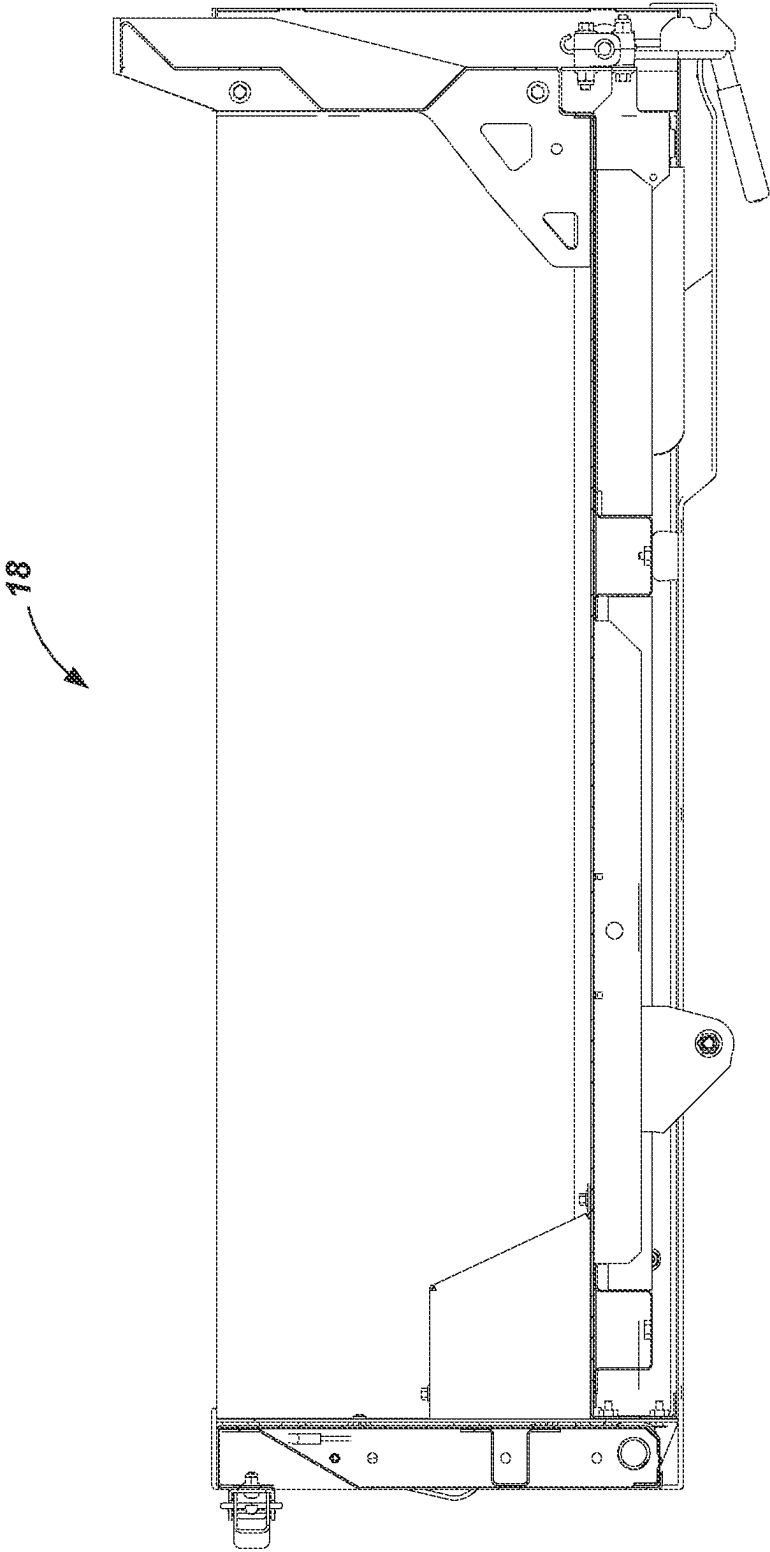


FIG. 119

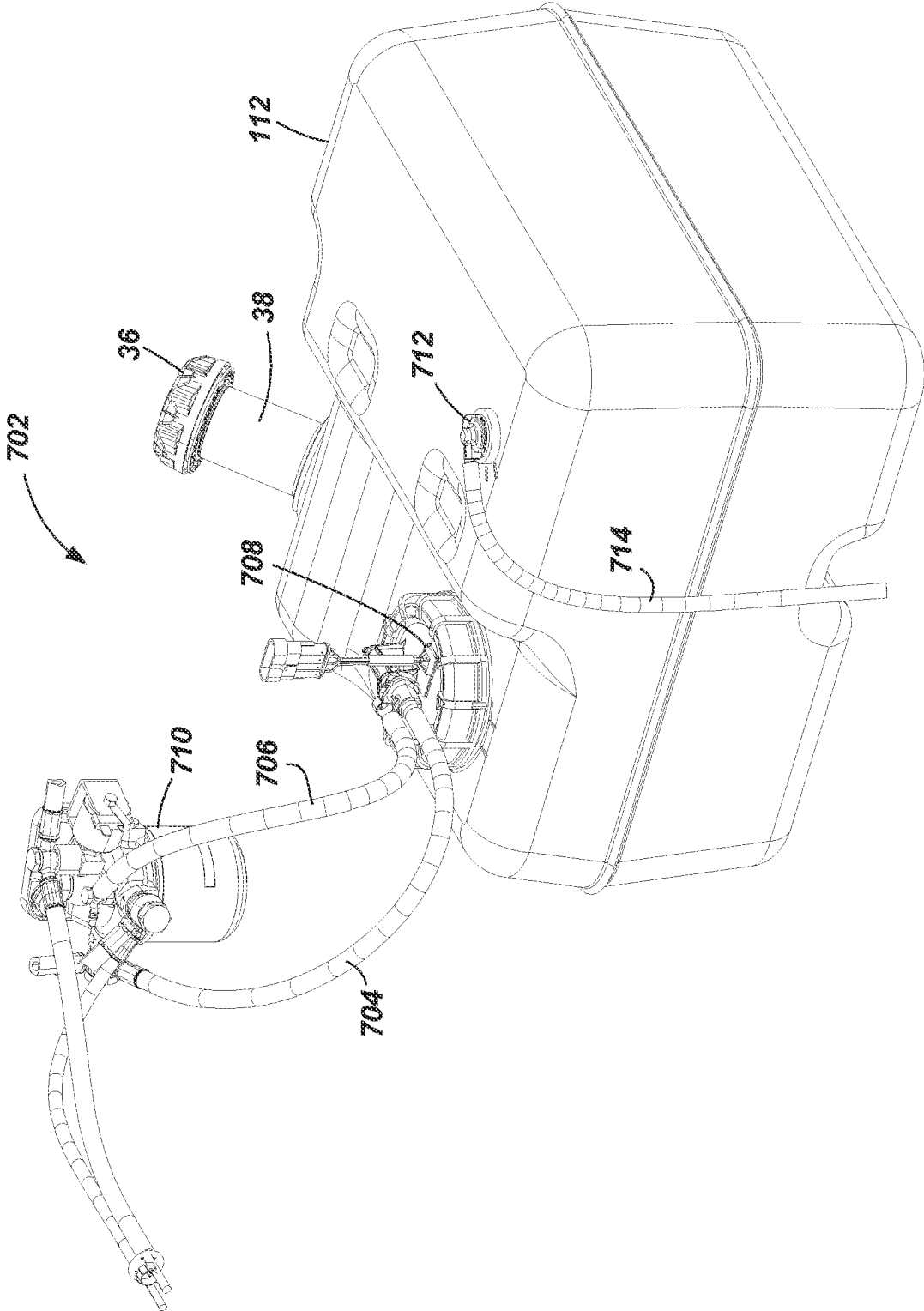


FIG. 120

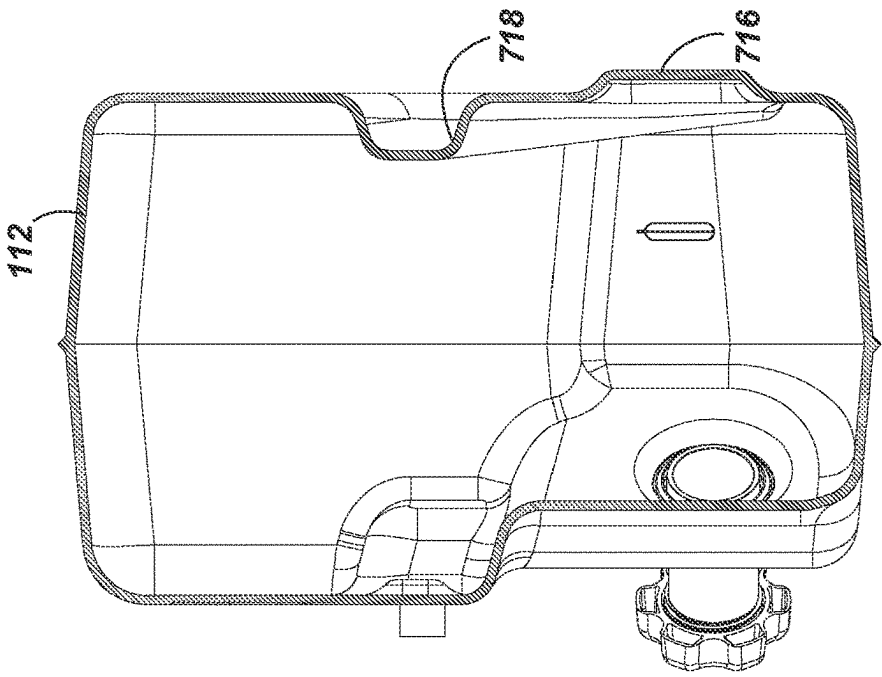


FIG. 122

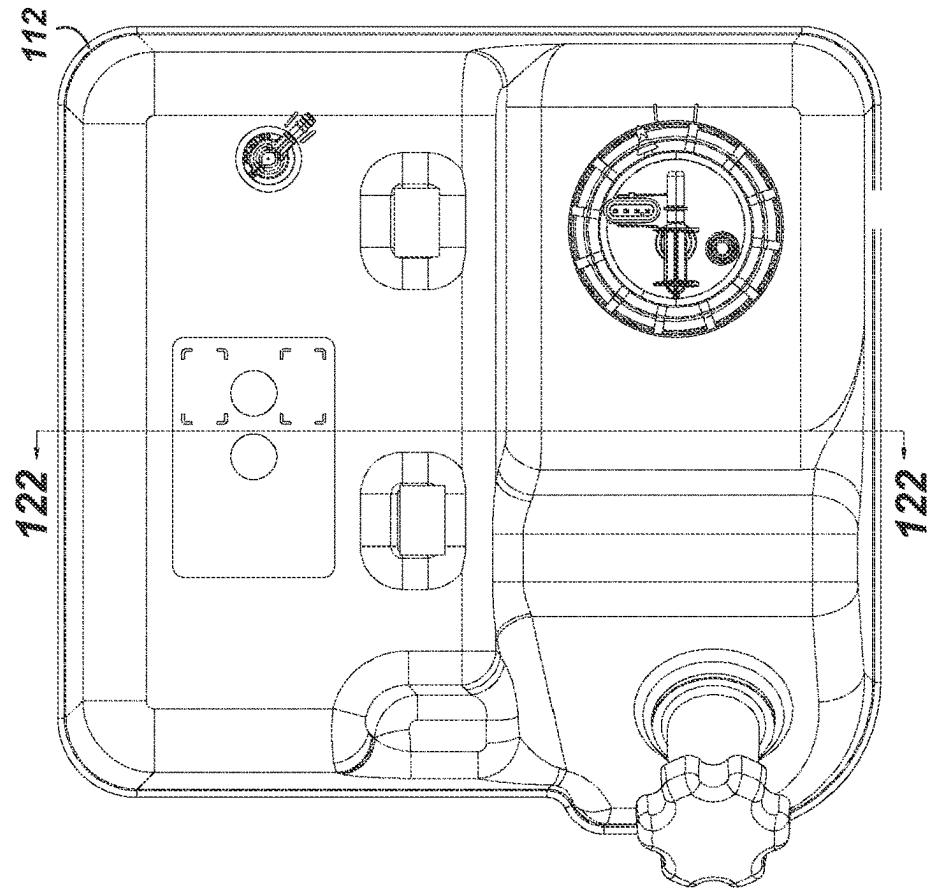
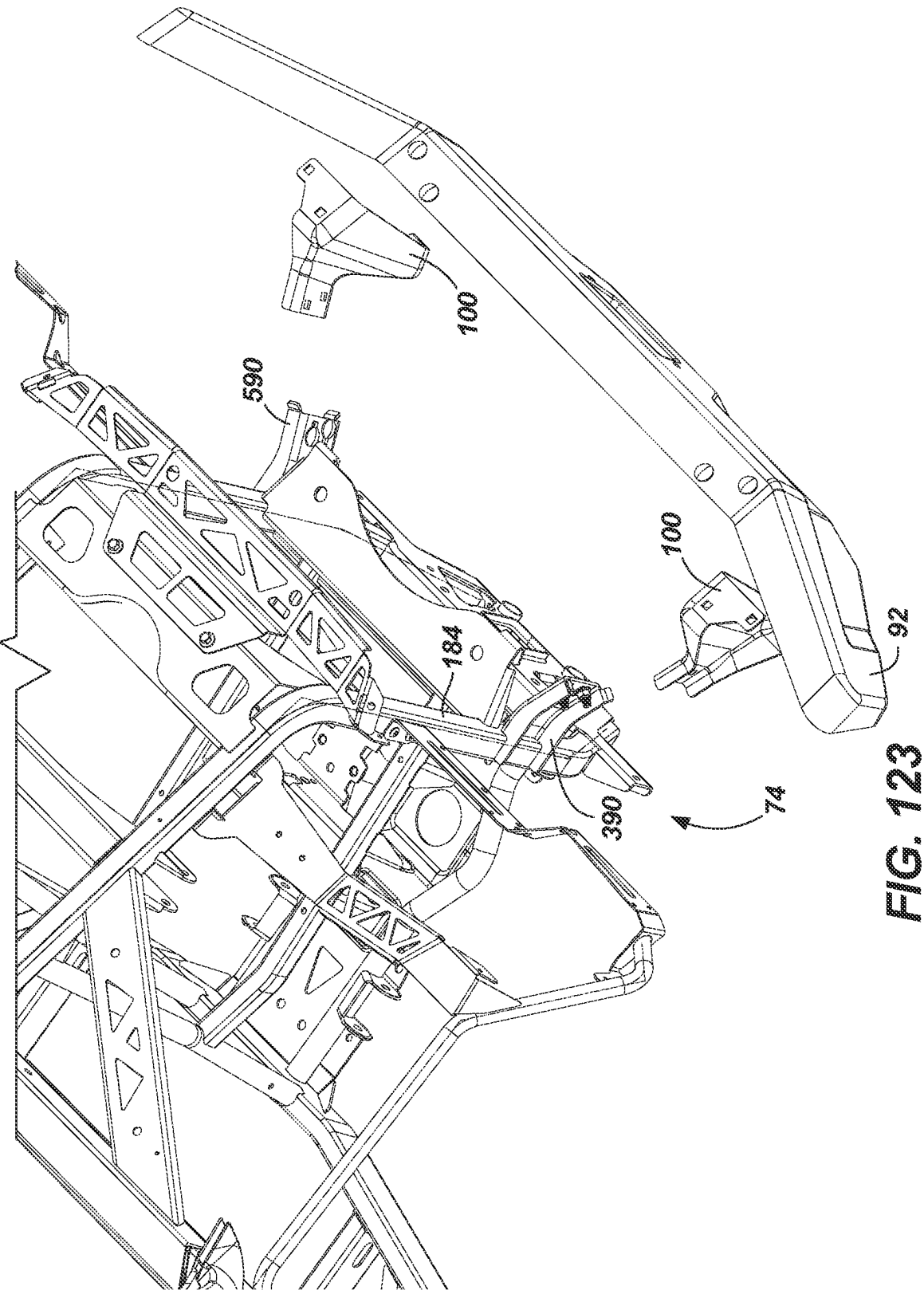


FIG. 121



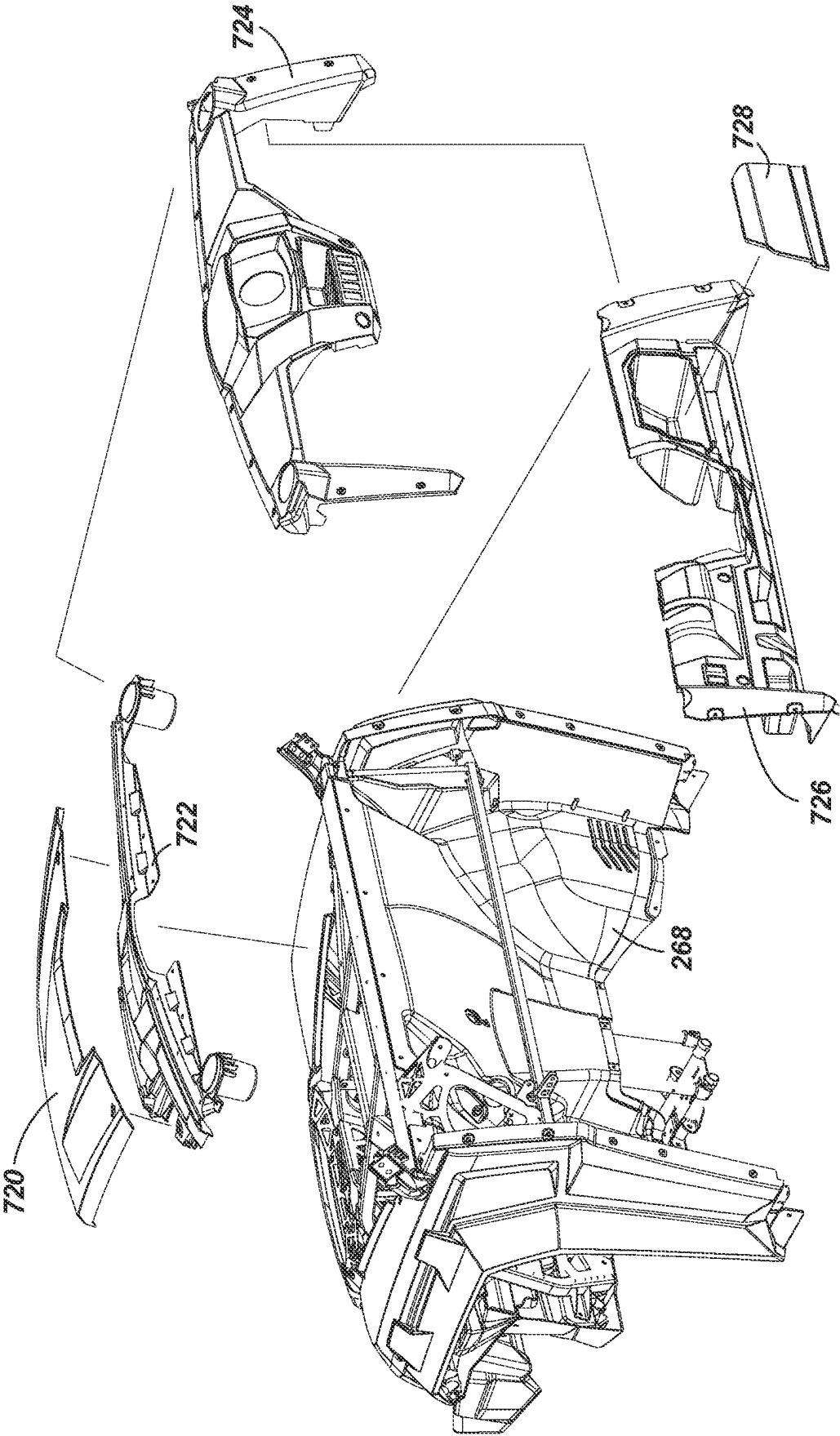
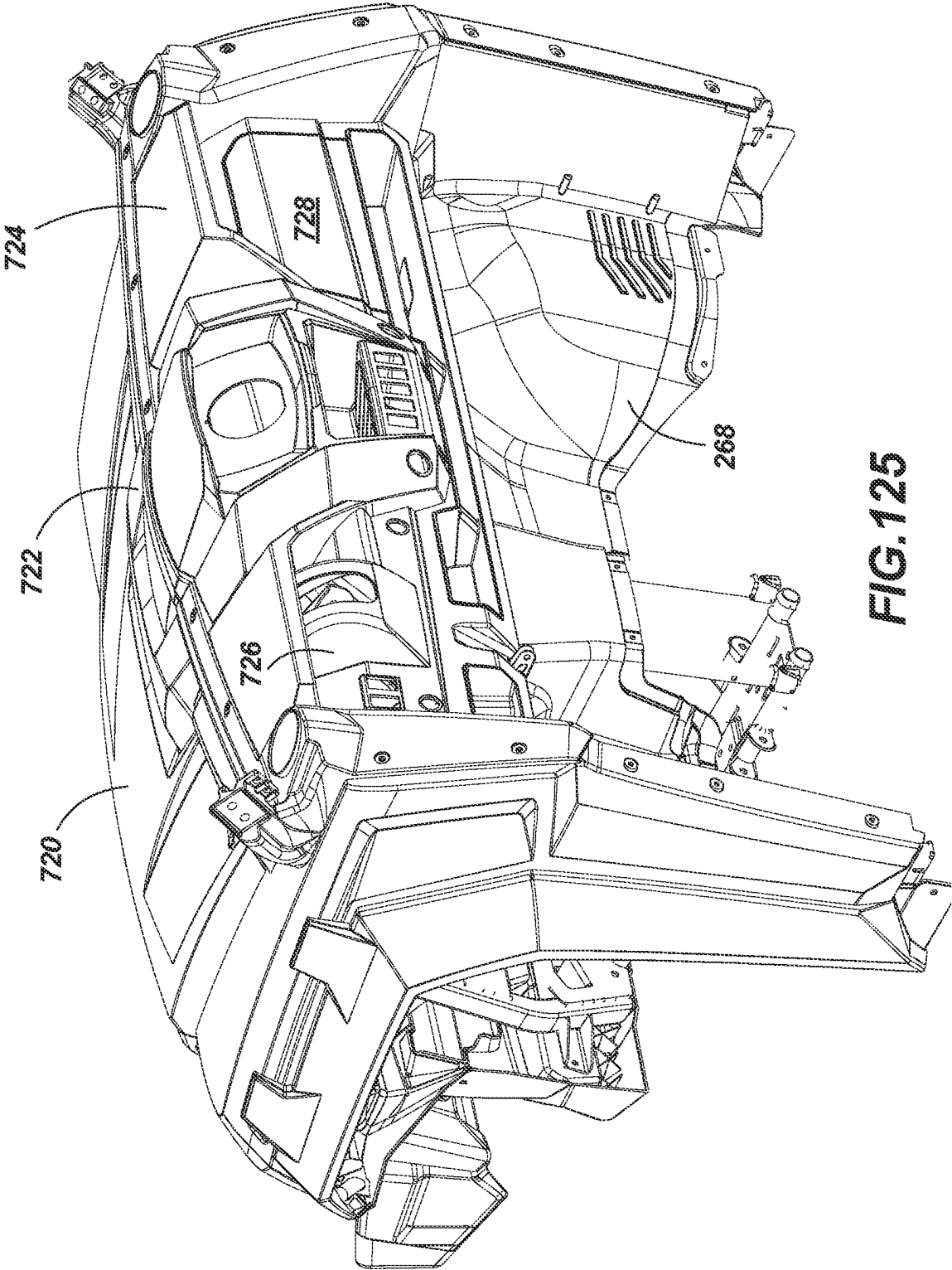
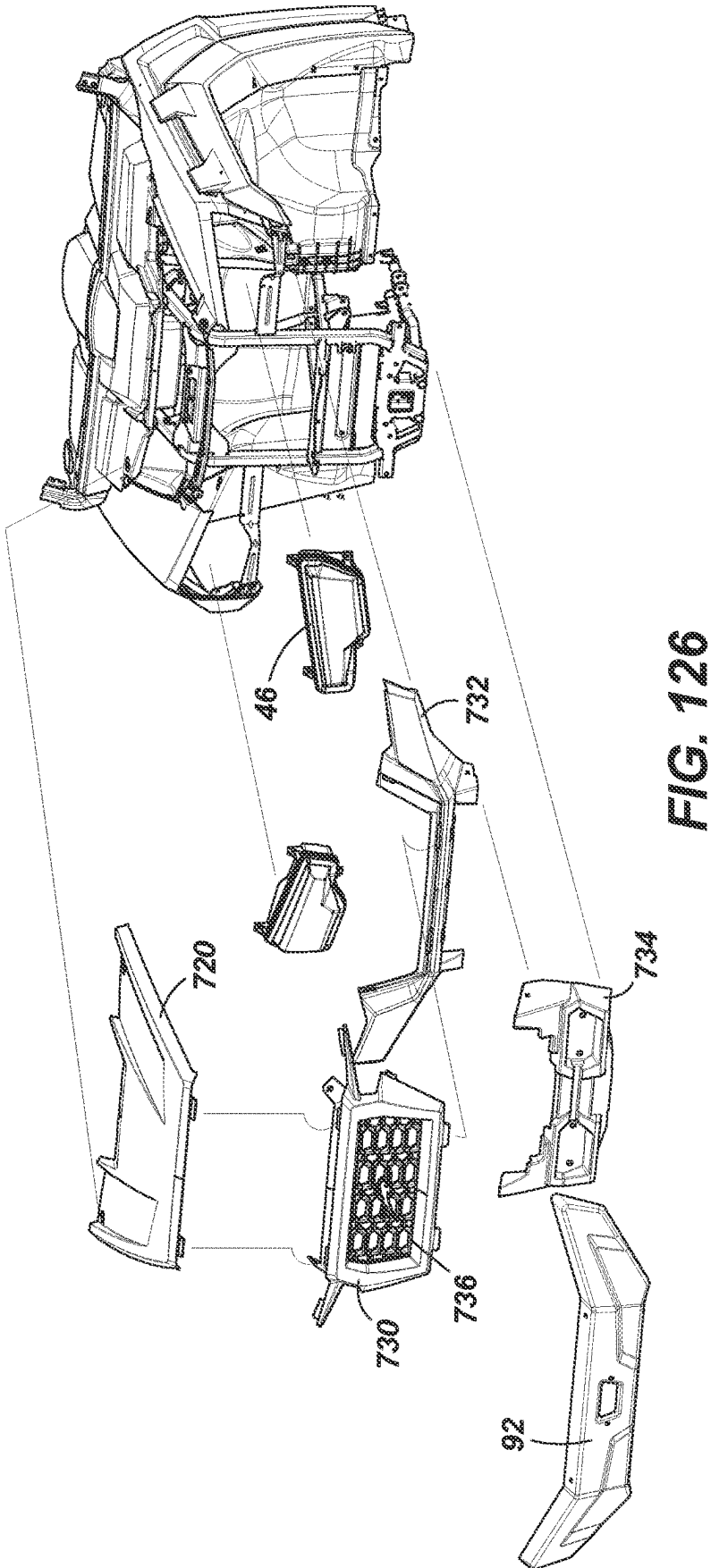


FIG. 124





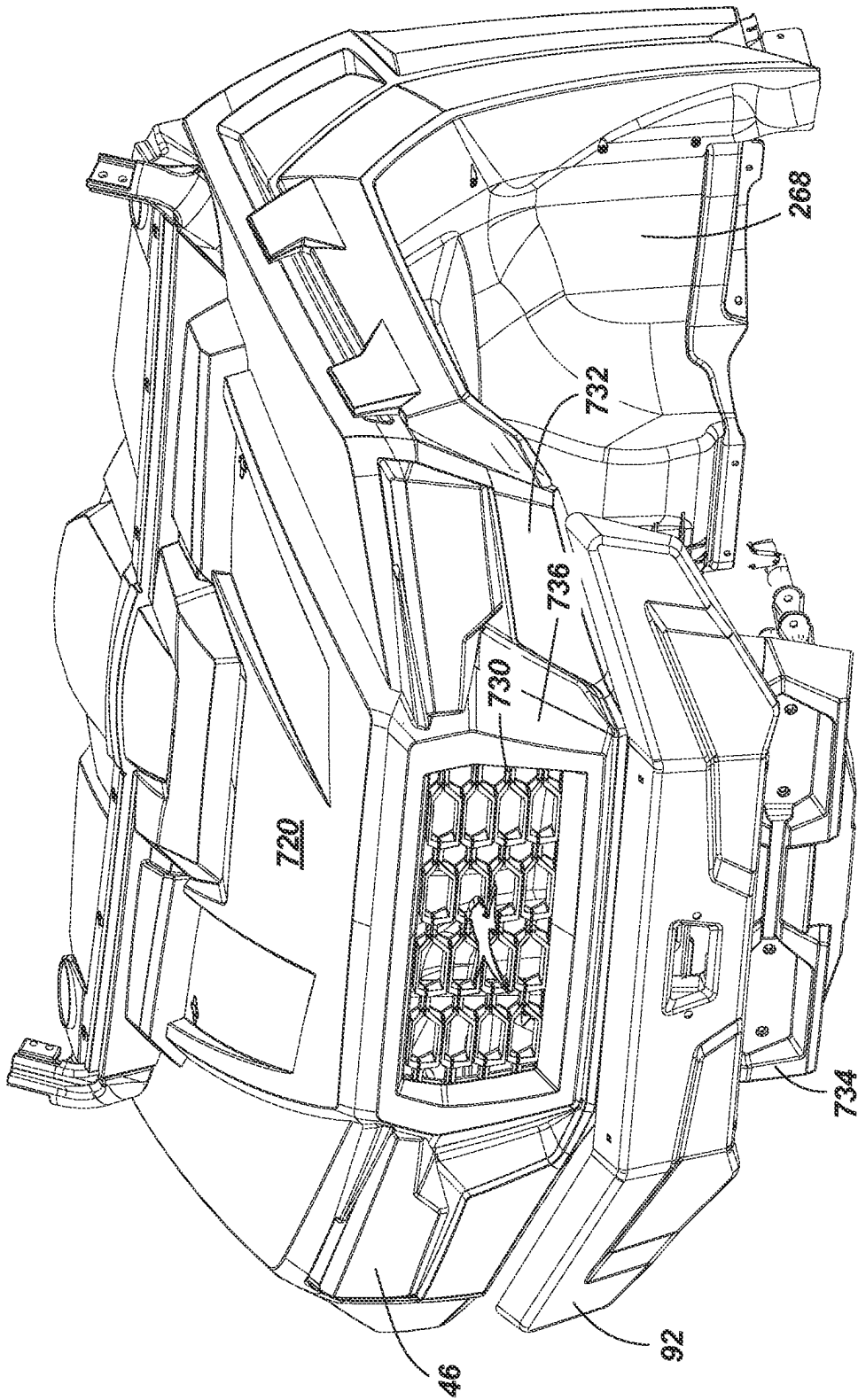
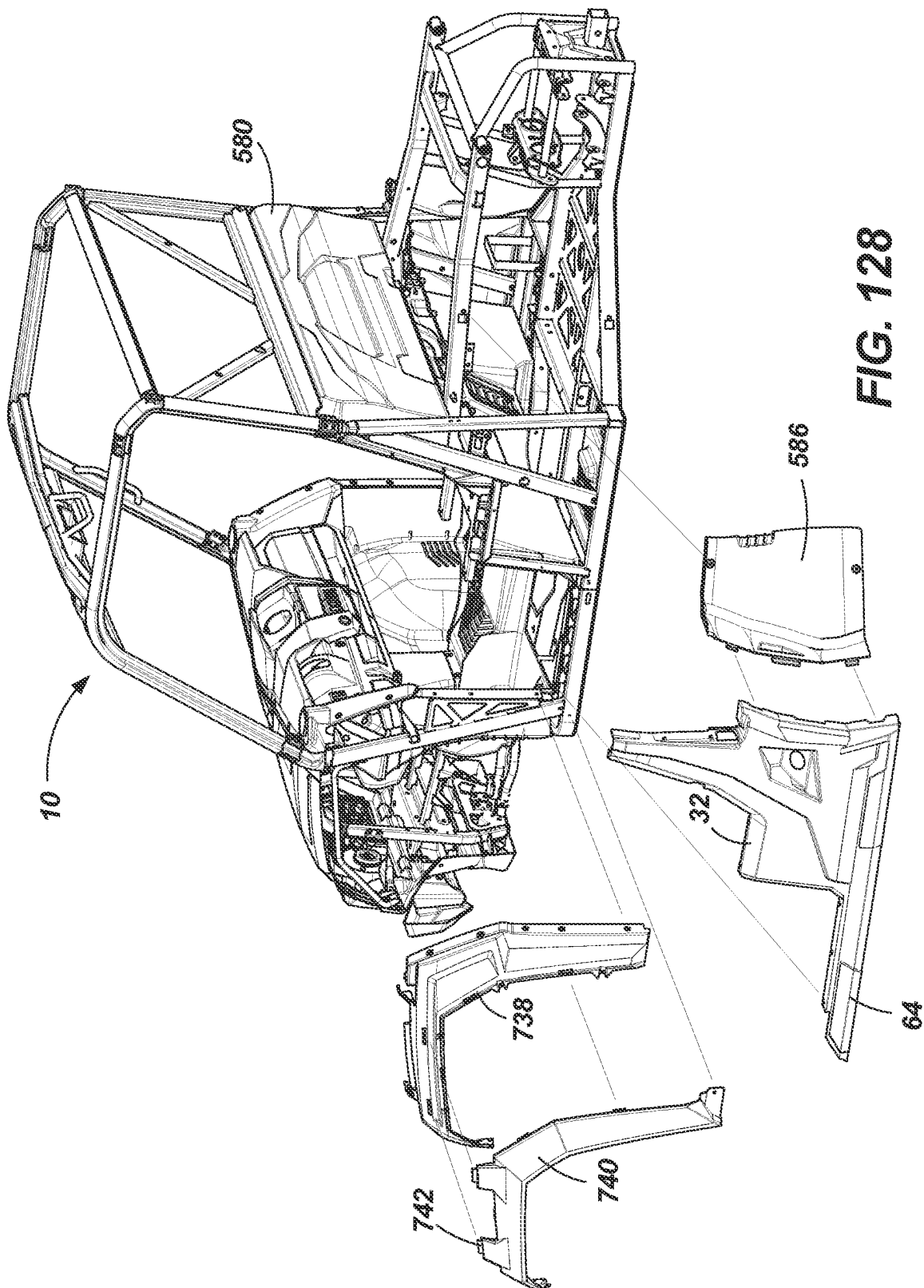


FIG. 127



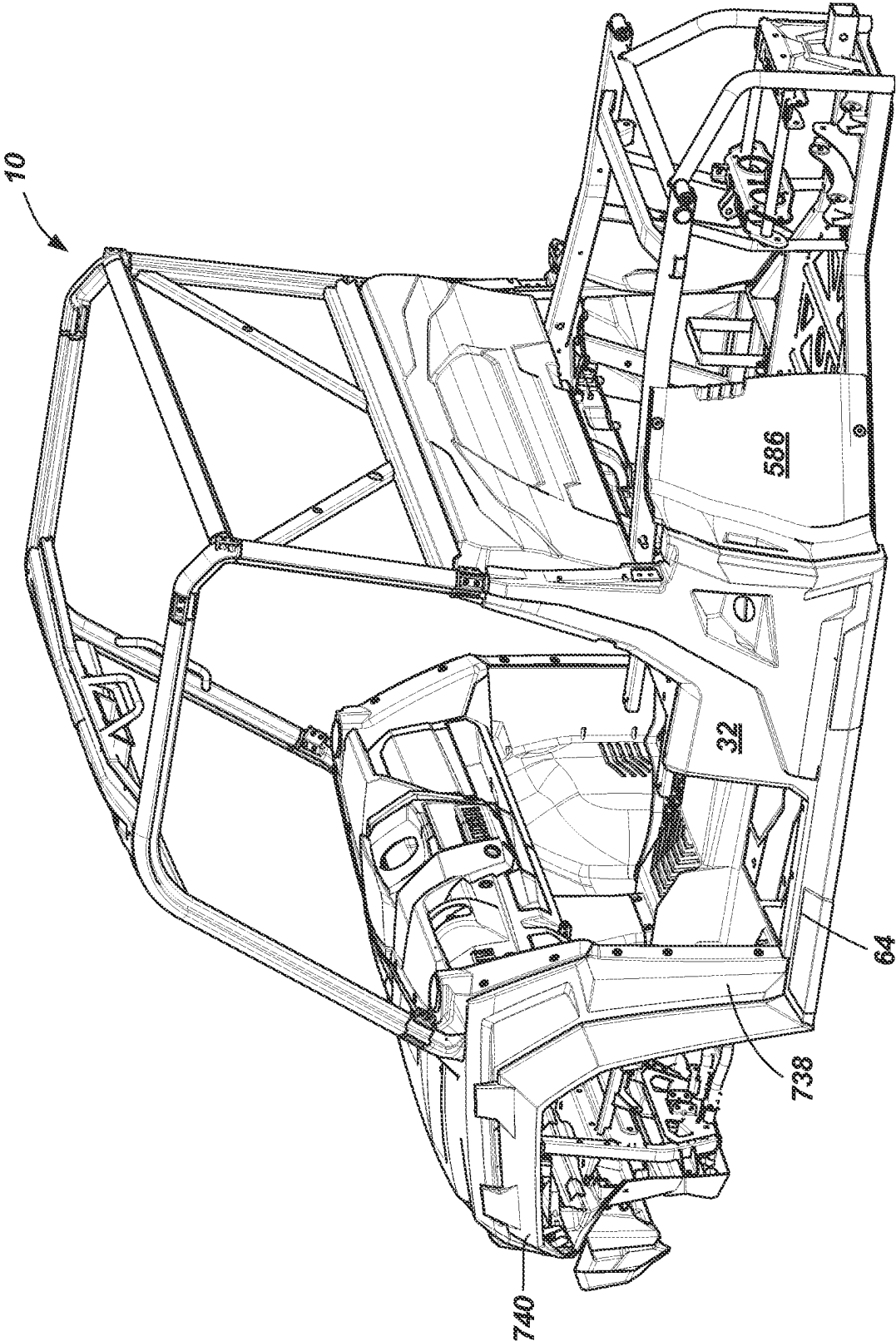


FIG. 129

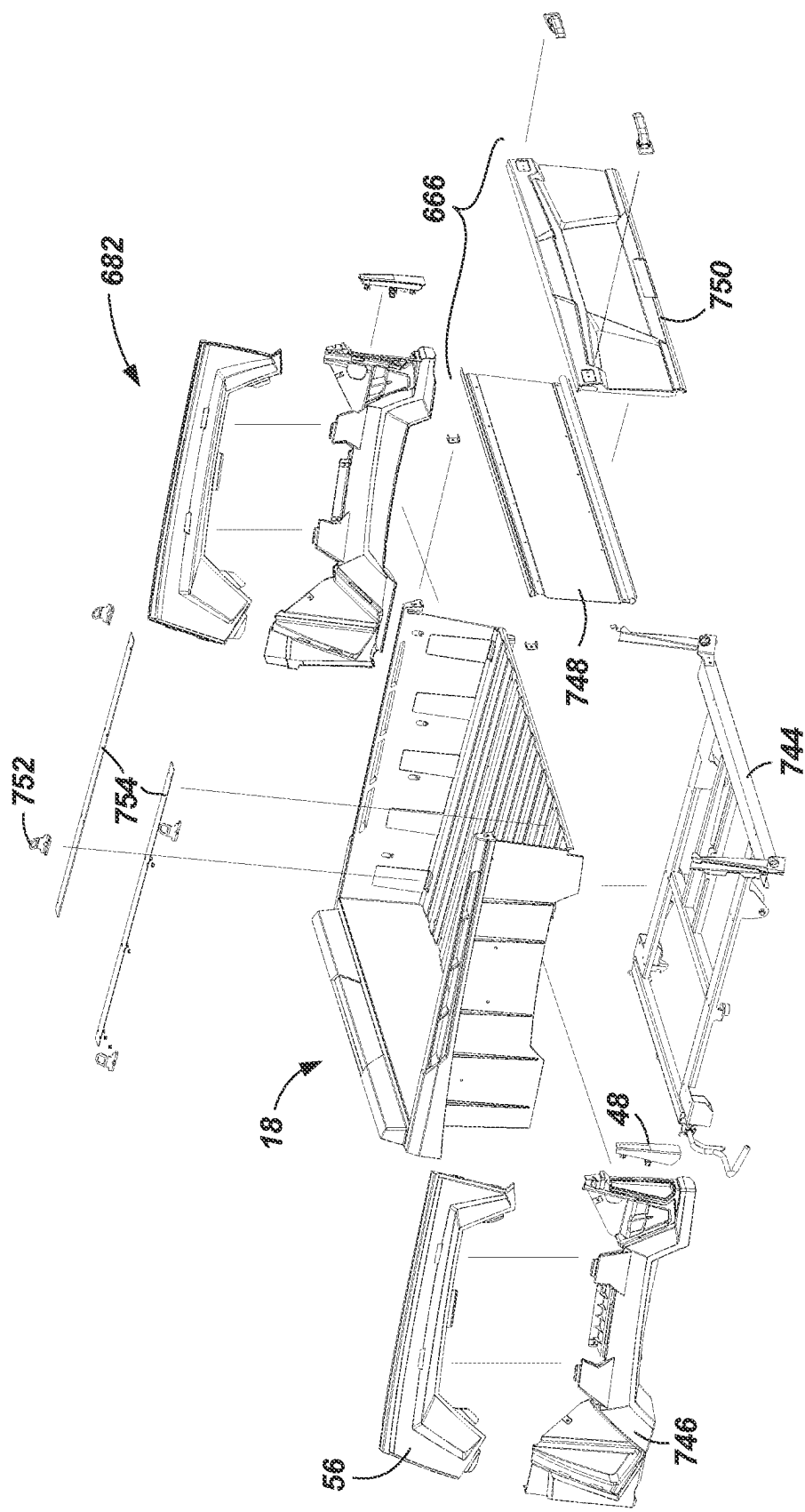


FIG. 130

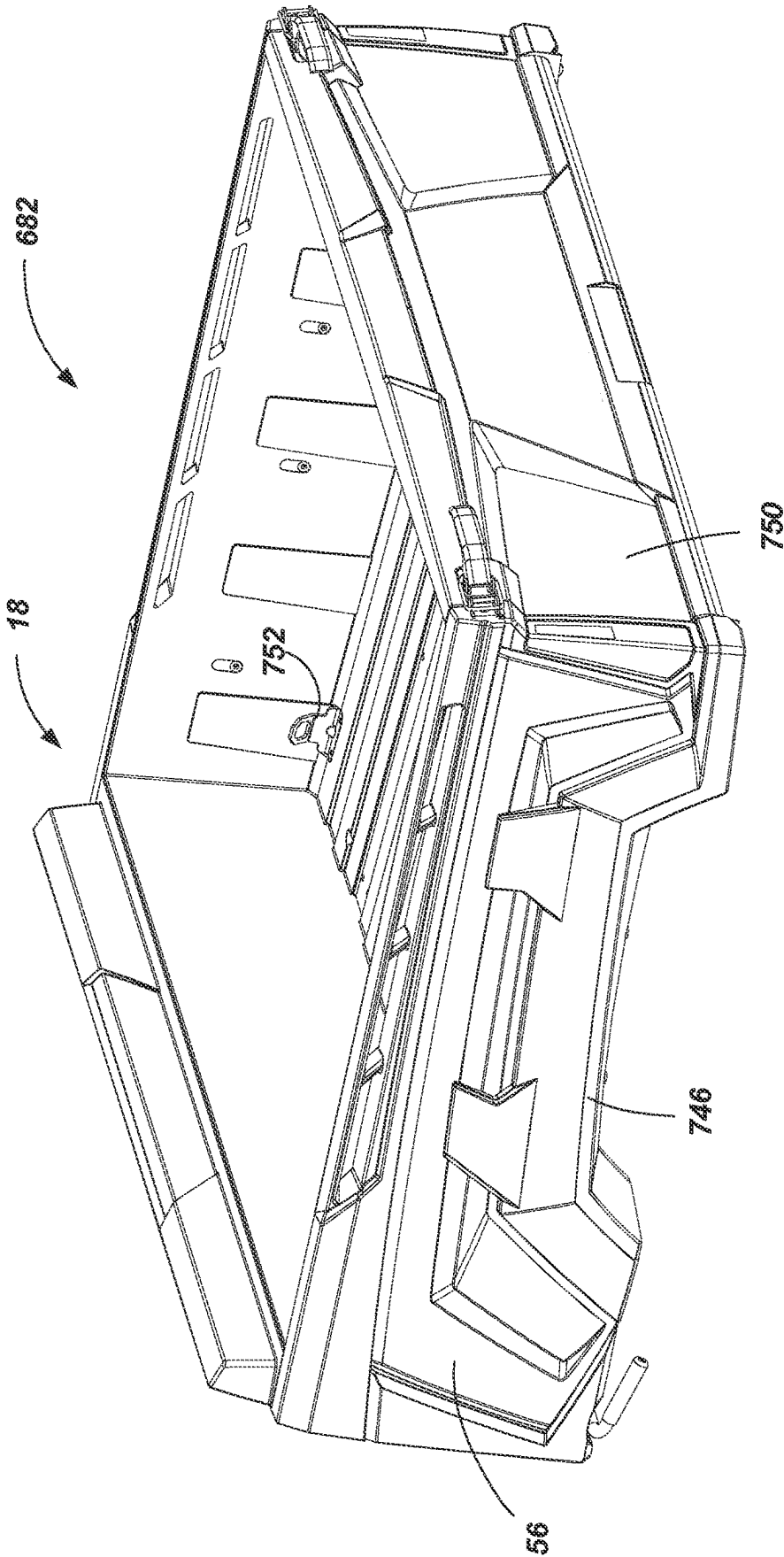


FIG. 131

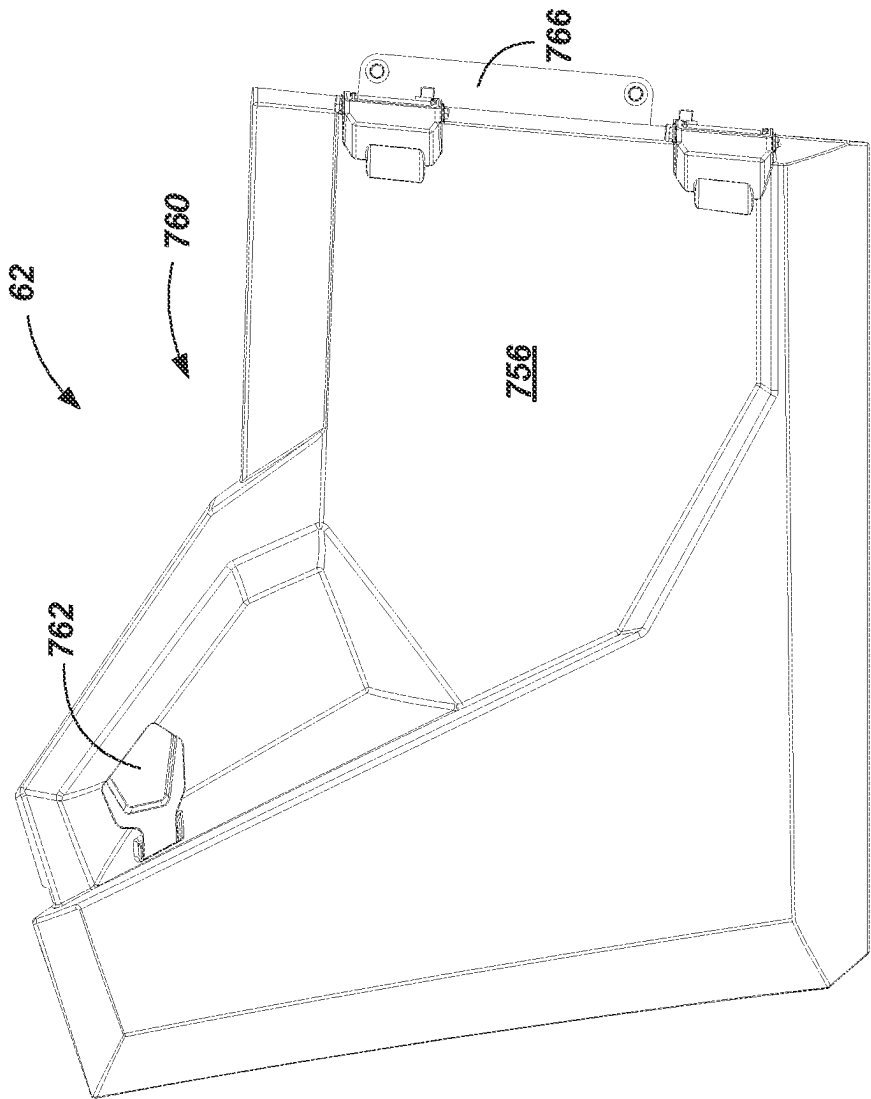


FIG. 132

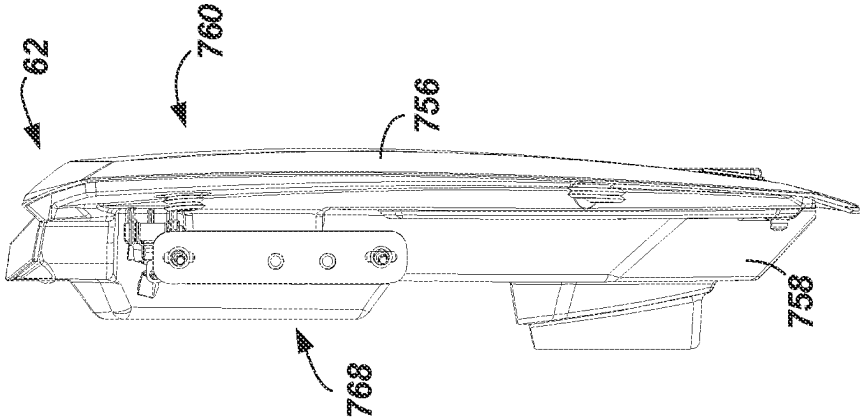


FIG. 133

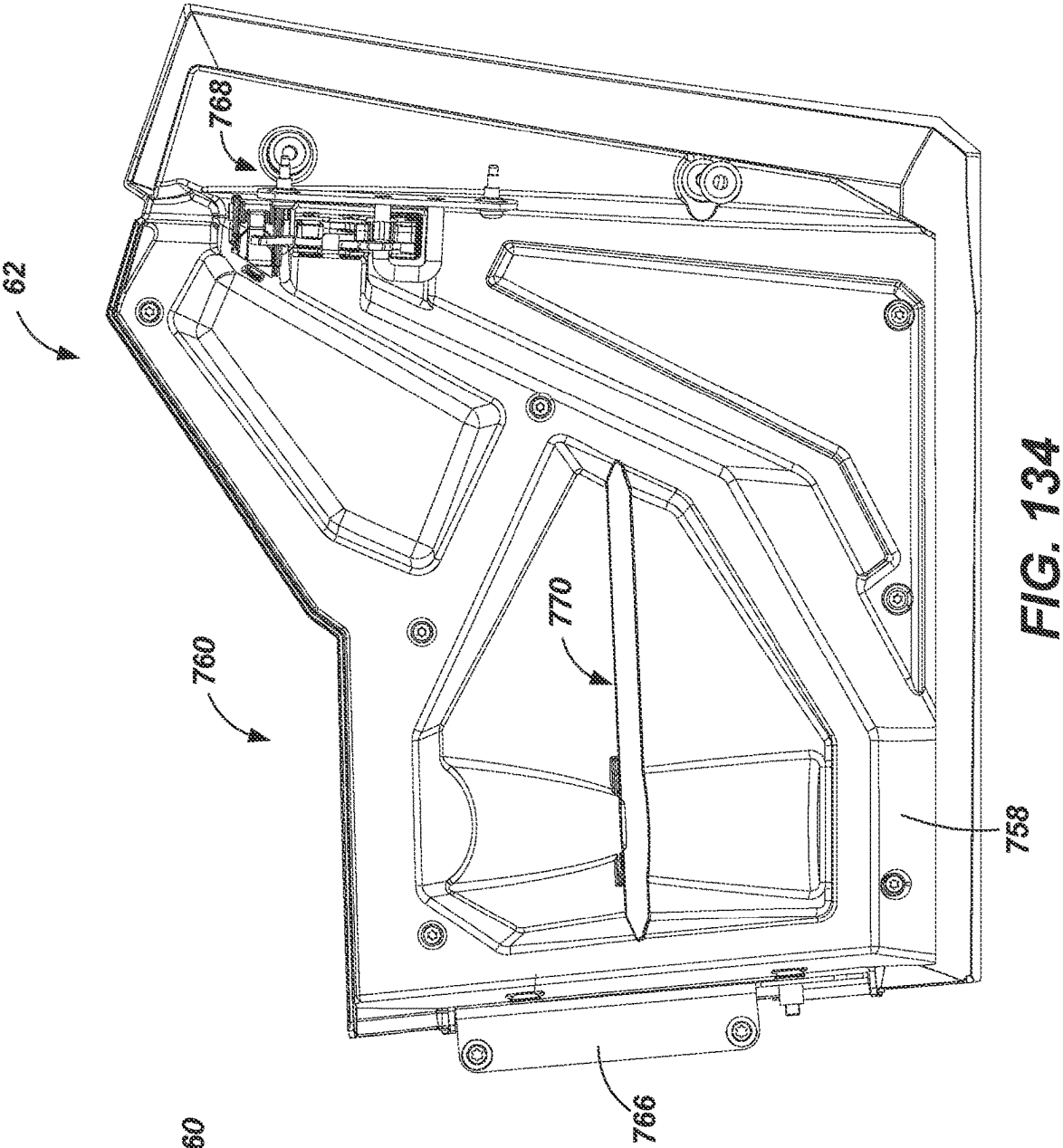


FIG. 134

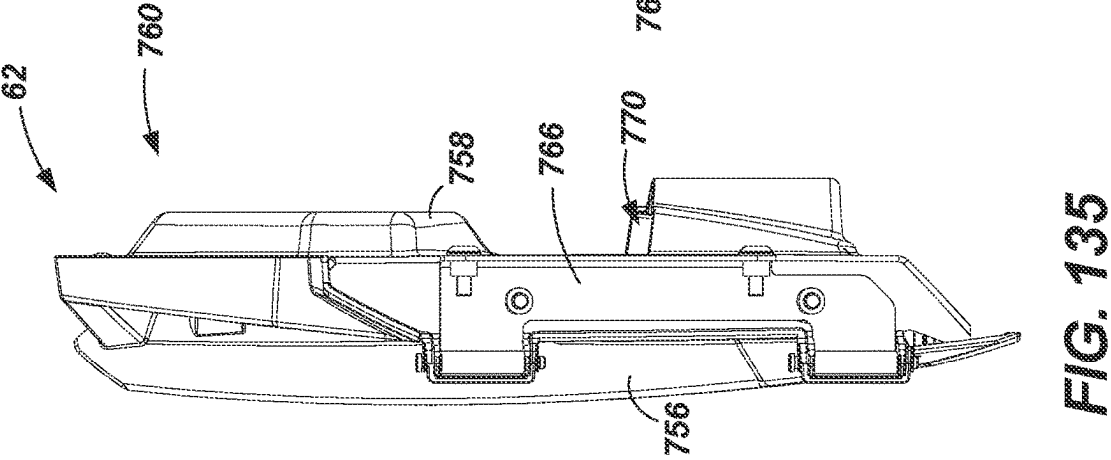


FIG. 135

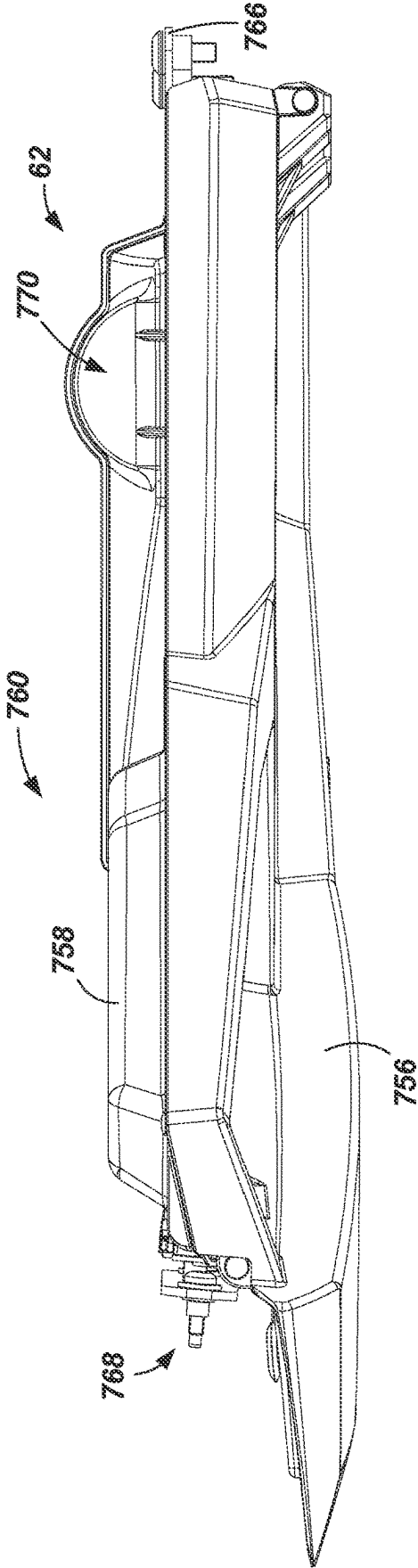


FIG. 136

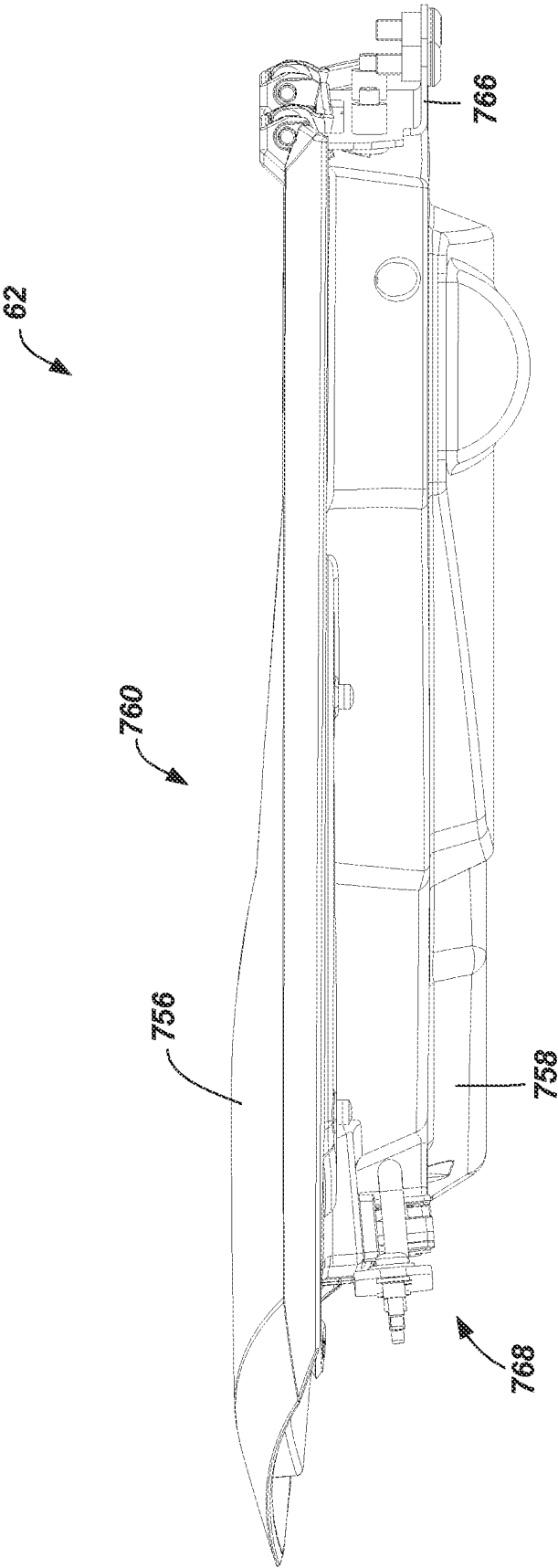


FIG. 137

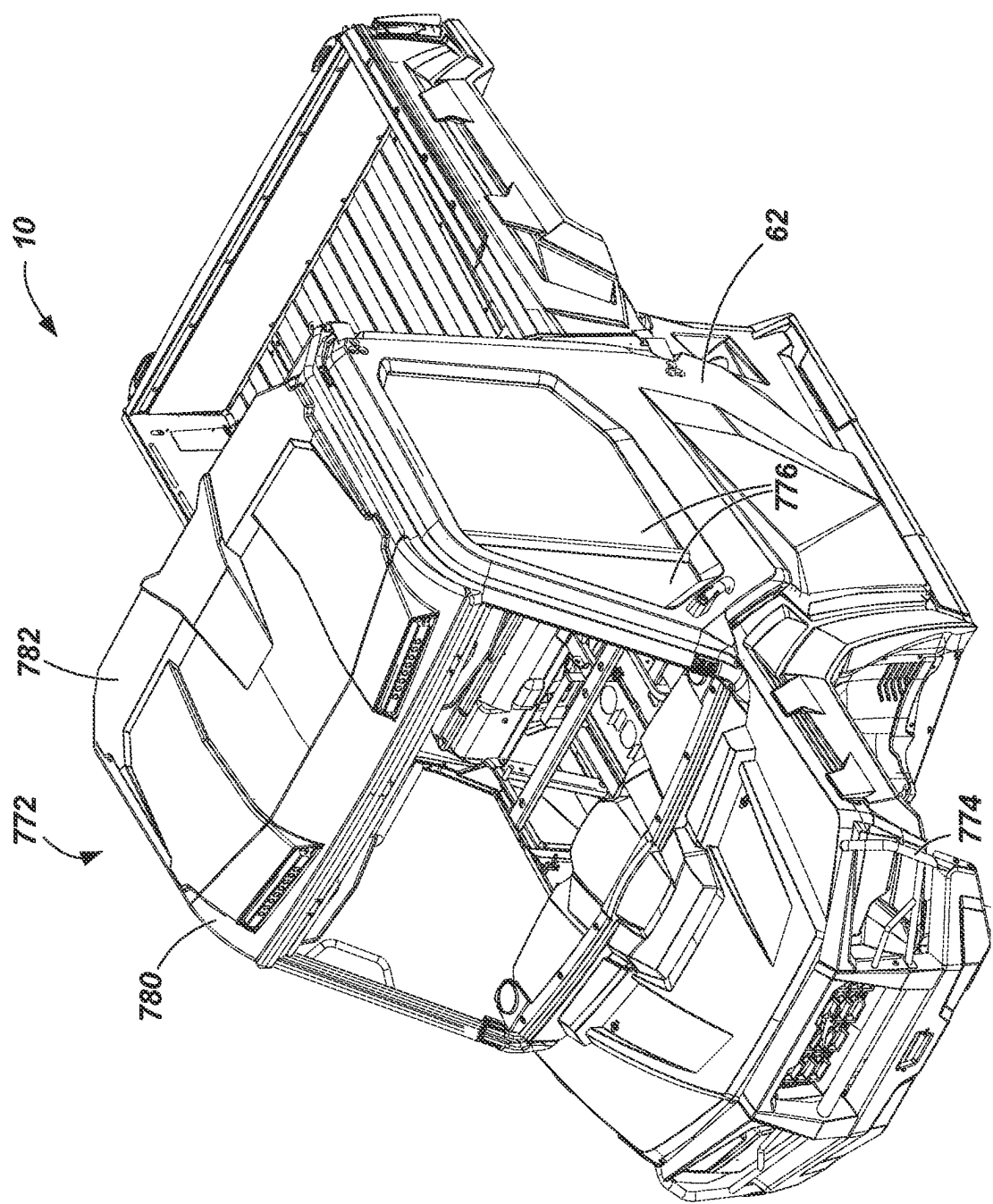
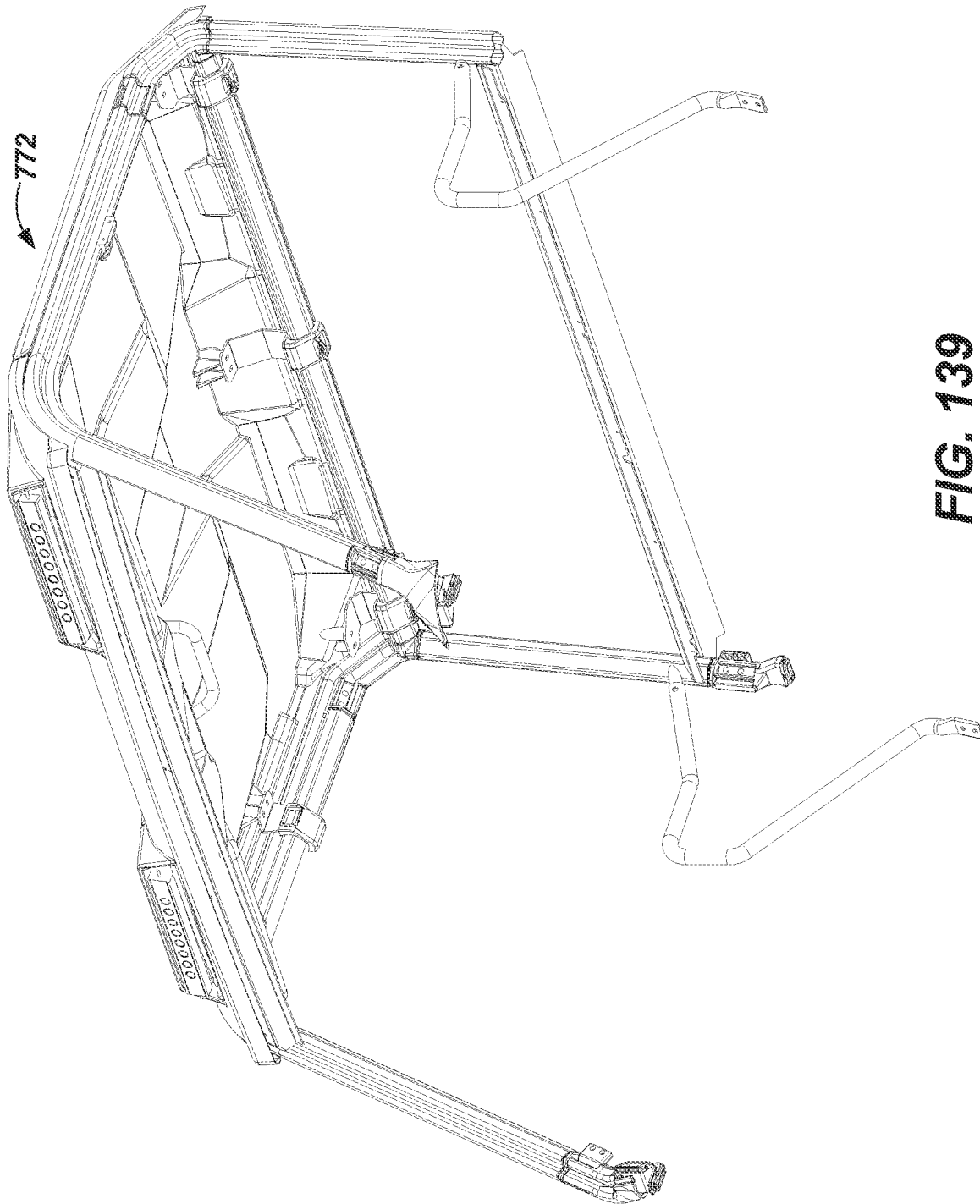


FIG. 138



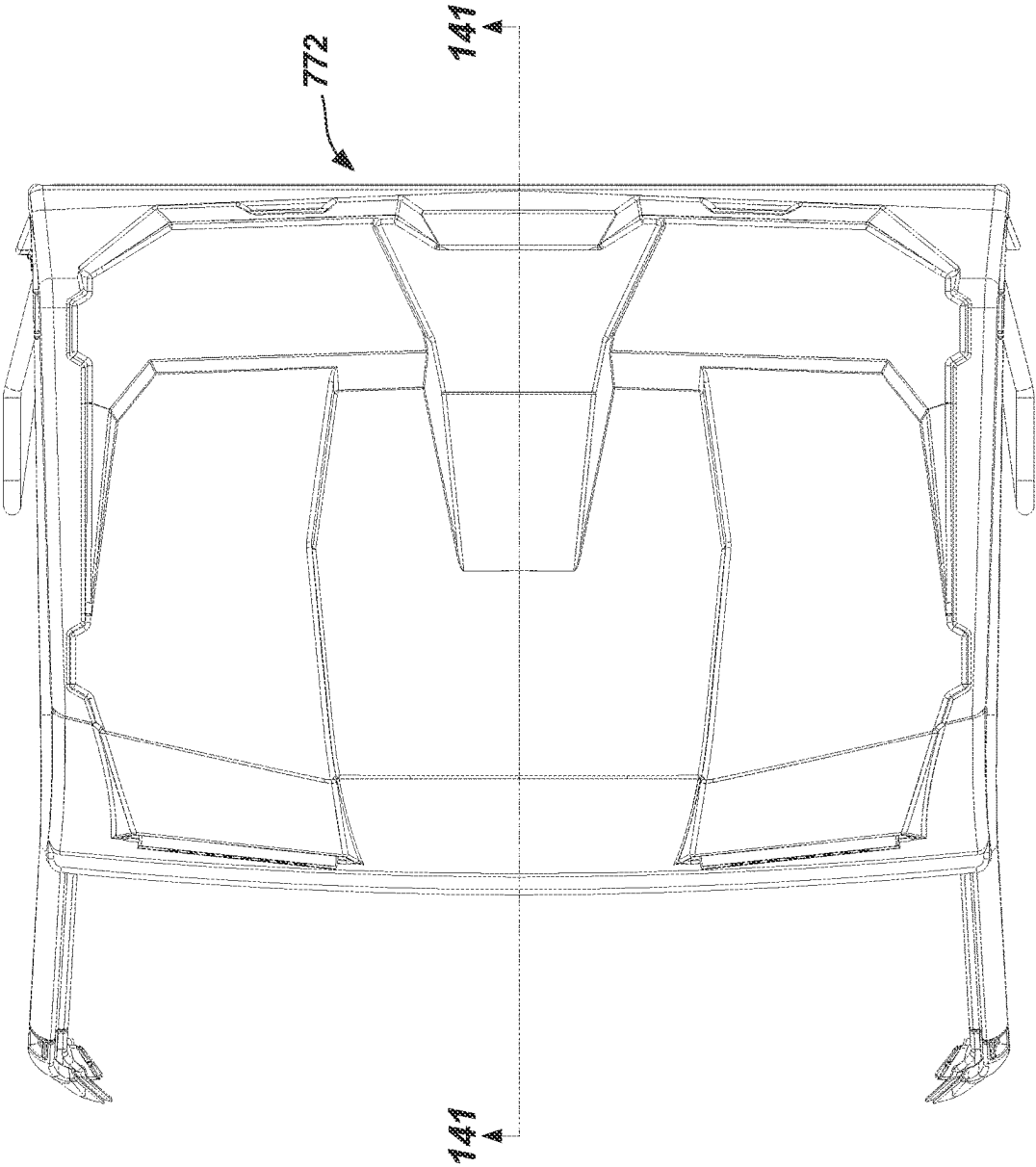


FIG. 140

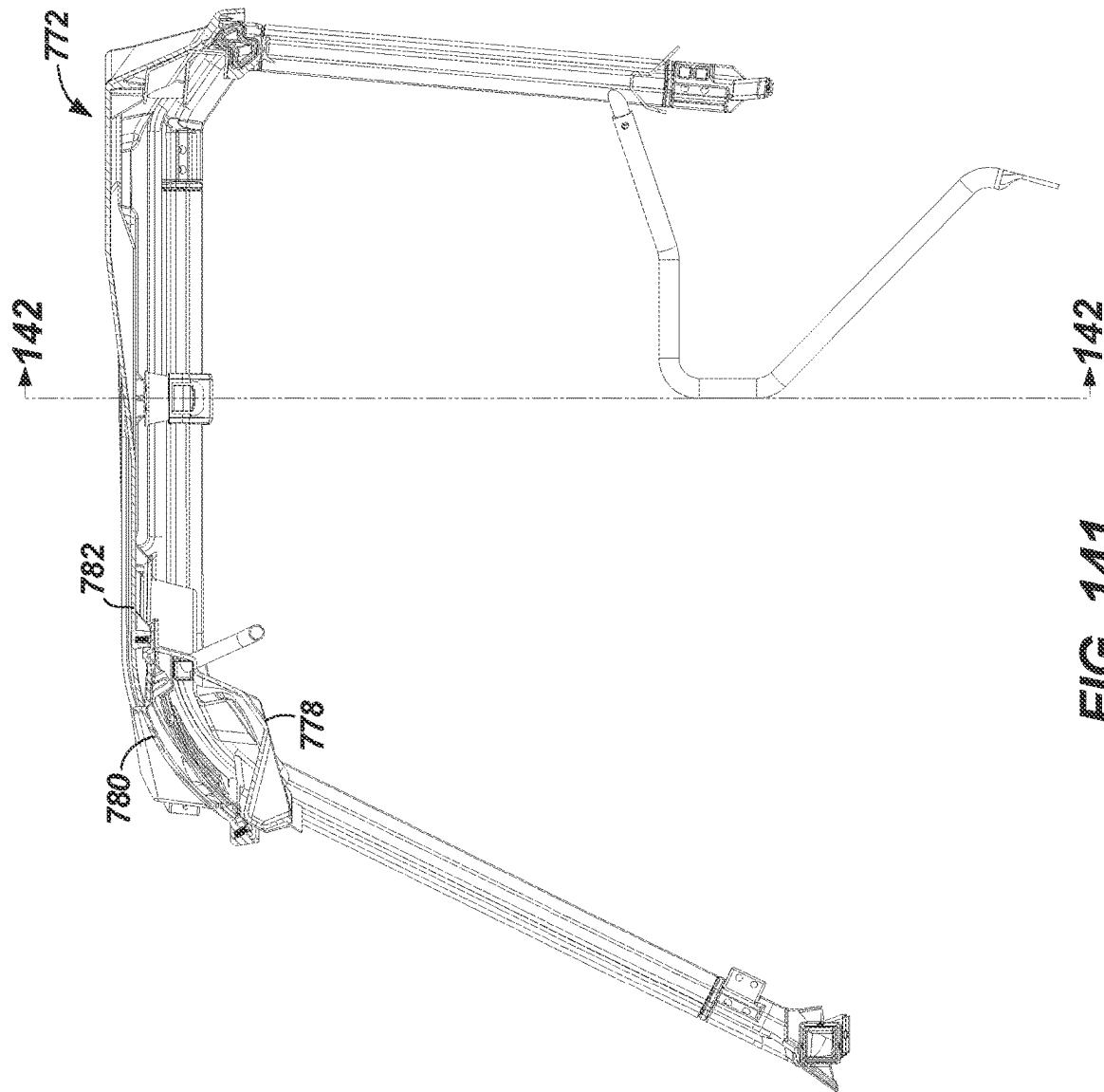


FIG. 141

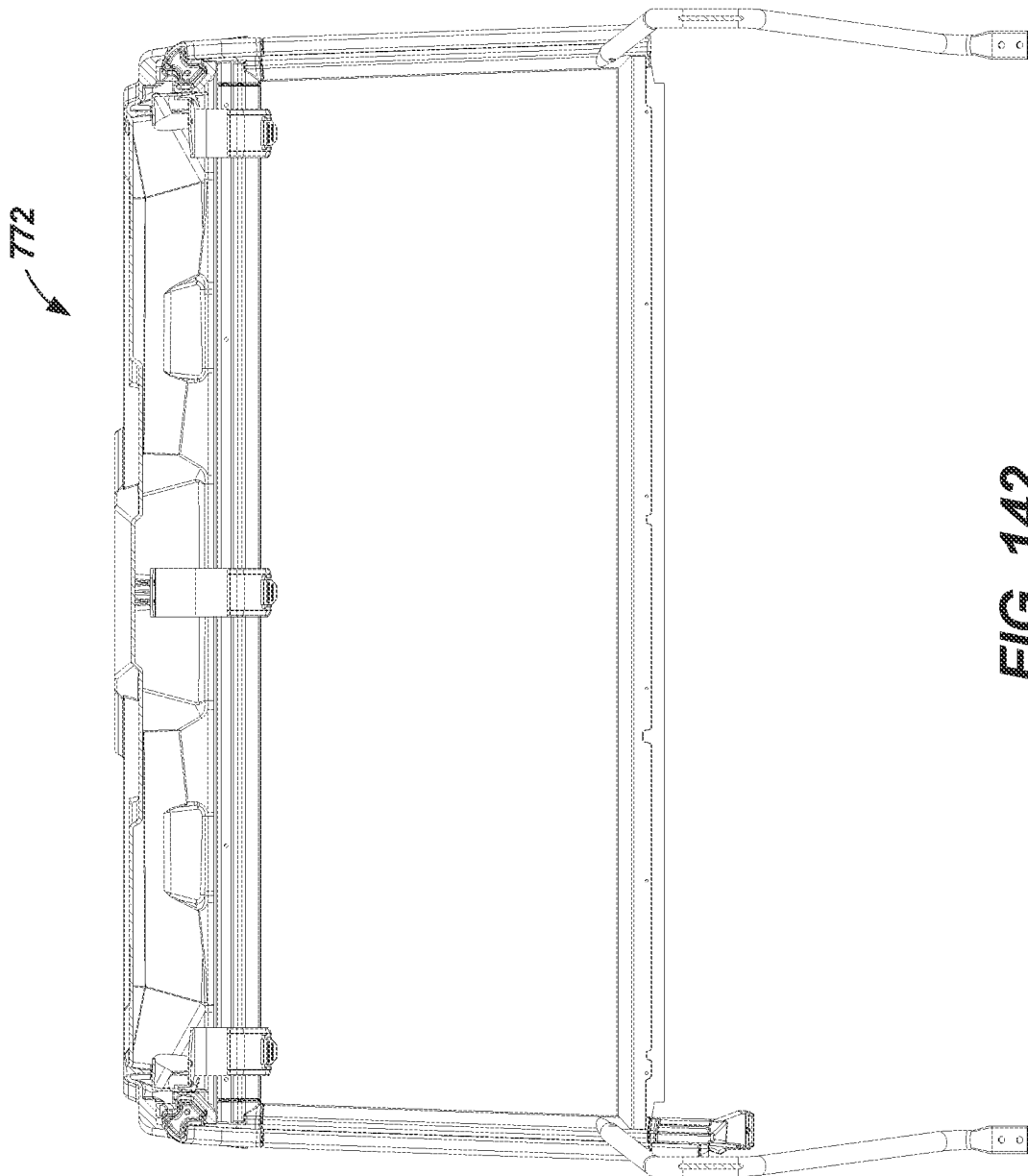


FIG. 142

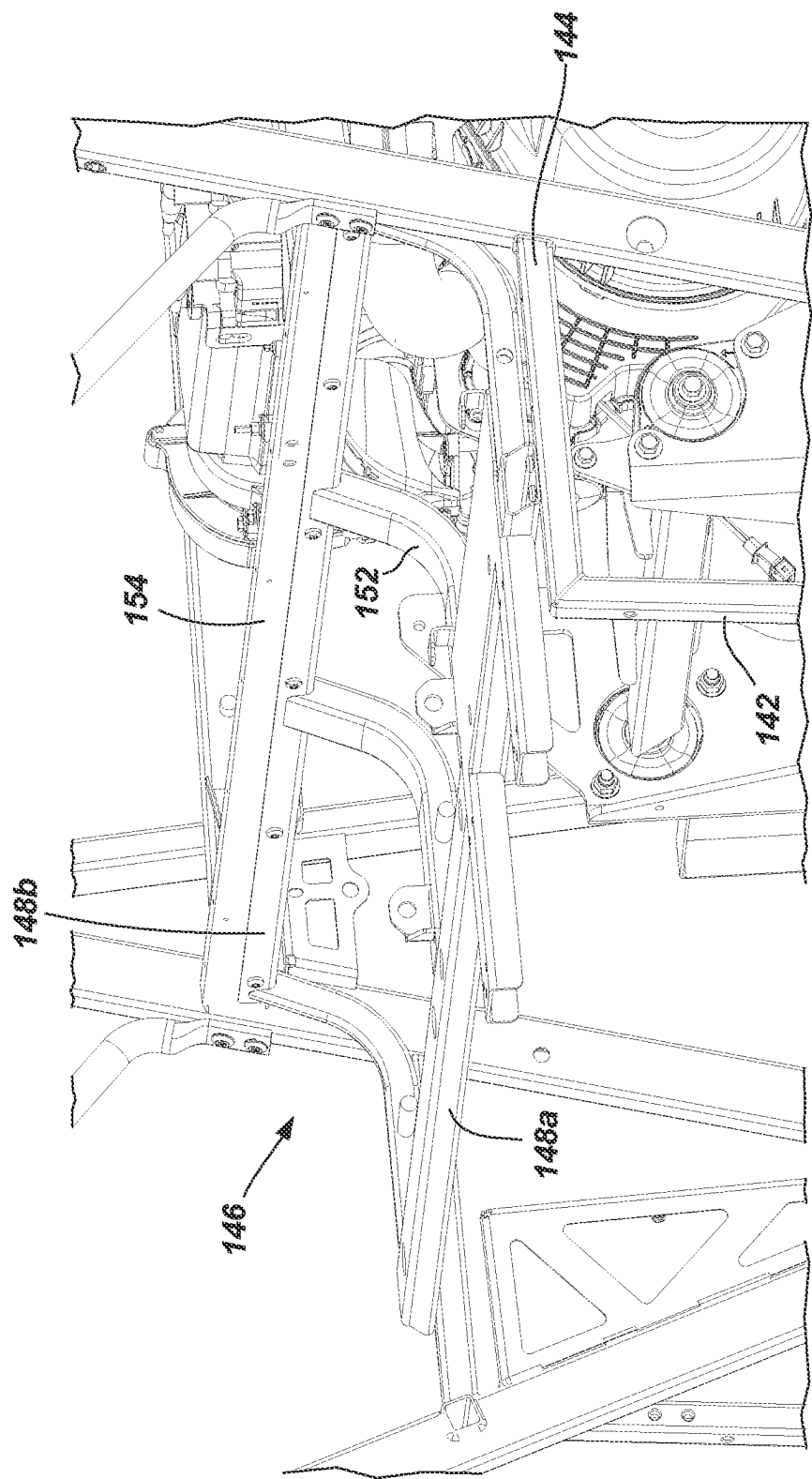


FIG. 143

1

OFF-ROAD UTILITY VEHICLE**CROSS-REFERENCE**

This application claims priority to U.S. Provisional Application No. 62/474,562, filed Mar. 21, 2017, and titled "OFF-ROAD UTILITY VEHICLE", which is incorporated by reference herein. This application also incorporates by reference the subject matter of co-filed application titled, "Cab and Fasteners for Vehicle Cab"; having inventors Leigh Raymond Collyer, Aaron Lee Swanson, and Matt Eastwood; and Application No. 62/474,556 filed on Mar. 21, 2017, the contents of which are herein incorporated by reference.

BACKGROUND

Side-by-side recreational off-highway vehicles ("ROVs") are quite capable in a wide variety of riding environments and situations, whether for sport or utility purposes. The ability of the vehicles to carry multiple occupants in a side-by-side seating arrangement makes them socially enjoyable to ride as well. The vehicles can be easy to enter and exit and easy to operate with controls and ergonomics somewhat similar to automobiles. However, unlike most automobiles, ROVs can be driven on harsh off-road terrain. The extent to which such terrain can be accessed depends on multiple factors, including the vehicle width, suspension, turning radius, under-carriage clearance, wheelbase, center of gravity, and power. The arrangement of these aspects and their interrelations can be important in determining the occupant ride characteristics, reliability, ease of maintenance, and terrain and cargo capabilities of the ROV.

SUMMARY

According to some embodiments, an off-highway recreational vehicle is disclosed having a longitudinal centerline and including a frame and a rear suspension. The rear suspension includes a left A-arm, wherein the left A-arm has a pivot axis, the pivot axis being on a right-hand side of the longitudinal centerline.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of the off-road utility vehicle according to some embodiments.

FIG. 2 is a side view of the off-road utility vehicle according to some embodiments.

FIG. 3 is a front view of the off-road utility vehicle according to some embodiments.

FIG. 4 is a rear view of the off-road utility vehicle according to some embodiments.

FIG. 5 is a front perspective view of the off-road utility vehicle, with body components removed to illustrate the frame, drivetrain, and front/rear suspension components of the utility vehicle according to some embodiments.

FIG. 6 is a rear perspective view of the off-road utility vehicle, with body components removed to illustrate the frame, drivetrain, and front/rear suspension components of the utility vehicle according to some embodiments.

FIG. 7 is a side view of the off-road utility vehicle, with body components removed to illustrate the frame, drivetrain, and front/rear suspension components of the utility vehicle according to some embodiments.

FIG. 8 is a front view of the off-road utility vehicle, with body components removed to illustrate the frame, drivetrain,

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and front/rear suspension components of the utility vehicle according to some embodiments.

FIG. 9 is a back view of the off-road utility vehicle, with body components removed to illustrate the frame, drivetrain, and front/rear suspension components of the utility vehicle according to some embodiments.

FIG. 10 is a top view of the off-road utility vehicle, with body components removed to illustrate the frame, drivetrain, and front/rear suspension components of the utility vehicle according to some embodiments.

FIG. 11 is a perspective view of the off-road utility vehicle, with body components, drivetrain and front/rear suspension components removed to illustrate frame components of the off-road vehicle according to some embodiments.

FIG. 12 is a side view of the off-road utility vehicle, with body components, drivetrain and front/rear suspension components removed to illustrate frame components of the off-road vehicle according to some embodiments.

FIG. 13 is a top view of the off-road utility vehicle, with body components, drivetrain and front/rear suspension components removed to illustrate frame components of the off-road vehicle according to some embodiments.

FIG. 14 is a perspective view of a roll over protection system (ROPS) according to some embodiments.

FIG. 15 is an exploded view illustrating the connection of a first ROPS connection member to a second ROPS connection member according to some embodiments.

FIG. 16 is a perspective view of a ROPS including a forward overhead member overlapping with a mating surface according to some embodiments.

FIG. 17 is a cross-sectional view of ROPS members according to some embodiments.

FIG. 18 is a side view of second ROPS connection member; and FIGS. 18a-18c are cross-sectional views of the second ROPS connection member according to some embodiments.

FIG. 19 is a perspective view illustrating an accessory attachment connected to the second ROPS connection member according to some embodiments.

FIG. 20 is a perspective view of second ROPS connection member having first, second and third connection portions according to some embodiments.

FIG. 21 is a side view of a brake pedal assembly according to some embodiments.

FIG. 22 is a top view of a brake pedal assembly according to some embodiments.

FIG. 23 is a front view of a brake pedal assembly according to some embodiments.

FIG. 24 is a bottom view of a brake pedal assembly according to some embodiments.

FIG. 25 is a front view of a brake pedal assembly formed from a single piece of metal according to some embodiments.

FIG. 26 is a side view of a brake pedal assembly formed from a single piece of metal according to some embodiments.

FIG. 27 is an exploded view of brake pedal assembly according to some embodiments.

FIG. 28 is an assembled view of brake pedal assembly according to some embodiments.

FIG. 29 is a side view of a front suspension system according to some embodiments.

FIG. 30 is a top view of a front suspension system according to some embodiments.

FIG. 31 is a front view of a front suspension system according to some embodiments.

FIG. 32 is a bottom view of a front suspension system according to some embodiments.

FIG. 33 is a side view of a front spring and front anti-roll bar (ARB) connected to front upper A-arm according to some embodiments.

FIG. 34 is a top view illustrating the connection of front ARB to front upper A-arm according to some embodiments.

FIG. 35 is a front view illustrating the connection of front ARB to front upper A-arm according to some embodiments.

FIG. 36 is a bottom view illustrating the connection of front ARB to front upper A-arm according to some embodiments.

FIG. 37 is an exploded view of the upper A-arm according to some embodiments.

FIG. 38 is an assembled view of the upper A-arm according to some embodiments.

FIG. 39 is a side view of a rear suspension system according to some embodiments.

FIG. 40 is a top view of a rear suspension system according to some embodiments.

FIG. 41 is a front view of a rear suspension system according to some embodiments.

FIG. 42 is a bottom view of a rear suspension system according to some embodiments.

FIG. 43 is a side view illustrating connection of a rear anti-roll bar (ARB) to rear upper A-arm according to some embodiments.

FIG. 44 is a top view illustrating connection of a rear anti-roll bar (ARB) to rear upper A-arm according to some embodiments.

FIG. 45 is a front view illustrating connection of a rear anti-roll bar (ARB) to rear upper A-arm according to some embodiments.

FIG. 46 is a bottom view illustrating connection of a rear anti-roll bar (ARB) to rear upper A-arm according to some embodiments.

FIG. 47 is a perspective view illustrating the relationship between upper A-arm and lower A-arm according to some embodiments.

FIG. 48 is a perspective view illustrating the connection of tie rods to a steering rack according to some embodiments.

FIG. 49 is a perspective view illustrating the connection of tie rods to a steering rack according to some embodiments.

FIG. 50 is a perspective view illustrating the connection of tie rods to a steering rack according to some embodiments.

FIG. 51 is a perspective view illustrating the connection of tie rods to a steering rack according to some embodiments.

FIG. 52 is a side views illustrating the connection of tie rods to a steering rack according to some embodiments.

FIG. 53 is a side views illustrating the connection of tie rods to a steering rack according to some embodiments.

FIG. 54 is a rear view illustrating the connection of tie rods to the steering rack according to some embodiments.

FIG. 55 is a top view illustrating the connection of tie rods to the steering rack according to some embodiments.

FIG. 55A is a perspective view of the front upper cross bracket according to some embodiments.

FIG. 55B is a perspective view of the front upper cross bracket according to some embodiments.

FIG. 56 is a perspective view illustrating the driveline according to some embodiments.

FIG. 57 is a top view of the engine and transmission according to some embodiments.

FIG. 58 is an exploded view of the continuously variable transmission (CVT) and transmission relative to the engine according to some embodiments.

FIG. 59 is an exploded view of front drive shaft and front drive assembly according to some embodiments, and FIG. 59A is a cross-sectional view taken along line 59A-59A according to some embodiments.

FIG. 60 is a perspective view of front drive assembly and CVT cover according to some embodiments.

FIG. 61 is a side view illustrating the connection of front drive shaft to the transmission according to some embodiments.

FIG. 62 is a side view illustrating the connection of front drive shaft to the transmission according to some embodiments.

FIG. 63 is a perspective view illustrating standoff members and flywheel attached to a prime mover according to some embodiments.

FIG. 64 is a perspective view of the continuously variable transmission (CVT) and associated CVT intake according to some embodiments.

FIG. 64A is a perspective view of the continuously variable transmission (CVT) and associated CVT intake according to some embodiments.

FIG. 64B is a perspective view of the flywheel located within the continuously variable transmission (CVT) according to some embodiments.

FIG. 65 is a perspective view of the CVT cover according to some embodiments.

FIG. 66 is an exploded view of the CVT cover assembly according to some embodiments.

FIG. 67 is a side view of CVT cover assembly according to some embodiments.

FIG. 68 is a rear view of CVT cover assembly according to some embodiments.

FIG. 69 is a magnified view of a portion of the CVT over and intermediate CVT cover highlighted in FIG. 68 according to some embodiments.

FIG. 70 is a perspective view illustrating the mounting of inner CVT cover to a mount plate adjacent to the engine according to some embodiments.

FIG. 71 is a side view of the steering column assembly and the gear selector assembly according to some embodiments.

FIG. 72 is a top view of the steering column assembly and the gear selector assembly according to some embodiments.

FIG. 73 is a perspective view of gear selector assembly according to some embodiments.

FIG. 74 is a perspective view of gear selector assembly according to some embodiments.

FIG. 75 is an exploded view of the steering column assembly according to some embodiments.

FIG. 76 is an assembled view of the steering column assembly according to some embodiments.

FIG. 77 is a front view of the seating area according to some embodiments.

FIG. 78 is a rear perspective view of the seating area according to some embodiments.

FIG. 79 is a rear perspective view of the seating area according to some embodiments.

FIG. 80 is a front perspective view of the seating area according to some embodiments.

FIG. 81 is a top view of the floorboards included in the seating area.

FIG. 82 is a cross-sectional view of the floorboards taken along line 82-82 according to some embodiments.

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FIG. 83 is a cross-sectional view of the floorboards taken along line 83-83 according to some embodiments.

FIG. 84 is a perspective view of the seating area that includes the seatback body panel and upwardly extending floorboard portion according to some embodiments.

FIG. 85 is an isometric view illustrating the coupling of the seatback body panel with the ROPS cross member according to some embodiments.

FIG. 86 is a rear view of seat backs according to some embodiments.

FIG. 87 is a front view of seat backs according to some embodiments.

FIG. 88 is a front view of seat bases and seat backs according to some embodiments.

FIG. 89 is a back view of seat bases and seat backs according to some embodiments.

FIG. 90 is a top view of a first seat base according to some embodiments.

FIG. 91 is a bottom view of a first seat base according to some embodiments.

FIG. 92 is a top view of a second seat base according to some embodiments.

FIG. 93 is a bottom view of a second seat base according to some embodiments.

FIG. 94 is a perspective view that illustrates rotation of the seat bases relative to the seat backs and removal of the seat bases from the seat support frame according to some embodiments.

FIG. 95 is a perspective view that illustrates rotation of the seat bases relative to the seat backs and removal of the seat bases from the seat support frame according to some embodiments.

FIG. 96 is a perspective view illustrating the connection of seat backs and seat bases to the seat support frame according to some embodiments.

FIG. 97 is a perspective view illustrating the connection of seat backs and seat bases to the seat support frame according to some embodiments.

FIG. 98 is a perspective view of the bumper assembly according to some embodiments.

FIG. 99 is a perspective view illustrating the cooling system in relation to various other components of the off-road utility vehicle according to some embodiments.

FIG. 100 is a perspective view illustrating the cooling system in relation to various other components of the off-road utility vehicle according to some embodiments.

FIG. 101 is a perspective view illustrating the cooling system in relation to various other components of the off-road utility vehicle according to some embodiments.

FIG. 102 is a perspective view illustrating the cooling system in relation to various other components of the off-road utility vehicle according to some embodiments.

FIG. 103 is a perspective view of the exhaust and intake systems in relation to a diesel engine according to some embodiments.

FIG. 104 is a perspective view of the intake system according to some embodiments.

FIG. 105 is a perspective view of the exhaust system according to some embodiments.

FIG. 106 is a perspective view of the of the exhaust and intake systems in relation to a gasoline engine according to some embodiments.

FIG. 107 is a perspective view of the intake system according to some embodiments.

FIG. 108 is a perspective view of the exhaust system in relation to the engine according to some embodiments.

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FIG. 109 is a perspective/exploded view of the exhaust system according to some embodiments.

FIG. 110 is a top view of the off-road utility vehicle according to some embodiments.

FIG. 111 is a cross-sectional view of the off-road utility vehicle shown in FIG. 110 taken along line 111-111 according to some embodiments.

FIG. 112 is a rear perspective view of frame components of the off-road utility vehicle according to some embodiments.

FIG. 113 is a side view illustrating tilting of the cargo box according to some embodiments.

FIG. 114 is a top perspective view of the cargo box according to some embodiments.

FIG. 115 is a bottom perspective view of the cargo box according to some embodiments.

FIG. 116 is an exploded view of panels associated with the cargo box according to some embodiments.

FIG. 117 is a top view of the cargo box according to some embodiments.

FIG. 118 is a sectional view of the cargo box taken along line 118-118 according to some embodiments.

FIG. 119 is a sectional view of the cargo box taken along line 119-119 according to some embodiments.

FIG. 120 is a perspective view of fuel system and fuel tank according to some embodiments.

FIG. 121 is a top view of the fuel tank according to some embodiments.

FIG. 122 is a cross-sectional view of the fuel tank taken along line 122-122 according to some embodiments.

FIG. 123 is a perspective view illustrating the attachment of the bumper to the main frame according to some embodiments.

FIG. 124 is an exploded view illustrating attachment of dash panels and hood panels to the main frame according to some embodiments.

FIG. 125 is a perspective view of dash panels and hood panels affixed to the main frame according to some embodiments.

FIG. 126 is an exploded view of the front fascia members attached to the main frame according to some embodiments.

FIG. 127 is a front perspective view of the front fascia members according to some embodiments.

FIG. 128 is an exploded view of the fender panels attached to the main frame according to some embodiments.

FIG. 129 is a rear perspective view of the fender panels according to some embodiments.

FIG. 130 is an exploded view of the cargo box panels affixed to the cargo box according to some embodiments.

FIG. 131 is a perspective view of the cargo box including affixed cargo box panels according to some embodiments.

FIG. 132 is an outer view of the cab door according to some embodiments.

FIG. 133 is a front view of the cab door according to some embodiments.

FIG. 134 is an inner side view of the cab door according to some embodiments.

FIG. 135 is a rear view of the cab door according to some embodiments.

FIG. 136 is a top view of the cab door according to some embodiments.

FIG. 137 is a bottom view of the cab door according to some embodiments.

FIG. 138 is a perspective view of roof assembly, brush guard and windows according to some embodiments.

FIG. 139 is a perspective view of the roof assembly affixed to the ROPS system according to some embodiments.

FIG. 140 is a top view of the roof assembly according to some embodiments.

FIG. 141 is a cross-sectional view of the roof assembly and ROPS system taken along line 141-141 as shown in FIG. 140 according to some embodiments.

FIG. 142 is a cross-sectional view of the roof assembly and ROPS system taken along line 142-142 as shown in FIG. 141 according to some embodiments.

FIG. 143 is a perspective view of the seat support frame according to some embodiments.

In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in difference views. The drawings illustrate generally, by way of example, but not by way of limitation, various embodiments discussed in the present document.

DETAILED DESCRIPTION

As shown in FIG. 1, a utility vehicle 10 has a body 12, a frame 14, and a plurality of ground engaging members 16 (e.g., tires, tracks). In at least some embodiments, the utility vehicle 10 includes a cargo carrying portion such as a cargo box 18. As shown in FIG. 1, the cargo box 18 is rearward of a seating area 20. The frame 14 includes structural members 22 which are coupled together (e.g., welded, bolted, glued). Further, the structural members 22 can be tubular steel or aluminum, stamped sheet metal (e.g., steel, aluminum), hydroformed, cast, forged, or formed in any other suitable manner. The utility vehicle 10 can be 2-wheel or 4-wheel drive. Further, it can have any suitable style of drive system. In some embodiments, the utility vehicle 10 is 4-wheel drive and includes a differential in both the front end and rear end of the vehicle 10. The differentials can include optional locking differentials or they can be open differentials, which can be manually selectable by an operator or engaged automatically in response to terrain conditions (e.g., wheel slip). In some embodiments, the vehicle has a limited slip differential (e.g., clutch pack, Quaife, Torsen) or any other suitable configuration (e.g., spool). In other embodiments, the utility vehicle is a 2-wheel drive system. The 2-wheel drive system may be a rear-wheel drive system or a front-wheel drive system.

As further illustrated in FIGS. 1 and 5, the seating area 20 includes one or more seats 24. The seats 24 can be arranged in a side-by-side configuration and can include bench seating or bucket seating. As shown in FIG. 1, for example, the seats 24 are arranged in a three-abreast configuration. The seats 24 can further be arranged in a 60/40 arrangement. Further, in some embodiments, one or more of the seat bottoms and/or seat backs is adjustable. In some embodiments, the driver's seat 24a is adjustable (e.g., forward and back) and one or more of the passenger seats (24b, 24c) are not adjustable. In some embodiments, both the driver's seat 24a and one or more of the passenger seats 24b, 24c are adjustable. In some embodiments, one or more of the seat base(s) 26a-26c (FIG. 10) are adjustable, for example vertically, horizontally, pivotally (e.g., tilt/recline), and/or laterally. In some embodiments one or more of the seat backs 28a-c are adjustable, for example vertically, horizontally, and/or laterally. The seat bases 26a-c can be adjustable and the seat backs 28a, b can be fixed or vice-versa. In some embodiments, at least some of the seats have adjustable backs and adjustable bases. In some embodiments, the seat backs have a lumbar support adjustment, for example via an

air filled bladder in a portion of the seat back 28. In some embodiments, the seat back 28b extends across two seat bases, for example 26a, 26b.

With further regard to FIG. 1, the utility vehicle 10 has a fuel filling port 30. In some examples, the fuel filling port 30 is located on the left-hand side of the utility vehicle 10, however it can be located on any appropriate side or location of the vehicle, for example rear, front, side, etc. In some embodiments, a rear cab-side body panel 32 (e.g., 32a, left-hand rear cab-side body panel) has a cutout to provide access to the filling port 30. As shown in FIG. 2, in some embodiments, a concavity 34 can be formed into the rear cab-side body panel 32 in order to provide access to the fuel cap 36 (FIG. 79). In some embodiments, fuel cap 36 is located inside the outermost surface of rear cab-side body panel 32, such that rear cab-side body panel 32 provides protection to fuel cap 36. The concavity 34 can further be formed provided in a body panel that is independent of the rear cab-side body panel 32. As illustrated in FIG. 120, the fuel cap 36, in turn, is fastened to a filler neck 38, as discussed in greater detail below.

With further regard to FIG. 6, the utility vehicle 10 includes a receiver hitch assembly 40. The receiver hitch assembly 40 can be sized for any suitably sized ball mount (not shown), for example using as a 2" receiver or 1 1/4" receiver. The receiver hitch assembly 40 further includes a hitch plate 42 which can include one or more rear suspension attachments 44, as illustrated in FIGS. 11 and 41.

Returning to FIGS. 1 and 2, the utility vehicle 10 further can include headlights 46 and taillights 48. In some embodiments, the utility vehicle 10 can further include brake lights 50 (FIG. 4) and turn signals 52. Any of the headlights 46, taillights 48, brake lights 50, and turn signals 52 can be located in any suitable location, for example on the front, rear, or sides of the vehicle, as appropriate. Moreover, the utility vehicle can also include fog lamps, hazard lights, high and low beam lights, light bars, integrated reflectors, etc., as desired. Any lights and reflectors can be placed in any suitable location, including on the front and rear of the cab 54 (FIG. 2).

As further shown in FIGS. 1-4, in some embodiments, the utility vehicle 10 includes one or more headrests 58. As shown, each of the headrests 58 is coupled to an angled cab member 60, for example with one or more fasteners (FIG. 4).

In some embodiments, the vehicle 10 has one or more (e.g., one on each side) doors 62. The doors 62 can be of any desirable configuration. In some embodiments, one or more of the doors 62 is positioned upwardly of a portion of the lower cab panel 64. As further shown in FIG. 2, in some embodiments, one or more of the doors 62 extends upwardly above the top of the seat base 26a. In some embodiments, one or more of the doors 62 extends upwardly to a height that is level with the top of the seat base 26a and, in some embodiments, one or more of the doors extends upwardly (e.g., vertically) to a height that is below the top of the seat base 26a. In some embodiments, one or more of the doors 62 include one or more hinges 766 (FIG. 133-137). As shown in FIGS. 5 and 133-137, in some embodiments, the hinges 766 are coupled to a door support member 68 which, in some embodiments, is also referred to as a first seating support member 142 (FIG. 12) and which extends vertically from an outer lower frame member 70. As discussed in greater detail below, in some embodiments, the lower cab panel 64 is integrally formed with the rear cab-side body panel 34, as shown for example in FIG. 128.

In some embodiments, the door support member **68** is a tubular member which is welded to the outer lower frame member **70**. In some embodiments, the door support member **68** has a rectangular cross-section, however, the door support member can have any suitable cross-section; it can be manufactured from any suitable material (e.g., steel, plastic, carbon-fiber). Further, it can be stamped, extruded, molded, cast, etc. As further shown in FIG. **5**, in some embodiments, the door **62** is hinged at its rear. Alternatively, the door **62** can be hinged at its front, so as to swing forwardly upon opening.

In some embodiments, the vehicle **10** further includes a ROPS (roll-over protection structure) **72**. In some embodiments, the ROPS **72** is attached to the main frame **74**. As used herein, the term “frame” **14** includes both the ROPS **72** and the main frame **74**.

With regard to FIGS. **2**, **5**, **11**, and **12**, in some embodiments, the vehicle **10** further includes at least one side restraint **76**. In some embodiments, the side restraint **76** is coupled to main frame **74** (e.g., rear outer frame member **78**), for example via one or more fasteners or welding. In some embodiments, the side restraint **76** is further coupled to a portion of the ROPS **72**, for example via one or more fasteners. In some embodiments, an upper end of the side restraint **76** is coupled to a portion of the ROPS **72** via a stub **80** that extends from an interior portion of the rear ROPS member **82**. In some embodiments, the stub **80** is welded to an interior portion of the rear ROPS member **82** and the side restraint **76** is attached to the stub via one or more fasteners. In some embodiments, the upper end of the side restraint **76** is attached to the ROPS **72** inwardly of the channel in the ROPS tubing, as shown in FIGS. **139** and **142**. In some embodiments, both ends of the side restraint **76** are coupled to the main frame **74**; in some embodiments, both ends of the side restraint **76** are coupled to the ROPS **72**.

In some embodiments, the vehicle **10** includes a steering wheel **84** which is coupled, for example via a steering linkage, to at least two of the ground engaging members **16**, for example the front ground engaging members **16a**. The steering wheel **84** is coupled to the front ground engaging members **16a** in any suitable way, for example by mechanical steering linkage, electric power steering (EPS), hydraulically assisted power steering, electric power steering without mechanical linkage (e.g., drive-by-wire), electric assisted power steering ((EPAS), e.g., including pull-drift compensation, active nibble control, etc.) or in any other suitable way. Further, in some embodiments, the steering can include variable ratio steering and it can be programmable such that the use can set the steering ratio (and rate-of-change of steering ratio, if it is variable) to illicit a steering response in accordance with the user's or manufacturer's desires (e.g., exhibiting understeer characteristics). In some embodiments, the steering wheel **84** is coupled to the rear ground engaging members **16b**, for example, via a steering linkage mechanical steering linkage, electric power steering (EPS), hydraulically assisted power steering, electric power steering without mechanical linkage (e.g., drive-by-wire), electric assisted power steering ((EPAS), e.g., including pull-drift compensation, active nibble control, etc.) or in any other suitable way. As illustrated in FIG. **7**, an electric power steering unit **102** is positioned between the steering wheel **84** and the steering rack **104**. In some embodiments, the electric power steering unit **102** includes a motor **106** (e.g., direct current or alternating current motor), at least a portion of which is positioned in a vertical plane **110** passing through the motor **106** and brake pedal

108, the vertical plane extending laterally orthogonal to the longitudinal axis of the vehicle **10**.

With further regard to FIG. **2**, in some embodiments, the steering wheel **84** tilts, shown via arrow **86**. A tilt assembly **88** includes a shock **90** to adjust the tilt configuration of the steering wheel **84**.

Turning to FIG. **3**, in some embodiments, the utility vehicle **10** has a front bumper **92** which, in some embodiments, is positioned in front of the front ground engaging members **16a**. Further, in some embodiments, the vehicle **10** has at least one hand-hold **94**. The hand-hold **94** can be coupled to a portion of the vehicle **10** in any suitable way and in any suitable location(s). By way of example, in some embodiments, the hand-hold **94** is coupled to a portion of the ROPS **72**, for example an A-frame member **96**. In some embodiments, the hand-hold **94** is coupled to a forward overhead member **98**. In some embodiments, the hand-hold **94** is coupled to a portion of the frame **14**, for example via welding or via one or more fasteners (e.g., bolts).

With regard to FIGS. **6** and **7**, in some embodiments, the front bumper **92** is coupled to the main frame **74**, for example via one or more bumper brackets **100**.

As shown in FIGS. **11** and **12**, in some embodiments, the frame **14** comprises a rear upper frame member **124**, a rear lower frame member **126**, a front outer frame member **128**, and an upper dash frame member **130**. In some embodiments, the rear upper frame member **124**, rear lower frame member, and an upper dash frame member **130** are horizontal frame members. As shown in FIG. **112**, in some embodiments, the frame **14** comprises a left-hand inner frame member **132** and a right-hand inner frame member **134**. In some embodiments, one or both of the left-hand inner frame member **132** and a right-hand inner frame member **134** includes one or more bends **136**. In some embodiments, the frame **14** comprises angled connecting member **138**; in some embodiments, the angled connecting member **138** is coupled to the outer lower frame member **70**. In some embodiments, the frame **14** includes a ROPS support member **140**, which, in some embodiments is rearward of the rear outer frame member **78**. In some embodiments, the frame includes a first seating support member **142** and a second seating support member **144**; in some embodiments, the first seating support member **142** is coupled to the outer lower frame member **70** and the second seating support member **144** is coupled to the rear outer frame member **78** (e.g., by welding or via one or more fasteners). As shown for example in FIGS. **11** and **12**, the first seating support member **142** extends vertically and slightly rearwardly, however the first seating support member **142** can have any suitable configuration. Further, as shown in FIGS. **11** and **12**, the second seating support member **144** extends horizontally. In some embodiments, the first seating support member **142** is coupled to the second seating support member **144**.

In some embodiments, a seat support frame **146** is coupled to the frame **14**, e.g., main frame **74**. In at least some embodiments, the seat support frame **146** includes a plurality of structural members. In some embodiments, the seat support frame **146** includes one or more lateral seat support frame members **148** and one or more longitudinal seat support frame members **150**. As shown in FIG. **143**, in some embodiments, the lateral seat support frame members **148** include first and second lateral support frame members **148a** and **148b**, respectively.

As shown in FIG. **143**, in some embodiments, the longitudinal seat support frame members **150** include a bent portion **152**. In some embodiments, the seat support frame **146** is coupled to the main frame **74** via one or more

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fasteners or weldments. In some embodiments, the seat support frame **146** is coupled to the second seating support member **144** via a plurality of fasteners. Further, in some embodiments, the seat support frame **146** is coupled to the lateral rear cab frame member **154** via a plurality of fasteners.

Returning to FIGS. **11** and **12**, in some embodiments, the frame **14** includes rear joining frame member **156**, which couples the rear upper cross member **178** and rear lower frame member **126**. In some embodiments, the frame **14** has an intermediate joining frame member **158**, which couples the box support member **160** and rear lower frame member **126** at a location forward of the rear joining frame member **156**. The box support member **160**, in turn, is coupled to left and right rear upper frame members **124a**, **124b** (FIG. **13**).

With further regard to FIGS. **12** and **13**, the frame **14** includes longitudinal intermediate rear frame member **162**, which couples the rear joining frame member **156** and intermediate joining frame member **158**. In some embodiments, the frame **14** additionally include a front upper rear-suspension support member **164**, a rear upper rear-suspension support member **166**, a front lower rear-suspension support member **168**, and a rear lower rear-suspension support member **170**. In some embodiments, the front upper rear-suspension support member **164** and rear upper rear-suspension support member **166** extend between opposing longitudinal intermediate rear frame members **162a** and **162b** (FIG. **13**). In some embodiments, the front lower rear-suspension support member **168** and rear lower rear-suspension support member **170** extend between opposing rear lower frame members **126a** and **126b** (FIG. **112**). In some embodiments, the frame **14** further includes an intermediate lower rear-suspension support member **172** which extends between the rear lower frame members **126a** and **126b**. As will be appreciated from certain of the drawings, frame members, body panels, etc., may be, where appropriate, referred to via a suffix 'a' or 'b'.

As further shown in FIG. **112**, in some embodiments, the frame **14** includes a third panel **174**. In some embodiments, one or more of the first panel **114**, second panel **122**, and third panel **174** includes one or more stiffening ribs **176**. The stiffening ribs can be formed via a stamping operation, for example. In some embodiments, the third panel **174** is welded (or otherwise coupled) to the main frame **74**. In some embodiments, one, two, or all of the first panel **114**, second panel **122**, and third panel **174** are formed from a metallic material such as steel or aluminum. Where the frame **14** is made of steel, for example, and one or more of the panels (**114**, **122**, **174**) are welded thereto, the panels (**114**, **122**, **174**) may also be formed of steel.

With additional reference to FIGS. **11** and **112**, in some embodiments, the frame **14** includes lower front frame member **180**, longitudinal intermediate front frame member **182**, front joining frame member **184**, front intermediate joining frame member **186**, intermediate dash support member **188**, front angled connecting member **190**, and front upper cross member **192**. In some embodiments, the lower front frame member **180** includes rear lower front-suspension support member **194**. In some embodiments, the lower front frame member **180** is coupled to a front frame bracket **196**. In some embodiments, the frame **14** includes a fourth panel **198**. In some embodiments, the fourth panel **198** extends laterally between the opposing lower front frame members **180a** and **180b**, forwardly to the front frame bracket **196**, and rearwardly to a portion of the lateral lower frame member **200**. In some embodiments, the fourth panel **198** is formed from a metallic material, such as steel.

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As shown in FIGS. **11** and **12**, an outer fender support member **202** is coupled (e.g., by one or more fasteners) to the front outer frame member **128** and front joining frame member **184**. In some embodiments, the front joining frame member **184** includes a bend such a portion of the front joining frame member **184** includes a vertical portion and a generally longitudinal portion.

In some embodiments, the utility vehicle **10** includes a fuel tank **112**. As shown for example in FIGS. **13** and **78**, in some embodiments, a first panel **114** (e.g., steel panel) supports the fuel tank **112**. In some embodiments, the first panel **114** is coupled (e.g., welded) to one or more adjacent frame members. In some embodiments, the first panel **114** includes one or more depressions **116**. The one or more depressions **116** can be formed, for example, via a sheet metal press. In some embodiments, one or more of the depressions **116** can be used to locate the fuel tank **112** relative to the first panel **114**. In some embodiments, the fuel tank **112** includes a protruding portion **716** (FIG. **122**) that matches one or more depressions **116**. In some embodiments, one of the depressions **116** corresponds to a low point of the fuel tank **112** where a fuel pick is located within the fuel tank **112**. In some embodiments, the one or more depressions **116** extend downwardly from the first panel **114**. The one or more depressions **116** can extend below the bottom of the adjacent outer lower frame member **70**. In some embodiments, however, for example as shown in FIG. **112**, the bottom of the outer lower frame member **70** is lower than the bottom of one or more of the depressions **116**. In this way, one or more skid plates (not shown) are coupled to the frame **14** below the one or more depressions **116**. Further, as shown in FIG. **12**, in some embodiments, a horizontal plane **120**, defined by the bottom of the frame **14**, is lower than the depressions **116**. Although shown on the left side of the vehicle **10** (e.g., driver's side), the fuel tank **112** can be located in any appropriate location and with any appropriate arrangement. By way of example, the fuel tank **112** can also be located on the passenger side of the vehicle **10**; further, in some embodiments, the vehicle **10** can have two or more fuel tanks **112**, which can be located in any appropriate location and with any appropriate arrangement.

With further regard to FIG. **13**, in some embodiments, the vehicle **10** has a second panel **122**. In some embodiments, the second panel **122** is welded to one or more of the adjacent frame members. In some embodiments, the second panel **122** is located on the right side of the vehicle **10**. The second panel **122** can support, for example, a battery or fuel tank; in at least some embodiments, the second panel **122** provides structural rigidity for the frame **14**. In some embodiments, the first panel **114** provides structural rigidity for the frame **14**. The first panel **114** and second panel **122** can include one or more cut-outs **124**, which can serve to lighten the respective panel.

Turning to FIG. **14**, in some embodiments, the ROPS **72** includes a rear lateral ROPS member **204**, longitudinal ROPS members **206** (**206a**, **206b**), forward overhead member **98**, A-frame members **96** (**96a**, **96b**), rear ROPS members **82** (**82a**, **82b**), and ROPS cross member **208**. In some embodiments, one or more of the ROPS members have a channel, as shown in FIG. **17**, in order to provide mating surfaces **210** for doors, a windshield, rear window, and roof. In some embodiments, forward overhead member **98** is formed from sheet steel and is stamped to provide a portion generally matching the contoured channel of the adjacent ROPS members (e.g., **206a**, **206b**, **96a**, **96b**). In some embodiments, an end edge **212** of the forward overhead member **98** overlaps a portion of the mating surface **210** of

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longitudinal ROPS member **206**, as shown in FIG. **16**, for example. Further, as shown in FIG. **1**, in some embodiments, an edge **212** of the forward overhead member **98** overlaps a portion of the mating surface **210** of the A-frame members **96**.

With regard to FIG. **15**, in some embodiments, the ROPS **72** includes a first ROPS connection member **214** and a second ROPS connection member **216**. In some embodiments, the first and second ROPS connection members **214**, **216** are formed from castings. In some embodiments, first ROPS connection member(s) is/are welded to an adjacent section of ROPS tubing. As shown in FIG. **15**, for example, in some embodiments, the first ROPS connection member **214** would be welded to the longitudinal ROPS member **206**. Further, in some embodiments, a portion of the first ROPS connection member **214** is inserted into the interior space of the adjacent ROPS tube such that an end of the ROPS tube abuts a ledge of the first ROPS connection member **214**, permitting the first ROPS connection member **214** to be easily located relative to the ROPS tube for welding.

As further illustrated in FIG. **15**, the first ROPS connection member **214** can be fastened to the second ROPS connection member **216**, for example via fasteners (e.g., nuts and bolts).

As shown in FIGS. **18-20**, the second ROPS connection member **216** includes a first connection portion **218**, a second connection portion **220**, and a third connection portion **222**. In some embodiments, the first connection portion **218** is fastened to a mating portion **234** of the first ROPS connection member **214**. In turn, in some embodiments, the second connection portion **220** is coupled to the rear ROPS member **82**. In some embodiments, the second connection portion **220** is welded to the rear ROPS member **82**, for example by fitting a portion of the second connection portion **220** into a portion of the ROPS member **82**, in a fashion similar to that discussed previously with respect to the first ROPS connection member **214** and longitudinal ROPS member **206**. In some embodiments, the third connection portion **222** is coupled to the rear lateral ROPS member **204**, for example by welding in a fashion similar to that discussed above with respect to the first ROPS connection member **214** and longitudinal ROPS member **206**. Of course, the second ROPS connection member **216** can be coupled to adjacent tubes/connection members in any suitable way, for example with fasteners, by welding, gluing, etc.

As shown in FIG. **19**, in some embodiments, the second ROPS connection member **216** includes an accessory attachment portion **224**. In some embodiments, an accessory can be attached thereto, for example, via fasteners (e.g., bolts) that are inserted into the one or more aperture(s) **226**. Further, nuts can be inserted into opening(s) **228** to mate with bolts inserted through aperture(s) **226**. The walls of the opening(s) **228** can capture sides of a nut such that a tool or wrench is not required to keep the nut from turning as the fastener (e.g., bolt) is tightened against the nut. In some embodiments, a door hinge is attached to the attachment portion **224**, as discussed in greater detail below. In some embodiments, a headache rack or other accessory can be attached to the accessory attachment portion **224**. Further still, in some embodiments, more than one accessory can be attached to the accessory attachment portion **224** at any given time.

Regarding FIGS. **18A-18C** show cross-sectional profiles of the second ROPS connection member **216** along portions thereof. With regard to FIG. **18C**, in some embodiments, one or more nuts (not shown) can be captured by channel **230**

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such that upon insertion of fasteners (e.g., bolts) through apertures **232** of the first connection portion **218**. In this way, a tool or wrench is not required to keep the nut from turning as the fastener (e.g., bolt) is tightened against the nut.

With regard to FIGS. **21-28**, an embodiment of a brake pedal assembly **236** is shown. The brake pedal assembly **236** is coupled to brake calipers, via brake lines, to stop or slow the vehicle **10** when the brake pedal **240** is depressed. In some embodiments, a portion of the brake pedal assembly **236** is coupled to pedal assembly mount **238**. As shown in FIGS. **25-27**, in some embodiments, the pedal assembly mount **238** is formed from a single piece of sheet steel which has bends **242**. The pedal assembly mount **238** can also be formed from multiple pieces of material and of any suitable material (e.g., plastic, composite, aluminum, alloy, etc.). As further shown, the pedal assembly mount **238** is coupled to the frame **14** and, in some embodiments, it is welded to the upper dash frame member **130** and the intermediate dash support member **188**, though it can be coupled to the frame **14** in any suitable way.

As shown in FIG. **27**, in some embodiments, the brake pedal assembly **236** has a brake pedal arm **244**, to which the brake pedal **240** is coupled (e.g., via one or more fasteners, welding, etc.), a collar member **246**, to which the brake pedal arm **244** is coupled (e.g., via one or more fasteners, welding, etc.), a shoulder bolt **248** around which the brake pedal arm **244** pivots, one or more bushings **250**, through which a portion of the shoulder bolt **248** extends, and a nut **252** (FIG. **25**), which is mated to shoulder bolt **248**. In some embodiments, the brake pedal assembly **236** further includes a master cylinder **254**, having therein a piston (not shown) which is coupled to rod **256**. The rod **256**, in turn, is coupled to fork **258**. The fork **258** is rotatably coupled to the brake pedal arm **244**. As shown in FIGS. **27-28**, the fork **258** is rotatably coupled to the brake pedal arm **244** via coaxial apertures in the fork **258** and brake pedal arm **244**, through which pivot pin **260** extends. Pivot pin **260** is held in place with a retaining clip (e.g., snap ring, c-clip, c-clip), nut, or other appropriate fastener.

In some embodiments, the brake pedal assembly **236** includes at least one pressure switch **262**. The at least one pressure switch **262** has an electrical output to activate the brake light(s) **50** (FIG. **4**) and/or provide a signal to the vehicle's CAN (controlled area network) bus (not shown), ECU (engine control unit), or other electrical component, when the brake pedal **240** is depressed.

In some embodiments, the brake pedal assembly **236** includes a first fluid reservoir **264**. In some embodiments, the first reservoir **264** is positioned vertically above the master cylinder **254** and rearwardly of the front splash panel **268** (FIG. **124**). In some embodiments, the brake pedal assembly **236** includes a second fluid reservoir **266**. In some embodiments, the second fluid reservoir **266** is positioned forwardly of the front splash panel **268** (FIG. **124**) and forwardly of the first fluid reservoir **264** and master cylinder **254**. In some embodiments, the second fluid reservoir **266** is in fluid communication with the first fluid reservoir **264** via fill tube **270**. In some embodiments, the fill tube **270** slopes downwardly and to the rear such that a user can fill the first fluid reservoir **264** via the second fluid reservoir **266**. The second fluid reservoir **266**, being remotely located relative to the first fluid reservoir **264**, allows the user to conveniently fill the brake fluid while the master cylinder **254** and first fluid reservoir **264** are positioned rearwardly of the front splash panel **268**.

In some embodiments, the master cylinder **254** has at least one piston (not shown) which is between 0.5 and 1.25 inches

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in diameter, though other sizes can also be utilized. In some embodiments, the master cylinder is 0.5-1.0 inches in diameter, and in some embodiments, the piston is 0.75 inches in diameter. Further, in some embodiments, each of the front calipers (not shown) can be single piston calipers or dual piston calipers, though other arrangements can also be used. In some embodiments, the front calipers are dual piston calipers, the pistons having a diameter of between 0.75 inches and 2.0 inches. In some embodiments, the front calipers are dual piston, each piston having a diameter of 1.375 inches. In some embodiments, each rear caliper is a single piston caliper, the pistons having a diameter of between 0.75 inches and 2.0 inches. In some embodiments, the rear calipers are single piston, each piston having a diameter of 1.375 inches. In some embodiments, the rear calipers are dual piston, though other arrangements can also be used.

The brake pedal assembly **236** can also include more than one master cylinder (e.g., two master cylinders), for example one master cylinder in fluid communication with front calipers and one master cylinder in fluid communication with the rear calipers. In some embodiments, the brake pedal assembly **236** includes an adjustable brake balancer, which can be adjusted when the vehicle is stopped or it can be adjusted dynamically from within the seating area during usage of the vehicle **10**.

In some embodiments, the brake pedal assembly **236** has a pedal ratio, which is ratio of distance moved by the center of the face of the brake pedal **240** relative to the distance moved by the pivot pin **260**. In some embodiments, the pedal ratio is between 4:1 and 7:1; in some embodiments, between 4.5:1 and 6.5:1; in some embodiments, between 5.5:1 and 6.5:1; in some embodiments, between 5.5:1 and 6:1. By way of example, in some embodiments, the pedal ratio is 5.8:1. In such an example, the pivot pin **260** moves 1 inch for every 5.8 inches moved by the center of the face of the brake pedal **240**. Other ratios can also be used, as appropriate, depending upon the size of the master cylinder(s), size of the brake caliper pistons, number of brake caliper pistons, brake bias, etc. In some embodiments, the brake pedal assembly **236** further includes power brakes, for example, via a brake booster.

Turning to FIGS. **29-38**, the utility vehicle **10** includes a front suspension **272**. In some embodiments, the front suspension **272** includes a front upper A-arm **274**, front lower A-arm **276**, front knuckle **278**, front damping member **280**, front anti-roll bar (ARB) **282**, front hanger **284**, and front ARB link **286**. In some embodiments, the front ARB link **286** is coupled to the front upper A-arm **274**, for example via a fastener such as a nut. Further, as shown at least in FIGS. **29** and **33**, in some embodiments, a portion of the front ARB link **286** extends through an aperture the front anti-roll bar **282** and a portion of the front ARB link **286** extends through front link aperture **288** (FIG. **37**) in the front upper A-arm **274**. Although the front ARB link **286** is shown as extending through an aperture in the front upper A-arm **274** (e.g., front link aperture **288**), the front ARB link **286** can, instead, be linked with the front lower A-arm **276**, for example via an aperture or in any appropriate way. Further, the front ARB link **286** can be linked with the front upper A-arm **274** in any appropriate way. In some embodiments, one or more portions of the front ARB link **286** extends through one or more elastomeric members **290**. In some embodiments, there are four elastomeric members **290**—one on each side of the aperture in the front anti-roll bar **282** (above and below the aperture) and one on each side of the front link aperture **288** in the front lower A-arm (above and

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below the front link aperture **288**). Thus, in some embodiments, the front suspension **272** includes eight total elastomeric members **290**. In some embodiments, the elastomeric members **290** an annular-cross sectional shape. In some embodiments, the front anti-roll bar **282** connects the left side of the front suspension with the right side of the front suspension in order to reduce roll.

By way of example, in a turn, the outside suspension compresses and the vehicle **10** tends to lean to the outside. The front anti-roll bar **282** connects the left side front suspension with the right side front suspension, thereby reducing roll and increasing the effective spring rate (as compared to the same suspension without the anti-roll bar) and roll stiffness. In some embodiments, however, the vehicle **10** does not include a front anti-roll bar **282**.

In some embodiments, the front anti-roll bar **282** includes a central portion **291** (FIG. **34**) and arm portions **292** (**292a**, **292b**). As shown, the arm portions **292** extend from the central portion **291**. In some embodiments, the central portion **291** undergoes torsional strain when one of the sides (left or right) of the front suspension is compressed or extended relative to the other side of the front suspension (left or right). The arm portions **292** can be rearwardly extending (as shown in FIG. **34**) or forwardly extending. Further, the central portion **291** can be located inside the frame **14**, outside the frame **14**, above an appropriate frame member, below an appropriate frame member, or otherwise packaged in any appropriate way. As will be appreciated, the material, length of the arm portions **292**, length of the central portion **291**, cross-section of the central portion **291**, cross-section of the arm portions **292**, wall thickness (if not formed from a solid piece of material), diameter, etc., of the anti-roll bar **282** can all be selected to achieve desired suspension characteristics.

In some embodiments, the front damping member **280** include a shock **294** and a spring **296**. In some embodiments, the damping member **280** comprises a coil-over shock, as shown in FIGS. **29-36**, however, other arrangements are also contemplated. By way of example, leaf springs, coils springs, torsion springs can also be used. Further, the shock **294** can be located remotely from the spring.

In some embodiments, the shock **294** is remotely adjustable, for example to adjust shock characteristics. In some embodiments, the shock **294** is adjustable via a mechanical adjuster (e.g., dial) or graphical user interface (GUI), which may employ a controller to adjust shock valving, etc.

In some embodiments, the damping member **280** includes a magnetorheological fluid, such that the shock characteristics can be dynamically adjusted by a microprocessor. Further still, in some embodiments, the utility vehicle **10** may not include a front anti-roll bar **282** (or rear anti-roll bar), as the microprocessor can react to changing conditions encountered by the vehicle, and vary the shock characteristics.

In some embodiments, for example as shown in FIG. **36**, the front lower A-arm **276a** (left hand side of the vehicle) is the same as the front lower A-arm **276b** (right hand side of the vehicle), such that these front lower A-arms are interchangeable. In some embodiments, the front lower A-arm **276** is stamped from a piece of sheet metal (e.g., sheet steel). In some embodiments, the front lower A-arm is welded together from tubular steel, however, the A-arms can be formed in any suitable way and from any suitable material or materials, alloys, etc. Further, in some embodiments, the front lower A-arm **276** is formed from two or more stamped components which are welded together.

With regard to FIG. 37, in some embodiments, the front upper A-arm 274 has bearings 298 which are pressed into respective receptacles 300. In some embodiments, the bearings 298 are spherical bearings, however other types of bearings or bushings can also be used. In some embodiments, the bearings 298 have an inner bearing member 302 and an outer bearing member 304. In some embodiments, the inner bearing member 302 includes a center portion having a convex spherical portion which is encircled by the outer bearing member 304. In some embodiments, the inner bearing member 302 has end portions 306, extending from the center portion. The end portions 306 are encircled by respective seals 309, which are also inserted into receptacles 300, to protect the bearing 298.

With further reference to FIG. 37, in some embodiments, the front upper A-arm 274 has a ball joint 308 which is retained in the front upper A-arm 274 via a retaining clip (e.g. snap ring 310), or in any other suitable way (e.g., press fit).

In some embodiments, the front lower A-arm 274 includes a mount 310, to which the front damping member 280 is coupled.

Although the bearing 298, receptacles 300, ball joint 308, etc., are discussed with respect to the front upper A-arm, these components are, in some embodiments, employed with the front lower A-arm 276, as well.

Turning now to FIGS. 39-47, the utility vehicle 10 includes a rear suspension 312. In some embodiments, the rear suspension 312 includes a rear upper A-arm 314, rear lower A-arm 316, rear knuckle 318, rear damping member 320, rear anti-roll bar (ARB) 322, rear hanger 324, and rear ARB link 326. In some embodiments, the rear ARB link 326 is coupled to the rear upper A-arm 314, for example via a fastener such as a nut. As further shown, in some embodiments, a portion of the rear ARB link 326 extends through an aperture in the rear anti-roll bar 322 and a portion of the rear ARB link 326 extends through rear link aperture 328 (FIG. 47) in the rear upper A-arm 314. Although the rear ARB link 326 is shown as extending through an aperture in the rear upper A-arm 314, the rear ARB link 326 can, instead, be linked with the rear lower A-arm 316, for example via an aperture or in any appropriate way. Further, the rear ARB link 326 can be linked with the rear upper A-arm 314 in any appropriate way. In some embodiments, one or more portions of the rear ARB link 326 extend(s) through one or more elastomeric members 290, in a fashion similar to that discussed above with respect to the front anti-roll bar.

By way of example, in a turn, the outside suspension compresses and the vehicle 10 tends to lean to the outside. The rear anti-roll bar 322 connects the left side front suspension with the right side front suspension, thereby reducing roll and increasing the effective spring rate (as compared to the same suspension without the anti-roll bar) and roll stiffness. In some embodiments, however, the vehicle 10 does not include a rear anti-roll bar 322.

In some embodiments, the rear anti-roll bar 322 includes a central portion 330 (FIG. 44) and arm portions 332 (332a, 332b). As shown, the arm portions 332 extend from the central portion 330. In some embodiments, the central portion 330 undergoes torsional strain when one of the sides (left or right) of the rear suspension is compressed or extended relative to the other side of the rear suspension (left or right). The arm portions 332 can be rearwardly extending (as shown in FIG. 39) or forwardly extending. Further, the central portion 330 can be located inside the frame 14, outside the frame 14, above an appropriate frame member,

below an appropriate frame member, or otherwise packaged in any appropriate way. As will be appreciated, the material, length of the arm portions 332, length of the central portion 330, cross-section of the central portion 330, cross-section of the arm portions 332, wall thickness (if not formed from a solid piece of material), diameter, etc., of the rear anti-roll bar 322 can all be selected to achieve desired suspension characteristics.

In some embodiments, the rear damping member 320 includes a shock 294 and a spring 296. In some embodiments, the rear damping member 320 comprises a coil-over shock, as shown in FIGS. 39-43, 45, and 46, however, other arrangements are also contemplated. By way of example, leaf springs, coils springs, torsion springs can also be used. Further, the shock 294 can be located remotely from the spring.

In some embodiments, the shock 294 is remotely adjustable, for example to adjust shock characteristics. In some embodiments, the shock 294 is adjustable via a mechanical adjuster (e.g., dial) or graphical user interface (GUI), which may employ a controller to adjust shock valving, etc.

In some embodiments, the rear damping member 320 includes a magnetorheological fluid, such that the shock characteristics can be dynamically adjusted by a microprocessor. Further still, in some embodiments, the utility vehicle 10 may not include a front anti-roll bar 282 (or rear anti-roll bar), as the microprocessor can react to changing conditions encountered by the vehicle, and vary the shock characteristics.

In some embodiments, the rear lower A-arm 316a (left hand side of the vehicle) is the same as the rear lower A-arm 316b (right hand side of the vehicle), such that these rear lower A-arms are interchangeable. In some embodiments, the rear lower A-arms 316 is stamped from a piece of sheet metal (e.g., sheet steel). In some embodiments, the rear lower A-arm is welded together from tubular steel, however, the A-arms can be formed in any suitable way and from any suitable material or materials, alloys, etc. Further, in some embodiments, the rear lower A-arm 316 is formed from two or more stamped components which are welded together. In some embodiments, the rear lower A-arms 316a, 316b are not the same.

In some embodiments, one or more of the rear upper A-arm 314, rear lower A-arm 316, front lower A-arm 276, utilize bearings 298 and seals 309, as discussed above with respect to FIG. 37.

With regard to FIG. 47, in some embodiments, the rear upper A-arm 314 includes a mount 334 to which the rear damping member 320 is coupled. The rear damping member 320 can also be coupled to the lower A-arm.

In some embodiments, at least a portion of the rear lower A-arm 316 extends through a vertically extending longitudinal plane 336. The vertically extending longitudinal plane 336 includes the centerline (shown in FIG. 42) of the vehicle 10. Plane 336 is shown illustratively in FIGS. 40 and 41. In some embodiments, at least a portion of the lower A-arm 316 is positioned above the lower rear frame member 126, as shown in FIG. 42. In some embodiments, the left lower A-arm 316a has a left lower A-arm pivot axis 338, around which the left lower A-arm pivots as it moves through its range of travel. As shown in FIG. 42, the left lower A-arm pivot axis 338 extends longitudinally and is positioned (in bottom view looking upwardly) between the centerline of the vehicle 10 and the right rear lower frame member 126b. Similarly, in some embodiments, the right lower A-arm pivot axis 340, around which the right lower A-arm 316b

pivots, is positioned, in bottom view, between the centerline of the vehicle **10** and the left rear lower frame member **126a**.

In some embodiments, respective upper and lower A-arms of left rear suspension, right rear suspension, left front suspension, right front suspension are parallel A-arms. In some embodiments, the respective arms are non-parallel A-arms. In some embodiments, the respective upper and lower A-arms are unequal in length. In some embodiments, the lower A-arm is longer than the upper A-arm, as shown in FIGS. 44-47.

Turning to FIGS. 48-55, in some embodiments, the steering rack **104** has tie rods **342** (**342a**, **342b**) coupled thereto. In turn, in some embodiments, the tie rods **342** have tie rod ends **344** (**344a**, **344b**) respectively coupled to the tie rods **342**, for example by threading the tie rods to threads on the tie rod ends **344** include steering ball joints **346** which, in some embodiments, are coupled to front knuckle **278** (FIG. 29) in order to facilitate steering of the vehicle **10**. In some embodiments, the steering ball joints **346** are coupled to respective front knuckles **278** via a castle nut **348** and cotter pin **350**.

In some embodiments, the steering rack **104** is positioned above at least a portion of a front drive assembly **352**. In some embodiments, the front drive assembly **352** includes a differential (not shown) which transmits power from the front drive shaft **354** to the front half-shafts **356**. In some embodiments, the steering rack **104** is coupled to rack bracket **358** (FIG. 54). In turn, in some embodiments, the rack bracket **358** is coupled to rack mount **360**. The rack mount **360** is coupled to a portion of the front upper cross-bracket **372**, as shown for example in FIG. 55. In some embodiments, the rack bracket **358** is coupled to the rack mount **360** via one or more fasteners; further, in some embodiments, the rack mount **360** is coupled to the front upper cross-bracket **372** via one or more fasteners.

Turning to FIGS. 52-56 and 59, in some embodiments, the front drive assembly **352** is coupled to an upper front drive assembly bracket **362** and a lower front drive assembly bracket **364**. In some embodiments, the housing of the front drive assembly **352** is coupled to the upper front drive assembly bracket **362** and lower front drive assembly bracket **364** via one or more fasteners (e.g., bolts). In turn, in some embodiments, one or both of the upper front drive assembly bracket **362** and lower front drive assembly bracket **364** are coupled to the frame **14**. In some embodiments, the upper front drive assembly bracket **362** is coupled, for example via one or more fasteners, to front upper cross-bracket **372** (FIG. 50). In some embodiments, the front upper cross-bracket **372** is formed from a piece of bent sheet metal (e.g., steel). In some embodiments, the front upper cross-bracket **372** includes one or more mounting locations for the front upper A-arm **274** and, in some embodiments, the front upper front-suspension support member **370** is integrally formed with the front upper cross-bracket **372** (FIG. 52). As shown in FIG. 52, in some embodiments, the front upper cross-bracket **372** includes tabs to which a front fork portion and rear fork portion, respectively, of the front upper A-arm **274** are rotatably mounted, to permit movement of the suspension. In some embodiments, a horizontal portion of the upper front drive assembly bracket **362** is positioned above a horizontal portion of the front upper cross-bracket **372**.

Returning to FIG. 32, in some embodiments, the front fork portion **414** of the front lower A-arm **276** is rotatably mounted to the front lower front-suspension support member **366** and the rear fork portion of the front lower A-arm **276** is rotatably mounted to a backer **418**. In some embodi-

ments, the backer **418** is piece of sheet steel that is welded to a portion of the front joining frame member **184** (FIG. 11). In some embodiments, a portion of the front upper A-arm **274**, can be rotatably mounted in a similar fashion, for example, to an appropriate backer. Portions of the rear suspension **312** can be rotatably mounted via an appropriate backer, as well.

With further regard to FIG. 52, in some embodiments, the frame **14** includes a rear upper front-suspension support member **368** and a front lower front-suspension support member **366**. In some embodiments, the rear upper front-suspension support member **368** and front lower front-suspension support member **366** are welded to the lower front frame member **180**. In some embodiments, the front upper cross bracket **372** and rear upper front-suspension support member **368** are formed from a single piece of bent sheet metal (e.g., steel), as shown for example in FIGS. 55A and 55B. In some embodiments, the front upper cross-bracket **372** is welded to one or more adjacent portions of the frame **14** (e.g., longitudinal intermediate front frame member **182**). In some embodiments, one or more portions of the front upper cross-bracket **372** are welded to one or more other portions of the front upper cross-bracket **372**. As shown in FIG. 55B, for example, angled portion **390** of the front upper cross-bracket **372** is welded at **392**.

In some embodiments, the lower front drive assembly bracket **364** is coupled to the fourth panel **198**, for example via one or more fasteners (e.g., bolts), as shown in FIG. 54.

Turning to FIGS. 56-63, one or more portions of the driveline **374** are shown. In some embodiments, the driveline **372** includes a prime mover **376** (e.g., engine, electric motor, etc.), transmission **378** (e.g., transaxle, which includes a rear differential), front drive shaft **354**, front drive assembly **352**, front half shaft(s) **356**, rear half shafts **380**, drive clutch **382** (FIG. 58), and driven clutch **384** (FIG. 58). In some embodiments, the prime mover **376** and transmission **378** are rigidly coupled to one another, for example via one or more coupling members **386** (FIG. 56). In turn, in some embodiments, the prime mover **376** and transmission **378**, as a unit, is isolated from the frame **14** (FIG. 11) via vibration isolators **388** (FIG. 56). As shown in FIGS. 58, 63, and 70, in some embodiments, the prime mover **376** is coupled to a lateral mount member **394**, for example via one or more brackets. In turn, the lateral mount member **394** is isolated, via vibration isolators **388**, from standoff members **396**. The standoff members **396** are coupled to portions of the frame **14**, for example, via one or more fasteners. In some embodiments, the vibration isolators include an elastomeric portion (e.g., rubber isolator), to isolate vibration caused by one or more of the driveline components (e.g., engine, transmission, etc.).

In some embodiments, the transmission **378** is coupled to vibration isolators **388** via one or more transmission isolator brackets **398**, as shown in FIGS. 56, 57, and 70.

With regard to FIGS. 58 and 61-63, in some embodiments, a portion of the front drive shaft **354** extends under the lateral mount member **394** and under a portion of the prime mover **376**. As shown in FIG. 63, in some embodiments, a portion of the front drive shaft **354** extends between an oil pan **400** and an inner CVT (continuously variable transmission) cover **402**.

With further regard to FIG. 58, in some embodiments, the CVT cover assembly **404** includes an inner CVT cover **402**, an intermediate CVT cover **406**, and an outer CVT cover **408**. In some embodiments, the drive clutch **382** and driven clutch **384** are housed within a space between the intermediate CVT cover **406** and outer CVT cover **408**. In some

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embodiments, the flywheel **410** (FIGS. **63** and **70**) is housed within a space between the inner CVT cover **402** and mount plate **412**. In some embodiments, the mount plate **412** also serves as a coupler member **386** to couple the prime mover **376** with the transmission **378**.

In some embodiments the front drive assembly **352** is coupled to one or more of the lower front drive assembly bracket **364**, upper front drive assembly bracket **362**, and frame **14** via one or more vibration isolators. Alternatively, in some embodiments, the front drive assembly **352** is rigidly coupled to one or more of the lower front drive assembly bracket **364**, upper front drive assembly bracket **362**, and frame **14**.

With regard to FIGS. **64**, an embodiment of a CVT cover assembly **404** is shown. Also shown in FIG. **64** is a CVT intake duct member **420**, CVT exhaust duct member **422**, and CVT air box **424**. In some embodiments, the outer CVT cover **408** is removable from the intermediate CVT cover **406** without the use of tools (i.e. with hands only). In some embodiments, the outer CVT cover **408** is removed from the intermediate CVT cover **406** by releasing CVT cover fasteners **426**. In some embodiments, CVT cover fasteners **426** are spring-type retainers.

Shown in FIG. **64A** is another embodiment of a CVT cover assembly **404** and CVT air box **424**. In some embodiments, the CVT air box **424** has one or more mounting tabs **428**. The mounting tabs **428** can support (or help support) the CVT air box **424** such that the CVT air box **424** can be coupled to an adjacent frame member or piece of material (e.g., sheet metal) which is welded or otherwise coupled to the frame. An example of the CVT air box **424** being coupled to an adjacent frame member is shown in FIG. **5**. In some embodiments, the CVT air box **424** is coupled to a frame member, mounting bracket, etc., via an air box fastener **430** (e.g., sheet metal screw, nut, etc.).

Turning to FIG. **64B**, the flywheel **410** is shown in position within the inner CVT cover **402**. In some embodiments, the flywheel **410** includes a plurality of projections **432** extending radially from the flywheel **410**. In some embodiments, the projections **432** are formed of the same piece of material as the flywheel **410**; in some embodiments, however, the projections **432** are individual members which are attached to the flywheel **410**; in still other embodiments, the projections **432** extend from an annular portion of a member that is coupled to the flywheel **410** via one or more fasteners.

With further regard to FIG. **64B**, in some embodiments, a sensor **434** (e.g., magnetic pickup) is positioned adjacent to the flywheel **410** such that upon rotation of the flywheel **410** the sensor **434** detects passage of the projections **432**. In this way, the engine speed is known and can be displayed on the dash of the vehicle. In this way, an operator will know how fast the engine is rotating. Such information can be displayed in revolutions per minute (RPM). In addition, the projections **432** can be arranged in a configuration that the sensor **434** can detect the position of the crankshaft of the engine. In some embodiments, an electrical lead **436** extends from the sensor **434**. The electrical lead **436** can transmit a signal to the CAN bus, ECU, etc. In some embodiments, however, the signal is transmitted wirelessly to the CAN bus, ECU, etc. such that the inner CVT cover **402** does not have a CVT cover cutout **438** for the electrical lead **436**. Stated differently, in some embodiments, the electrical lead **436** extends from an interior of the inner CVT cover **402** to the exterior of the inner CVT cover **402**. Thus, where a signal is transmitted wirelessly from within the inner CVT cover **402**, the CVT cover cutout **438** may not be necessary.

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In some embodiments, the inner CVT cover **402** is coupled to the mount plate **412** (FIG. **70**) via one or more fasteners. Further, in some embodiments, an inner CVT gasket **442**. In some embodiments, the CVT gasket **442** resides in an inner CVT cover gasket channel **444** (FIG. **68**). In some embodiments, the CVT gasket **442** is pressed against the mount plate **412** to provide seal therebetween.

With regard to FIGS. **65**, **66**, **68**, and **69**, in some embodiments, one or more of the CVT covers (e.g., inner CVT cover **402** and intermediate CVT cover **406**) are coupled together via a CVT cover retainer **440**. In some embodiments, the CVT cover retainer **440** comprises a snap ring. In some embodiments, the inner CVT cover **402** comprises one or more retaining tabs **446**. As illustrated in FIGS. **65** and **69**, the CVT cover retainer **440** is positioned between a portion of the intermediate CVT cover **406** and retaining tabs **446**. In this way, the intermediate CVT cover **406** is secured to the inner CVT cover **402**. In order to disassemble the intermediate CVT cover **406** from the inner CVT cover **402**, in some embodiments, the CVT cover retainer **440** is expanded (in diameter) and removed from the first groove **448** (FIG. **65**). Thereafter, the intermediate CVT cover **406** can be removed from the inner CVT cover **402**.

With further reference to FIG. **69**, in some embodiments, the inner CVT cover **402** and intermediate CVT cover **406** include ridges **450**. The ridges **450** of the inner CVT cover **402** and the ridges **450** of the intermediate CVT cover **406** mate with each other in order to provide a positive interaction between the inner CVT cover **402** and the intermediate CVT cover **406**, and to facilitate locating the inner CVT cover **402** and the intermediate CVT cover **406** relative to one another during assembly. Further, the ridges **450** can provide a water-tight seal between the respective CVT covers. Although the ridges **450** are shown between the inner CVT cover **402** and intermediate CVT cover **406**, they can be utilized at an interface with any of the CVT covers.

Returning to FIG. **66**, in some embodiments, an outer CVT gasket **452** is positioned in a second groove **454** (FIG. **68**). The outer CVT gasket **452** provides a watertight seal between the outer CVT cover **408** and the intermediate CVT cover **406**. In some embodiments, the outer CVT gasket **452** is formed from an elastomeric material. In some embodiments, the outer CVT gasket **452** is formed from an extruded foam material or molded rubber material. In some embodiments, the outer CVT gasket **452** has a circular cross-section.

With additional reference to FIG. **66**, in some embodiments, the inner CVT gasket **442** is formed from an elastomeric material. In some embodiments, the inner CVT gasket **442** is formed from an extruded foam material or molded rubber material. In some embodiments, the inner CVT gasket **442** includes a plug portion **456**. As shown in FIG. **64B**, the plug portion **456** fits in the CVT cover cutout **438**. In some embodiments, the plug portion **456** includes a slit **458** and/or plug portion aperture **460**. During assembly or disassembly of the inner CVT cover **402**, the slit **458** can facilitate passage of the electrical lead **436**. As is shown in FIG. **70**, in this way, the sensor **434** can remain coupled to the mount plate **412** (for example via an appropriate mounting bracket) upon removal of the inner CVT cover **402**. Further, in some embodiments, the electrical lead **436** extends through plug portion aperture **460** when the inner CVT cover **402** is assembled to the mount plate **412**. In some embodiments, the plug portion **456** is integrally formed with the inner CVT gasket **442**, for example in a molding process.

As also shown in FIG. **66**, in some embodiments, a CVT filter **462** (e.g., foam) is positioned within a cavity of the

CVT air box **424**. The CVT filter **462** can be of any suitable type, such as paper filter, foam filter, centrifugal filter, etc.

In some embodiments, air enters the CVT cover assembly **404** via CVT air box **424** and is then routed into the inner CVT cover **402**. The air then flows from the inner CVT cover **402**, past the inner CVT cover opening **466** and into a space between the intermediate CVT cover **406** and outer CVT cover **408** (a portion of which space is, in some embodiments, also occupied by the drive clutch **382** (FIG. **58**)). Thereafter, the air flows into a narrowing region in the space between the intermediate CVT cover **406** and outer CVT cover **408** and exists through the exit port **464** (FIG. **64B**) which leads into CVT exhaust duct member **422**. The air is then exhausted out into the atmosphere or is used to cool an oxygen sensor or other sensor, for example as disclosed in U.S. Pub. No. 2016/0061088, titled, "Oxygen Sensor Cooling Duct," the contents of which are herein incorporated by reference.

Turning to FIGS. **71-76**, in some embodiments the vehicle **10** includes a steering column assembly **468**. In some embodiments, the steering column assembly **468** includes a gear selector assembly **470** and a tilt assembly **88** (also shown in FIG. **2**). In some embodiments, the gear selector assembly **470** includes a cable **472** (e.g., a push-pull cable); the cable **472** extends to the transmission **378**, thereby permitting the operator to select a desired gear (e.g., park, high, low, neutral, reverse). In some embodiments, the cable **472** is surrounded by a sheath **526** (FIG. **73**) to protect the cable **472**.

In some embodiments, the steering column assembly **468** further includes an upper steering shaft **474**, first steering shaft bushing **476**, and a second steering shaft bushing **478**. In some embodiments, the steering column assembly **468** includes a first steering shaft washer **480** (e.g., a wave washer), a second steering shaft washer **482**, and a steering shaft retaining clip **484** (e.g., c-clip, e-clip, snap ring, etc.). Although shown with a steering shaft retaining clip **484** in FIG. **75**, other fasteners can also be used, such as a nut. In some embodiments, the upper steering shaft **474** has a splined portion **486** and a threaded portion **488**. In some embodiments, the threaded portion **488** has a hole extending therethrough such that a castle nut (not shown) and cotter pin (not shown) can be used to fasten the steering wheel **84** (FIG. **2**) to the upper steering shaft **474**. In some embodiments, the upper steering shaft **474** includes a U-joint **490** at a lower end thereof.

In some embodiments, the tilt assembly **88** (shown without shock **90** in FIGS. **71-76**) includes a barrel member **492**, through which a portion of the upper steering shaft **474** extends. Further, in some embodiments, the first steering shaft bushing **476** and second steering shaft bushing **478** are positioned within a portion of the barrel member **492**. In some embodiments, the tilt assembly **88** further includes a horseshoe member **494**, which may be coupled to barrel member **492**, for example by welding or via one or more fasteners. In some embodiments, the horseshoe member **494** and barrel member **492** are formed from a single piece of material (e.g., by casting). In some embodiments, the tilt assembly **88** further includes an inner tilt bushing **496**, an outer tilt bushing **498**, a tilt pivot member **500** (e.g., bolt), and tilt fastener **502** (e.g., nut). In some embodiments, for example as shown in FIG. **75**, the tilt assembly **88** includes two of each of the inner tilt bushing **496**, outer tilt bushing **498**, tilt pivot member **500**, and tilt fastener **502**. In some embodiments, the tilt assembly **88** includes an upper shock

mount **504**. In some embodiments, the upper shock mount **504** is coupled to the barrel member **492**, for example with one or more fasteners.

With regard to FIGS. **73-76**, in some embodiments, the gear selector assembly **486** includes a shift tube **510**, shift handle **506**, shift arm **508**, jam nut **514**, torque arm **512**, first gear selector bushing **516**, and second gear selector bushing **518**. The first and second gear selector bushings **516**, **518** are positioned over first and second collar portions **520**, **522**, respectively, of the barrel member **492**, as shown in FIG. **75**. In some embodiments, at least one of the first and second gear selector bushings **516**, **518** is formed from a polyacetyl resin. In some embodiments, at least one of the first and second gear selector bushings **516**, **518** has a hardness of at least 80 shore D. In some embodiments, the jam nut **514** is utilized to adjust the position length and position of the shift handle **506**.

In some embodiments, the shift arm **508** is coupled to the shift tube **510**, for example by welding, however the shift arm **508** can be coupled to the shift tube **510** in any other suitable manner (e.g., one or more fasteners, adhesive, etc.). In some embodiments, the shift arm **508** has a threaded end onto which the shift handle **506** and jam nut **514** are threaded. In some embodiments, the jam nut **514** is tightened against the shift handle **506** to prevent the shift handle **506** from loosening.

In some embodiments, the gear selector assembly **486** further includes a cable bracket **524**. In some embodiments, an end of the cable sheath **526** (FIG. **73**) includes a threaded member that extends through a slot or aperture **530** in the cable bracket **524**. In this way, the cable **472** can move within the cable sheath **526**, thereby permitting the operator to select the desired gear by rotating the shift handle **506**. As further shown in FIG. **73**, in some embodiments, a cable boot **528** extends over a portion of the cable **472**.

In some embodiments, the shift tube **510** includes a shift tube opening **532** (FIG. **74**). The shift tube opening **532** permits rotation of the shift tube **510** over the upper shock mount **504**.

Although components of the steering column assembly **468** may be described within the context of the gear selector assembly **470** or tilt assembly **88**, it will be appreciated that some components can be grouped into more than one of these assemblies. Further, it will be appreciated that the steering/tilt/gear select components discussed in the preceding paragraphs can be aptly included in the context of any one of the steering column assembly **468**, gear selector assembly **470**, and/or tilt assembly **88**.

Turning to FIGS. **77-84**, in some embodiments, the vehicle **10** includes a floorboard **534**. In some embodiments, the floorboard **534** extends the width of the seating area **20**; in some embodiments, it extends forwardly to the front splash panel **268** (FIG. **124**); and, in some embodiments, it extends rearwardly to a forward end of the rear outer frame member **78** (FIG. **11**). In some embodiments, one or more portions of the floorboard **534** include one or more gutters **536**. The one or more gutters **536** direct water (or other liquid or debris) to the one or more drains **538**. In some embodiments, the gutters **536** are connected such that water (or other liquid or debris) can flow downhill when the vehicle **10** is situated on level ground. In some embodiments, the floorboard **534** is sloped to permit flow via the gutters **536** for exit via the drain(s) **538**. With regard to FIGS. **82** and **83**, in some embodiments, some of the gutters **536** have a greater depth, x_1 , than other gutters **536**, x_2 . In this way, fluid (or other debris) will flow via the gutters **536**, in combination, even where the floorboard **534** is not oth-

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erwise sloped. In some embodiments, the depth of gutters **536** can change along the length of the gutter to facilitate flow. In some embodiments, certain of the gutters **536** are wider (or narrower) than others.

As shown for example in FIG. **82**, in some embodiments, the floorboard **534** includes one or more traction-enhancing features, for example projections **540** (which can also be indentations, texturing, etc.) which, in some embodiments, are arranged in a pattern. With regard to FIGS. **77**, **78**, and **81**, in some embodiments, the floorboard **534** includes a footrest **542** (e.g., dead-pedal) which in some embodiments comprises a sloped portion of the floorboard **534** on which the operator can rest a foot (e.g., left foot). In some embodiments, the floorboard **534** includes a raised tunnel portion **544**. The raised tunnel portion **544** can provide a space for a portion of the driveline **374**, such as the front drive shaft **354** (FIG. **56**).

As shown in FIG. **80**, in some examples, the floorboard **534** includes one or more removable covers **546**. In some embodiments, the removable cover **546**, as shown in FIG. **80**, snaps into place. As further shown in FIG. **80**, the removable cover **546** can be removed to provide access to one or more components (not shown in FIG. **80**) behind the removable cover **546**. By way of example, the removable cover **546** can be removed to access a dipstick, oil filter, or other engine component. In some embodiments, the removable cover **546** includes a cover handle **548**, which can be gripped by an operator to facilitate removal or replacement of the removable cover **546**.

With regard to FIGS. **81**, **83**, and **84**, in some embodiments, the floorboard **534** includes one or more upwardly extending portions **550**. As shown in FIGS. **83** and **84**, the upwardly extending portion **550** is integrally formed with the floorboard **534**, for example during a molding process. In some embodiments, the floorboard **534** includes a plateau portion **552**. The plateau portion **552** extends longitudinally and laterally, for example above at least a portion of the fuel tank **112** (FIG. **7**).

In FIGS. **88** and **89**, an embodiment of a driver's seat **24a** is shown. In some embodiments, the driver's seat **24a** includes one or more seat structural portions **554a** (e.g., frame members). The seat structural portions **554a** can comprise any suitable material, for example metal (e.g., steel), composite, plastic, glass-filled nylon, etc. In some embodiments, the seat base **26a** includes a cushion **556a**. In some embodiments, the seat back **28a** includes a cushion.

In FIGS. **90** and **91**, an embodiment of a passenger seat base **26b** is shown. In some embodiments, the passenger seat base **26b** is removably coupled to one or more portions of the seat support frame **146** (FIGS. **94-97**). In some embodiments, the passenger seat base **26b** includes one or more structural portions **554b** (e.g., frame members) and a cushion **556b**. The seat structural portions **554b** can comprise any suitable material, for example metal (e.g., steel), composite, plastic, glass-filled nylon, etc. In some embodiments, the passenger seat base **26b** includes one or more seat attachment members **558b**. The seat attachment members **558b** can facilitate attachment of the seat base **26b** to one or more portions of the seat support frame **146** (FIGS. **94-97**). In some embodiments, the seat attachment members **558b** comprise one or more hooks. The one or more hooks, in turn, are matingly coupled to seat support pins **560b** (FIGS. **95-97**). Although shown with the seat base **26b** having hooks and the seat support frame **146** having seat support pins **560b**, the seat base **26b** can alternatively comprise supports pins and the seat support frame **146** can have hooks. Further, the seat support frame **146** and seat base **26b** can comprise

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any suitable mating, interlocking, coupling structure(s). In some embodiments, the seat base **26b** (e.g., seat structural portion **554b**) includes one or more seat alignment members **562b**. The seat alignment members **562b** mate with seat receiver members **564b** (FIGS. **94-97**). In some embodiments, the seat receiver members **564b** facilitate alignment of the seat base **26b** with respect to the seat support frame **146** when the seat base **26b** is placed on the seat support frame **146** (e.g., upon returning the seat base **26b** to a seating configuration, as discussed in greater detail below). In some embodiments, the seat receiver members **564b** also support the seat base **26b**. Although shown in FIGS. **94-97** with the seat alignment members **562b** having a male portion and the seat receiver members **564b** having a female portion, these roles can be reversed such that a male portion is positioned on the seat support frame **146** (or is an integral part thereof) and a female portion is positioned on the seat structural portion **554b** (or is an integral part thereof). Other arrangements and structures can also be employed.

With regard to FIGS. **92** and **93**, an embodiment of a passenger seat base **26c** is shown. In some embodiments, the passenger seat base **26c** is removably coupled to one or more portions of the seat support frame **146** (FIGS. **94-97**). In some embodiments, the passenger seat base **26c** includes one or more structural portions **554c** (e.g., frame members) and a cushion **556c**. The seat structural portions **554c** can comprise any suitable material, for example metal (e.g., steel), composite, plastic, glass-filled nylon, etc. In some embodiments, the passenger seat base **26c** includes one or more seat attachment members **558c**. The seat attachment members **558c** can facilitate attachment of the seat base **26c** to one or more portions of the seat support frame **146** (FIGS. **94-97**). In some embodiments, the seat attachment members **558c** comprise one or more hooks. The one or more hooks, in turn, are matingly coupled to seat support pins **560c** (FIGS. **95-97**). Although shown with the seat base **26c** having hooks and the seat support frame **146** having seat support pins **560c**, the seat base **26c** can alternatively comprise supports pins and the seat support frame **146** can have hooks. Further, the seat support frame **146** and seat base **26c** can comprise any suitable mating, interlocking, coupling structure(s). In some embodiments, the seat base **26c** (e.g., seat structural portion **554c**) includes one or more seat alignment members **562c**. The seat alignment members **562c** mate with seat receiver members **564c** (FIGS. **94-97**). In some embodiments, the seat receiver members **564c** facilitate alignment of the seat base **26c** with respect to the seat support frame **146** when the seat base **26c** is placed on the seat support frame **146** (e.g., upon returning the seat base **26c** to a seating configuration, as discussed in greater detail below). In some embodiments, the seat receiver members **564c** also support the seat base **26c**. Although shown in FIGS. **94-97** with the seat alignment members **562c** having a male portion and the seat receiver members **564c** having a female portion, these roles can be reversed such that a male portion is positioned on the seat support frame **146** (or is an integral part thereof) and a female portion is positioned on the seat structural portion **554c** (or is an integral part thereof). Other arrangements and structures can also be employed.

FIGS. **94-97** show progressive views for removing and storing the seat bases **26b** and **26c**. Reference is further made to FIGS. **86** and **87**. As illustrated in FIG. **94**, front portions of the seat bases **26b** and **26c** are rotated upwardly about an axis of the seat support pins **560** such that the seat alignment members **562** are free and clear of the seat receiver members **564**. With regard to FIG. **95**, the seat bases **26b**, **26c** are

removed from the seat support frame 146. As further shown in FIGS. 95-97, in some embodiments, the seat backs 28b, 28c are supported by one or more lower seatback supports 566. In some embodiments, the lower seatback supports 566 are coupled to the lateral rear cab frame member 154 (FIG. 111), for example via one or more fasteners. In some embodiments, the lower seatback supports 566 comprise hook members into which a portion of the seat backs 28b, 28c extends. In some embodiments, the lower seatback supports 566 have an axis which is collinear with a seatback axis 568 (FIG. 95), discussed below.

Referring to FIG. 96, in some embodiments, the seat backs 28b, 28c are rotated forward. In some embodiments, the seat backs 28b, 28c are rotated about seatback axis 568 (FIG. 95). Referring to FIGS. 86 and 87, in some embodiments, the seat backs 28b, 28c are coupled together via one or more seatback bridging members 570, for example an upper seatback bridging member 572 and a lower seatback bridging member 574. In some embodiments, the lower seatback bridging member 572 comprises a metallic member (e.g., tubular member) that is bent or formed into an appropriate shape. The lower seatback bridging member 574 can be formed from any suitable material (e.g., metal, plastic, composite, etc.) and in any suitable way. In some embodiments, the lower seatback bridging member 574 is coupled to one or more structural portions of the seat backs 28b, 28c. In some embodiments, the lower seatback bridging member 574 is coupled to structural portions of each of the seat backs 28b, 28c in three locations, as illustrated in FIG. 86, for example via fasteners (e.g., screws, bolts, etc.) (not shown).

In some embodiments, the upper seatback bridging member 572 comprises a metallic member that is bent or formed into an appropriate shape. In some embodiments, the upper seatback bridging member 572 is formed from a piece of sheet metal (e.g., sheet steel) which is bent. In some embodiments, the upper seatback bridging member 572 has one or more holes which receive fasteners (e.g., to couple the upper seatback bridging member 572 to one or more structural portions of each of the seat backs 28b, 28c) (e.g., screws, bolts, etc.) (not shown). In some embodiments, the upper seatback bridging member 572 includes one or more upper seatback receiver members 576, which mate with the seatback alignment members 578 (FIG. 85). Although shown with the female member (e.g., upper seatback receiver member(s) 576) being a portion of (or coupled to) upper seatback bridging member 572 and the male member (e.g., seatback alignment member(s) 578) being a portion of (or coupled to) the seatback body panel 580, the male member can be a portion of (or coupled to) the upper seatback bridging member 572 and the female member (e.g., seatback alignment member(s) 578) can be a portion of (or coupled to) the seatback body panel 580. Other configurations, fasteners, etc. are also suitable. In some embodiments, the seatback body panel 580 is coupled (e.g., fastened via one or more fasteners (e.g., screws, bolts, etc.)) to the ROPS cross member 208, as shown in FIG. 85.

Returning to FIG. 86, in some embodiments, the upper seatback bridging member 572 includes one or more seatback retainer apertures 582. Further, as shown in FIG. 85, one or more seatback fastening members 584 are coupled to the seatback body panel 580 such that the seatback fastening member(s) 584 matingly engages with the one or more seatback retainer apertures 582. In some embodiments, the seatback fastening member 584 is a quarter-turn fastener, as shown.

Returning now to FIG. 96, the seat bases 26b, 26c are stowed forward of the rear portion of the seatback body

panel 580 (see also FIG. 111). Thereafter, as illustrated in FIG. 97, the seat backs 28b, 28c are rotated rearwardly into an upright configuration and the seatback fastening member 584 is oriented (e.g., turned ¼ turn) to retain the upper seatback bridging member 572. As shown in the cross-section of FIG. 111, the seat base 26b is shown in both configurations: (1) in a stowed configuration, wherein the seat base 26b is positioned forward of the rear portion of the seatback body panel 580 and rearward of the seat back 28b, and (2) in a seating configuration, wherein the seat base 26b is situated such that an operator can sit on it.

In order to orient the seat bases 26b, 26c from a stowed configuration to a seating configuration, the process is reversed. Specifically, after orienting the seatback fastening member 584 in an appropriate fashion, the seat back 28b, 28c is rotated forwardly about seatback axis 568 (FIG. 95). Thereafter, the seat bases 26b, 26c are removed from their stowed location, positioned relative to the seat support pins 560, and then rotated about an axis defined by the seat support pins 560 until the seat alignment members 562 are received within the seat receiver members 564, as shown in FIGS. 93-96.

In some embodiments, the seat back 28b, 28c can be removed by orienting the seatback fastening member 584 (e.g., rotating ¼ turn to unlock), rotating a top portion of the seat back 28b, 28c forwardly about seatback axis 568 (FIG. 95), and lifting the seatback 28b, 28c such that the lower seatback bridging member 574 (FIG. 86) is free of the lower seatback supports 566 (FIG. 95).

Returning to FIGS. 78 and 79, in some embodiments, the vehicle 10 includes a rear splash panel 586. In some embodiments, the vehicle includes left (586a) and right (586b) rear splash panels. In some embodiments, one or both of the rear splash panels 586a, 586b include rear splash panel attachment tabs 588. In some embodiments, the rear splash panel attachment tabs 588 couple the rear splash panels 586a, 586b to the adjacent body panel or frame member, as discussed in greater detail below. In some embodiments, the splash panel attachment tabs 588 include barbs to facilitate a positive connection to maintain hold between the panels.

Turning now to FIGS. 98 and 123, in some embodiments, the bumper 92 is coupled to the main frame 74, for example via one or more bumper brackets 100. In some embodiments, the bumper brackets 100 are coupled to portions of the main frame 74 for example via one or more fasteners (e.g., bolts). In some embodiments, the bumper brackets 100 are coupled, via fasteners, to frame bumper mounts 590. In some embodiments, the frame bumper mounts 590 are welded to front joining frame members 184, respectively. In some embodiments, the bumper 90 is formed from more than components that are fastened together, for example via fasteners. In some embodiments, the bumper 92 is coupled to the bumper brackets 100 via one or more fasteners.

Referring specifically to FIG. 98, in some embodiments, the bumper 92 has a winch guide 592 coupled thereto, for example via one or more fasteners. In some embodiments, a winch 594 is positioned rearwardly of at least a portion of the bumper 92. In some embodiments, at least a portion of the bumper 92 is positioned longitudinally between the winch guide 592 and the winch 594.

With regard to FIGS. 99 and 100, in some embodiments, a cooling system 596 is shown, for example, in relation to a diesel engine. As illustrated, in some embodiments, the cooling system 596 includes a radiator 598, fill tank 600, coolant feed conduit 602, and coolant return conduit 604. In some embodiments, the fill tank 600 also serves as an

overflow tank. With regard to FIG. 99, in some embodiments, the fill tank 600 is positioned rearwardly of the core support member 606.

With regard to FIGS. 101 and 102, in some embodiments, the cooling system 596 is shown, for example, in relation to a gasoline engine. As illustrated, in some embodiments, the cooling system 596 includes a radiator 598, fill tank 600, coolant feed conduit 602, and coolant return conduit 604. In some embodiments, the fill tank 600 also serves as an overflow tank. With regard to FIG. 101, in some embodiments, the fill tank 600 is positioned rearwardly of the core support member 606.

Turning now to FIGS. 103-105, an embodiment of one or more portions of exhaust and intake systems are illustrated in respective figures. In FIG. 103, portions of illustrative intake and exhaust systems are shown in relation to a diesel engine. In some embodiments, an intake system 608 includes (FIG. 104) an air filter assembly 610, an engine intake duct member 612 (e.g., rubber hose), filter support member 614, first engine intake conduit 616 (e.g., angled rubber hose), second engine intake conduit 618 (e.g., plastic tube, aluminum tube), and third engine intake conduit 620 (e.g., angled rubber hose). In some embodiments, the third engine intake conduit 620 is coupled to an intake manifold or throttle body, for example via a hose clamp.

With regard to FIG. 105, in some embodiments, an exhaust system 622 includes a first exhaust gasket 624, a first exhaust pipe 626, an exhaust donut 628, one or more exhaust retaining members 630 (e.g., springs), a second exhaust pipe 632, a second exhaust gasket 634, a muffler 636, and one or more exhaust support members 638. In some embodiments, the first exhaust pipe 626 is coupled (e.g., via fasteners such as nuts) to an exhaust manifold. In some embodiments, the exhaust retaining members 630 couple the first and second exhaust pipe 626, 632 together. In some embodiments, the exhaust support members 638 are welded to portions of the muffler 636 and, as shown in FIG. 103, support the muffler 636 (or other component of the exhaust system 622) via one or more exhaust isolators 640 (e.g., rubberized inserts).

In FIGS. 106-109, an embodiment of one or more portions of exhaust and intake systems are illustrated in respective figures. In FIG. 106, portions of illustrative intake and exhaust systems are shown in relation to a gasoline engine, however, the type of engine is only illustrative. Further, the engine can be an HCCI (homogeneous charge compression ignition), HCSI (homogeneous charge spark ignition), SCCI (stratified charge compression ignition), RCCI (reactively controlled compression ignition), etc., with any appropriate fuel type (gas, propane, diesel, etc.).

With regard to FIG. 107, in some embodiments, an intake system 608 includes an air filter assembly 610, an engine intake duct member 612 (e.g., rubber hose), filter support member 614, first engine intake conduit 616 (e.g., angled rubber hose), second engine intake conduit 618 (e.g., plastic tube, aluminum tube), and third engine intake conduit 620 (e.g., angled rubber hose). In some embodiments, the third engine intake conduit 620 is coupled to an intake manifold or throttle body, for example via a hose clamp. As further shown in FIG. 107, in some embodiments, the intake system 608 includes a breather tube 642. As shown FIG. 106, in some embodiments, the breather tube 642 is in fluid communication with a portion of the prime mover 376 (e.g., engine), such as valve cover 644. The breather tube 642 can provide ventilation for the engine.

With regard to FIGS. 108 and 109, in some embodiments, an exhaust system 622 includes a first exhaust gasket 624, a

first exhaust pipe 626, an exhaust donut 628, one or more exhaust retaining members 630 (e.g., springs), a second exhaust pipe 632, a second exhaust gasket 634, a muffler 636, and one or more exhaust support members 638. In some embodiments, the first exhaust pipe 626 is coupled (e.g., via fasteners such as nuts) to an exhaust manifold 646 (FIG. 108). In some embodiments, the exhaust retaining members 630 couple the first and second exhaust pipe 626, 632 together. In some embodiments, the exhaust support members 638 are welded to portions of the muffler 636 and, as shown in FIG. 103, support the muffler 636 (or other component of the exhaust system 622) via one or more exhaust isolators 640 (e.g., rubberized inserts). In some embodiments, the exhaust system 622 further includes a bung 648 for an oxygen sensor 650 (O₂ sensor).

FIG. 110 shows a top view of an embodiment of a vehicle 10, while FIG. 111 shows the vehicle of FIG. 110 in cross-section. As illustrated in FIG. 111, in some embodiments, the cargo box 18 tilts with respect to the frame 14, for example to dump cargo out of the cargo box 18. In some embodiments, the cargo box tilts about tilt axis 652 (also shown in FIG. 118). In some embodiments, the rear joining frame member 156 includes a sloped portion 654, which provides clearance for the cargo box 18 when it is tilted, as shown in FIG. 113. In some embodiments, a portion of the cargo box 18 extends forwardly of a portion of the seatback body panel 580, as illustrated in FIG. 111. Moreover, in some embodiments, when the seat base 26b (or 26c) is in the stowed configuration, a portion of the cargo box 18 is forward of at least a portion of the seat base 26b (or 26c).

With regard to FIG. 113, in some embodiments, the vehicle 10 includes a hip restraint 656. Although the vehicle 10 is shown in FIG. 113 with both a side restraint 76 and a hip restraint 656, the vehicle 10 will have, in some embodiments, either a side restraint 76 or a hip restraint 656, but not both. As illustrated in FIG. 112, in some embodiments, the hip restraint 656 is coupled to the rear outer frame member 78 and the first seating support member 142.

Referring to FIGS. 114-119, in some embodiments, the cargo box 18 has a first lateral side 658, a second lateral side 660, a front side 662, and a rear side 664. In some embodiments, the rear side 664 comprises a tailgate 666. In some embodiments, the front side 662 includes a portion that extends upward away from cargo box 18. This upward extending portion is located adjacent to ROPS cross member 208 (shown in FIG. 14) to act as a noise barrier between cargo box 18 and passengers. In some embodiments, the upward extending portion directly abuts ROPS cross member 208. In other embodiments, the upward extending portion is situated to allow cargo box 18 to tilt as shown in FIG. 113.

In some embodiments, the vehicle 10 includes a cargo box assembly 682. In some embodiments, the cargo box assembly 682 includes the cargo box 18, a tilt release member 668 (FIG. 115), and one or more tilt release catches 670. In some embodiments, the tilt release member 668 is formed from piece of metallic (e.g., steel) rod or tubing that is formed into the appropriate shape. In some embodiments, the one or more tilt release catches 670 have a latch portion that releasably catches on latch pin 672 (FIG. 78). In some embodiments, the tilt release member 668 further tilt handle 674. The tilt handle 674 is pulled upwardly to release the catch from the latch pin 672 (FIG. 78) such that the cargo box 18 can be tilted. Although shown as having a circular cross section, the latch pin 672 can have any desirable shape. For example, it can be formed from square tubing, round tubing, a bracket, etc. In some embodiments, the tilt release

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member 668 interfaces with one or more biasing members 676 (e.g., springs, torsion springs), such that tilt release member 668 is biased into a latched configuration.

With regard to FIG. 115, in some embodiments, the cargo box assembly 682 has a tilt shock 678. In some embodiments, the tilt shock 678 resists quick raising or lowering of the cargo box 18. In some embodiments, the tilt shock 678 is coupled, for example via a fastener, to one or more box supporting ribs 680 or channel members.

Referring to FIG. 116, in some embodiments, the cargo box assembly 682 further includes at least one protective cover 684, one or more tailgate handles 686, one or more handle latches 688 (for example, over center latches), one or more latch receivers 690, one or more tailgate hinge members 692, one or more tailgate hinge bushings 694, one or more tailgate support members 696 (e.g., cables), one or more bumper members 698, and one or more tilt release mounts 700. In some embodiments, the protective cover 684 protects a portion of the wiring harness (not shown), for example a portion of the wiring harness that connects to a brake light. In some embodiments, a portion of the wiring harness (not shown) is routed between opposing lateral sides of the protective cover 684. In addition, hardware (e.g., fasteners such as nuts, bolts, screws, etc.) can be utilized to assemble various components, as illustratively shown in FIG. 116. In some embodiments, the bumper members 698 are formed from an elastomeric material (e.g., rubber). In some embodiments, a bottom surface of the bumper members 698 rests on a portion of the frame 14, as shown in FIG. 111, for example when the cargo box 18 is in a lowered configuration. In some embodiments, the bumpers 698 rest on the rear upper frame members 124 (FIG. 111).

In some embodiments, the cargo box 18 is formed from a metallic material, such as steel or aluminum. In some embodiments, the cargo box 18 is formed from one or more sheets of steel that are bent into an appropriate shape, as illustratively shown in FIGS. 117-119, to form the floor of the cargo box 18, one or more portions of the first lateral side 658, second lateral side 660, front side 662, rear side 664, and tailgate 666. In some embodiments, one or more plastic body panels are attached to exterior portions of the bent metallic sheets which form structural portions of the cargo box 18.

Turning to FIGS. 120-122, in some embodiments, a fuel tank 112 and portions of a fuel system 702 are shown. The fuel system 702 includes the fuel tank 112, filler neck 38, fuel cap 36, fuel supply line 704, fuel return line 706, fuel level sender 708, fuel filter 710, pressure release valve 712, and vent tube 714. In some embodiments, the fuel level sender 708 communicates with a fuel gauge on the dash to display the level of fuel in the fuel tank 112 operator. In some embodiments the pressure release valve 712 can vent pressure inside or outside the tank in order to equalize pressure inside the tank with respect to the atmosphere. In some embodiments, the pressure release valve 712 is a rollover valve, such that, in the event the fuel tank 112 is tipped, liquid fuel does not leak out of the fuel tank 112.

With regard to FIG. 122, in some embodiments, the fuel tank 112 includes a protruding portion 716 (discussed previously). In some embodiments, the fuel tank 112 also includes a recessed portion 718, which can facilitate locating the fuel tank 112 relative to a frame member, panel (e.g., first panel 114), body panel, etc.

With regard to FIGS. 124-131, one or more panels (e.g., plastic body panels) are shown. In FIGS. 124 and 125, in some embodiments, the vehicle 10 has a hood 720, bridging dash member 722, upper dash member 724, and lower dash

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member 726, and glovebox door 728. In some embodiments, the hood 720 is removable, hingeable, etc. to facilitate access to a compartment, accessory, electrical, radiator, etc. which may be located under the hood 720.

With regard to FIGS. 126 and 127, in some embodiments, the vehicle 10 includes an upper fascia member 730, intermediate fascia member 732, and lower fascia member 734. In some embodiments, the upper fascia member 730 includes a grille 736, which can be integrally formed with the remainder of the upper fascia member 730 or it can be formed separately and coupled (e.g., with tabs, fasteners, etc.) to the remainder of the upper fascia member 730. In some embodiments, the radiator is located rearwardly of the grille 736.

As shown in FIGS. 128 and 129, in some embodiments, the vehicle 10 includes a fender panel member 738 and a fender trim member 740. In some embodiments, the fender trim member 740 is coupled to the fender panel member 738 via one or more trim member attachment tabs 742, which fit into tab apertures in the fender panel member 738.

Turning to FIGS. 130 and 131, in some embodiments, one or more portions of the cargo box assembly 682 are formed from one or more plastic or composite materials. In some embodiments, the cargo box 18 is formed in a single molding process, for example from a high-density polyethylene (HDPE) with a foaming agent. In some embodiments, the cargo box assembly 682 further includes a box sub-frame 744, upper cargo box body panel 56, and lower cargo box body panel 746. In some embodiments, the tailgate 666 includes an inner tailgate member 748, and an outer tailgate member 750. In some embodiments, the inner tailgate member 784 is formed from a sheet of stamped metal (e.g., stamped sheet metal). In some embodiments, the outer tailgate member 750 is formed from an HDPE material. In some embodiments, the upper cargo box body panel 56 is formed from a thermoplastic olefin (TPO) or an ionomer resin such as DuPont's Surlyn. In some embodiments, the lower tailgate member 748 is formed from HDPE. These materials are only illustrative, and other materials such as thermoplastics, thermosets, metals, metal alloys, composites, etc. can also be used. In some embodiments, the box sub-frame 744 is formed from one or more steel components, which may be welded or otherwise fastened together (e.g., using fasteners).

In some embodiments, the cargo box assembly 682 further comprises one or more tie-down members 752. The tie-down members 752 are coupled, for example via one or more fasteners, to the cargo box 18, as illustratively shown in FIG. 131. In some embodiments, the cargo box assembly 682 includes one or more box retainers 754. In some embodiments, the box retainers 754 secure the floor of the cargo box 18 to one or more portions of the box sub-frame 744, for example via fasteners such as bolts (e.g., carriage bolts), nuts, and/or screws, etc.

Turning to FIGS. 133-137, in some embodiments, the one or more doors 62 comprises a door which vertically spans only a portion of the cab opening. For example, in some embodiments, an upper edge of the door 62 extends upwardly to a position that is 1/2-4" above the top of the seat base 26a, as illustratively shown in FIG. 2. The upper edge of the door 62 can also be below the top of the seat base 26a, level with the seat base 26a, or any other suitable height. In some embodiments, the doors 62 include an outer door panel 756 and an inner door panel 758. In at least some embodiments, the door 62 is part of a door assembly 760. In some embodiments, the door assembly 760 further includes an outer door handle 762, a door hinge 766, and a door latch

assembly 768. One or more portions of the door latch assembly 768 are coupled to the frame 14 (FIG. 5), for example via one or more fasteners (e.g., nuts, bolts, screws, etc.). Further, one or more portions of the door hinge 766 may be coupled to the frame 14, seat support frame 146, etc., for example via one or more fasteners (e.g., nuts, bolts, screws, etc.). In some embodiments, a portion of the door hinge 766 is coupled to the first seating support member 142 (FIG. 12).

In some embodiments, actuation of the outer door handle 766 unlatches the door latch assembly 768, permitting the door 62 to open. Although not shown in FIGS. 133-137, in some embodiments, the door assembly 760 also includes an inner door handle, by which an operator can unlatch the door assembly 768 to open the door 62.

Referring to FIGS. 134-135, in some embodiments, the inner door panel 758 includes a door pocket 770. In some embodiments, the door pocket 770 is at least partly defined by a space between the inner door panel 758 and outer door panel 756.

Turning now to FIGS. 138-142, in some embodiments, the vehicle 10 includes a roof assembly 772, a brush guard 774 (FIG. 138), and one or more side windows 776. In some embodiments, the vehicle 10 further includes front and rear windshields, which may have one or more transparent portions that pivot outwardly, slide (e.g., rear sliding windows), include one or more accessories (e.g., wiper and washer assemblies), etc. As shown in FIG. 141, in some embodiments, the roof assembly 772 includes an inner front panel 778, an outer front panel 780, and an outer main panel 782. In some embodiments, for example as shown in FIG. 141, the inner front panel 778 and outer front panel 780 are coupled to one another, for example via one or more fasteners (e.g., clips, screws, bolts, nuts, etc.). In some embodiments, the roof assembly 772 include a headliner (not shown), roof console, and/or one or more roof mounted controls (e.g., light switch), roof mounted storage areas, etc.

In some embodiments, one or more of the body panels can be formed from a composite material, HDPE material, TPO material, or other suitable material. In some embodiments, one or more of the panels (e.g., rear cab-side panel 32, upper cargo box body panel 56, lower cab panel 64, front splash panel 268, seatback body panel 580, rear splash panel 586, hood 720, bridging dash member 724, lower dash member 726, glovebox door 728, upper fascia member 730, intermediate fascia member 732, lower fascia member 734, fender panel member 738, fender trim member 740, lower cargo box body panel 746, inner tailgate member 748, outer tailgate member 750, outer door panel 756, inner door panel 758, inner front panel 778, outer front panel 780, outer main panel 782) can be formed from a composite material having a resin and fibers. In some embodiments, the fibers have an average fiber length of 0.2-2 inches. In some embodiments, the fibers have an average fiber length of 0.5-1.5 inches, and in some embodiments, the fibers have an average fiber length of 0.75-1.25 inches. In some embodiments, the fibers have an average fiber length of 1 inch. In some embodiments, one or more of the panels (e.g., rear cab-side panel 32, upper cargo box body panel 56, lower cab panel 64, front splash panel 268, seatback body panel 580, rear splash panel 586, hood 720, bridging dash member 724, lower dash member 726, glovebox door 728, upper fascia member 730, intermediate fascia member 732, lower fascia member 734, fender panel member 738, fender trim member 740, lower cargo box body panel 746, inner tailgate member 748, outer tailgate member 750, outer door panel 756, inner door panel 758, inner front panel 778, outer front panel 780, outer main

panel 782) is formed via a Reaction Injection Molding process (RIM), which can utilize IMP (In-mold-paint). In some embodiments, the composite utilizes a polyurethane. In some embodiments, polyurethane based composites include chopped glass reinforcement fibers, structural foam, and elastomers. In some embodiments, the composite is a dicyclopentadiene (DCPD) composite or a fiberglass/polyester composite.

In some embodiments, the composite uses a 2-part thermoset polyurethane [Iso/poly] material. In some embodiments, one or more of the panels (e.g., rear cab-side panel 32, upper cargo box body panel 56, lower cab panel 64, front splash panel 268, seatback body panel 580, rear splash panel 586, hood 720, bridging dash member 724, lower dash member 726, glovebox door 728, upper fascia member 730, intermediate fascia member 732, lower fascia member 734, fender panel member 738, fender trim member 740, lower cargo box body panel 746, inner tailgate member 748, outer tailgate member 750, outer door panel 756, inner door panel 758, inner front panel 778, outer front panel 780, outer main panel 782) has a high gloss (class "A") on one or more of the surfaces (e.g., exterior surface). In some embodiments, one or more of the panels (e.g., rear cab-side panel 32, upper cargo box body panel 56, lower cab panel 64, front splash panel 268, seatback body panel 580, rear splash panel 586, hood 720, bridging dash member 724, lower dash member 726, glovebox door 728, upper fascia member 730, intermediate fascia member 732, lower fascia member 734, fender panel member 738, fender trim member 740, lower cargo box body panel 746, inner tailgate member 748, outer tailgate member 750, outer door panel 756, inner door panel 758, inner front panel 778, outer front panel 780, outer main panel 782) has color on only one side of the panel. In some embodiments, one or more of the panels (e.g., rear cab-side panel 32, upper cargo box body panel 56, lower cab panel 64, front splash panel 268, seatback body panel 580, rear splash panel 586, hood 720, bridging dash member 724, lower dash member 726, glovebox door 728, upper fascia member 730, intermediate fascia member 732, lower fascia member 734, fender panel member 738, fender trim member 740, lower cargo box body panel 746, inner tailgate member 748, outer tailgate member 750, outer door panel 756, inner door panel 758, inner front panel 778, outer front panel 780, outer main panel 782) is formed via compression molding, using fiberglass reinforced plastics, and/or resin transfer molding.

In some embodiments, one or more of the roof panels (e.g., inner front panel 778, outer front panel 780, outer main panel 782) provides falling object protection, for example when is formed using fibers having an average fiber length of 0.5-1.5 inches.

In some embodiments, one or more of the panels (e.g., rear cab-side panel 32, upper cargo box body panel 56, lower cab panel 64, front splash panel 268, seatback body panel 580, rear splash panel 586, hood 720, bridging dash member 724, lower dash member 726, glovebox door 728, upper fascia member 730, intermediate fascia member 732, lower fascia member 734, fender panel member 738, fender trim member 740, lower cargo box body panel 746, inner tailgate member 748, outer tailgate member 750, outer door panel 756, inner door panel 758, inner front panel 778, outer front panel 780, outer main panel 782) includes one or more composite materials, which can include carbon fiber, fiberglass, and Kevlar. In some embodiments, one or more of these materials can be utilized to provide light armament for the vehicle 10, for example small-arms fire ballistic protection. Additionally, nomex, titanium, steel, epoxy, or polyes-

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ter resins can be utilized. Further still, additional panels can be added, for example, between inner and outer door panels, which are formed from a material which provides ballistic protection (e.g., Kevlar composites, as previously discussed).

In some embodiments, driveline can also include a differential between the front and rear differentials. Thus, the utility vehicle **10** can have one, two, three, or more differentials. The differentials can be oriented laterally or longitudinally. For example, a differential coupling the front and rear differentials can be oriented longitudinally, while front and rear differentials can be oriented laterally.

What is claimed is:

1. An off-road utility vehicle comprising:

a frame;

an engine;

a transmission;

first and second rear half-shafts connected between the transmission and first and second rear wheels; and

a rear suspension that includes first and second upper arms, wherein the first and second upper arms are connected to the frame at a forwardmost upper arm connection point and a rearwardmost upper arm connection point, respectively, wherein the forwardmost upper arm connection point and the rearwardmost upper arm connection point are rearward of a connection of the first and second half-shafts to the transmission, and wherein the rear suspension includes a damping member attached to the frame at a point rearward of the forwardmost upper arm connection point.

2. The off-road utility vehicle of claim **1**, wherein the rear suspension further includes first and second upper suspension support members that connect the first and second upper arms, respectively, to the frame.

3. The off-road utility vehicle of claim **2**, wherein the first and second upper arms form an upper A-arm assembly.

4. The off-road utility vehicle of claim **1**, wherein the first and second upper arms are connected to the frame in a horizontal plane that is vertically above the connection of the first and second half-shafts to the transmission.

5. The off-road utility vehicle of claim **1**, wherein the rear suspension further includes first and second lower arms are connected to the frame rearward of the connection of the first and second half-shafts to the transmission.

6. The off-road utility vehicle of claim **5**, wherein the rear suspension further includes first and second lower suspension support members that connect the first and second lower arms, respectively, to the frame.

7. The off-road utility vehicle of claim **5**, wherein the first and second lower arms form a lower A-arm assembly.

8. The off-road utility vehicle of claim **5**, wherein the first upper arm is located forward of the second upper arm and the first lower arm is located forward of the second lower arm.

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9. The off-road utility vehicle of claim **8**, wherein the first upper arm is located forward of the first lower arm.

10. The off-road utility vehicle of claim **8**, wherein the second upper arm is located rearward of the second lower arm.

11. An off-road utility vehicle comprising:

a frame;

an engine;

a transmission;

first and second rear half-shafts connected between the transmission and first and second rear wheels; and

a rear suspension comprising:

an upper A-arm having first and second upper arms connected to the frame via first and second upper suspension support members; and

a lower A-arm having first and second lower arms connected to the frame via first and second lower suspension support members, wherein the first and second upper suspension support members and first and second lower suspension support members are located rearward of a connection of the first and second rear half-shafts to the transmission.

12. The off-road utility vehicle of claim **11**, wherein the first upper arm is located forward of the second upper arm and the first lower arm is located forward of the second lower arm.

13. The off-road utility vehicle of claim **12**, wherein the first upper arm is located forward of the first rear arm.

14. The off-road utility vehicle of claim **13**, wherein the second upper arm is located rearward of the second lower arm.

15. An off-road utility vehicle, comprising:

a frame;

an engine;

a transmission;

a first rear half-shaft connected to a second rear-half shaft between the transmission and first and second rear wheels; and

a rear suspension including first and second upper arms, and first and second lower arms, wherein the first and second upper arms are connected to the frame rearward of a connection of the first and second half-shafts to the transmission, wherein a first connection of the first upper arm to the frame is forward of a second connection of the second lower arm to the frame.

16. The off-road utility vehicle of claim **15**, wherein the first upper arm is forward of the second upper arm and wherein the first lower arm is forward of the second lower arm.

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